within the primary care medical home.

DEVELOPMENTAL AND BEHAVIORAL SURVEILLANCE

The American Academy of Pediatrics (AAP) has published a policy statement on developmental surveillance, screening, and evaluation, and a clinical report on behavioral screening. Developmental and behavioral surveillance is a longitudinal process of developmental and behavioral monitoring over time within the primary care medical home. Developmental and behavioral surveillance should occur at every well child visit, and it helps to provide parent education, assists in supporting healthy development and behavior, and is useful in identifying children who may be at an increased risk of developmental and behavioral differences. Developmental surveillance relies on the clinical judgment and accumulated experience of pediatric medical providers, and it consists of five key components: (1) specifically asking about parental developmental-behavioral concerns, (2) identifying neurobiologic (eg, prematurity) and psychosocial (eg, maternal depression) risk and protective factors, (3) obtaining a developmental and behavioral history, (4) observing the child’s development and behavior during the clinic visit, and (5) maintaining an accurate documentation of findings. If concerns are raised through developmental-behavioral surveillance, the child should be referred to his or her local early intervention (birth through 3 years of age) or early childhood special education (3 to 5 years of age) program.

DEVELOPMENTAL AND BEHAVIORAL SCREENING

With the goal of identifying developmental and behavioral disorders as early as possible, so that interventions can begin at the youngest possible age when the central nervous system has increased capacity for plasticity, pediatric medical providers should implement developmental and behavioral screening during early childhood. Developmental and behavioral screening involves the administration of standardized and validated measures to parents/caregivers, as research has shown that parents reliably report on their children’s development and behavior. Important screening test properties include sensitivity, specificity, predictive validity, reliability, and standardization sample. These screening tools are most commonly parent questionnaires, and developmental and behavioral screening tests are designed to identify children at significant risk for a developmental or behavioral problem who require further evaluation. The AAP recommends, in addition to surveillance at every well-child visit, that all children be screened using a standardized screening tool during well-child visits at the specified ages of 9 months, 18 months, and 24 or 30 months. These specific ages were chosen at time points when specific developmental disorders are most likely to be identified. The AAP also recommends that all children be specifically screened for autism spectrum disorder (ASD) at the 18-month and 24-month well-child visits. Developmental-behavioral screening tests may be categorized as general, domain-specific, or disorder-specific screening tests. Examples of commonly used screening tools are presented in Table 84-1. When a screening test is positive, indicating that a child is at an increased risk for a developmental or behavioral problem, the child should be referred to their local early intervention or early childhood special education programs.

TABLE 84-1 DEVELOPMENTAL SCREENING TESTS

General Screening:
Ages and Stages Questionnaire (ASQ)
Parents' Evaluation of Developmental Status (PEDS)
Pediatric Symptom Checklist (PSC)
Domain-Specific:
Communication and Symbolic Behavior Scales (CSBS)
Ages and Stages Questionnaires: Social-Emotional (ASQ: SE)
Strengths and Difficulties Questionnaire (SDQ)
Disorder-Specific (Autism):
Modified Checklist for Autism in Toddlers (M-CHAT)
Screening Tool for Autism in Toddlers and Young Children (STAT)
Social Communication Questionnaire (SCQ)

DEVELOPMENTAL EVALUATION

For children who are identified as being at risk for developmental or behavioral differences through developmental-behavioral surveillance and screening, the AAP recommends that a diagnostic developmental evaluation be conducted in order to confirm the presence of a developmental disorder. Given the prevalence of developmental-behavioral disorders in the pediatric population and the scarcity of subspecialists (such as developmental-behavioral or neurodevelopmental disability pediatricians) to whom to refer, primary pediatric medical providers can perform developmental evaluations on all children determined to be at risk through surveillance or screening.

The developmental evaluation serves to confirm suspicions about possible developmental or behavioral disorders raised by surveillance or screening. The purpose of the developmental evaluation is to identify the presence of a specific developmental-behavioral diagnosis, to determine the severity and possible long-term prognosis, and to initiate specific therapeutic and/or educational interventions. If direct developmental evaluations are not completed by the primary pediatric medical provider to confirm concerns raised by surveillance or screening, the primary care provider needs to work closely with local early intervention or early childhood special education teams, who can perform the developmental evaluation. If the developmental evaluation performed by the early intervention or early childhood special education team confirms a developmental-behavioral disorder, the primary pediatric medical provider will need to proceed with further medical workup in an attempt to establish an etiologic diagnosis to account for the developmental-behavioral disorder.

Developmental evaluations may include direct evaluations of cognitive, communicative, adaptive, motor, and social skills. Developmental evaluations are accomplished through the use of standardized developmental evaluation tools that are validated and have acceptable psychometric properties. In developmental evaluation, multiple tests may be needed in order to obtain a thorough understanding of a child's developmental strengths and challenges. Factors such as examiner-child rapport and the child's level of interest, cooperation, and effort are also important to consider. Also, with very young children, having the caregiver present in the examination room may be important to help the child feel comfortable. In older children, it may be helpful to allow the child to determine whether he or she would like for the parent to remain in the examination room during the developmental evaluation. For infants, toddlers, and preschoolers, the developmental evaluation generally focuses on motor, nonverbal/visual-motor problem-solving, language, and social skills. For school-aged children, pediatric medical providers might consider brief standardized tests that evaluate visual-motor, language, and academic achievement to provide data to confirm the need for further school psychoeducational testing or to confirm findings from school psychoeducational testing. See **Table 84-2** for examples of standardized developmental evaluation tools that can be administered by licensed primary pediatric medical providers.

TABLE 84-2 DEVELOPMENTAL EVALUATION TOOLS THAT

CAN BE ADMINISTERED BY PEDIATRIC MEDICAL PROVIDERS
Capute Scales: Cognitive Adaptive Test/Clinical Linguistic and Auditory Milestone Scale (CAT/CLAMS)
Parents' Evaluation of Developmental Status: Developmental Milestones, Assessment Level (PEDS:DM, Assessment Level)
Gesell Developmental Observation-Revised
Mullen Scales of Early Learning
Battelle Developmental Inventory
Peabody Picture Vocabulary Test
Expressive Vocabulary Test
Beery-Buktenica Developmental Test of Visual-Motor Integration
Kaufman Brief Intelligence Test (K-BIT)
Wide Range Achievement Test

CONCLUSION

Given that developmental-behavioral disorders are among the most prevalent chronic medical problems in pediatrics, developmental surveillance, developmental screening, and developmental evaluation are important within the scope of primary care pediatric practice. Accumulated clinical experience and clinical judgment about typical and atypical child development and behavior, combined with knowledge and experience in administering and interpreting standardized developmental screening and evaluation measures, provides the best opportunity for earlier identification and earlier intervention, with the ultimate goal of improved long-term developmental-behavioral outcomes.

SUGGESTED READINGS

American Academy of Pediatrics Council on Children with Disabilities, Section on Developmental-Behavioral Pediatrics, Bright Futures Steering Committee, Medical Home Initiatives for Children with Special Needs Project Advisory Committee. Identifying infants and young children with developmental disorders in the medical home: an algorithm for developmental surveillance and screening. *Pediatrics*. 2006;118:405-420.

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85 Basics of Management of Common Behavior Problems

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INTRODUCTION

Behavior problems are among the most common presenting complaints that parents and caregivers have when seeking pediatric medical care, as up to 20% to 25% of the pediatric population may experience behavioral or emotional problems. Behavior problems can range from mild actions with minimal consequences to more significant behavior disorders with potentially severe outcomes. Evaluating behavior problems in children must take into account normal child development, temperament, parenting styles, and environmental circumstances.

TEMPERAMENT, PERSONALITY, AND PARENTING STYLES

Temperament

Temperament refers to the inborn, biologically based behavioral style with which an individual interacts with his or her environment. Drs. Alexander Thomas (1914–2003) and Stella Chess (1914–2007), who were American child psychiatrists, expanded upon the pioneering work of American pediatrician Dr.

Herbert Birch (1918–1973) in the study of child temperament and behavior. Thomas, Chess, and Birch conducted a longitudinal study of child behavior and temperament and proposed nine behavior characteristics (**Table 85-1**). These nine characteristics (activity, rhythmicity, approach/withdrawal, adaptability, threshold of sensory responsiveness, intensity of reaction, quality of mood, distractibility, and attention/persistence) combine into three temperamental categories: easy, difficult, and slow to warm up.

TABLE 85-1 TEMPERAMENTAL BEHAVIOR CHARACTERISTICS

Behavior Characteristic	Description
Activity	The level and extent of motor activity
Rhythmicity	The degree of regularity of functions such as eating, elimination, and the cycle of sleeping and wakefulness
Approach or withdrawal	The response to a new object or person, in terms of whether the child accepts the new experience or withdraws from it
Adaptability	The adaptability of behavior to changes in the environment
Threshold of responsiveness	Sensitivity to a stimulus
Intensity of reaction	The energy level of responses
Quality of mood	The child's general mood or "disposition," whether cheerful or given to crying, pleasant or cranky, friendly or unfriendly
Distractibility	The degree of the child's distractibility from what he is doing
Attention span or persistence	The span of the child's attention and his persistence in an activity

Children with an easy temperament (40% of children) are described as having regular biological rhythms (eg, eating, sleeping, toileting), adapt easily to change and new environments, and are in a generally pleasant mood. These children do not become extremely agitated when they do not get their own way or their expectations are not met.

Children with a slow-to-warm-up temperament (15% of children) are often described as being shy. These children tend to adapt slowly to new people and environments. They will watch a group of children play while staying close to their parents. When given time, they will gradually join the group of children playing once they feel comfortable.

Children with difficult temperaments (10% of children) may be challenging for their parents to manage. Their biological rhythms are unpredictable, and they may not sleep, eat, or toilet at desired times. Children with difficult temperaments are more likely to have an irritable mood and react very negatively to unexpected changes in their routines or when their expectations are not met.

Personality

Personality is the combination of temperament and life experience. Personality can be shaped by the influence of parenting styles, culture, environment, and underlying temperament.

Goodness of Fit

Goodness of fit refers to how a particular trait or characteristic interacts with the environment and the people within the environment. Goodness of fit is based on an individual's temperamental characteristics and the specific aspects of the child's environment. The term is used to describe the interaction between the child's temperament and the parent's temperament.

Parenting Styles

Parenting style refers to the manner in which a parent interacts and provides guidance to the child. The four most common parenting styles are authoritarian, passive, neglectful, and authoritative.

An authoritarian parenting style is also described as a strict parent. The parent tends to be demanding and does not respond to the individual needs of their children. There is little negotiation and dialogue between parent and child. Authoritarian parents do not offer warmth and nurturing to their children. The authoritarian parenting style is more likely to produce children who have low self-esteem, are insecure, are inappropriately shy, and have difficulty in social situations.

A passive parenting style is characterized by parents who are permissive toward their children. Rules and expectations are not clear, and there is often blurring of roles between the parent and child. The permissive parenting style can produce children who are insecure, have conflict with authority figures, have poor social skills, and are self-centered.

A neglectful parenting style is characterized by a parent who fails to meet the child's physical, emotional, and/or developmental needs. This is the most harmful parenting style. Children reared by neglectful parents are likely to have significant difficulty establishing appropriate trusting relationships with peers and others. This parenting style needs immediate intervention with mental health professionals and possibly child welfare authorities.

An authoritative parenting style is considered to be the healthiest parenting style. Authoritative parents tend to be flexible and nurturing toward their children. They offer rules and boundaries and discuss their expectations with their children. Authoritative parents are more willing to negotiate with their children when appropriate. The authoritative parenting style is more likely to produce confident, well-adjusted children.

PRINCIPLES OF BEHAVIOR MANAGEMENT

Behavior management requires a systematic evaluation of (1) the description of the behaviors, (2) temperament and goodness of fit, (3) the developmental level of the child, (4) parental expectations and whether these are appropriate for the child's developmental abilities, (5) parenting styles, and (6) appropriate interventions to redirect the undesired behaviors. Anticipatory guidance and parent education about normal child development and behavior are important aspects of behavior management education for parents. Also important is the distinction between discipline and punishment. While these terms are often incorrectly used synonymously, there is an important difference between discipline and punishment. Discipline refers to a series of boundaries, rules, and guidance designed to encourage desired behaviors in children, while punishment is designed exclusively to reduce unwanted behaviors in children.

Behavior Categories

When evaluating behaviors, it may be helpful to separate behaviors into categories. While there are a number of rubrics for evaluating behaviors, the following will explore one rubric involving three types of behaviors: internalizing behaviors, externalizing behaviors, and disruptive behaviors.

Internalizing behaviors generally refer to behaviors that may not be immediately noted by the parent but are experienced by the child. Internalizing behaviors may include feeling nervous or anxious, overwhelmed, or saddened. These behaviors may be exhibited in a variety of ways, such as resisting suggestions or directions from an adult, running away from an overwhelming situation, or isolating oneself from others.

Externalizing behaviors are easily observed by the parent or caregiver. Common externalizing behaviors include hyperactivity, poor impulse control, failure to follow directions, and becoming easily distracted. Externalizing behaviors can cause problems with how the child and/or the family function depending on where the behaviors occur and the circumstances around the behaviors. For example, the family may have a difficult time going on family outings because a child is impulsive and runs away from the parents in public places.

Disruptive behaviors are behaviors that may be potentially harmful or dangerous and more significantly interfere with how the child functions. Disruptive behaviors include aggression, self-injurious behaviors, destructive behaviors, and excessive tantrums.

Purpose of Behavior

Behavior has meaning. It is a form of communication. One purpose of a behavior may be for the child to avoid an aversive or painful experience. For example, an adolescent may have an algebra test at school and does not feel adequately prepared for the exam. The teen may feign an illness to stay home from school, thus avoiding taking the algebra test and possibly failing it.

Another purpose of behavior may be to escape an aversive stimulus. In this case, a child may engage in a behavior to get out of an unpleasant situation. For example, if it is a child's turn to wash dishes, and she does not like to wash dishes and does not want to do this chore, she may start a fight with a sibling or disobey a family rule. The parent reprimands the child and sends her to her bedroom as a punishment. Thus, the child escapes the aversive stimulus of having to wash the dishes.

Some children misbehave in order to gain attention from parents, caregivers, or teachers. When children gain attention by engaging in positive behaviors (eg, following instructions, completing an assignment), the child is more likely to continue to display the positive behavior. The positive behavior is positively reinforced by adding positive attention to the child when the child displays the desired behaviors. Conversely, some children only gain attention by engaging in inappropriate behaviors, such as throwing tantrums or breaking family rules. In this case, a parent may inadvertently ignore the child's good behaviors and preferentially give negative attention (eg, verbal reprimands, punishment) when the child engages in undesirable behaviors. As a result, the parent is reinforcing negative behaviors by inadvertently rewarding undesired behaviors with attention. In general, children seek the attention of the adults around them, whether good or bad, and parents need to make an effort to provide positive attention to desired behaviors and to ignore undesired, attention-seeking behaviors (that are not potentially harmful) in order to reinforce desired behaviors and extinguish undesired behaviors in their children.

Evaluating Behaviors

Evaluating a child's behavior requires a systematic approach. One way to evaluate behavior is to assess the ABCs of behavior: antecedents, behavior, and consequences.

ABCs of Behavior: Antecedents, Behavior, and Consequences Antecedents refer to the circumstances in the environment that occurred just before the child engaged in the behavior being observed. Antecedents can be triggers for undesirable behaviors. Clinicians should not ignore antecedents when taking a behavior history because many behavior problems can be addressed simply by understanding the circumstances that trigger a behavior. For example, a child is playing his favorite video game when a parent tells him to shut off the game and go complete his homework. The child throws a tantrum that includes crying, yelling at the parent, and refusing to comply with the command. In this case, the antecedent was engaging in a preferred activity that was unexpectedly interrupted and terminated.

Behaviors are the specific actions that the child displays that are undesirable to the parent. Behaviors should be described in detail because parents have different thresholds for what behaviors they consider to be problematic. For example, a "tantrum" may mean one thing to one parent and something completely different to another. The parent should describe the specific behaviors the child displays including verbal responses (eg, crying, screaming, name-calling, threatening), physical actions (eg, hitting, kicking, falling on the floor, self-injurious behaviors, running away), and what the child does with objects (eg, breaking toys, throwing objects, slamming doors). The clinician should also determine the frequency, duration, and location of the behavior outbursts.

Consequences are the circumstances that occur as a result of or reaction to the child's behaviors. Consequences may include any combination of reinforcers or punishments and can be natural or logical. A natural consequence occurs as an automatic result of the child's behavior. For example, if a child pulls a cat's tail and the cat scratches the child, this would be considered a natural consequence. Most natural consequences allow a child to learn from his own experiences.

Some natural consequences may be dangerous (eg, the child rides his bike into the street and gets hit by a car), and such consequences should obviously be avoided.

Logical consequences are those that logically follow an undesired behavior. For example, if a student has a major temper tantrum at school that includes destroying school property, verbal aggression, and physical aggression, the other students in the class may ostracize the child and refuse to play with her because of her disruptive behaviors. This would be a logical consequence or punishment that is the result of the student's disruptive behaviors. If the child desires to make friends with her classmates, she will not engage in these disruptive behaviors, understanding that her behaviors can lead to being ostracized by peers. There are times when adults' responses to a child's behavior can reinforce the child's undesired behaviors. For example, when shopping at the grocery store, a parent walks down the toy aisle. The child screams and begs for a toy. The parent tells the child "no" and tries to distract the child. The child continues to scream and demand the toy. Customers in the grocery store begin to look at the parent and child. Embarrassed, the parent gives the child a toy to stop the tantrum. The child takes the toy and stops screaming. In this case, the parent positively reinforced tantrum behaviors on the toy aisle, so whenever the child wants a toy, he will have a tantrum to gain his desired object. Giving the child the toy acted as a negative reinforcer for the parent because giving the toy stopped the tantrum, which was aversive to the parent.

Behavior Rating Scales

There are times when behavior rating scales can be administered to a parent/caregiver or teacher to gain a better understanding of a child's behaviors. A number of behavior rating scales address behaviors such as symptoms of attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorders (ASDs), anxiety, depression, and sleep, and there are also general behavior rating scales that evaluate a number of behaviors, such as attention, mood, adaptive behaviors, and resiliency. Research suggests that adults often accurately rate externalizing behaviors in children, while children are better able to rate their own internalizing symptoms. When children are 8 years old, they are capable of completing behavior rating scales that contain a Likert scale (ie, never, sometimes, often, always). A number of behavior rating scales have forms for parents/caregivers and teachers and a self-report form for the child (for ages 8 years and older). In order to gain a more comprehensive perspective of a child's behavior, it may be helpful for the clinician to have the parent, teacher, and child (if appropriate) complete behavior rating scales.

Behavior Management

Clinicians can offer parents guidance for addressing their child's behavior needs. Three principles to consider when evaluating behaviors and providing guidance for behavior management are the following: (1) behavior has meaning and is a form of communication, (2) learned behaviors can be unlearned, and (3) people do what they do as long as it works.

Behavior Has Meaning Many times children engage in undesired behaviors as a way of communicating a thought or need to their parent/caregiver or teacher. Sometimes children may engage in inappropriate behavior when they desire the attention of an adult. At other times, a behavior may reveal the child is distressed or overwhelmed. Behaviors could also demonstrate that a child has difficulty regulating himself. It is important to note when inappropriate behaviors are a sign of more significant underlying psychopathology or psychosocial stress, such as abuse, anxiety disorders, mood disorders, neglect, or emerging psychoses.

Learned Behaviors Can Be Unlearned Parents train children and children train parents. There is generally a reciprocal give-and-take in a parent-child relationship. Effective parents adjust their parenting behaviors to meet the needs of their children. Children adjust their behaviors to meet parental expectations. As previously described, there are times when parents inadvertently reward and encourage undesired behaviors by reinforcing them. Over time, the child learns to engage in the undesired behavior more often than the desired behavior. Conversely, children learn to manipulate their parents by engaging in behaviors that are likely to give them what they want. A parent will need to unlearn ineffective parenting strategies in order to develop more effective parenting strategies. Children can unlearn inappropriate behaviors as they gain mastery of more appropriate behaviors.

People Do What They Do as Long as It Works It is common for people to engage in behaviors that meet their own needs. In most cases, these behaviors are appropriate. In some cases, these behaviors may be maladaptive and interfere with healthy interpersonal relationships. When children frequently engage in undesirable behaviors, they may continue to do so because a particular desire or need is being met by this negative behavior. As long as an undesired behavior is being rewarded or reinforced, the behavior is likely to continue. However, once the undesired behavior is no longer rewarded, it is likely to reduce in frequency and intensity. When the child learns that their inappropriate behavior is no longer effective in meeting their desires or needs, the child will need to learn a more appropriate way of behaving to meet their own needs. Therefore, people do what they do as long as it works; when it stops working, the behavior fades.

Principles for Behavior Management Children often function best when there is structure and routine, clear rules and expectations, and predictability and consistency in the home. Chaotic home environments can produce chaotic behaviors in children. Clinicians can encourage parents to develop structure in the home to include a basic routine and schedule, so the child can know what to expect and what his role is in the family. It is crucial that parents be consistent in how they establish rules and boundaries in the home. When parents are inconsistent in the way that discipline is applied in the home, the child is more likely to display inappropriate behaviors. Clinicians can encourage parents to sit down with their children to establish a few (3–5 maximum) basic, non-negotiable family rules (eg, Use your words; don't hit others. Sit on furniture; do not jump on it.). The family rules may need to be clearly displayed in the family home and reviewed on a regular basis in order to maintain family behavior expectations. Parents and children can establish agreed-upon rewards for following the rules and consequences for breaking the rules. When parents remove something from a child, it should be replaced with something else the parent prefers. For example, if the parent wants to reduce a child's TV time, replace it with outdoor play time. Parents should be encouraged to catch children being good by rewarding and reinforcing desired behaviors with praise, smiles, high-fives, hugs, and special attention. They should also be encouraged to ignore undesired attention-seeking behaviors, so as not to reinforce them. Consequences and punishments should be appropriate for undesired behavior that is potentially harmful and should encourage the child to learn self-control and self-regulation. A consequence of punishment should not injure a child's self-esteem or self-worth. If behavior concerns are more significant, parent behavior management training with a mental health professional may be warranted.

COMMON BEHAVIOR PROBLEMS

Parents commonly seek pediatric care when they have concerns about their child's behavior problems. Behavior problems must be evaluated in the context of the setting, the child's developmental level, potential triggers, parenting style, the child's temperament, and parental expectations. Most behavior problems are benign and easily addressed. Some behavior problems are more significant and can be signs for diagnosable behavior disorders that require more extensive treatment and intervention. In this section, some common behavior problems will be reviewed.

Temper Tantrums

Temper tantrums are outbursts of anger or frustration in a child. Temper tantrums are most common in children between the ages of 18 months and 3 years. Tantrums tend to occur more frequently in children with limited expressive language. Once language skills improve, and a child's ability to explain wants, needs, and opinions improves, tantrums tend to be reduced in frequency. Tantrums can take many forms, from quietly sulking to being physically and verbally aggressive with destruction of property or self-injurious behaviors.

When evaluating a child for temper tantrums, clinicians should take several factors into account, including a specific description of the tantrum behaviors (eg, crying, hitting, head banging, throwing objects), the antecedents or triggers (eg, being hungry or tired, being told "no," having a preferred activity interrupted, unexpected changes), the frequency (ie, how many times a day or week), the duration of the tantrums (eg, 15 minutes, 45 minutes), and the location where tantrums occur (eg, at home, at school, in the community). Taking a thorough history of behavior concerns sometimes requires a longer office visit (45–60 minutes). However, a few basic questions can provide a general idea of the characteristics of the tantrum behaviors:

- What are the potential triggers for the tantrums? Hunger or fatigue? Being denied his own way? Being told "no"? Being overstimulated or overwhelmed?
- What do the tantrums look like? Please describe a typical tantrum.
- Where do tantrums typically occur? At home? During family outings? At school or daycare? Are they more frequent in one setting than in others?
- How long does a typical tantrum last? Less than 5 minutes? Longer than 20 minutes?
- How frequently do they occur? Daily? Weekly?
- What makes the tantrum better or worse? Ignoring the behavior? Reprimanding the child? Walking away from the child? Trying to reason with the child?

Parents can be reassured that tantrums are common in children between the ages of 18 months and 3 years. The child's developmental level must also be taken into account. Behavior expectations should be appropriate for the child's developmental level, not necessarily the child's chronological age. A child who has developmental delays or intellectual disability may have a developmental level between 18 months and 3 years. In this case, tantrums may be expected in a child of this developmental level, despite the chronologic age of the child. Sometimes children with developmental delays have tantrums because adult expectations or demands are too complex for them to understand. In this case, adult expectations should be adjusted accordingly to meet the child's developmental level.

Tantrums can often be reduced by teaching a young child to communicate better to describe his or her feelings in simple words (eg, "I don't like it" or "I'm mad!"). An adult cannot reason with a child in the midst of a tantrum, but once the tantrum subsides, the child can be taught to describe his feelings with words rather than actions. Parents can ignore mild tantrums by avoiding eye contact with the child and walking away from the child in the midst of a tantrum (as long as the child is safe). Once the child realizes that no one is paying attention to his behavior, the tantrum will often subside, as the child runs after the parent. The parent can then compliment the child for using his words instead.

Sleep Problems

Sleep behavior problems are one of the most common specific behavior problems parents report. When a child struggles to fall asleep or remain asleep, it can be quite disrupting to the functioning of the family. Behavior-related sleep problems can be divided into two main categories: sleep initiation and sleep maintenance problems. Problems with sleep initiation or maintenance can include inappropriate sleep associations, poor limit-setting, or delayed sleep phase.

Inappropriate Sleep Associations Sleep associations are the conditions needed for an individual to fall asleep. They are common because each individual prefers to fall asleep under different sets of conditions, such as temperature, lighting, or bedding, and can be inappropriate when the conditions are not sustained throughout the night. For children, sleep associations may involve a parent and accoutrements such as stuffed animals, music, lighting, and other items. The brain remembers the sleep environment as the person falls asleep. During the sleep cycle, the brain goes from deep sleep to light sleep and back. As the brain transitions from deep sleep to light sleep, the brain experiences a partial arousal, which is a break in the sleep cycle. The eyes may open to survey the environment. As long as the environment remains consistent, the person will immediately return to sleep. If the environment is different, the brain will awaken the person to re-create the environment that occurred at bedtime. A classic example is a 4-year-old child who likes for her father to read a book a bedtime, turn on a lamp in her room, and rub her back until she falls asleep. Once the child is asleep, the father turns off the light and leaves his daughter's bedroom. About 2 hours later, the child wakes up and notices that her bedroom is dark, and she is alone. The child calls out for a parent or gets out of bed and goes into her parents' room. Her father escorts her back to her bedroom, turns on the lamp, and rubs her back until she falls asleep again. This cycle continues throughout the night. To help children who have inappropriate sleep associations, the associations can be gradually reduced until a child learns to fall sleep without the "extras." One rule for bedtime success is to put a child to sleep when he is too sleepy to be awake and not too awake to go to sleep. When inappropriate sleep associations significantly impair the way a child sleeps, the child may be diagnosed with a sleep-onset association disorder and may require treatment with a sleep or behavior specialist.

Poor Limit-Setting Children with more difficult temperaments may have more problems with bedtime. Parents who do not consistently set limits and expectations may have difficulty getting a child to comply with any request, including mealtimes, bedtimes, and other activities. Poor limit-setting in the parent-child relationship can occur when the child refuses to go to bed and frequently makes demands on the parent, and the parent gives in to these demands. Inappropriate sleep associations may occur as a result of poor limit-setting. If a parent struggles with setting appropriate limits with their child, the parent may need help from a behavior psychologist, who can help re-establish natural parental authority in the parent-child relationship.

Delayed Sleep Phase Delayed sleep phase occurs when the natural circadian sleep rhythm shifts to a later time. This condition is common in children going through puberty because melatonin is secreted at a later time during puberty. Delayed sleep phases can also occur during weekends and school holidays, when the child's schedule is not as structured. For example, on weekends, a child may be allowed to stay awake until 11:00 PM and sleep until 9:00 AM the next morning. Then when school starts again on Monday, the child is expected to go to sleep by 9:00 PM and wake up for school at 6:00 AM. This would be a 3-hour shift in the child's sleep schedule, which can make it difficult for the child to awaken for school on Monday morning. Many cases of delayed sleep phase are more significant, where a child's sleep cycle is on a completely different time zone than the time zone in which the child lives (eg, going to sleep at 2:00 AM and waking up at noon). In order to address a delayed sleep phase, the child's weekend wake up time should be within 1 hour of the weekday wakeup time. It is important for the child to not sleep in on weekends and be allowed to take a short nap (less than 2 hours) in the middle of the day if needed on weekends. More significant sleep concerns can be addressed by a pediatric sleep specialist.

Feeding Problems

Parents can become distressed when children do not eat as much or as often as the parent expects. It is estimated that 25% of typically developing children can have feeding problems. Feeding behaviors can range from mild sensory sensitivities to severe feeding disorders.

Many feeding and eating behaviors can be expected based upon normal child development. When children are toddlers, they are developing a sense of independence and may prefer to feed themselves. They will also refuse to eat nonpreferred foods. By preschool, children should be able to make themselves a simple meal (eg, a bowl of cereal, a sandwich). Preschoolers may prefer to eat the foods they can make for themselves, such as a peanut butter and jelly sandwich. By elementary school, a child should be able to help select the dinner menu, help pack his own lunch, and order his own food at a restaurant. The elementary school–aged child may refuse to take home-cooked meals to school for lunch, instead preferring to eat the cafeteria food that the other students are eating. By adolescence, a child is developing more independence from parents and establishing relationships outside the family. A teen may prefer to eat away from home with friends.

Some children refuse to try new foods. They prefer a select number of items and will not even taste a new food that is introduced. Some children do not like to try new foods because of the way the food looks, smells, or tastes. Others may be highly sensitive to certain food textures and are reluctant to put certain foods in their mouths. There are also situations where poor limit-setting may contribute to food refusals in a child who does not comply with adult requests. Finally, a child who is a bit anxious may be hesitant to try something new. The approach to helping a child try a new food should be based on the underlying reason for the refusal—sensory sensitivity, poor limit-setting, aversions, or anxieties. Treatment may include gradual introduction of very small portions (pea-sized) of the new food, with positive reinforcers provided when the child tolerates the new food. For example, if the child tolerates one green bean sitting on the side of his plate while he eats his other foods, the child earns one sticker. For children with more significant food aversions or severe sensory sensitivities, consultation with a pediatric occupational therapist or pediatric speech pathologist may be warranted.

SUMMARY

Behavior problems are among the most common chief complaints in pediatric practice. Most behavior problems are mild and easily addressed through anticipatory guidance. Some behavior problems are more significant and begin to interfere with how a child functions and with the parent-child relationships. When evaluating behavior problems, clinicians should take into account normal development, child temperament, parenting styles, and environmental influences. Using a comprehensive approach to addressing common behavior problems is most effective. Consultation with a behavior specialist to address more severe behavior disorders may be warranted.

SUGGESTED READINGS AND RESOURCES

Suggested Readings

Barkley RA, Benton CM. *Your Defiant Child: Eight Steps to Better Behavior*. 2nd ed. New York, NY: Guilford Press; 2013.

Barkley RA. *Taking Charge of ADHD: The Complete and Authoritative Guide for Parents*. 3rd ed. New York, NY: Guilford Press; 2013.

Suggested Resources

Empowering Parents: www.empoweringparents.com

Center on the Developing Child: http://developingchild.harvard.edu

Kids Health: www.kidshealth.org

National Institute of Mental Health (NIMH): www.nimh.nih.gov

Understood: www.understood.org

86 Early Intervention and Special Education

Dinah L. Godwin, Marcia Berretta, and Marie Turcich

WHO REQUIRES EARLY INTERVENTION OR SPECIAL EDUCATION SERVICES?

This chapter will discuss the types of early intervention and special education services available, the referral process, and what pediatricians and other pediatric medical providers can do to help their families learn to navigate these important systems. Some red flags that should alert pediatric medical providers to the potential need for early intervention supports for infants and toddlers include high-risk neurobiologic (eg, prematurity) or psychosocial (eg, teenaged parents, maternal depression, poverty) conditions and failure to attain developmental milestones as expected. Red flags for preschool-aged children to alert pediatric medical providers to the potential need for special education preschool supports include high-risk neurobiologic or psychosocial conditions, new medical or developmental diagnoses, failure to attain developmental milestones as expected, and associated maladaptive behaviors. For school-aged children, red flags to alert pediatric medical providers to the potential need for special education supports include high-risk neurobiologic or psychosocial conditions, new medical or developmental diagnoses, failure to attain developmental milestones and academic skills (ie, school failure), and associated maladaptive behaviors.

Role of Primary Care Pediatric Medical Providers

The role of primary care pediatric medical providers can be pivotal, not only in the medical outcomes for a child with special needs, but in the child's educational outcomes as well. The primary care pediatric medical provider can be both a source of information and advocacy for children and their families and an expert resource for educational staff to understand complex medical conditions, treatment plans, and their integration with educational services from preschool through adolescence. Children with neurodevelopmental disabilities, in particular, typically have chronic, lifelong conditions that will need varying degrees of intervention and treatment. The American Academy of Pediatrics (AAP) directs pediatricians to provide a medical home for these patients, to help coordinate medical care, and to act as a resource for families. While such patients may have ongoing needs for medical procedures or therapies, they are also likely to need additional educational services and supports.

Early intervention and special education are mandated by federal law to provide free, appropriate public education (FAPE) services to children with disabilities either through the Individuals with Disabilities Education Act (IDEA) or through Section 504 of the Rehabilitation Act of 1973. IDEA is the federal law that supports special education and related service programming for children and youth with disabilities. It was originally known as the Education of All Handicapped Children Act, passed in 1975, which provided special education services for children from 5 to 21 years of age. In 1986, this law was extended to include special education preschool support for children from 3 to 5 years of age. In addition, the 1986 law established the Grants for Infants and Families program (Part C of IDEA), which is a federal grant program that assists states in operating a comprehensive statewide program of early intervention services for infants and toddlers from birth through 3 years of age.

EARLY INTERVENTION

Key Concepts of Early Intervention Programs

Each state's Part C program establishes a statewide, family-centered, multidisciplinary interagency system of early intervention (EI) services for delivery in the local or regional area. Two categories of children are served by EI: (1) those with a diagnosed physical or mental condition with a high likelihood of developmental delays (eg, Down syndrome), or (2) those with a developmental delay in one or more of five developmental domains (cognitive, motor, communication, social/emotional, adaptive). Some states may also elect to serve infants at risk for delay because of biological or psychosocial/environmental factors.

Part C eligibility is determined by each state's definition of developmental delay and whether it includes children at risk for disabilities in the eligibility formula. States have been given some discretion for determining eligibility for entry into their programs. Each state can determine what the eligibility requirements are, as well as how much funding will be allotted to their program. An important part of the evaluation process for infants and toddlers (ages 0–36 months) includes informed clinical opinions of professionals experienced with the development of very young children. Although a referral from a pediatrician or other pediatric medical provider is not required, clinical information from pediatric medical providers may be helpful in determining a child's eligibility for EI services. There are a number of medically diagnosed conditions that will qualify an infant or toddler for EI services. A list of these conditions can be accessed by an online search using the term "early intervention" and the name of a state. The state's EI site will also provide information about the referral process and eligibility process for that specific state.

Some key principles of EI include: (1) infants and toddlers learn best through everyday experiences and interactions with familiar people in familiar contexts;

(2) all families, with the necessary supports and resources, can enhance their children's learning and development; and (3) Individualized Family Service Plan (IFSP) outcomes must be functional and based on children's and families' needs and family-identified priorities.

Early Intervention Referral Process

Pediatricians and other pediatric medical providers who identify that a child under the age of 3 years has a developmental delay or disability are legally mandated to make a referral to EI services. Federal regulations implementing Part C of IDEA require that children under 3 years of age be referred *no more than 7 days* after being identified as having a developmental delay or disability.

A referral to a state's EI program can be made by a physician or other healthcare provider, a family member, or a teacher. Referral forms are available through an online search using the term "early intervention" and the name of a state. An interdisciplinary team will conduct a comprehensive evaluation to determine a referred child's eligibility for EI. If this evaluation determines that a child is eligible for services, the interdisciplinary team, including the child's parents, develops an IFSP that focuses on involving the family in therapeutic interventions and builds on the settings and routines familiar to the child.

Examples of Early Intervention Services

Early intervention services include licensed or credentialed professionals who provide (1) speech-language therapy, (2) physical therapy, (3) occupational therapy, (4) audiology and services for the hearing impaired, (5) services for the visually impaired, (5) nutrition services, (6) specialized skills training, and (7) case management. Part C of IDEA requires that eligible infants and toddlers with disabilities receive services in natural environments to the maximum extent appropriate. Natural environments include the family's home and/or community settings in which children without disabilities participate (such as childcare centers).

Value of Early Intervention

Early intervention plans and services for infants and toddlers are based on research that demonstrates that learning occurs between intervention sessions. During a session, the EI provider utilizes his or her professional knowledge, skills, and expertise to share information with the child's regular caregivers. The caregivers then provide the intervention within the child's daily routines. Studies have found that children who participate in high-quality EI programs tend to have less need for special education and have greater language abilities, improved nutrition and health, and experience less child abuse and neglect. Economic analyses demonstrate that programs that intervene early to improve child outcomes have returns on investment (ROI) from $2.50 to $17.07 for every dollar spent on EI services.

How Are Early Intervention Services Funded?

Early intervention programs receive federal, state, and local funds, as well as collecting Medicaid, Children's Health Insurance Program (CHIP), private insurance, and payments from families. However, even though EI does collect payments from families, no family will be denied services due to an inability to pay.

Transition from Early Intervention Programs

When a child approaches the age of 3 years (no less than 3 months prior to the child's third birthday), the EI team works with the family to develop a plan for transition from EI services. There is a strong evidence base for children with and without special needs to participate in high-quality preschool programs, although preschool and even kindergarten are not mandatory in all states, and some families do opt to keep their children at home after EI services terminate. Options for children ages 3 years and up may include transitioning to a public school district's special education preschool program (for children from 3 to 5 years of age), a Head Start program, a private preschool, or a combination of public and private services. For the purposes of this chapter, we will focus on the public school programs for children with disabilities, as these services are mandated by federal law and are available to all children who meet eligibility criteria, regardless of variables such as geographic location or family income.

SPECIAL EDUCATION

Key Concepts in Special Education

Children who have or are suspected of having developmental delays, learning problems, sensory deficits (such as hearing or visual impairment), or health conditions that affect their ability to learn may be eligible for special education services beginning at age 3 years. As with EI services, special education services are guided by IDEA, the federal law that delineates the requirements for school districts to provide educational services and supports to children with disabilities. The federal Office of Special Education and Rehabilitative Services

(OSERS) oversees IDEA and funds special education programs to the states, with each state education agency responsible for administering IDEA in their state and distributing funds for special education programs. State boards of education and local school boards have some discretion in how services are provided, but every state, district, and individual school must comply with the mandates set forth in IDEA.

The underlying principle governing special education law is that all children, regardless of their disabilities, have the right to a “free and appropriate public education” in the “least restrictive environment.” This terminology is important, as it underscores the expectations that children with disabilities have the right to be educated with their typically developing peers to the greatest extent possible.

The basic provisions of IDEA that relate to special education are as follows: (1) find and identify students who have a disability, (2) involve parents in decision making, (3) evaluate students in a nondiscriminatory way, (4) develop an individual education program (IEP) for each eligible student, (5) provide special instruction and supplementary aids and services, (6) provide services in the least restrictive environment, (7) maintain education records/files, and (8) provide processes for resolving parent complaints and grievances.

The Role of Pediatricians and Other Pediatric Medical Providers

Pediatricians and other primary care pediatric medical providers are often the first professionals to whom parents express their concerns about their child’s development, learning, or behavior. Parents assume that pediatricians, in their role as child advocates, have expert knowledge about the educational system and can help guide them toward appropriate school-based services. Consequently, it is important for pediatricians and other pediatric medical providers to have a basic understanding of the process for accessing special education evaluations and services, so that they will be able to effectively advise families.

Pediatricians and other pediatric medical providers also have a critical role in early identification and referral for children whose families have not yet recognized a need for special services. When parents raise concerns about their child not making progress in school, having behavioral outbursts in the classroom, or experiencing difficulties with attention or focus, these are red flags that may indicate a need for evaluation for additional special educational services or supports. In addition to their medical evaluation and treatment recommendations, pediatricians and other medical providers should remember to address the child’s school environment and to consider recommending a special educational evaluation as part of their treatment plan. Pediatricians and other pediatric medical providers should also be available to communicate with school district personnel to provide medical information and to discuss services, after obtaining parental consent as appropriate.

Brief Overview of Special Education Services

Beginning at age 3 years, children with known or suspected disabilities may be eligible for special education preschool services. This may include half-day or full-day preschool in a classroom setting with a low student-to-teacher ratio (usually no more than 4:1). Other services may be offered based on the child's evaluation and identified needs. These may include speech/language therapy, occupational therapy, physical therapy, behavioral therapy, transportation, and specialized services for children with visual and/or hearing impairments. Children with speech and language delays but with no other identified disabilities may qualify for speech/language therapy services through the school district, even if they are not enrolled in special education preschool or are attending a private preschool.

Special education preschool services are offered from age 3 years through kindergarten. For older children, special education services are offered in a variety of settings and formats, including self-contained special education classrooms, "resource" or specialized instruction classrooms for academic subjects, assistance from a paraprofessional in a mainstream classroom, or some combination of the above. The specific services provided will be based on the child's needs and, as noted above, should be offered in the least restrictive environment. Special education is available to children with disabilities through the age of 21 years, and for older adolescents, a plan for transitioning to postsecondary services that are appropriate for the adolescent's developmental level should be included in the IEP.

How to Request Special Education Services

Parents who have a concern about their child's development in one or more developmental domains or about their child's academic progress have the right under IDEA to request a full individual evaluation (FIE) of their child. Pediatricians and other pediatric medical providers should recommend that the parent request this FIE in writing to the school principal, who is responsible for ensuring that the campus is compliant with special education law. If a child is new to the school district and has not yet been enrolled in school, the parent should address the letter to the principal of the school to which the child is zoned. The family can contact their school district to obtain the contact information for their zoned school.

In the letter, parents should state the child's name and date of birth, their concerns and their reason for requesting the evaluation, and the contact information for the parents. The letter may include any supplemental information that may be helpful, such as the child's diagnosis or results from any outside evaluations that the child may have completed. Parents should keep a copy of the letter for their records. Parents may also wish to request a dated receipt from the school confirming that the letter was received. A sample letter for families to request a special education evaluation can be viewed at http://www.disabilityrightstx.org/resources/education.

The Evaluation Process

Once the school has received the parents' letter, the administration must respond to the parents within 15 school days. The school has the option to decline the parents' request, but they must decline in writing and state the reasons why they do not believe an evaluation is necessary. If the school does plan to evaluate the child, they must request that the parent sign consent forms giving them permission to complete an evaluation. Once these forms are received by the school, the testing process can begin, and the school has 45 school days in which to complete the FIE. Depending on the concerns identified, the evaluation may consist of any or all of the following: parent/caregiver interview, child interview, observation of the child, cognitive testing, speech and language evaluation, fine motor evaluation, gross motor evaluation, hearing testing, vision testing, psychological evaluation, and behavioral assessment.

What Happens After Evaluation?

After the FIE has been completed, the school must provide a copy of the evaluation report to the parents and must convene a meeting of the evaluation team and the family in order to review the results and recommend services as appropriate. This meeting is usually called an Individualized Education Program (IEP) meeting; in some states, it may also be referred to as an Admission, Review, and Dismissal (ARD) meeting. This meeting may include the following individuals: school principal, school district diagnostician who completed the evaluation, special education or general education teachers, the EI specialist (if a child is transitioning from EI), school district personnel, the parents, and a language interpreter, if needed. At the IEP meeting, the results of the evaluations will be reviewed, and the school will inform the parents whether the child qualified for special education services and what services are being recommended. The parents will have an opportunity to ask questions and provide input. At the end of the meeting, all parties present will be asked to sign the meeting report to indicate agreement with the plan.

IEPs must be reviewed at a formal IEP meeting at least annually, but parents have the right to request an IEP meeting at any time, if they have concerns about their child's services. In addition, children receiving special education services must have a new FIE completed every 3 years.

Children may qualify for special education services under the following 13 educational classifications: (1) autism; (2) deafness; (3) deaf-blindness; (4) emotional disturbance; (5) hearing impairment; (6) intellectual disability; (7) multiple disabilities; (8) orthopedic impairment; (9) other health impaired (including attention-deficit/hyperactivity disorder [ADHD]; (10) specific learning disability (including dyslexia, dysgraphia, dyscalculia, nonverbal learning disability); (11) speech or language impairment; (12) traumatic brain injury; and (13) visual impairment.

SECTION 504

Section 504 is a civil rights law with assistance and enforcement through the federal Office for Civil Rights (OCR). Section 504 is an antidiscrimination law that does not provide any type of funding; its purpose is to protect individuals with disabilities from discrimination in all federally assisted programs and activities. Section 504 prohibits recipients of federal funds, such as public school districts, from discriminating against individuals with disabilities. Students with disabilities are to be afforded the same opportunities to participate in academic, nonacademic, or extracurricular activities as non-disabled students.

Students who are identified as eligible for special education and related services under IDEA are included under provisions of Section 504. However, not all students who qualify for Section 504 are eligible for special education services under IDEA because Section 504 has a broader definition of the term "disability." Eligibility under 504 is not limited to certain disability categories, as it is under IDEA. Section 504, as a civil rights law, includes persons who have a "physical or mental impairment that substantially limits a major life activity," such as caring for oneself, performing tasks, walking, seeing, hearing, speaking, breathing, reading, writing, calculating math problems, concentrating, interacting with others, learning, or working. An impairment does not automatically mean that a student has a disability under Section 504; the impairment must substantially limit one or more major life activities in order to be considered a disability under Section 504. A student is not considered as an individual with a disability if the impairment is minor or transitory (ie, with an actual or expected duration of 6 months or less). An impairment that is episodic or in remission may be considered a disability if it substantially limits a major life activity when active. It is important for pediatricians and other pediatric medical providers to help families understand that a child's medical diagnosis does not automatically mean that the child is eligible to receive services under Section 504; an illness or impairment must cause a substantial limitation of the child's ability to learn or of another major life activity. Medical diagnoses are not sufficient to serve as an evaluation for the purposes of providing a "free and appropriate public education," but diagnoses may be considered among other sources of information in evaluating a student with an impairment. In addition to a medical diagnosis, other necessary sources of information include achievement/aptitude test results, teacher recommendations, physical condition, social/cultural background, and adaptive behavior.

Students with disabilities under Section 504 are eligible to receive accommodations, modifications, and supplementary aids and services in school, so that their individual educational needs are met as adequately as nondisabled peers. Students with disabilities under Section 504 must also be educated with their nondisabled peers "to the maximum extent appropriate." Placement in separate classrooms or special education programs can only occur following an individual evaluation that determines that a student cannot benefit from education in regular classes, even with the use of supplementary aids and services. Thus, services for students with disabilities under Section 504 regulations may include education in regular classrooms, education in regular classes with supplementary services, or special education and related services.

Primary care pediatric medical providers may advise parents to request Section 504 services when their patient has a chronic medical condition, such as diabetes or heart disease, which is not covered under IDEA. Section 504 may also be appropriate when a child with a disability is covered by IDEA—such as a child with ADHD, a mild emotional disorder, or an orthopedic impairment—but does not require IDEA/special education services to benefit from their school program and can remain in regular education with accommodations or modifications. Educational accommodations or modifications vary based on the child's specific needs; examples include shortened assignments, preferential seating, use of a tape recorder, or accessible building and transportation accommodations. Under IDEA, the ARD committee develops an IEP for the student, while a Section 504 committee develops a 504 plan for the student.

There are some similarities between Section 504 and IDEA procedures that involve identification, evaluation, provision of services, and individual plan and procedural safeguards, although mandatory procedures are not identical, and school districts generally have greater flexibility under Section 504. It is important to keep in mind that although a child with a disability may be not eligible for special education services within the public school system, Section 504 accommodations or modifications, as well as supplementary aids and services, are available to ensure that the child's educational needs are met as adequately as those of nondisabled students.

WHAT IF THE CHILD DOES NOT QUALIFY?

If a child does not meet eligibility criteria for IDEA or Section 504 services but nonetheless has difficulty meeting learning goals or passing benchmark examinations, he or she may receive other types of services that do not fall under the auspices of special education. Children who are struggling with academics may be offered Response to Intervention (RTI), a tiered system of evidence-based interventions for children who are not meeting grade-level expectations. RTI may involve supports such as tutoring, individual and group targeted instruction, or behavioral interventions. Children in RTI should be monitored frequently and consistently to ensure that the interventions are effective. RTI should not be implemented in lieu of a special education evaluation if a child appears to have a disability that may qualify them for special education services.

When disagreements arise between parents and school districts regarding eligibility and services under IDEA or Section 504, either party may use due-process hearing procedures to seek resolution. If a pediatric medical provider has a patient whose family has tried to obtain special education services or additional assistance for their child in school, but the school has not been responsive, the provider should recommend that the family seek help from community-based advocacy organizations, such as the Arc, which provides detailed educational advocacy and has chapters in all 50 states and the District of Columbia. Local chapters provide materials, training, and consultation to guide families through the special education system and to help families understand how to initiate complaints or grievances when their concerns are not resolved. For more information, families can contact their local Arc (http://www.thearc.org/find-a-chapter). There are also professional advocates, including attorneys and educational specialists, who are available for hire to help represent families in their conflicts with schools. State boards of education are also entrusted with the responsibility to ensure that local school districts and schools are in compliance with IDEA and other related laws, and each state board has established a process to investigate and respond to complaints and concerns.

WHAT CAN PEDIATRIC MEDICAL PROVIDERS DO TO HELP FAMILIES WITH SCHOOL ISSUES?

Primary care pediatricians and other pediatric medical providers and parents of children with disabilities can obtain additional information about available services under Section 504 and IDEA by contacting their campus principal, district special education director, or the district 504 coordinator. These educators are knowledgeable about the federal laws and local school services that are available. Parents of students with special needs are often ill-equipped to navigate the complexities of the public school system and will benefit from healthcare professionals who have a knowledge base of the interplay of health care and children's educational programs.

Primary care pediatricians and other pediatric medical providers can have a significant positive impact on a child's educational services by providing personalized medical recommendations on behalf of their patient at an ARD or 504 committee meeting through written recommendations or participation in a conference call before or during the meeting. Pediatricians and other pediatric medical providers can also educate parents and school personnel regarding various medical conditions and their implications for educational programming and supports needed in the home and school environments. Opportunities for primary care pediatric medical providers to educate parents and teachers are widespread and can vary in complexity and duration, depending on the needs of the patient. Effective communication by healthcare professionals with parents and educators may involve brief phone calls or office visits or could include more extensive interactions, such as a series of community programs about a variety of common medical conditions found in schools, such as ADHD, neurological problems (eg, seizures), autism spectrum disorders, learning and intellectual disabilities, or adolescent health issues for students with special needs. Public school districts typically provide in-service training for their staff at scheduled times throughout the year and welcome presentations by medical professionals regarding children with developmental disabilities.

In its 2015 report on IDEA and children with special educational needs, the AAP strongly encourages pediatricians and other pediatric healthcare providers to develop an understanding of the laws governing early intervention and special education and to advise and support families whose children require these services. Through their advocacy, guidance, and support, pediatric medical providers can help ensure that children with disabilities have access to programs and services that will help to maximize their educational and functional potential.

SUGGESTED READINGS AND WEB SITES

Readings

Anderson W, Chitwood S, Hayden D, Takemoto C. *Negotiating the Special Education Maze*. 4th ed. Bethesda, MD: Woodbine House; 2008.

Lipkin PH, Okamoto J; Council on Children with Disabilities; Council on School Health. The Individuals with Disabilities Education Act (IDEA) for children with special educational needs. *Pediatrics*. 2015;136:e1650-e1662.

Wilmshurst, L, Brue AW. *The Complete Guide to Special Education: Expert Advice on Evaluations, IEPs, and Helping Kids Succeed*. 2nd ed. San Francisco, CA: John Wiley & Sons, Inc; 2010.

Wright P, Wright P. *From Emotions to Advocacy*. Hartfield, VA: Harbor House Law Press; 2002.

Web Sites

The Arc (http://www.thearc.org/what-we-do/public-policy/policy-issues/education): An organization dedicated to the welfare of all children and adults with intellectual and developmental disabilities.

Center for Parent Information and Resources (http://www.parentcenterhub.org/): General information for parents of children with special needs, including links to state parent centers.

Council for Exceptional Children (http://www.cec.sped.org/): A professional organization dedicated to improving the educational success of individuals with disabilities.

International Dyslexia Association (https://dyslexiaida.org/)

LD Online (http://www.ldonline.org/): An educators' guide to learning disabilities and ADHD.

Understood (https://www.understood.org/en): Resources for parents of children struggling with learning and attention issues.

U.S. Department of Education – Information from the Office of Special

Education and Rehabilitative Services – (http://www2.ed.gov/about/offices/list/osers/index.html?src=mr).

U.S. Department of Education Office of Special Education and Rehabilitative Services (http://www2.ed.gov/about/offices/list/ocr/504faq.html): Information from the Office of Civil Rights on Section 504.

Wrightslaw (http://www.wrightslaw.com/): Information about special education law, education law, and advocacy.

87 Global Developmental Delay and Intellectual Disability

Sarah Risen and Sherry Sellers Vinson

NEURODEVELOPMENTAL STREAMS

Observation over centuries confirms that infants and children learn developmental skills in a predictable order within a predictable age range. These skills occur in the following neurodevelopmental streams: neuromotor (including gross motor, fine motor, and oral motor), neurocognitive (including language, social, nonverbal/visual-motor problem solving, and adaptive), and neurobehavior (including attention and activity/impulsivity). Gross motor skills include rolling over, crawling, and walking, and progressing to running, jumping, and skipping; fine motor skills include picking up an item, first with a full fist and then progressing to doing the same with a mature pincer grasp; oral motor skills include speech articulation and feeding. Nonverbal/visual-motor problem-solving skills include determining how to obtain an item that is out of reach or determining which of two containers is smaller. Expressive language skills include cooing, babbling, and progressing to saying words, phrases, and sentences. Receptive language skills include following a one-step command given with a gesture and progressing to following nongestured commands. Social skills include reciprocally smiling and progressing to interactive play with others.

One of the primary responsibilities of a primary care pediatric medical provider is the monitoring of each patient's skill attainment in each developmental stream to make certain that each patient acquires age-expected skills, which is considered meeting developmental milestones. Pediatric medical providers need to partner with their patients' parents and caregivers in this monitoring responsibility, since parents and caregivers observe their children daily, which allows them to report skill attainment, lack of skill attainment, or skill loss in a timely manner.

Global Developmental Delay

Developmental delay occurs when a child gains a skill at an age beyond the expected age range for that skill's acquisition. Given that the most common etiologies for developmental delay tend to affect the brain diffusely (eg, chromosomal/genetic abnormalities), global developmental delay is more prevalent than focal or dissociated delay. Global developmental delay is defined by a significant delay in two or more developmental domains. While a mild or borderline delay is usually defined by developmental quotient, or DQ (developmental age/chronologic age × 100), below 85%, a significant delay is usually defined as a DQ of less than 70%. Thus, an 18-month-old child whose nonverbal/visual-motor problem-solving skills are at a 12-month level (DQ = 12 months/18 months × 100 = 67%) and whose language skills are at a 10-month level (DQ = 10 months/18 months × 100 = 56%) would be considered to have global developmental delay.

Infants and toddlers with global developmental delay who have delays involving both nonverbal/visual-motor problem solving and receptive language development are at significant risk for intellectual disability. The more severe the intellectual disability is, the younger the age a child is when it can be reliably identified. However, mild intellectual disability may be more difficult to confirm at early ages, particularly in children who have not received appropriate developmental stimulation. A hearing and seeing child with exposure to developmentally stimulating activities who has acquired developmental skills in nonverbal/visual-motor problem solving and receptive language at less than 70% the expected rate for age by 6 years of age would be predicted to score in a range of intellectual disability on a standardized intelligence test and on at least two adaptive behaviors on a standardized caretaker adaptive behavior questionnaire, with both of these measures using standard scores rather than DQs.

STANDARD SCORES ON IQ TESTS

When considering intellectual disability and scores on standardized intelligence tests (which report intelligence quotient, or IQ scores) and measures of adaptive behavior, one must understand the meaning of standard scores. IQ tests are designed so that scores fall along a normal or bell-shaped curve for each age, with a mean IQ score of 100 and a standard deviation of 15 points. Intellectual disability is defined as intelligence that scores more than 2 standard deviations (ie, more than 30 IQ points) below the mean. IQ scores from 90 to 109 are considered average (accounting for 50% of the population), from 110 to 119 are considered high average (accounting for 16.1% of the population), from 120 to 129 are considered superior (accounting for 6.7% of the population), and scores of 130 and above are considered very superior (accounting for 2.2% of the population). IQ scores from 80 to 89 are considered low or below average (accounting for 16.1% of the population), from 70 to 79 are considered borderline (accounting for 6.7% of the population), and scores of 70 and below are considered in the range of intellectual disability (accounting for 2.2% of the population). To be diagnosed with an intellectual disability, in addition to scoring more than 2 standard deviations below the mean on IQ testing, the individual must also score more than 2 standard deviations below the mean on a standardized parent/caregiver completed questionnaire measuring adaptive behavior (which includes activities of daily living and other social/practical skills), and the onset of the disability must occur prior to 18 years of age. It is important to note that although they do not exhibit intellectual disability, children in the borderline and low average ranges of intelligence typically have significant difficulty keeping up in regular classrooms, and combined, they are often referred to as "slower learners." Such slower learners make up 22.8% of the population, and thus, slower learning is the most prevalent neurodevelopmental cause of school failure.

LEVELS OF INTELLECTUAL DISABILITY

The levels of intellectual disability are (1) mild, with IQ scores between 50 and 69 (accounting for 85% of individuals with intellectual disability); (2) moderate, with IQ scores between 35 and 49 (accounting for 10% of individuals with intellectual disability); (3) severe, with IQ scores between 20 and 34 (accounting for 3% to 4% of individuals with intellectual disability); and (4) profound, with IQ scores less than 20 (accounting for 1% to 2% of individuals with intellectual disability). IQ tests generally provide a full-scale IQ score as well as IQ scores on subdomains of intelligence, such as verbal comprehension, nonverbal reasoning, working memory, and processing speed. Most commonly, individuals with intellectual disability score in the same range or adjacent ranges for each component of IQ and adaptive behavior testing. For example, an individual may score in the mild range of intellectual disability for all components of their IQ and adaptive behavior testing. Another individual may score in the moderate range of intellectual disability for verbal comprehension, working memory, and processing speed on IQ testing and in communication on adaptive behavior testing, but he or she may score in the mild range of intellectual disability for nonverbal reasoning on IQ testing and in daily living skills on adaptive behavior testing. However, a child with fine-motor incoordination accompanying the intellectual disability may have a processing speed standard score in the severe range of intellectual disability that drags down the child's other standard scores, which may be in the mild range. Knowing the standard scores for each component of the patients' IQ testing allows primary care pediatric medical providers to assist their patients' families to recognize their children's strengths to use when helping them learn in their weaker areas, determine appropriate assistive technology for circumventing significantly lower ability areas, and make lifespan plans.

ADULT OUTCOMES

Years of evidence indicates that individuals with intellectual disability continue to gain new skills within the breadth of complexity of their learning abilities across their lifespans; however, the depth of the complexity of what they are able to learn plateaus in adulthood. The adult abilities of individuals with intellectual disability can be generally predicted as follows. The depth of complexity of learning of individuals with mild intellectual disability generally plateaus at an approximate 8- to 11-year-old level. This means that individuals with mild intellectual disabilities should have late 2nd- to early 6th-grade–level academic abilities as an adult, and most individuals with mild intellectual disability can live independently as adults, although they typically require intermittent support (such as when making financial decisions). The depth of complexity of learning for individuals with moderate intellectual disability generally plateaus at a 5- to 8-year-old level. Thus, individuals with moderate intellectual disability should have kindergarten to late 2nd-grade–level academic abilities, and individuals with moderate intellectual disabilities typically require limited support throughout the day (to navigate daily situations) as adults. The depth of complexity of learning for individuals with severe intellectual disability generally plateaus at a 2- to 5-year-old level; thus, individuals with severe intellectual disability typically require extensive support. Finally, the depth of complexity of learning for individuals with profound intellectual disability generally plateaus at less than a 2-year-old level; thus, individuals with profound intellectual disability typically require pervasive support. A single individual with intellectual disability usually has adult abilities scattering from the lowest age to the highest age in the predicted age range for his/her predicted adult ability range, with skills requiring more abstract reasoning being at the lowest age and skills requiring more concrete reasoning being at the highest age. Understanding these predicted adult outcomes allows primary care pediatric medical providers to help their patients' parents advocate for appropriate educational interventions to help their children reach their predicted adult outcomes and to plan for their children's transition to adulthood. Such advocacy and planning can begin early, since formal intellectual testing can be performed in a child's early years of elementary school.

RESPONSIBILITIES OF THE MEDICAL HOME

Global developmental delay and intellectual disability are descriptive developmental diagnoses, and it is the primary care pediatric medical provider's role to attempt to determine an etiologic diagnosis to account for the global developmental delay or intellectual disability. The more severe the developmental delay or intellectual disability, the more likely an underlying etiologic diagnosis will be made. Unfortunately, since the vast majority (85%) of individuals with intellectual disability have mild intellectual disability, an etiologic diagnosis will not be discovered for the majority of individuals; however, recent advances in genetic testing are resulting in more etiologic diagnoses being made currently than ever before.

All individuals with global developmental delay or intellectual disability should undergo a chromosome microarray analysis and DNA testing for fragile X syndrome (the most common cause of inherited intellectual disability). Metabolic testing (eg, plasma amino acids, ammonia, lactate, acylcarnitine profiles; urine organic acids; mucopolysaccharides) should be considered in individuals with global developmental delay/intellectual disability who present with developmental regression, episodic encephalopathy, seizures, ataxia, involuntary movements, organomegaly, coarse facial features, decompensation with mild illness, episodic vomiting, or other metabolic signs. Other causes of intellectual disability include complications of prematurity, prenatal toxin exposure (eg, fetal alcohol syndrome), congenital infections, postnatal infections (encephalitis), hypoxia/ischemia, traumatic brain injury, and psychosocial deprivation.

Pediatric medical providers need to realize that the diagnosis of an intellectual disability is devastating to parents, as this diagnosis predicts that their child will need to live under some level of adult support across the lifespan. To provide an appropriate medical home for a child with an intellectual disability, pediatric medical providers and their staff should (1) assist parents in understanding public school special education law; (2) assist parents in recognizing appropriate expectations on which to base educational interventions/assistive technology requests; (3) monitor for mental health disorders (as all psychiatric disorders are more prevalent in individuals with intellectual disability compared to those without intellectual disability); (4) provide parents information about the availability of post-secondary job training and appropriate supervised jobs; (5) educate parents on the need for extracurricular/recreational activities and where to find them; (6) counsel parents on the importance of establishing adult guardianship before the child reaches 18 years of age and planning for the child's future past the parents' lives; (7) provide families information on Medicaid waivers and disability supports; and (8) transition their patients to adult medical/dental care.

It is critical for pediatric medical homes of children and adolescents with intellectual disability to be skilled in assisting parents in understanding their children's ability levels, so that they may establish developmentally appropriate expectations for their children, so as to avoid anxiety and frustration derivative of a mismatch between expectations and abilities, which may result in secondary social, emotional, or behavioral problems. If parents' expectations exceed a child's developmental abilities, in addition to frustrating the child and potentially causing low self-esteem, the parents will remain chronically disappointed in the child's inability to meet expectations and feel like a failure as a parent. On the other hand, if parents have appropriate expectations based on the ability level of the child, they can experience pride in their child's accomplishments, which validates their parenting skills.

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88 Delays in Motor Development and Motor Disabilities

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INTRODUCTION

Given that the most common causes of developmental delay affect the brain diffusely (eg, chromosomal/genetic syndromes), delays in motor development are most commonly a component of more globally delayed developmental milestones across neuromotor, neurocognitive, and neurobehavioral streams of development. Thus, when a child presents with a chief complaint of delays in motor development, a thorough developmental history is needed to uncover possibly more subtle delays in language, nonverbal/visual-motor problem-solving, social, or adaptive development. However, there is a spectrum of dissociated and deviated motor development, where motor development is discrepantly delayed compared to other domains in development, and in more severe cases, where motor skills are acquired in a developmentally deviated manner (see Chapter 83). This chapter will review the motor disabilities contained within this spectrum of discrepant (or dissociated and deviated) motor delay.

Embryologically, the neural system is one of the earliest to begin to develop and the last to be completed. This system develops in a predictable manner of progression from cephalad to caudal and proximal to distal, as do motor skills after birth. A deviation from this predicted pattern may represent the earliest signs of a future motor delay or disability. Motor disorders can be central or peripheral in origin. Understanding patterns of typical development will aid the clinician in appropriately diagnosing and treating the underlying condition.

Motor skills are not as correlated as other domains of development are to overall cognitive function, but when delayed, they are the most evident early on. When severe, these delays not only are obvious to clinicians but also are evident to family members. However, subtle differences in motor development may or may not predict an actual disorder. Therefore, it is important to remember that there is a wide range of normal variation in motor development, and even typically developing children may achieve motor milestones at different times.

Since a motor delay can be particularly anxiety provoking to family members, it is imperative to approach these concerns in a caring manner, be able to explain the observations in family-centered language, and provide reassurance when appropriate.

In addition to identifying a discrepant delay (dissociation) or deviation in motor development, it is imperative that a clinician assess the nature of the disorder and evaluate for potential associated medical conditions and complications. Even when a definitive diagnosis cannot be obtained early on, timely identification of delays should prompt referrals to therapeutic services, which may improve the ultimate function and quality of life for children with motor disabilities.

PATHOGENESIS AND EPIDEMIOLOGY

While motor development most often occurs in a predictable pattern, a delay, dissociation, or deviation from this pattern is most often, but not always, indicative of an underlying diagnosis. While motor delay is required for a motor disability diagnosis, deviation from the typical pattern of development does not always represent a problem (eg, walking before crawling). However, if the degree of this delay, dissociation, or deviation is severe, this usually represents an underlying motor disability. As motor pathways mature very rapidly in the first year of life, motor disabilities often present early on. These signs may be present, depending on severity, in utero (possibly leading to a difficult delivery) or at birth (eg, tone abnormalities and abnormal head growth). In mild cases, a deviation from the typical patterns may not present until later on and be most noticeable at school age. A clinician's familiarity with the sequence of typical motor development will aid in identification of an abnormality (see Chapter 82).

The progression of motor development includes the integration of primitive reflexes (eg, the asymmetric tonic neck reflex, the tonic labyrinthine extensor reflex, the positive support reflex) and the synchronous appearance of postural reactions, followed by the development of voluntary motor skills. This represents a shift from involuntary brain stem control to voluntary higher cortical control. The integration of most primitive reflexes occurs by 6 months of age. Early postural reactions are required prior to the appearance of other motor milestones, such as rolling. After this, motor development typically occurs in a stepwise progression from rolling to sitting, crawling, cruising, and ultimately to walking (**Table 88-1**). While it is important to be familiar with the typical timetable of motor milestones, it is also important to be familiar with the qualitative progression of motor coordination. For example, a 12-month-old child with left hemiparesis may show a voluntary release of objects (with his or her right hand) as expected at 12 months of age and thus does not exhibit a developmental delay in this skill, but careful observation will reveal that this child cannot accomplish this feat with either hand and tends to flex the left arm and fist the left hand. Thus, the best way to diagnose a motor disability is to carefully obtain and consider the developmental history and combine this with a full physical examination and detailed neurologic and neurodevelopmental examinations.

TABLE 88-1 MOTOR MILESTONES

Milestone (Gross Motor, Fine Motor)	Age Expected
Holds chin up in prone	1 month
Holds chest up in prone	2 months
Supports self on elbows in prone	3 months
Rolls prone to supine; hands mostly open	4 months
Rolls supine to prone; transfers objects	5 months
Sits with anterior propping; rakes object with hand and grasps	6 months
Crawls; holds one object in each hand	8 months
Cruises; immature pincer	9 months
Walks with one hand held; mature pincer	11 months
Walks without assistance; voluntary release of objects	12 months
Walks down stairs with one hand held; scribbles spontaneously	18 months
Jumps off ground; imitates stroke	24 months
Performs broad jump; draws circle	36 months

The spectrum of motor disabilities is wide and can be divided into central and peripheral in nature. Central motor disorders include a spectrum from the high-prevalence, low-morbidity developmental coordination disorders (eg, gross-motor dyspraxia and fine-motor dysgraphia) to the low-prevalence, high-morbidity presentation of cerebral palsy (CP). The estimated prevalence of CP is approximately 2 per 1000. CP is a static encephalopathy caused by a brain insult occurring in the prenatal, perinatal, or postnatal time periods, and it is nonprogressive. The most significant risk factor for CP is prematurity (< 32 weeks gestation, < 1500 g), and nearly half of children with CP are born preterm. Prenatal risk factors include genetic abnormalities and in utero strokes or exposure to viral infections, teratogens, or drugs. Perinatal risk factors include intrapartum asphyxia resulting in neonatal hypoxic-ischemic encephalopathy, which accounts for less than 10% of cases of CP. Postnatal risk factors include severe neonatal infections, strokes, trauma, or metabolic disease. Postnatal risk factors account for 10% of children with CP.

Peripheral causes of motor disability include both neuromuscular and structural defects. Peripheral causes of motor disability also represent a spectrum and include such disorders as spinal muscular atrophy, Charcot-Marie-Tooth disease, muscular dystrophy, and clubfoot.

CLINICAL MANIFESTATIONS

Motor disorders are not usually evident at birth, and therefore, children with identifiable risk factors should be closely monitored longitudinally. However, many children who will go on to be diagnosed with a motor disability will have a history of poor feeding in the neonatal period. Those with severe motor disabilities may be detected in the first year of life, while those with mild disabilities may not be noted until later ages, when more complex motor activities are expected.

Cerebral Palsy

Children with CP most often present with delays and deviation in their acquisition of motor milestones combined with abnormalities in their neurological exams. Abnormalities in muscle tone may present as poor head control, opisthotonus, or scissoring of the lower extremities. Children with CP may exhibit persistence of primitive reflexes beyond the usual age of integration and brisk deep tendon reflexes or clonus.

CP is clinically described by the type and location of impaired motor function. Spastic CP presents with hyperreflexia and increased muscle tone due to muscle spasticity, and over time, children with spastic CP tend to develop muscle contractures. Spastic CP has been attributed to damage to the corticospinal or pyramidal tracts. Extrapyramidal CP includes dyskinetic and ataxic forms. Dyskinetic CP presents with uncontrolled, involuntary movements, such as dystonia or choreoathetosis, and it is related to damage to the basal ganglia. Ataxic CP presents with poor balance or coordination and gait instability, and it is related to damage to the cerebellum. CP is also described by the affected limbs: monoplegia (1 limb), hemiplegia (ipsilateral arm and leg), diplegia (bilateral lower extremities), triplegia (1 limb being unaffected or minimally affected when compared to the other 3; typically involves diplegia with a superimposed hemiplegia), and quadriplegia (all limbs affected, usually along with the trunk).

CP often presents with multiple medical comorbidities related to the underlying brain damage and poor motor control and coordination. Typically, the more severe the motor impairment, the more severe the comorbidities. Problems related to growth and feeding are attributable to oral-motor dysphagia, and 7% of children with CP require tube feeding. Gastroesophageal reflux is present in 50% and constipation in up to 70% of children with CP, and these gastrointestinal complications also contribute to poor feeding and growth. Orthopedic complications of spastic CP include joint dislocations, bone deformities, scoliosis, osteopenia, fractures, and muscle contractures. Sensory deficits are common, with visual impairments present in 30% and hearing deficits present in 10% to 20% of children with CP. Seizures occur in 25% to 40% of children with CP. Cognitive deficits or intellectual disability occur in 50% of children with spastic CP, with severity often correlated to the severity of the motor impairment. Children with CP without intellectual disability are at increased risk for learning disabilities. However, it is important to recognize that accurate cognitive testing is often challenging in children with limited motor control and verbal output.

Progressive Neurodegenerative Disorders

Children with progressive neurodegenerative disorders present with motor impairments that worsen over time. Children in this category of disorders may have early typical motor development that is then noted to slow, plateau, and then regress or deteriorate. Duchenne muscular dystrophy (an X-linked disorder) will often be diagnosed at 4 to 6 years of age, when the child is noted to be clumsier than his peers, with difficulty running, jumping, and climbing stairs. A classic physical exam feature is Gowers' sign, which is seen when a child uses his hands to push off the floor in order to attain upright posture. Children with Duchenne muscular dystrophy often die in their early 20s.

Myotonic dystrophies often present in adulthood, but they can be quite severe and have congenital or juvenile presentations with poor prognosis. Spinal muscular atrophy types 2 to 4 have adult-onset muscle weakness; however, type 1 (Werdnig-Hoffmann disease) is the most common and the most severe. This type typically presents in the neonatal period with rapidly progressive weakness leading to respiratory failure and poor prognosis. Most children with this type of myotonic dystrophy often die by 2 to 3 years of age. Charcot-Marie-Tooth disease is a collection of hereditary peripheral neuropathies. These neuropathies primarily affect distal sensory and motor nerves, resulting in lower-extremity weakness, foot deformities such as pes cavus or hammer toes, and sensory deficits. Later changes include atrophy of intrinsic hand and foot muscles.

Although progressive neurodegenerative disorders are rare and represent a small number of children who present with motor delays, accurate diagnosis is crucial in order to predict the rate of progression, provide genetic counseling to family members, and deliver intervention for maximizing function and improving quality of life.

DIAGNOSIS

The diagnosis of CP is often suggested by a child's clinical and neurological presentation, and evaluation should begin with detailed medical and developmental histories and physical and neurodevelopmental examinations. Particular attention should be paid to a child's prenatal and neonatal history to detect risk factors, age of attainment of developmental milestones and the presence of dissociated delays in motor development and deviated acquisition of motor milestones, review of the newborn screen, and careful review of the family history to look for relatives with neurological conditions that may suggest a genetic cause for CP. A detailed physical and neurologic exam that includes evaluation of tone, strength, balance, posture, and coordination will help to classify the type of CP. Regression or deterioration of milestones, atrophy or sensory changes, worsening of symptoms with illness, or hypotonia with weakness are suggestive of a neurodegenerative disease or a metabolic disorder.

Neuroimaging should be offered for all children with a new diagnosis of CP. Magnetic resonance imaging (MRI) is preferred over other imaging techniques, as it has been shown to have a higher diagnostic yield and can help determine etiology and timing of injury. MRI abnormalities can be seen in > 80% of children who present with an abnormal neurologic examination. Metabolic and genetic testing should be offered to those with atypical presentations, with suggestive family histories, or with normal neuroimaging or neuroimaging suggesting a developmental brain abnormality (such as lissencephaly/schizencephaly) or for those with no other etiologies identified by history. Metabolic testing should include serum concentrations of glucose, bicarbonate, creatine kinase (CK), ammonia, lactate, pyruvate, plasma amino acids, and urine organic acids. Genetic testing including chromosomal microarray analysis or whole-exome sequencing has been shown to determine a cause of CP in 15% to 30% of children who previously lacked an identifiable etiology. An electroencephalogram (EEG) may also be helpful if there is concern for seizures.

TREATMENT

Treatment of a motor disability is tailored to the principal diagnosis. The core components of the longitudinal management of motor disabilities include (1) timely family-centered care; (2) family involvement in the therapeutic process; (3) maximization of functional independence; and (4) support of transition to adult medical care.

Primary care pediatric medical providers are key to timely family-centered care, as their early, frequent, and longitudinal relationship with children and their families allows for initial identification of parental concerns and early recognition of early signs of existent or evolving motor disorders. This prompt identification will allow for referral to appropriate therapy services and medical subspecialists as needed. In managing children with motor disabilities, the primary care pediatric medical provider should assume a leadership role of a multidisciplinary team (that starts with the child and his or her family and may also include physical therapists, occupational therapists, speech/language therapists, orthotists, dieticians, and medical subspecialists, including physiatrists, neurologists, developmental-behavioral pediatricians, neurosurgeons, and orthopedic surgeons) and assume coordination of care among the team members.

Throughout the duration of treatment, the focus should remain on optimizing developmental potential, decreasing medical comorbidities, and maximizing functional independence and quality of life. This is singularly important for those conditions expected to lead to significant lifelong disability or mortality (eg, muscular dystrophy, myotonic dystrophy, spinal muscular atrophy). From pharmacological, to physical, to surgical, there is a wide array of therapies to address issues associated with motor disabilities. Yet, regardless of the approach that the underlining condition may dictate, the family's active involvement in the therapeutic process is critical, significantly influencing the ultimate outcome. The primary care pediatric medical provider does not necessarily need to become an expert in every facet of care for all of the conditions that may lead to motor disability; however, having a basic understanding of the numerous aspects of care will allow mindful coordination of care and comanagement of comorbidities and complications with subspecialists.

In order to maximize functional independence over a lifetime, interventions must be commenced early on. These interventions can be initially provided through early intervention or early childhood special education programs and then through special education programs at school. Early intervention or school-based therapeutic services can be supplemented by additional private therapy services. Early intervention services may include feeding therapy, developmental therapy, speech therapy, occupational therapy, and physical therapy, and as the child ages, the need for more tailored therapies may emerge, such as augmentative communication therapy or robot-assisted walking therapy. Braces and splints to support positioning, sitting, or ambulation are often needed, as is ambulation equipment, such as walkers, standers, and wheelchairs. Adaptive modifications to bathrooms, bedrooms, access doorways, and automobiles may also be required.

At the center of an effective treatment approach for children with motor disabilities is the management of spasticity, pain, and musculoskeletal deformity. Spasticity can be isolated or generalized; such delineation will help guide management. Isolated spasticity may be treated with injections of onabotulinumtoxinA (Botox) directly into the affected muscle. Generalized spasticity can be addressed through the use of oral muscle relaxants such as dantrolene, diazepam, and baclofen or through neurosurgical implantation of an intrathecal baclofen pump.

Although surgical interventions have historically been aimed at diminishing muscle imbalance and preventing bony and soft-tissue deformity, recent advances in surgical approaches are providing innovative solutions to physiological drivers of motor disabilities. One treatment, dorsal rhizotomy, involves the selective separation of portions of the dorsal roots of the spinal cord, which can diminish lower-extremity spasticity.

No treatment algorithm is complete without a clear pathway to supporting patients through their transition to adult medical care. Rarely is a transition successful without providing the family with accurate information regarding community resources, adult healthcare providers, and information on patient and family support groups. Thus, the primary care office should have access to such information and help the family connect with these resources as part of comprehensive care.

PREVENTION

While most cases of motor disabilities cannot be prevented currently, acquired cases of CP can be prevented. Measures to decrease meningitis (through appropriate immunizations), nonaccidental trauma, motor vehicle accidents, and stroke may contribute to a decrease in incidence of motor disability. Counseling families about unproven alternative therapies is also important in morbidity prevention. Clinicians should be prepared to discuss therapies for which no evidence base exists and to caution families when they are considering alternative therapies that may pose safety or financial burdens.

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89 Delays in Speech/Language or Nonverbal

Development and Learning Disabilities

Sonia Monteiro and Noel Mensah-Bonsu

INTRODUCTION

Language and nonverbal/visual-motor problem solving are components of the neurocognitive stream of development. Because the most common causes of neurodevelopmental disability affect the brain diffusely (eg, genetic disorders), delays in language development are most commonly accompanied by delays in nonverbal/visual-motor problem-solving development and vice versa. Thus, the most prevalent mild neurodevelopmental difficulty that results in school failure is “slower learning,” where both language and nonverbal/visual-motor problem solving are delayed, resulting in intelligence quotients in the borderline (IQ = 70–79) to low average (IQ = 80–89) range. The globally delayed pattern of slower learning occurs in 22.8% of the population, while the dissociated pattern of learning disabilities occurs in only approximately 5% to 10% of the population. In the dissociated pattern of learning disabilities, there is a discrepancy between cognitive abilities and academic achievement, typically with discrepant or dissociated delays in language relative to nonverbal/visual-motor problem-solving development (as observed in language-based learning disabilities) or discrepant or dissociated delays in nonverbal/visual-motor problem solving relative to language development (as observed in nonverbal learning disabilities). It is also important to note that inadequate academic instruction is a common cause of learning problems. This chapter will review the spectrum of dissociated delays in language and nonverbal/visual-motor problem-solving development, including specific language-based and nonverbal learning disabilities.

DEFINITIONS OF SPEECH AND LANGUAGE

Language is defined as a method of both spoken and written communication. Language is made up of multiple components: phonology, morphology, syntax, semantics, and pragmatics. Phonology involves speech sounds (phonemes) that make up words, and morphology involves the units of language (morphemes) that make up words. Syntax is defined as the rules regarding how words can be combined into sentences, including verb tense, word order, and sentence structure. Semantics is defined as the meaning of words in context or when combined into sentences. Pragmatics is defined as the rules for social communication. Language is separated into expressive and receptive components. Expressive language refers to what a child is able to communicate verbally or through the use of signs or gestures. Receptive language refers to what a child is able to understand.

Speech includes the following components: articulation, fluency, and voice. Articulation involves the production of speech sounds and affects the intelligibility of speech. Fluency refers to the flow of sounds, syllables, and words together to form sentences. Voice includes the anatomical function of the vocal folds as well as airflow to produce sounds.

SPEECH AND LANGUAGE DEVELOPMENT

Speech and language development can be divided into 3 periods during early childhood (birth to 2 years of age). The first is the *prespeech* period, which begins at birth. In the first few months of life, an infant will typically progress from alerting to sound to responding to and seeking familiar voices. An infant's cry will start to differentiate based on his or her needs. Cooing (the production of vowel sounds) occurs at around 3 months of age. The infant will begin to combine vowel sounds at around 5 months of age. This will be followed by the production of single consonant sounds, and by 6 months of age, vowel and consonant sounds will be combined together in the form of babbling. By 9 months of age, gestures including reaching to be picked up and waving "bye-bye" emerge.

The second period, known as the *naming* period, occurs between 10 and 18 months of age. It begins with the ability to identify caregivers and objects by name, and this ability progresses to the verbal expression of single words. Prior to 1 year of age, most children begin to use a specific "Mama" and "Dada" as their first words, and by 1 year of age, 1 or 2 additional single words will be used. Most children will start to comprehend words frequently used by caregivers (eg, "bath," "bottle") and will start to follow simple gestured commands by 12 months of age. By 14 months of age, a child may combine sounds or mimic conversational sounds with varying intonation in the form of immature jargon. By 18 months of age, a child may begin to mimic words that they hear (echolalia). Also by 18 months of age, a child should have a vocabulary of about 10 words. Gestures progress during this period with the emergence of pointing. By 12 months of age, a child will start pointing to indicate a want or need (protoimperative pointing), and by 14 months of age, the child will start pointing to obtain the attention of an adult to share something of interest (protodeclarative pointing).

Between 18 and 24 months of age, children will experience a dramatic increase in vocabulary, both receptively and expressively, which marks the *word combination* period. Total expressive language can include up to 50 to 100 words by the end of this period, and a child's receptive vocabulary is usually larger. Also by the end of this period, a child should be combining words into 2-word sentences. From a receptive language standpoint, by 18 months of age, children will also begin to identify body parts and point to pictures when named. By 2 years of age, a child's speech articulation should be at least 50% intelligible to others.

By 30 months of age, echolalia should cease, children should be using pronouns appropriately, and they should be able to distinguish just 1 item from a greater number. By 3 years of age, a child should communicate in 3-word sentences, and their speech articulation should be 75% intelligible. Receptively, a 3-year-old child should be able to follow 2-step directions including prepositions and pay attention for longer periods of time when read to. At 3 years of age, children may begin to ask "what" and "where" questions. By 4 years of age, speech should be completely (100%) intelligible. A child at 4 years of age should be able to relate experiences verbally using complex syntax and speak fluently. Red flags for speech and language delay are shown in **Table 89-1**.

TABLE 89-1 RED FLAGS FOR SPEECH AND LANGUAGE DELAY[a]

Age	Receptive Language	Expressive Language
12 mo	Does not respond to name; does not gesture (wave, point)	Does not babble
16 mo		Does not use single words
18 mo	Cannot follow simple commands ("give me," "come here")	Does not use at least 8–10 words
24 mo	Does not follow a 2-step direction; cannot point to a picture	Does not use any 2-word combinations; speech is not at least 50% intelligible
30 mo		Continues to use echolalia; continues to confuse pronouns
36 mo	Cannot answer "who/what/where" questions	Does not use 3- to 4-word sentences; speech is not 75% intelligible; leaves beginning or ending sounds off words

[a]Loss of speech, babbling, or social skills is a red flag at any age.

PEDIATRIC ASSESSMENT OF SPEECH AND LANGUAGE

Speech and language milestones should be regularly assessed as part of routine developmental surveillance at every well child visit. Surveillance involves asking parents questions and documenting their responses regarding attainment of speech and language milestones, reviewing risk factors (including prematurity, previous concerns on prior evaluation, or concerns about appropriate environmental language stimulation). Parental concerns should be elicited and taken seriously. Surveillance also involves the provider's direct observation of the child's speech and language skills in the office visit.

Standardized screening tests, such as the Ages and Stages Questionnaire (ASQ) and Parents' Evaluation of Developmental Status (PEDS), are commonly used developmental screeners that assess multiple areas of development, including speech and language, and identify children who require further evaluation. Standardized screening tests should be regularly implemented at the 9-month, 18-month, and 24- or 30-month well child visits. Parents of older children should be asked if their child has any difficulty with verbal expression or with following directions.

Once a concern is identified (either through screening or surveillance), a referral for a speech and language evaluation should be made. This can be performed through early intervention or early childhood special education programs or by a private speech/language pathologist. In addition to referral for a speech and language evaluation, an audiology evaluation needs to be completed for every child with delayed speech and language skills to rule out hearing loss. Finally, it is important to note that boys with Klinefelter syndrome (XXY) tend to present with language disorders and language-based learning disabilities.

The differential diagnosis for a child presenting with concerns about delayed speech and language development includes language disorder, speech disorder, speech and language disorder, inadequate environmental stimulation, global developmental delay, social communication disorder, autism spectrum disorder, and hearing loss.

PROMOTION OF LANGUAGE DEVELOPMENT

Developmental milestones in language cannot be attained without appropriate language stimulation in the environment. Previous research has emphasized the importance of exposing children to language from an early age. Parents are encouraged to speak frequently to their infants and describe what is going on around them. It is also critical that parents read to and share books with their children starting in infancy. It has been established that the number of words that children are exposed to can vary significantly based on socioeconomic factors. Children from lower-income homes hear far fewer words than children in middle- or high-income settings. In addition to these gaps, the quality of parent–child interaction in language development, regardless of socioeconomic status, is critically important to language development. Parents should respond to their infants' sounds and attempts at communication to encourage progression in language development. As their children become older, parents should be encouraged to have frequent conversations with their children.

SPECTRUM OF DISSOCIATED SPEECH DELAY: SPEECH DISORDERS

Dissociated delays in speech development are defined as speech disorders. Speech disorders involve difficulties with sounds required in the production of speech and include voice disorder, speech fluency disorder, phonologic disorder, apraxia of speech, and dysarthria.

Voice Disorder

Deficits in pitch, loudness, or vocal quality define a voice disorder. There may also be a hyper- or hyponasal quality to speech that may be secondary to anatomical differences (such as velopharyngeal insufficiency or adenoid hypertrophy).

Fluency Disorder

In children with fluency disorders, the flow of speech is interrupted secondary to repetition of sounds or words and changes in the rate or rhythm of speech. Children between the ages of 2 and 3 years may have pauses, sound prolongations, or may repeat parts of words. This is considered to be a normal stage of language development and may be secondary to the dramatic increase in speech production during this time period. Usually these problems resolve without intervention by 4 years of age. If the dysfluency continues, stuttering is more likely, and referral for evaluation and treatment is indicated.

Phonologic Disorder

Many children will make errors in pronunciation when using new words. Each speech sound has an age by which it should be mastered, and errors in the production or pattern of these sounds result in speech articulation that is difficult to understand. Simple consonant sounds (/b/, /p/, /m/) as well as all vowel sounds are usually mastered by 2 years of age. More complex consonant sounds (/j/, /r/, /s/) as well as blends (/ch/, /sh/) are mastered later on. Children should be able to produce all sounds in the English language by 8 years of age.

Childhood Apraxia of Speech

Childhood apraxia of speech (CAS) is thought to be secondary to deficits in central nervous system mechanisms involved in organizing the motor movements necessary for the production of speech. It is not secondary to muscle weakness or paralysis. Children with CAS have difficulties with spontaneous production of sounds, syllables, and words. Most have a history of delayed cooing or babbling during infancy and delayed production of first words. Compared to children with a phonologic disorder who make consistent errors in speech production, children with CAS have irregular and inconsistent speech production.

Dysarthria

Dysarthria is a motor disorder that is the result of impairment of muscles used in speech production. Muscles may be weak or paralyzed, or there may be problems with coordination. This limits jaw, lip, and tongue movement, resulting in a slow rate of speech, poor articulation, change in voice quality or pitch, and speech that sounds "mumbled" or "slurred." Dysarthria is most commonly observed in children with cerebral palsy (CP).

SPECTRUM OF DISSOCIATED LANGUAGE DELAY

When children exhibit discrepant delays in their language development relative to their development in other developmental streams, they are described as exhibiting a dissociation in their language development. Of all the streams of development, the language stream is the best single predictor of cognitive potential. In addition, the ability to express and understand language has an impact on social functioning. Language ability predicts school readiness, and deficits in early language development are directly correlated to future language-based learning disabilities and decreased academic achievement.

Language Disorders (Specific Language Impairment)

A language delay occurs when the progression of language development is occurring in the correct sequence, but at a rate slower than typical. Language delays occur more frequently in boys and when there is an established family history of language delay, speech/language disorders, or language-based learning disabilities. Additional risk factors include prematurity, lack of appropriate developmental stimulation, and lower socioeconomic status. Language delays should not be attributed to birth order (having siblings who "speak for" the child) or to the child being exposed to multiple languages. By 4 years of age, many children with a history of speech and language delays, especially those who have experienced a lack of appropriate language stimulation before receiving appropriate early intervention and early childhood special education services, will "catch up" and not have persistent language difficulties. However, many children will continue to exhibit persistent language delays. Preschool-aged children with dissociated delays in their language development currently meet DSM-5 criteria for a diagnosis of language disorder. However, children with persistent language delays as they enter school are at significant risk for developing language-based learning disabilities.

Children with a language disorder, also known as a specific language impairment, struggle to use language to express themselves and have difficulties understanding the messages of others. Children with language disorders can have discrepant delays in all or just 1 of the components of language (phonology, morphology, syntax, semantics, and pragmatics). These problems with language development are not secondary to hearing loss or to a language delay that is observed as a component of a global developmental delay. Children with language disorders by definition are not globally delayed, and most have typical nonverbal abilities, adaptive functioning, and social skills. Children with language disorders may initially present with a chief complaint of "delayed speech." However, once they start to use words, they may continue to struggle with following directions, being able to form sentences or use verb tenses correctly, or having a conversation.

Language disorders can involve both receptive and expressive language or just expressive language. They cannot involve only receptive language with spared expressive language, as a child can never express what he or she does not understand. A child with a receptive language disorder will have difficulty with understanding what is said and with following directions. Children with an expressive language disorder present with problems expressing their thoughts or needs. Their sentences may be shorter and less complex than what would be expected for their age or developmental level. They may mix up the order of words in sentences, leave words out, or make errors in verb tense. Most children with language disorders will have deficits in both expressive and receptive language. Children with a language disorder are at increased risk of language-based learning difficulties, even those children who appeared to have progressed to having typical spoken language. Children with both speech and language disorders are also at increased risk of emotional and behavioral problems.

Language-Based Learning Disabilities

Definitions of Learning Disability A learning disability (LD) has been classically defined as an unexpected discrepancy in academic performance compared to intellectual potential; however, this definition has been challenged and remains controversial. The *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition (DSM-5) characterizes "specific learning disorders" as being evidenced by academic skills that are substantially and quantifiably below those expected for an individual's chronologic age, which have persisted for at least 6 months despite the provision of targeted interventions, and that are not due to intellectual disability, uncorrected vision or hearing deficits, other mental or neurological disorders, psychosocial adversity, lack of proficiency in the language of academic instruction, or inadequate academic instruction. Thus, children with specific learning disorders must now show a lack of response to intervention before being considered to have a learning disability by many school districts. Previously, most children with slower learning (IQs between 70 and 90), who often struggled to keep up with average and above-average peers in regular classroom placements, did not qualify for any special education services. This is because the majority of children with slower learning, despite having academic achievement scores that are below grade level, have academic achievement abilities that are commensurate with their cognitive expectations, and thus show no discrepancy between cognitive abilities and academic achievement. However, the new DSM-5 definition qualifies students under the specific learning disorder diagnosis whose academic achievement is below that expected for chronologic age. Under this definition, a majority of children with slower learning may now qualify for special education services. Although these children do not have academic achievement that is below expected for their cognitive abilities, their academic achievement is certainly below that expected for their chronologic age. Hopefully, this change in definition will provide children who learn more slowly the same remedial special education services received by children with specific learning disabilities, particularly as research has shown these interventions to be effective in improving academic outcomes no matter whether an individual has a discrepancy between cognitive abilities and academic achievement or not.

Spectrum of Language-Based Learning Disabilities Children with preschool language disorders are at significant risk of developing language-based learning disabilities as they enter school. Children with language disorders can have discrepant delays in all or just 1 of the components of language (phonology, morphology, syntax, semantics, and pragmatics). Thus, the mild end of the spectrum of dissociated language delay would include discrepant delays in 1 component of language (eg, phonology), while the severe end would include discrepant delays in multiple components of receptive and expressive language, which is associated with discrepant verbal reasoning compared to nonverbal reasoning on IQ testing (**Fig. 89-1**).

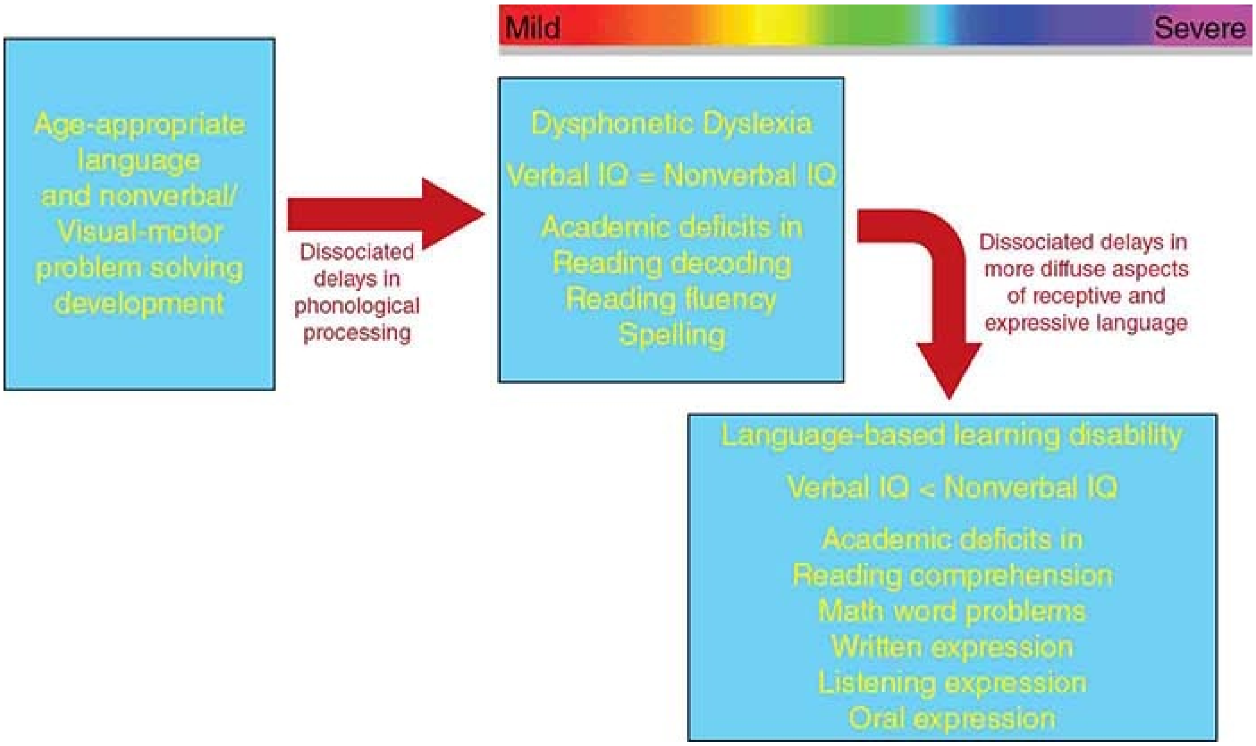

FIGURE 89-1 Spectrum of language dissociation.

Thus, the mild end of a spectrum of dissociated language delay includes individuals with dissociated difficulties in phonology, resulting in phonological processing disorders (Fig. 89-1). Children with difficulty with phonology have trouble learning to associate written symbols with the sounds they make, leading to dysphonetic dyslexia. Children with dysphonetic dyslexia have trouble sounding out words, and thus, they have difficulty with reading decoding, reading fluency, and encoding (spelling). However, children with dysphonetic dyslexia have intact nonverbal abilities, and they can memorize sight words. This skill may allow a child with dysphonetic dyslexia to compensate early on at school, but their reading disability becomes more apparent in advancing grades, when the focus shifts from "learning to read" to "reading to learn." In addition, despite their difficulties with phonology, children with dysphonetic dyslexia have otherwise intact language abilities. Thus, while they may not be able to obtain information through reading, they can understand and obtain information through being read to or through listening to books on tape. Similarly, while their difficulties with encoding negatively impact the written expression required of written reports, they can give oral reports without significant incident.

As more components of receptive and expressive language become dissociated, one moves toward the more severe end of the spectrum of dissociated language delay, and children experience language-based learning disabilities. These more severe delays—which may diffusely affect phonology, morphology, syntax, and semantics—negatively impact verbal reasoning, and children with language-based learning disabilities typically exhibit verbal IQ scores that are dissociated from nonverbal IQ scores. Rather than simply affecting reading decoding and encoding, language-based learning disabilities result in difficulties in reading comprehension, math word problems, written expression, listening comprehension, and oral expression. Children with language-based learning disabilities would have similar difficulties with comprehension whether attempting to read themselves or when being read to or listening to books on tape.

Children suspected of having dyslexia or language-based learning disabilities need to undergo a full and individual evaluation through their local public schools and be provided with direct language therapy services, remedial academic instruction, and maximal accommodations and modifications of all assignments, materials, texts, pacing, and grading when included in regular classroom settings.

NONVERBAL/VISUAL-MOTOR PROBLEM-SOLVING DEVELOPMENT

Nonverbal/visual-motor problem-solving development involves the development of visual perceptual, visual-spatial, and visual-motor skills. In the first 3 months of life, this domain of development focuses on visual tracking, as children should be able to visually track through 360 degrees by 3 months of age. As this domain of development matures, skills include reaching for and transferring objects at 5 months, using an immature pincer grasp to pick up pellet-sized objects by 9 months, uncovering hidden toys by 10 months, and intentionally releasing objects into containers by 12 months. After 12 months, drawing and block construction skills emerge, with making a crayon mark at 12 months, scribbling spontaneously and stacking 3 blocks at 18 months, imitating a horizontal and vertical stroke and building a horizontal train of blocks at 24 months, and drawing a circle and building a 3-block bridge at age 3 years. Many adaptive skills also rely on nonverbal/visual-motor skill development, such as spoon feeding by 14 months, unzipping by 21 months, unbuttoning by 3 years, buttoning by 4 years, and tying shoes by 5 years.

PEDIATRIC ASSESSMENT OF NONVERBAL/VISUAL-MOTOR PROBLEM SOLVING

Nonverbal/visual-motor problem-solving milestones should be regularly assessed as part of routine developmental surveillance at every well child visit. It may be more difficult to obtain a developmental history of nonverbal/visual-motor problem-solving skill acquisition from families; for example, they may not have exposed their children to crayons or blocks. Thus, the nonverbal/visual-motor problem-solving developmental history may need to focus on activities of daily living that rely on visual-motor problem-solving skills, such as feeding and dressing. Fortunately, while it may be difficult to elicit a developmental history in this domain, nonverbal/visual-motor problem solving is the domain that is easiest to observe in the office, as the items used in this domain are fun for children to complete (drawing, building with blocks, completing puzzles).

Standardized screening tests, such as the ASQ and PEDS, assess multiple areas of development, including nonverbal/visual-motor problem solving, and identify children who require further evaluation. Standardized screening tests should be regularly implemented at the 9-month, 18-month, and 24- or 30-month well child visit. Older children can be asked to draw pictures of a person or to provide a handwriting sample.

Once a concern is identified (either through screening or surveillance), a referral for an occupational therapy evaluation should be made. This can be performed through early intervention or early childhood special education programs or by a private occupational therapist. In addition to referral for an occupational therapy evaluation, a vision screen or ophthalmology evaluation needs to be completed for every child with delayed nonverbal/visual-motor problem-solving skills to rule out a vision impairment. Finally, children with specific medical diagnoses appear at higher risk for nonverbal/visual-motor problem-solving disorders and nonverbal learning disabilities, including children with Turner syndrome (XO), hydrocephalus/spina bifida, and velocardiofacial syndrome (deletion of chromosome 22q11.2).

SPECTRUM OF DISSOCIATED NONVERBAL/VISUAL-MOTOR PROBLEM-SOLVING DEVELOPMENT

Nonverbal/Visual-Motor Problem-Solving Disorders

Children may exhibit dissociated delays in nonverbal problem solving in multiple domains, including visual discrimination, visual figure-ground discrimination, visual sequencing, visual-motor processing, visual memory, visual closure, and visual-spatial relationships. Children with nonverbal/visual-motor problem-solving disorders will evidence dissociated delays in the milestones reviewed above. School-aged children with these delays will have difficulty with drawing, writing, right/left orientation, and completing puzzles, and may get lost in familiar places.

Spectrum of Nonverbal Learning Disabilities

When children with persistent nonverbal/visual-motor problem-solving disorders enter school, they are at increased risk of developing nonverbal learning disabilities. At the mild end of the spectrum of nonverbal learning disabilities are children whose dissociated delays in nonverbal development involve primarily their orthographic processing (**Fig. 89-2**). Orthographic processing is the use of the visual system to form, store, and recall written letters and words and also involves using the visual system to process punctuation and capitalization. Children with orthographic processing deficits are at risk for developing dyseidetic dyslexia. Children with dyseidetic dyslexia have problems memorizing the pattern of letters (eg, they may confuse “b” and “d”) and difficulty memorizing sight words. Given their preserved phonological processing, children with dyseidetic dyslexia need to laboriously sound out the same word over and over, and they tend to misspell phonetically.

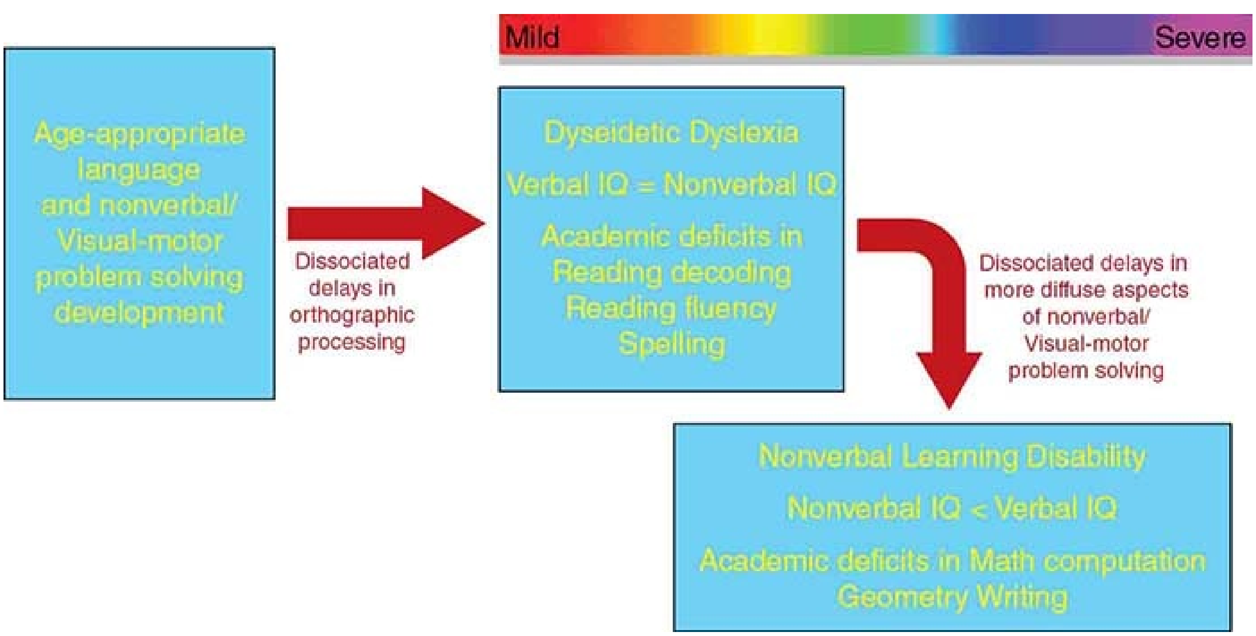

FIGURE 89-2 Spectrum of nonverbal/visual-motor problem-solving dissociation.

As more components of nonverbal/visual-motor problem solving become dissociated, one moves toward the more severe end of the spectrum of dissociated nonverbal/visual-motor problem-solving delay, and children experience nonverbal learning disabilities. These more severe delays, which may diffusely affect visual discrimination, visual figure-ground discrimination, visual sequencing, visual-motor processing, visual memory, visual closure, and visual-spatial relationships, negatively impact nonverbal reasoning, and children with nonverbal learning disabilities typically exhibit nonverbal IQ scores that are dissociated from verbal IQ scores. Rather than simply having difficulty with reading sight words, nonverbal learning disabilities result in difficulties in writing (dysgraphia), drawing, right/left discrimination, completing puzzles, math computation and processing (dyscalculia), geometry, geography, and getting lost in familiar places, and may involve understanding spatial relations, including social reasoning. (The most severe end of the spectrum of nonverbal dissociation + deviation includes the social communication disorder typically observed in children previously described as having Asperger disorder, but currently described as having autism spectrum disorder without a language disorder; see Chapter 83 and Chapter 91.)

Children suspected of having dyseidetic dyslexia or nonverbal learning disabilities need to undergo a full and individual evaluation through their local schools and be provided with direct occupational therapy services, remedial academic instruction, and maximal accommodations and modifications of all assignments, materials, texts, pacing, and grading when included in regular classroom settings.

COMORBIDITIES OF LEARNING DISABILITIES

In the continuum of developmental-behavioral diagnoses, dissociated delays in 1 stream of development are often associated with dissociated delays in other streams of development (see Chapter 83). Thus, children with the dissociated neurocognitive developmental profile of learning disabilities often also have the dissociated neurobehavioral developmental profile of attention-deficit/hyperactivity disorder (ADHD) and the dissociated neuromotor developmental profile of speech articulation disorders, handwriting difficulties (dysgraphia), and gross motor incoordination (dyspraxia). Up to 50% of children with ADHD have comorbid learning disabilities in reading (dyslexia), math (dyscalculia), or written expression.

Secondary social, emotional, or behavioral comorbidities are particularly concerning in children with learning disabilities. An unrecognized learning disability may result in school failure. Signs and symptoms of academic distress secondary to an unrecognized learning disability include increased time and effort to complete classroom assignments, anxiety or avoidance of school, acting out, failing grades, or grade retention. It is critical for primary care pediatric medical providers to know that grade retention has not been found to improve a child's educational outcome, and those who have been retained are more likely to have behavioral issues and to eventually drop out of school. Children who have been recognized by their schools to have learning disabilities require remedial special educational instruction in their areas of disability. However, just as importantly, they require maximal accommodations and modifications in order to be successfully included in regular classroom activities. Children with learning disabilities should not be expected to compete with similarly aged peers who do not share their specific learning disabilities in a regular classroom without maximal accommodations and modifications of all assignments, teaching materials, texts, testing, pacing, and grading. Children with learning disabilities should not be expected to attend to academic material that they do not understand or to complete assignments that are beyond their current level of ability. Maximal accommodations and modifications are required in order to ensure that demands and expectations for academic performance in the regular classroom are made commensurate with underlying abilities. Without such accommodations and modifications, demands for a child's performance will exceed his or her abilities, and this could produce secondary anxiety and frustration and lead to social, emotional, or behavioral difficulties, including low self-esteem, low self-confidence, school negativity, social withdrawal, task-avoidant and passively resistant behaviors (secondary inattention), and potentially school dropout. Such a mismatch between demands and expectations and performance also can result in attention-seeking, oppositional, or acting-out behaviors (that can be misperceived as secondary impulsivity or hyperactivity), which can lead to detention or school suspension. The regular classroom teachers of children with learning disabilities need to work very closely with their schools' learning disabilities specialists to ensure that maximal accommodations and modifications are being made in all regular classes. Examples of such accommodations and modifications may include being given extended time to complete shortened and maximally modified assignments, untimed tests, being provided audio textbooks or a calculator, and being allowed to give oral reports and oral answers to essay questions rather than written reports and essays. It is also very important for children with learning disabilities to participate in extracurricular activities that they enjoy, and in which they feel they are successful, in order to serve as a source of self-esteem building, socialization, and peer interaction. Extracurricular activities at school are particularly encouraged, so that school can remain a rewarding experience.

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90 Attention-Deficit/Hyperactivity Disorder

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INTRODUCTION

In the spectrum and continuum of developmental-behavioral diagnoses, attention-deficit/hyperactivity disorder (ADHD) represents a dissociation between cognitive abilities and the neurobehavioral domains of attention, impulse control, and motor activity. Individuals with ADHD have developmentally inappropriate levels of inattention and/or impulsivity/hyperactivity (ie, their levels of attention and impulse control/activity are discrepantly delayed or dissociated from their underlying cognitive abilities) that are impairing their functioning across settings.

ADHD is a frequently diagnosed but often misunderstood neurodevelopmental condition. The balance of the core features of inattention and/or hyperactivity and impulsivity combined with the complex array of socialization difficulties, impaired executive functions, and emotional dysregulation make ADHD a great masquerader for a variety of developmental/behavioral and psychiatric diagnoses. Therefore, understanding the many facets of this disorder can enhance diagnostic veracity and allow for a targeted approach to treatment.

EPIDEMIOLOGY

ADHD is a disorder that begins in childhood, and symptoms are present from a young age. Several studies have been published on the prevalence of ADHD across populations and cultures, with most citing a prevalence between 4% and 12%. This relatively wide range reflects variations in diagnostic criteria, characteristics of the studied population, the number of sources used to establish the diagnosis, and the lack of biological markers for the disorder. Studies utilizing the diagnostic criteria of any edition of the *Diagnostic and Statistical Manual of Mental Disorders* have shown prevalence to be closer to 6% to 8%. ADHD is more prevalent in males than females, with a ratio of 3:1. Males are more likely to evidence the externalizing behaviors of hyperactivity and impulsivity (5:1 compared to females) and are therefore more likely to be identified early on, such as in preschool and early elementary school. Females tend to present with inattentive symptoms (2:1 male-to-female ratio) that may go undetected until later in childhood, such as middle to late elementary school. By early adolescence, hyperactivity mostly manifests as fidgetiness, restlessness, or impatience, while inattention persists.

ETIOLOGY

Organic causes of ADHD can be found in 20% to 25% of cases and may include a history of very low birthweight (< 1500 grams), very preterm birth (gestational age < 32 weeks), fetal alcohol spectrum disorders, in utero drug exposure, environmental toxins (eg, lead, nicotine), and central nervous system trauma or infection. In the majority of cases, however, ADHD is thought to be multifactorial in etiology, drawing from genetic and environmental effects on neurobiology. Studies have found inherent differences in the structure and function of several areas of the brain, notably the basal ganglia (contributes to inhibition of automatic responses), the cerebellar vermis (contributes to regulation of motivation and coordination), and the prefrontal cortex of the frontal lobe. The latter in particular controls the executive functions of organization and regulation. Organization functions include attention, "filtering" (attending to relevant while blocking out irrelevant stimuli), planning, sequencing, rule acquisition, abstraction, and cognitive flexibility required for problem solving. Functions of regulation include initiation, inhibition, emotional regulation, monitoring (of internal and external stimuli), moral reasoning, and decision-making. Working memory and processing speed are 2 cognitive functions commonly measured on intelligence batteries that are used as surrogate indicators for executive function.

On a molecular basis, dopamine and norepinephrine imbalances have been implicated in the pathogenesis of ADHD. Both of these neurotransmitters may enhance the inhibitory effects of the frontal cortex on subcortical structures. Evidence also exists for a delay in cortical maturation (particularly in the lateral prefrontal cortex) in individuals with ADHD compared to the general population.

ADHD is polygenic with a significant component of heritability, as demonstrated in identical twin studies with > 50% concordance, and clinically by increased occurrence in first-degree relatives by 2- to 8-fold compared to the general population. Gene associations have been found with the dopamine transporter gene (*DAT1*), the D4 receptor gene (*DRD4*), and the human thyroid receptor-beta gene.

CLINICAL MANIFESTATIONS

The hallmark characteristics of ADHD are developmentally inappropriate and persistent difficulties with inattention and/or hyperactive and impulsive behaviors that cause impairment in daily functioning across numerous settings. *Inattention* is characterized by a lack of focus, perseverance, or mental or physical presence for an expected, nonpreferred task or situation. Teachers may report that a student “stares off” in class or does not seem to know what is going on when called on. Inattentive students may be unable to filter out background noises to attend to the most salient stimulus (eg, the teacher). Socially, individuals with ADHD may be perceived as careless, disorganized, disinterested, or unmotivated.

Although *hyperactivity* is often equated with excessive motor activity (such as running about or climbing), it can also manifest as talkativeness or fidgetiness. Individuals may be perceived as being restless, “high-energy,” “the class clown,” or “a chatterbox.” *Impulsivity* may refer to a predilection for rash decisions or actions, which may result in unintended harm to oneself or others. Parents may report younger children running off from them in a parking lot or acting out physically when they do not get their way. Socially, impulsive individuals may be perceived as intrusive (eg, interrupting conversations), attention-seeking, “bossy,” short-sighted (ie, poor comprehension of both long-term rewards and consequences of decisions), and impatient. Impulsive individuals may also have “no filter,” meaning that they may make inappropriate remarks or say the first thing that comes to mind without regard for how the comments will be perceived. They may be labeled as “lacking common sense.”

It is important to note that symptoms of ADHD can vary in intensity depending on the setting. Symptoms tend to be more apparent in unstructured settings or when the child is bored. A child who is engaged in a highly motivating activity (such as a videogame) may not display hyperactive, impulsive, or inattentive behaviors and may instead seem to “hyper-focus” on the game. This highlights the important point that ADHD is not a disorder of complete lack of attention; it is a disorder of *attention allocation*, ie, difficulty focusing for a nonpreferred task. Symptoms may also be diminished when receiving 1-on-1 attention and/or positive reinforcement for desired behaviors, when in a new setting, or when otherwise engaged in something of interest.

DIAGNOSIS

The *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition (DSM-5) lists the current clinical criteria for diagnosis of ADHD and its subtypes in children and adolescents younger than 17 years of age. The predominantly inattentive subtype of ADHD requires that at least 6 of 9 symptoms of inattention be present over a 6-month time span, while the hyperactive/impulsive subtype of ADHD requires at least 6 of 9 corresponding symptoms. The combined type presentation requires at least 6 in both categories of symptoms. The requirement of symptoms being present for at least 6 months is meant to eliminate or mitigate contributory effects of acute factors, such as life changes (eg, new school, divorce) or early stages of chronic illness. Although meant to be a list of essential diagnostic features, the DSM-5 checklist is certainly not a substitute for understanding a child's behavior in the broader context of developmental level, which represents the crucial clinical judgment piece necessary to make an ADHD diagnosis.

The DSM-5 specifies that symptoms of ADHD must be present to a significant degree prior to 12 years of age. This represents an expansion of the previous DSM-IV's age cutoff of 7 years to account for later detection of symptoms. The DSM-5 notes the importance of understanding a child's developmental level (ie, level of cognitive function) when considering a child's behavior. In other words, a 10-year-old child with a cognitive level of a 5-year-old will behave as a 5-year-old, even though that child might be perceived as hyperactive, impulsive, and inattentive for chronological age. Likewise, presenting a child with a challenge above that child's level of ability (eg, placing a "slower learner" in a typical classroom) may result in behaviors that can be misconstrued as symptoms of ADHD.

In order to make a diagnosis of ADHD, the presenting child should also experience a significant degree of impairment. Since the clinical and developmental histories are imperative to making the diagnosis, balancing perceptions against actual performance via feedback from multiple sources (eg, parents, teachers, therapists, report cards) can help assess this criterion. Several clinical tools exist to allow providers to collect data from several observers (eg, National Institute for Children's Health Quality [NICHQ] Vanderbilt Assessment Scales, online initial and follow-up versions available at http://www.nichq.org/childrens-health/adhd/resources/vanderbilt-assessment-scales). As a rule, symptoms must be developmentally inappropriate, persistent, pervasive, and impairing in order to meet criteria for ADHD.

Impairment due to ADHD symptoms is most often recognized when school performance diminishes, behavioral problems in the classroom interfere with the child's and with other students' learning, or when a child seems to be underperforming academically compared to his or her ability. Children with ADHD are more likely to be identified as "a behavior problem," leading to less time in the classroom and more time in the principal's office or at home due to suspension. Alternatively, a child may be placed in a behavior classroom where academics are emphasized less. Deficits in learning may result, even in the absence of a diagnosed learning disability, which may require special education intervention. Children with ADHD are also often made to repeat grades due to their behavioral immaturity and associated poor academic performance, and this further contributes to the increased rate of school attrition in teens with ADHD.

Social impairment due to rejection and neglect by peers may be less apparent, especially if the child with ADHD is blamed for lack of social acceptance due to intrusive or inconsiderate behaviors. Children with ADHD are at increased risk for teasing and bullying and may be targeted because of impulsive overreactions. Impairment in family function can occur if the child with ADHD has to constantly be redirected, prompted, reminded, or supervised. Medically, impairment may manifest as increased frequency of ER visits due to accidental injuries secondary to impulsive and hyperactive behaviors.

There may be some instances in which impairment is reported but is unsupported by feedback from other reporters. For instance, a cognitively average child in a high-achieving family may be viewed as "lazy" or "unmotivated." New parents may report that a toddler is too active and difficult to control. A teacher may report that a very bright child is inattentive in a typical classroom, when in fact the child is merely bored. In these instances, education of parents about developmentally appropriate behaviors, and discussion with educators about developmentally appropriate classroom demands and expectations, may be all that is needed. Cultural interpretations of behavior should also be explored.

DIFFERENTIAL DIAGNOSIS AND COMORBIDITIES

Though the DSM-5 specifies that ADHD symptoms must not occur "solely as a manifestation of oppositional behavior, defiance, hostility or failure to understand tasks or instructions," all of these can occur as comorbidities. **Table 90-1** lists comorbid conditions, many of which must also be considered in the differential when making a diagnosis of ADHD. For instance, children with learning disabilities and speech/language disorders may be misconstrued as having ADHD, when in fact they are struggling with language comprehension and/or language production. An unidentified intellectual disability or autism spectrum disorder may be incorrectly perceived as evidence of behavioral immaturity or defiance. Victims of neglect and/or violence may be perceived as inattentive and unmotivated in the classroom; conversely, children with ADHD are at increased risk of victimization. This again speaks to the importance of obtaining feedback from multiple reporters.

TABLE 90-1 ADHD DIFFERENTIAL DIAGNOSIS AND POSSIBLE COMORBIDITIES

Medical
Anemia
Chronic medical illness (multiple absences or fatigue)
Malnutrition/insufficient caloric intake
Medication side effects (antiepileptic drugs, antihistamines, antihypertensive drugs, bronchodilators)
Sleep disturbance (including sleep-disordered breathing)
Thyroid dysfunction (hypo- or hyperthyroidism) or thyroid medications
Neurodevelopmental
Autism spectrum disorder
Developmental coordination disorder
Epilepsy
Hearing impairment
Intellectual disability/global developmental delay
Learning disabilities or slow learning
Language disorders (including articulation and speech sound disorders)
Neurodegenerative disorders
Tourette syndrome/tic disorder
Visual impairment
Behavioral/Mental Health
Adjustment disorders/reactive emotional states
Anxiety disorders
Bipolar and related disorders
Conduct disorder
Depressive disorders
Dissociative disorders
Obsessive-compulsive and related disorders
Oppositional-defiant disorder/disruptive behavior disorder
Personality disorders
Post-traumatic stress disorder
Somatic disorders
Thought disorders (including schizophrenia and psychotic disorders)
Psychosocial
Inconsistent parenting practices (including lack of parenting due to parental psychopathology or mental health issues)
Lack of structure in the home/unregulated environment
Poor limit setting
Poverty
Substance abuse (including parental use)
Trauma (abuse, neglect, violence, victim of bullying)

Common comorbid conditions with ADHD include oppositional-defiant disorder (up to 35%, though closer to 50% for combined type and 25% for predominantly inattentive type), conduct disorder (up to 25%), anxiety disorders (up to 25%), depressive disorders (up to 20%), and learning disorders (up to 50%). Motor coordination disorders (ranging from general "clumsiness" to dysgraphia) and tics also commonly co-occur with ADHD. Many children with ADHD also suffer from sleep disturbance, often due to an inability to calm their minds for sleep. Finally, children with ADHD may experience low self-esteem.

Difficulties with emotional regulation can result in overreactions to situations and then difficulty de-escalating. For instance, low frustration tolerance when a demand is placed on the child or when a desired object is denied can quickly lead to oppositional and defiant behaviors. Children with ADHD may be perceived as being "moody" or "irritable," even in the absence of a diagnosed mood disorder. Those with comorbid mood and/or conduct disorders, particularly in the context of substance abuse, are at increased risk for suicide attempts by early adulthood. Diagnostic surveys that include symptoms of common comorbidities, such as the NICHQ Vanderbilt Assessment Scale or the Conners 3rd Edition of the Comprehensive Behavior Rating Scales, can help identify co-existing conditions.

In-office evaluation should include vital signs, anthropometrics, hearing and vision screens, a comprehensive medical and developmental-behavioral history, a complete physical examination, a thorough neurological examination, and direct developmental evaluation. Further evaluation may include an audiological examination (or at least a hearing screen) to rule out a previously undetected hearing impairment and an ophthalmologic consultation (or at least a vision screen) to rule out a previously undetected visual impairment. A thorough medical history, including birth history, past medical history, current medications, review of systems, family history, and social history can help guide any further medical testing. A history concerning for staring spells or other types of seizure activity would necessitate a sleep-deprived EEG and possibly brain MRI and neurological consultation, especially with any abnormal neurological exam findings. It is important to note that children with ADHD often evidence motor overflow and difficulty with rapid alternating movements (neurological "soft signs"). Dysmorphic features could indicate the potential need for genetic testing (such as chromosome microarray analysis and DNA testing for fragile X syndrome), as there are genetic syndromes that are highly associated with ADHD (eg, Klinefelter syndrome, Turner syndrome, 22q11.2 deletion [velocardiofacial] syndrome, neurofibromatosis 1) and with ADHD in a setting of intellectual disability (eg, fragile X syndrome, Williams syndrome, fetal alcohol syndrome). It is also important to consider inborn errors of metabolism, though medical history, developmental history, and review of newborn screening should help guide appropriate studies. In a primary care setting, several visits may be necessary to accomplish a full evaluation and to accumulate instrumental data.

Referral for psychological assessment or speech/language evaluation may be necessary to identify an underlying cognitive or learning disability. These assessments can be performed by school psychologists or through providers in the community. A mental health assessment is recommended for a suspected underlying or comorbid psychiatric disorder. Social work consultation can provide insight into the psychosocial situation and family dynamic within which the child is functioning.

TREATMENT

Treatment of ADHD requires a multipronged, multidisciplinary approach. Demystifying the ADHD diagnosis and clarifying any misconceptions for families is the first step. ADHD is a neurodevelopmental (ie, brain-based) disorder and not the result of poor parenting or intentional misbehavior on the part of the child. Exploring mistaken beliefs about ADHD may help to relieve guilt, avoid blame, and alleviate stress and frustration, as well as increase the likelihood of adherence to a treatment plan. Anyone who has a role to play in any aspect of the child’s daily functioning—family/caregivers, educators, the school’s diagnostician or psychologist—needs to be a part of the treatment plan developed with the clinician. Treatment plans incorporating medication management, behavioral modification across settings, and development of life-long ADHD management strategies produce the best results.

Though medications can greatly decrease symptoms of ADHD, they are only effective when taken and are by no means curative. Many parents are wary of medications and would prefer a nonpharmacologic approach, making it all the more important to emphasize targeting the behavioral and cognitive aspects of ADHD, especially because this is what produces long-term benefits. The child with ADHD must learn coping strategies appropriate to developmental level at every stage of childhood and adolescence, so as to be well-prepared in adulthood. Early on, coping strategies will need to be modeled and provided externally until the child is able to internalize them. For instance, behavioral modification techniques, such as the use of praise, reward charts, and token economies (eg, earning stars that can be exchanged for tangible rewards), can help a child develop a new skill and reinforce a desired behavior until that behavior becomes routine. Praise should be frequent and specific, and it is highly effective when least expected (eg, "I like the way you are sitting quietly"). Understanding natural consequences (eg, not being able to watch a cartoon in the morning before school because it took too long to get dressed), losing privileges for undesired behaviors, and time-out also help modify behavior and are punctuated by maintaining a mostly positively reinforcing environment. Adults should model and help facilitate appropriate reactions to disappointment, problem-solving, and resolution of conflict. Having a daily report card that passes between home and school lets the child know that good behaviors are important across settings. Consistency within a setting and across settings is key for behavior management.

Parents and teachers need to be aware that they need the child with ADHD to look at them when giving directions; asking the child to repeat back the directions confirms what was heard. Keeping directions simple and brief, being clear about expectations and rules, and setting time limits for tasks can also be helpful. Breaking up large tasks into smaller tasks is imperative because of deficits in executive functions. For instance, making a calendar of what to accomplish week by week for an upcoming project, posting it in the child's room, and checking off accomplished items can avoid procrastination. Picture schedules for daily tasks (eg, steps for what to do in the shower), a checklist for chores, establishing a homework-only folder, and going through a child's backpack every day and getting it ready for the next day are all simple things that can be done to help a child function more successfully at home and school; these responsibilities can be passed on to the child as she or he gets older. Overall, behavior management can help to reduce oppositional and noncompliant behaviors, decrease internalizing symptoms (eg, feeling like a failure), increase positive self-image, and foster a feeling of being in control of symptoms—all of which may allow for lower doses of medication and long-lasting self-management benefits. Likewise, involvement in extracurricular activities that accentuate the child's strengths (eg, creativity, motoric abilities, intrepidness) and develop the child's natural interests can also help to build a positive self-concept and provide an outlet for energy that may not exist in the academic setting.

Parents should also be informed of Section 504 of the Rehabilitation Act of 1973, which addresses learning needs. Section 504 prohibits any program receiving federal funds from discriminating against an individual with a disability, defined as a “physical or mental impairment which substantially limits 1 or more major life activities, having a record of such an impairment, or regarded as having such an impairment.” “Major life activities” are further defined as “caring for oneself, walking, seeing, hearing, speaking, breathing, working, performing manual tasks, and learning.” Section 504 specifies that “reasonable accommodations” must be made. However, since schools do not receive extra funds to provide these services, 504 accommodations are not well-defined or regulated. What is offered may vary from district to district, school to school, and even classroom to classroom. Parents often report that the child’s teacher for a new school year is not aware that the child is on a Section 504 plan, so it is important to encourage parents to establish good rapport with the classroom teacher early in the year, remind the teacher that the child is on a Section 504 plan, and follow up frequently with the teacher for feedback. Section 504 accommodations can include extra time for classwork and tests, shortened or reduced assignments, testing in a quiet environment, use of a study carrel, permission to stand while doing seat work, preferential seating close to the teacher, special cues the teacher gives the student to get attention, and “movement breaks,” to name a few. It is important to emphasize that “extra time” for assignments and tests should not interfere with recess or lunch, as these are important socialization opportunities and built-in breaks that children with ADHD need to have as much as their peers. In fact, social skills groups at school, such as a “lunch bunch,” can greatly help to address this area of deficit.

For children with moderate to severe symptoms of ADHD, it is recommended that the family seek services under the Individuals with Disabilities Education Act (IDEA). IDEA allows for services for individuals with ADHD under a designation of other health impairment (OHI), which accounts for “limited strength, vitality or alertness, including a heightened alertness to environmental stimuli, that results in limited alertness with respect to educational environment” and adversely affects the child’s educational performance. Specific accommodations and modifications are then put into place via an individualized education program (IEP) and monitored over time according to established goals and objectives. The IEP is a legal document binding the school to these services and follows the student year to year. Although poor academic performance and a request for services under OHI should automatically initiate a full and individual evaluation (FIE), which includes cognitive, achievement, and speech/language testing in addition to behavioral assessment via observation and multi-rater checklists, this is not always the case, and private psychological assessment may be sought instead. Due to the high correlation between ADHD and learning disabilities and the susceptibility to mistake one for the other, a thorough psychological assessment is recommended.

Medication

Though developing behavioral and cognitive strategies is an important part of ADHD management and may mitigate symptoms initially, medication is an important adjunctive therapy that oftentimes can help a child be more receptive to environmental modifications. Methylphenidate, mixed amphetamine salts, and dextroamphetamine all have US Food and Drug Administration (FDA) indications for treatment of ADHD in children. Atomoxetine, long-acting guanfacine, and long-acting clonidine are nonstimulant medications that also have FDA indications for treatment of ADHD.

Stimulants Stimulant medications are considered the first-line treatment of ADHD symptoms because they address all 3 of the core symptoms of ADHD, primarily by increasing dopamine and norepinephrine activity through inhibiting reuptake most notably in the caudate nucleus and prefrontal cortex. They have demonstrated safety and efficacy, with usage dating back to the 1930s, when amphetamines were prescribed to increase alertness in fighter pilots. Methylphenidate was released for general use in 1957 and was marketed for a number of conditions, including fatigue, narcolepsy, and to counteract the sedating effects of other drugs. Their role in treatment of "hyperkinesis" was not recognized until the 1960s.

Stimulants are effective in up to 70% to 80% of children at reducing the core symptoms of ADHD, when the medication is "on-board". By reducing impulsivity, stimulants can consequently also reduce oppositional/defiant and aggressive behaviors. Methylphenidates primarily block reuptake of dopamine and norepinephrine and secondarily facilitate their direct release from the neuron. Amphetamines, on the other hand, mostly increase the release of both of these neurotransmitters from storage sites into the synapse. This difference in principal mechanism of action may explain why some individuals respond better to 1 class of stimulant than the other.

Methylphenidate (best known under the trade name Ritalin but also including Metadate, Daytrana, and Quillivant) is racemic and only active in the dextro-isomer. The half-life is 2 to 3 hours, with a duration of action of generally 4 hours. Methylphenidate comes in short-acting (3- to 6-hour duration), intermediate-acting (6- to 8-hour duration), and long-acting (8- to 12-hour duration) preparations (see **Tables 90-2** and **90-3**). Dexmethylphenidate is marketed as Focalin and contains solely the active isomer, and it comes in a short-acting and long-acting version. Therefore, 10 mg of methylphenidate

(Ritalin) is equivalent to 5 mg of dexmethylphenidate (Focalin). The methylphenidates have FDA approval for use in children as young as 6 years of age.

TABLE 90-2 STIMULANT PREPARATIONS

	MPH	D-MPH	MAS	D-AMPH	L-DAMPH
Short-acting (3–6 hours)	Ritalin Methylin	Focalin	Adderall	Dexedrine Dextrostat ProCentra	
Intermediate-acting (6–8 hours)	Ritalin SR Ritalin LA Metadate CD			Dexedrine Spansules	
Long-acting (8–12 hours)	Concerta Daytrana (patch) Quillivant	Focalin XR	Adderall XR		Vyvanse

TABLE 90-3 STIMULANT DOSING

Generic Name	Brand Name	Recommended Starting Dose	Prescribing Schedule	Maximum Daily Dose
MPH	Ritalin	2.5–5 mg	2.5–20 mg BID to TID	60 mg
	Methylin	2.5–5 mg	2.5–20 mg BID to TID	60 mg
	Ritalin SR	10 mg	10–60 mg daily	60 mg
	Ritalin LA	10 mg	10–60 mg daily	60 mg
	Metadate CD	10 mg	10–60 mg daily	60 mg
	Concerta	18 mg	18–72 mg daily	72 mg
	Daytrana	10 mg	10–30 mg daily	30 mg
	Quillivant	10 mg	10–60 mg daily	60 mg
D-MPH	Focalin	2.5–5 mg	2.5–10 mg BID or 2.5–5 mg TID	20 mg
	Focalin XR	5 mg	5–30 mg daily	30 mg
MAS	Adderall	5 mg	5–20 mg BID	40 mg
	Adderall XR	10 mg	10–30 mg daily	30 mg
D-AMPH	Dexedrine/Dextrostat	5 mg	5–20 mg BID or 5–10 mg TID	40 mg
	ProCentra	5 mg	5–20 mg BID or 5–10 mg TID	40 mg
	Dexedrine Spansules	5 mg	5–40 mg daily	40 mg
L-DAMPH	Vyvanse	10 mg	10–70 mg daily	70 mg

BID, twice daily; D-AMPH, dextroamphetamine; D-MPH, dexmethylphenidate; L-DAMPH, lisdexamfetamine; MAS, mixed amphetamine salts; MPH, methylphenidate; TID, 3 times daily.

The mixed-amphetamine salts (Adderall and Adderall XR) and the dextroamphetamines (Dexedrine) have FDA approval for use in children as young as 3 years. Adderall has a duration of action of 3 to 6 hours, while the XR formulation can last up to 8 to 12 hours. While only the dextro-isomer is active, the levo-isomer converts to the dextro in vivo. This is important to know for converting doses from 1 class of stimulant to another. Dextroamphetamine has a 3- to 6-hour duration of action. Lisdexamfetamine, which is marketed as Vyvanse, is a long-acting preparation (lasting up to 12 hours). It is a pro-drug, which is broken down in vivo into lysine and dextroamphetamine.

Stimulants are dosed by titrating to effect, not by a mg/kg model. Therefore, it is advisable to "start low and go slow"—ie, start at a low dose and titrate upward to maximize benefits and avoid side-effects. This underscores the need for frequent and reliable feedback from parents, caregivers, and teachers. Effects of stimulant medication can be seen within 20 to 60 minutes of taking the medication, although benefits may be modest initially before the dose is optimized. It is recommended that stimulant medication be taken after a meal or snack to avoid stomach upset and to provide calories before possible impending appetite suppression.

The most common side effects of stimulants in addition to appetite suppression and stomachache are headache, sleep disturbance, and jitteriness. Stomachache and headache typically abate after a week or so of initiating treatment. Appetite suppression may be substantial, so emphasizing a healthy but high-calorie diet is crucial. The higher the dose is and the longer the duration of action is, the greater the chance is for significant side effects. Higher doses also have the potential to create a “zombie” effect, which results from hyperfocusing and being perceived by parents as too quiet, sad, or having too much of a change in personality or typical vivaciousness. Twice-a-day or 3-times-a-day dosing of short-acting stimulant preparations allows the stimulant to nadir and appetite to return—eg, dosing the stimulant after breakfast, lunch, and late afternoon snack. However, this type of dosing does not allow for consistent coverage all day long, and troughs tend to occur during least-structured times of day (eg, bus ride to and from school, lunchtime, at daycare). Additionally, adherence to multiple doses can be an issue, and stigmatization among peers and school policies may make it difficult or impossible to take medications during the school day. Older children, who tend to have a significant amount of homework, studying, or extracurricular activities, as well as adolescents who may also have after-school jobs, may benefit most from 12-hour preparations, but they are at risk for sleep disturbance if taken too late in the day.

Understanding which medications can be opened and sprinkled versus which have to be swallowed whole is important when counseling families on options for stimulant medication treatment. Of the methylphenidate derivatives, Focalin XR, Ritalin LA, and Metadate CD are all capsules that can be opened and sprinkled on food (eg, applesauce). Focalin XR uses microbead technology, with 50% of the dose as immediate-release beads and 50% as enteric-coated for delayed release. Metadate CD has a biphasic release bead-delivery system, with 30% available for immediate release and 70% for delayed release. This differs from Ritalin LA, which has a bimodal release system, with 50% available for immediate release and 50% for delayed release. Methylphenidate ER (Concerta) utilizes an osmotic controlled-release oral delivery system (OROS) technology.

The overcoat delivers an immediate release dose of methylphenidate, while the rest of the medication gradually permeates via osmotic pressure. It must be swallowed whole. Concerta 18 mg is equivalent in effect to Ritalin 5 mg TID. Quillivant XR (long-acting) is a liquid formulation, and Methylin (short-acting) is also available in liquid but also comes in a chewable formulation. The tablet versions of methylphenidate also can be crushed. Daytrana is a methylphenidate preparation that comes in a patch for children who will not take any form of medication by mouth. When applied for 9 hours, its effects last for 12 hours. Of the mixed amphetamine salts, Adderall XR uses bead technology with 50% immediate and 50% delayed release, and can be opened and sprinkled on food, while short-acting Adderall can be crushed. Of the dextroamphetamines, Dexedrine Spansules can be opened and sprinkled on food, ProCentra comes in a liquid, and Vyvanse can be opened and the powder dissolved in water. It is important that children not chew the beads of any of these stimulants, since they are differentially coated for immediate versus extended release. The *ADHD Medication Guide*, published by Dr. Andrew Adesman, is a free, downloadable chart that contains an updated listing of ADHD medications, how they can be administered, and their FDA indications (www.ADHDMedicationGuide.com).

There is no evidence for development of tolerance to treatment with stimulant medication. There is also no evidence for increased risk of substance abuse; on the contrary, stimulant use has been correlated with reduced risk-taking behaviors, such as drug and alcohol abuse. However, short-acting stimulants have "street value" due to their potential for abuse when taken intranasally, and prescribers must be vigilant for drug diversion. There has been variable data about stimulants lowering the seizure threshold, but children with epilepsy are also at increased risk for ADHD. Communication with the managing neurologist is essential. Tics are another area of concern with stimulants, and although they do not cause tics, stimulants can unmask them in someone who is prone to tics. It is important to weigh the risks and benefits of stimulant use in children with tics, particularly given that tics naturally wax and wane.

Stimulants are considered to have minimal effects on vital signs and have not been found to increase the risk of adverse cardiac events in patients without cardiac disease. However, due to concerns over reports of sudden cardiac death and prolonged QT interval in patients with known cardiac conditions, a careful cardiac history and exam are required prior to starting treatment with stimulant medication. Before starting stimulants, patients should be screened for the following risk factors: history of congenital or structural heart abnormality, history of rheumatic fever, heart murmur, high blood pressure, chest pain, fainting, dizziness, unusual heart beat/racing heart, seizures, exercise intolerance, or use of medications or supplements known to prolong the QT interval or have cardiovascular effects. Likewise, family history should be screened for children or young adults with history of sudden cardiac death, sudden death of unknown origin, heart attack, or event requiring resuscitation; hypertrophic cardiomyopathy, dilated cardiomyopathy, long QT syndrome, cardiac arrhythmias (eg, Wolff- Parkinson-White syndrome), catecholaminergic paroxysmal ventricular tachycardia, Brugada syndrome, arrhythmogenic right ventricular dysplasia, or Marfan syndrome. Electrocardiography and possible cardiology referral should be considered for positive risk factors. Ongoing cardiovascular monitoring is advised for any patient taking stimulant medication.

Nonstimulants Some parents feel that nonstimulant medications are a "safer" option due to the vilification of stimulants in the media. However, it is important to review with families that none of the nonstimulants target all 3 of the core symptoms of ADHD, as the stimulants were designed to do. Furthermore, unlike stimulants, which can be used as needed because they only work when taken, nonstimulant medications need to be taken on a daily basis, including weekends, to maintain a therapeutic level. Atomoxetine (Strattera), long-acting guanfacine (Intuniv), and long-acting clonidine (Kapvay) all have FDA indications for treatment of ADHD in children as young as 6 years of age.

Atomoxetine is a selective norepinephrine reuptake inhibitor that acts by blocking the presynaptic norepinephrine transporter at the level of the prefrontal cortex. It is effective at addressing inattention. Dosing is recommended on a mg/kg basis, with an initial dose of 0.5 mg/kg/day that can then be titrated every 4 to 7 days to a maximum of 1.4 mg/kg/day. The capsule must be swallowed whole. Common side effects include appetite suppression (though not to the same extent as stimulants), gastrointestinal upset, sedation, and irritability. Atomoxetine has also been linked to elevated aminotransferases in a small percentage of patients (0.5%) that reversed upon stopping the medication and to acute liver failure in very rare instances. It can also potentially prolong the QT interval. Suicidal ideation has also been reported in some patients, usually within the first month of starting the medication, and atomoxetine carries an FDA black box warning. Its once-a-day dosing makes it convenient, though the dose can be delivered twice a day to minimize side effects. It can be given at night, which also helps to reduce side effects. Although its effect is less than that observed with stimulants, atomoxetine is an alternative for stimulant-wary parents and children prone to significant weight loss or tics on stimulants. Its main drawback is onset of effects, which may take weeks. Dosages are 10, 18, 25, 40, 60, 80, and 100 mg. Atomoxetine can be discontinued without weaning.

The alpha-2 adrenergic agonists are antihypertensives that have calming effects on the central nervous system. In fact, clonidine is often used to induce sleep. The alpha-2 adrenergic agonists mainly target hyperactivity and impulsivity and can be an option for children with ADHD and tics. Long-acting clonidine (Kapvay), which comes in 0.1- and 0.2-mg tablets, can be dosed every 12 hours, with a maximum dose of 0.4 mg/day. Long-acting guanfacine (Intuniv), which comes in 1, 2, 3, and 4 mg tablets, is meant to last 24 hours, with a maximum dose of 4 mg/day. Both of these medications must be swallowed whole. The "start low and go slow" principle applies here as well. The side-effect profile of the alpha-2 adrenergic agonists includes sedation, fatigue, anorexia, dry mouth, hypotension, and low mood. It is important to monitor orthostatic vital signs while titrating these medications. Since these medications may be used in conjunction with stimulants, it is imperative to ask about cardiac risk factors, as sudden cardiac death has been reported with combination treatment. These medications must also be weaned slowly due to risk of rebound hypertension.

The MTA Study In 1992, the National Institute of Mental Health and 6 teams of investigators began a multisite clinical trial known as the Multimodal Treatment of Attention Deficit Hyperactivity Disorder (MTA) study. Across 6 study sites, 579 children, aged 7 to 9+ years, were randomly assigned to 1 of 4 study groups for a 14-month intervention: (1) intensive medication management (MedMgt); (2) intensive behavioral treatments (Beh); (3) combined MedMgt and Beh (Comb); and (4) study diagnostic procedures followed by routine care in the community (CC). Medication management was initiated with short-acting methylphenidate; nonresponders were managed with an alternative stimulant or nonstimulant. The intensive behavioral treatments were based on known best practices at the time and involved the children, their parents, and their teachers.

MedMgt was found to be superior to routine CC in addressing core symptoms of ADHD, even though more than two-thirds of the children receiving CC were also being treated with stimulant medication. This was because the children in the MedMgt group were on significantly higher doses of the stimulant determined by optimum symptom control without limiting side effects. Though Comb treatment did not yield significantly greater benefits than MedMgt alone for core ADHD symptoms, similar outcomes could be achieved in the Comb group with lower medication doses. Comb treatment was also superior to MedMgt for many noncore ADHD symptoms, such as oppositionality, aggression, mood symptoms, social skills, parent-child relations, and aspects of academic functioning.

Benefits persisted across medication groups at 24 months but not at 36 months, regardless of original intervention group assigned. This was felt to reflect a possible age-related decline in ADHD symptoms, changes in intensity of medication management, starting or stopping medications altogether, or as yet unidentified reasons. Factors associated with worse outcomes at 36 months were severe symptoms, male gender, oppositional and disruptive behaviors, parent history of ADHD, and socioeconomic factors.

Follow-up studies at 6 to 8 years have shown that the type or intensity of the 14-month intervention did not predict subsequent functioning; rather, early ADHD symptom trajectory regardless of treatment type was prognostic. The implication is that children with behavioral and sociodemographic advantages, with the best response to any treatment, will have the best long-term prognosis. Long-term studies have found that stimulants are efficacious for at least 72 months and that children with the combined type presentation show the greatest impairment in adolescence. In summary, further studies are needed to understand the natural course of ADHD, but access to conscientious, responsive, and consistent medication management combined with ongoing behavioral interventions may produce the best long-term effects.

Alternative "Treatments" Often parents ask about over-the-counter supplements that can be used in lieu of medications. Popular supplements include essential fatty acids (linoleic and linolenic acids—omega-6 fatty acids), omega-3 fatty acids (eicosapentaenoic acid, docosahexaenoic acid), zinc, antioxidants, megavitamins, and herbs. There is no evidence that essential fatty acids or omega-3 fatty acids (marketed as "brain food") have direct benefit with regard to ADHD symptoms. Though nutrient deficiency (such as a diet poor in zinc, vitamins, or healthy foods in general) may produce sluggishness or conversely lack of attention due to hunger or related physical ailments, megadoses of vitamins and zinc supplementation in the absence of deficiency have not been proven to address symptoms of ADHD and may result in toxicity. Antioxidants (such as ginkgo biloba, ginseng, pycnogenol, carotenoids, and enzymatic and synthetic versions of the body's naturally occurring antioxidants) and herbs (such as chamomile, valerian root, water hyssop, wild oat extract, kava hops, lemon balm, and gotu kola) also have no clear medical evidence to support their use in the treatment of ADHD and may also cause toxicity. It is important to counsel parents that supplements are regulated by the FDA as "food," not "drugs," and are therefore not held to the same standard with regard to safety or efficacy, especially for children. Likewise, parents should be counseled against using drugs with no FDA indication for treatment of ADHD. For example, FDA-approved as antifungal treatments, nystatin and ketoconazole have also been recommended by some complementary and alternative medicine providers under the mistaken belief that yeast overgrowth in the gut causes disruptive behaviors. Instead of giving their children supplements, some parents may place their children with ADHD on elimination diets. Popular diets include those that eliminate refined sugars and added sugars, additives, preservatives, dyes, salicylates, gluten, and/or casein. Though none of these diets have proven efficacy in the reduction of ADHD symptoms, it is important to discuss healthy eating options and opportunities for physical activity/exercise with families.

SUGGESTED READINGS

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HELPFUL WEBSITES AND RESOURCES

ADHD Medication Guide: www.ADHDMedicationGuide.com

ADD Warehouse: www.addwarehouse.com

Children and Adults with Attention-Deficit/Hyperactivity Disorder: www.chadd.org

National Institute for Children's Health Quality: www.nichq.org

91 Autism Spectrum Disorder

Robert G. Voigt

AUTISM: IT'S NOT JUST A CHECKLIST

Children with autism spectrum disorder have impairments in reciprocal social interaction and social communication and restricted, repetitive interests and behaviors. The Centers for Disease Control and Prevention have reported that the estimated prevalence of autism spectrum disorder is 1 in 68 children aged 8 years; thus, autism spectrum disorder is 1 of the more common chronic medical conditions observed in pediatric medical practice. Unfortunately, rather than a focus on clinical judgment to diagnose and manage this common pediatric problem, in the era of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM), which is periodically updated by the American Psychiatric Association, autism has become a checklist. Autism did not appear as a diagnosis in the DSM until the DSM-III was published in 1980. Subsequently, the DSM checklist criteria for autism have changed over time. In the DSM-IV-TR, which was published in 2000, to meet criteria for a diagnosis of "autistic disorder," a child had to meet at least 6 out of a total of 12 checklist symptoms, with at least 2 criteria from a checklist of impairments in social interaction, at least 2 criteria from a checklist of impairments in communication, and at least 1 criteria from a checklist of restricted, repetitive, and stereotyped behaviors. A separate diagnosis of "Asperger disorder" was added to the DSM-IV (published in 1994), which included the same checklists for impairments in social interaction and restricted, repetitive, and stereotypic behaviors, but not the checklist of impairments in communication, as individuals with Asperger disorder were felt to have no delays in language development. However, the diagnosis of Asperger disorder was deleted from the DSM-5 (published in 2013). Currently, in the DSM-5, there is a single diagnosis of "autism spectrum disorder." To meet DSM-5 criteria for "autism spectrum disorder," a child must meet all 3 criteria from a checklist of persistent deficits in social communication and social interaction and at least 2 out of 4 criteria from a checklist of restricted, repetitive patterns of behavior, interests, or activities. The DSM-5 also requires (1) specification as to whether the autism spectrum disorder occurs with or without a language disorder, with or without an intellectual disability, or whether it is associated with a known medical or genetic condition or environmental factor; and (2) a rating of the severity level for both the impairment in social communication and the restricted, repetitive behaviors (level 1: requiring support; level 2: requiring substantial support; level 3: requiring very substantial support).

While this checklist approach to diagnosing autism spectrum disorder is considered by some to be the state of the art, the goal of this chapter is for pediatric medical providers to understand autism spectrum disorder within the entire spectrum and continuum of developmental-behavioral diagnoses, not as a simplified checklist. While autism itself presents with a wide spectrum of symptom severity and comorbidity, ranging from individuals with superior intelligence and mildly autistic behaviors to those with intellectual disability and severe autistic behaviors, it also occurs in continuity within a broader spectrum of developmental-behavioral diagnoses. Autism spectrum disorder exhibits significant overlap with other developmental-behavioral diagnoses, including language disorder, language-based learning disability, nonverbal learning disability, social communication disorder, attention-deficit/hyperactivity disorder (ADHD), and intellectual disability. It is often difficult, for example, to determine whether the social skills exhibited by a child with a severe intellectual disability are commensurate with underlying cognitive abilities or whether the child has an intercurrent autism spectrum disorder. Similarly, it is often difficult to distinguish whether the social skills deficits exhibited by a child with a language disorder and associated ADHD are secondary to the language disorder (hesitance to interact due to language difficulties) and ADHD (inattention to social cues) or are better accounted for by an autism spectrum disorder diagnosis. In each of these difficult cases, clinical judgment and a knowledge of the entire spectrum and continuum of developmental-behavioral diagnoses, rather than a simple checklist approach, is required to make the appropriate diagnosis. A checklist approach implies that anyone armed with an autism checklist can be an autism expert. However, to truly understand autism spectrum disorder and its differential diagnosis, pediatric medical providers need the clinical judgment to appreciate the entire spectrum and continuum of developmental-behavioral diagnoses (see Chapter 83) and to recognize autism spectrum disorder's place within this spectrum and continuum.

PEDIATRIC DEVELOPMENTAL ASSESSMENT

In Chapter 82, a medical model for evaluating chief complaints about developmental-behavioral concerns is presented. If a family expresses concern that their child might have autism spectrum disorder, or if a child fails an autism screen (the American Academy of Pediatrics recommends autism screening at 24 and 30 months of age), pediatric medical providers should obtain a comprehensive and detailed developmental history of the temporal pattern of developmental milestone acquisition across developmental streams to identify developmental delay, dissociation, and deviation. Early warning signs of a possible autism spectrum disorder may include concerns about a child's poor eye contact; speech/language delay; deviation in acquisition of language milestones; lack of response to his or her name being called ("acting deaf"); lack of showing, pointing, or sharing interests; or any loss of language and/or social skills (a loss of language and/or social skills is reported by about one-third of parents with children with autism spectrum disorder).

In children with autism spectrum disorder, the developmental history typically uncovers a scattered, uneven, developmentally dissociated and deviated pattern of development. As reviewed in Chapter 83, in children with developmental delays, it is most common for the delays to be more global in nature; developmental dissociation (differences in rates of development between developmental streams) and/or deviation (nonsequential acquisition of developmental milestones within a developmental stream) are therefore considered atypical compared to more globally delayed developmental milestones. While it is certainly true that every individual, whether he or she exhibits developmental delays or has no developmental delays, exhibits unique strengths and weaknesses across developmental abilities, it is most common in the general population to have relatively evenly developed developmental abilities, and large discrepancies (or dissociations) between abilities in different developmental domains can be considered atypical. For example, any IQ test manual contains tables documenting the percentage of the population with differences between different IQ subscores (such as verbal and nonverbal IQ), and the greater the difference is between scores, the fewer people in the population there are who exhibit this difference. Thus, individuals with significant dissociations in their cognitive development can be considered to process information in an atypical way compared to the general population (who do not exhibit such dissociation in abilities). Contributing to this atypical information processing is developmental deviation, or nonsequential milestone acquisition, which is atypical at any age. It should follow from this that individuals with significant developmental dissociation and deviation, who process information in an atypical way compared to the majority of the population who do not exhibit dissociation or deviation, are more likely to exhibit atypical behavior.

A key neurodevelopmental principle presented in Chapter 83 is that the more delayed, dissociated, and deviated the neurocognitive development is, the more atypical the associated neurobehavior is expected to be. Thus, when using the popular checklist approach to autism diagnosis, the symptoms of autism derive from a "black box" without explanation; when using a pediatric developmental assessment process, the symptoms of autism may be better explained to families. When a developmental history indicates a scattered, uneven, atypical pattern of developmental dissociation and deviation in the neurocognitive stream of development (which includes social development), it is expected that such an atypical pattern of cognitive development would most likely be associated with a similarly atypical pattern of behavior.

SPECTRUM OF NEUROCOGNITIVE DISSOCIATION AND DEVIATION

The neurocognitive stream of development includes the language and nonverbal/visual-motor problem-solving streams, which come together to form the social and adaptive streams of development (see **Fig. 91-1** and **Chapter 83**). The social stream of development is primarily communicative and includes social reciprocity and developing social relationships; the adaptive stream of development primarily relies on nonverbal/visual-motor problem solving and includes activities of daily living, such as feeding and dressing. The developmental history obtained from parents of children with autism spectrum disorder typically includes a history of dissociation and deviation in the neurocognitive stream of development, and this dissociation and deviation is usually confirmed on neurodevelopmental examination.

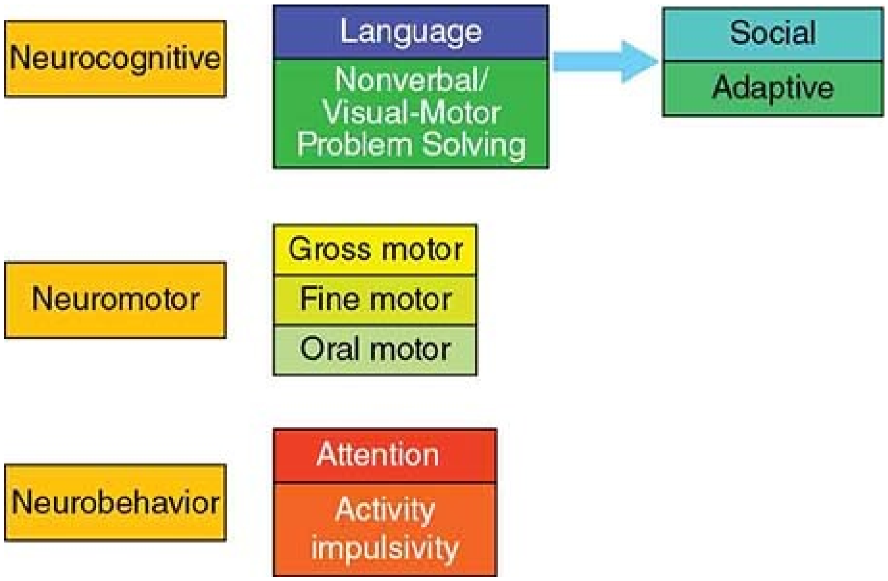

FIGURE 91-1 Streams of development.

This dissociation in neurocognitive development may occur along the spectrum of dissociated/deviated language development or along the spectrum of dissociated/deviated nonverbal/visual-motor problem-solving development (Fig. 91-2). In either case, there is a significant discrepancy or dissociation between language and nonverbal abilities, and as reviewed above, such a discrepancy between these cognitive domains can be considered atypical compared to more global developmental delays. In addition, developmental deviation in acquisition of milestones within each of these streams adds to the atypicality in neurocognitive development observed in children with autism spectrum disorders.

In the past, autistic disorder and Asperger disorder were considered to be separate disorders. From a neurocognitive standpoint, children with autistic disorder and Asperger disorder both exhibited atypical, dissociated, and deviated developmental profiles. However, the dissociations in these 2 disorders were mirror images of one another: Children with autistic disorder generally exhibited a strength in nonverbal/visual-motor abilities and dissociated delays in language, while children with Asperger syndrome exhibited a relative strength in language abilities and dissociated delays in nonverbal/visual-motor abilities. However, as children with both of these disorders exhibited significant dissociation (and deviation) in their cognitive (including social) development, consistent with the continuum of developmental-behavioral diagnoses, their atypical neurocognitive profiles were associated with atypical neurobehavior. However, reflecting the difference in their underlying neurocognitive dissociations, the atypical repetitive behaviors exhibited by children with autistic disorder (who had discrepantly stronger nonverbal/visual-motor problem-solving development) tended to reflect their area of strength, with visual perseveration on spinning objects, flashing lights, lining up objects, etc. On the other hand, the atypical repetitive behaviors exhibited by children with Asperger disorder (who had discrepantly stronger language abilities) tended to reflect their area of strength, with verbal perseveration on factual information about restricted interests, such as train schedules, dinosaurs, or baseball statistics. However, with the elimination of the diagnosis of Asberger disorder in the DSM-5, a single category, autism spectrum disorder, was created. Those children who received an autistic disorder diagnosis in the DSM-IV are now diagnosed with autism spectrum disorder with a language disorder in the DSM-5, and those children who received a diagnosis of Asperger disorder in the DSM-4 are now diagnosed with autism spectrum disorder without a language disorder in the DSM-5. Despite these changes in DSM checklist diagnoses, in children with autism spectrum disorder, these atypical mirror image patterns of neurocognitive dissociation/deviation that are associated with social impairment continue to be associated with atypical neurobehavioral manifestations within the continuum of developmental-behavioral diagnoses.

Figure 91-2 illustrates the spectrum of neurocognitive dissociation and deviation. As reviewed in Chapter 83, there are 2 components to this spectrum, depending on whether the dissociated delays in development are due to language delays (the spectrum of dissociated and deviated language development, the top part of Fig. 91-2) or due to nonverbal/visual-motor problem-solving delays (the spectrum of dissociated and deviated nonverbal development, the bottom part of Fig. 91-2). At the mild end of the spectrum of dissociated language development, children have more mildly dissociated language delays, involving aspects of receptive and/or expressive language. In the preschool-aged population, these children are diagnosed with language disorders, and in the school-aged population, these children are diagnosed with language-based learning disabilities. However, as one proceeds to the more severe end of this spectrum of dissociated and deviated language development, more diffuse aspects of language/communication development (including both verbal and nonverbal pragmatic or social language development) and social interaction (including social reciprocity and developing social relationships) are impaired, and more deviation in the acquisition of language milestones is observed. This results in a neurocognitive diagnosis of a social communication disorder with an associated language disorder. The language deviation observed at this severe end of the spectrum of language dissociation and deviation is actually often an early red flag to suspect an autism spectrum disorder diagnosis. As an example of such language deviation in a younger child, parents might report in the developmental history that a child has a 50-word vocabulary, as expected at 2 years of age, but he or she does not yet use a specific “Mama” or “Dada” to refer to his or her parents, as expected at 10 months of age. This inflated vocabulary is most often not used functionally and consists primarily of labels, rote memorized scripts, or echolalia. In older children, language deviation may be observed by a vocabulary that is too numerous to count (> 3 years), use of multiword sentences (> 3 years), and an ability to label pictures, while at the same time, the child continues to use echolalia (echolalia usually ceases by 30 months of age) and to confuse pronouns (pronoun confusion usually ceases by 30 months). Children at this severe end of the spectrum of language dissociation and deviation who have autism spectrum disorder often also have developmentally deviated strengths or “splinter skills” in aspects of nonverbal/visual-motor problem solving. Consider as an example a 36-month-old child whose language development is at a 12-month level but who was able to visually recognize all the letters of the alphabet (a 5½-year-old–level skill) at 18 months of age, who recognizes shapes such as a pentagon and octagon, and who completes puzzles designed for 4-year-olds, but who does not yet draw a circle (a 36-month-old skill). More extreme dissociation, or “autistic savant skills,” in art, math calculation, music, memory, or word reading (hyperlexia) also occur in approximately 10% of individuals with autism spectrum disorder.

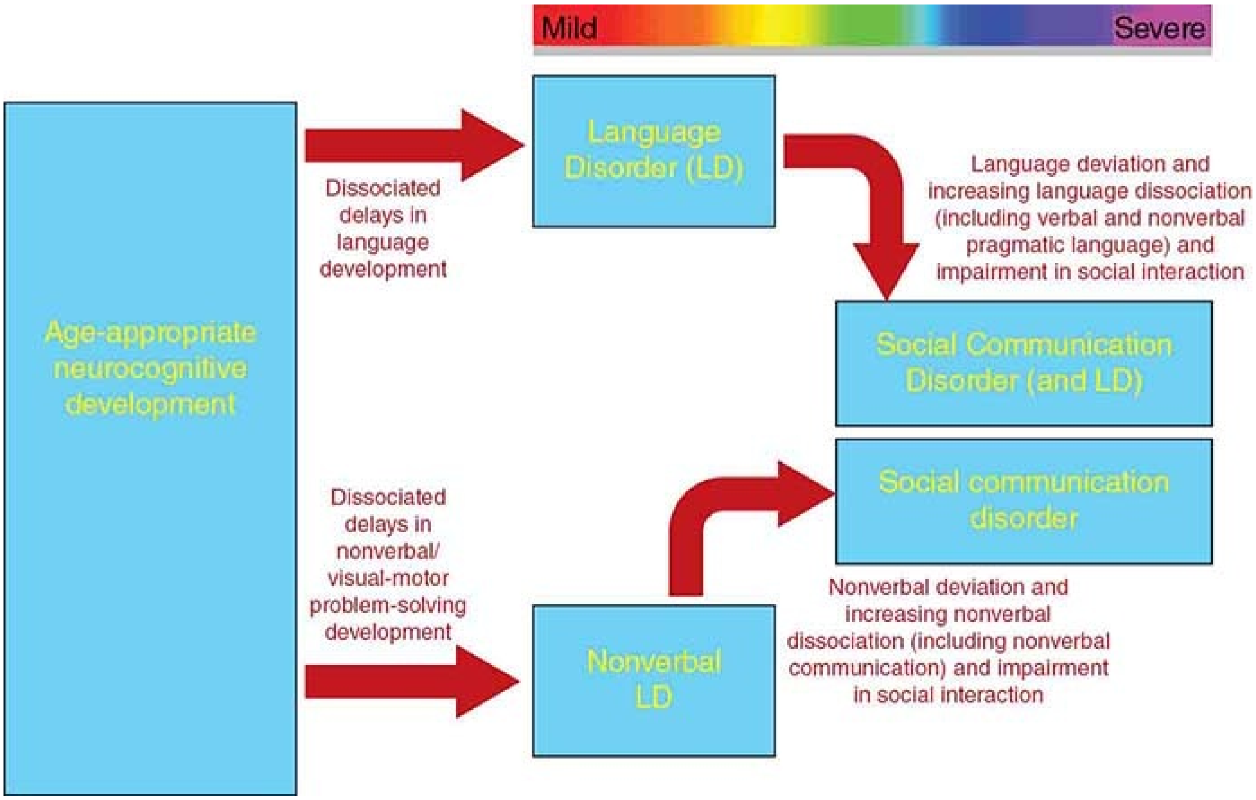

FIGURE 91-2 Spectrum of neurocognitive dissociation and deviation.

The bottom part of Figure 91-2 illustrates the spectrum of nonverbal dissociation and deviation. In the mild end of this spectrum, children have more mildly dissociated nonverbal delays, involving aspects of visual perceptual, visual spatial, and/or visual motor abilities. In the preschool-aged population, these children are typically described as having a visual-perceptual, visual-spatial, and/or visual-motor deficit, and in the school-aged population, these children are diagnosed to have nonverbal learning disabilities. However, as one proceeds to the more severe end of this spectrum of dissociated nonverbal development, more diffuse aspects of nonverbal development, including nonverbal aspects of social communication and social interaction, are impaired, and more deviation in the acquisition of nonverbal milestones is observed. This again results in a neurocognitive diagnosis of social communication disorder, but without an associated language disorder.

Social communication disorder has been included in the DSM-5 as a stand-alone diagnosis. This diagnosis describes children who have pragmatic and social communication impairments similar to those observed in autism spectrum disorder, but they do not exhibit the associated restricted, repetitive, or stereotypic behaviors observed in children with autism spectrum disorder. Thus, a diagnosis of autism spectrum disorder is not made exclusively based on this atypical social communication disorder neurocognitive profile. To make a diagnosis of autism spectrum disorder within the continuum of developmental-behavioral diagnoses, this atypical neurocognitive profile must be accompanied by the atypical neurobehavioral profile observed at the severe end of the spectrum of neurobehavioral dissociation and deviation.

SPECTRUM OF NEUROBEHAVIORAL DISSOCIATION AND DEVIATION

Figure 91-3 illustrates the spectrum of neurobehavioral dissociation and deviation. At the mild end of this spectrum, children exhibit dissociated delays in their attention spans and/or levels of impulse control and motor activity compared to their underlying cognitive abilities. In other words, they exhibit developmentally inappropriate levels of inattention and/or hyperactivity-impulsivity, and if these symptoms are impairing their functioning across settings, they are diagnosed as having ADHD. However, as one proceeds to the more severe end of this spectrum of dissociated and deviated neurobehavioral development, increased developmental deviation results in behaviors that are atypical at any age. Rather than simple inattention, children at the severe end of this spectrum exhibit atypical attention; while they may lack the attention (and social interest) to sustain eye contact, they may overfocus or perseverate on restricted interests and routines and engage in repetitive play (eg, lining, sorting, spinning). They may also show atypicalities in attention to other sensory stimuli. For example, they may not react to painful stimuli but overreact to certain sounds, or they may overfocus on tight-fitting clothes, tags on shirts, or seams on socks. Rather than the hyperactivity or fidgetiness observed at the mild end of this spectrum, at the severe end of this spectrum of neurobehavioral dissociation and deviation, more atypical stereotypic motor mannerisms (eg, hand flapping, body rocking or spinning, toe-walking) may be observed.

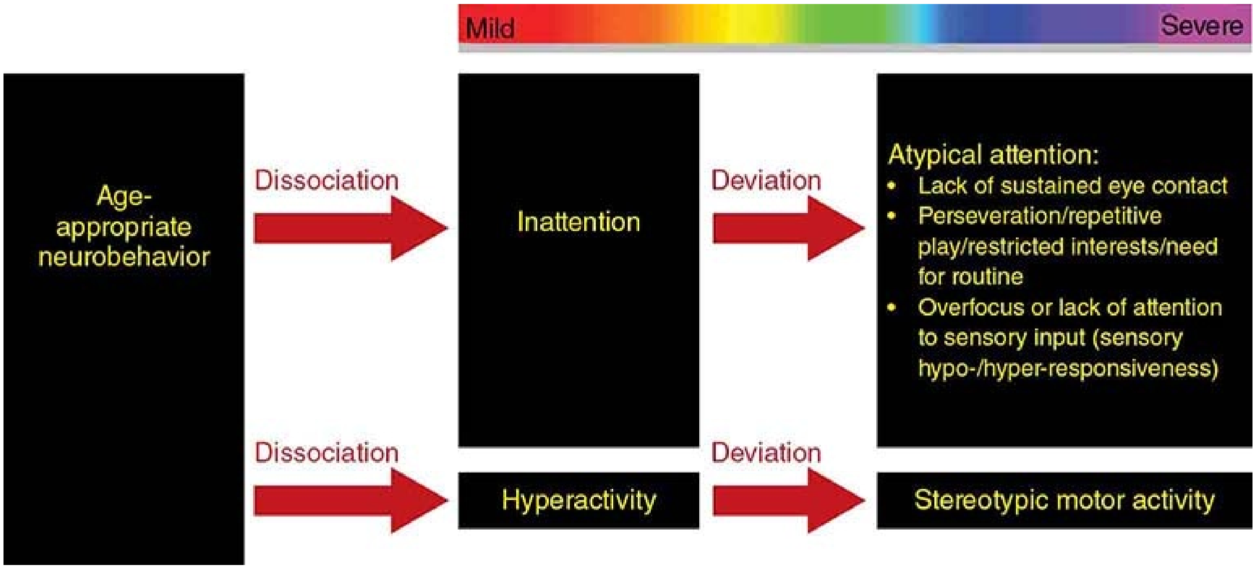

FIGURE 91-3 Spectrum of neurobehavioral dissociation and deviation.

CONTINUUM OF NEUROCOGNITIVE AND NEUROBEHAVIORAL STREAMS OF DEVELOPMENT

As discussed in Chapter 83, a key neurodevelopmental principle is that there is a continuum of developmental-behavioral disability across neurodevelopmental streams. Thus, delays in 1 stream of neurodevelopment are most likely to occur in association with delays in other streams of neurodevelopment. This continuum of developmental-behavioral disability can be observed among children with dissociated and deviated development and behavior, and it is the continuum between the severe end of the spectrum of neurocognitive dissociation and deviation (social communication disorder) and the severe end of the spectrum of neurobehavioral dissociation and deviation (restricted, repetitive, and stereotyped behavior) that defines autism spectrum disorder. **Figure 91-4** illustrates this continuum of neurocognitive and neurobehavioral disability. At the mild end of this continuum, one encounters children with the common combination of learning disabilities (from the spectrum of neurocognitive dissociation/deviation) and ADHD (from the spectrum of neurobehavioral dissociation/deviation). As one proceeds across to the severe end of this continuum of neurocognitive and neurobehavioral disability, one encounters children with a combination of social communication disorders (from the spectrum of neurocognitive dissociation/deviation) and the atypical attention (lack of sustained eye contact, perseveration on restricted interests, need for routine, repetitive play, sensory hypo- or hyper-responsiveness) and stereotypic motor mannerisms observed at the severe end of the spectrum of neurobehavioral dissociation and deviation. Children with this combination of neurocognitive and neurobehavioral symptomatology (social communication disorders + atypical repetitive and stereotypic behavior) are diagnosed with autism spectrum disorder.

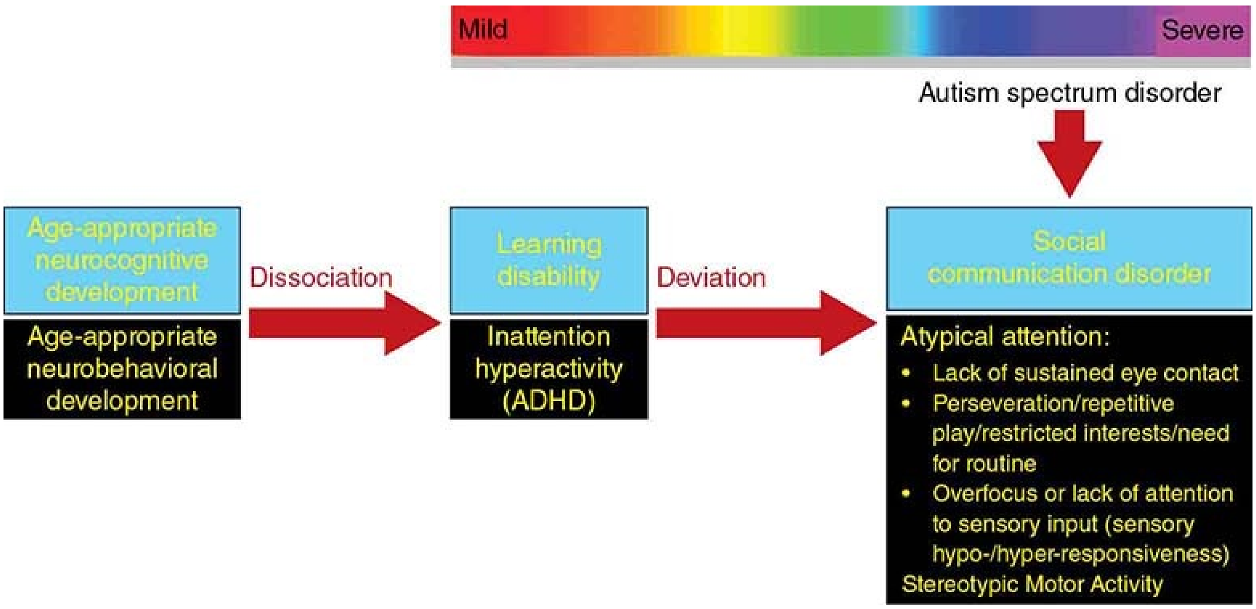

FIGURE 91-4 Continuum of neurocognitive and neurobehavioral dissociation and deviation.

With an understanding of the entire spectrum and continuum of developmental-behavioral diagnoses, clinical judgment becomes the gold standard for autism spectrum disorder diagnoses. The developmental history and observations made during the neurodevelopmental examination should confirm that a child meets DSM-5 checklist criteria for an autism spectrum diagnosis. However, other standardized measures are necessary to confirm clinical concerns about autism spectrum disorder. These measures are more typically administered by psychologists or school district Child Study Teams, and include the Childhood Autism Rating Scale (CARS) and the Autism Diagnostic Observation Schedule (ADOS).

DIFFERENTIAL DIAGNOSIS

Given that developmental-behavioral diagnoses exist along a spectrum and continuum, it is often difficult to make categorical diagnoses. The differential diagnosis for autism spectrum disorder includes social communication disorders, language disorder/language-based learning disabilities, nonverbal learning disabilities, ADHD, and intellectual disabilities. Children with social communication disorders share the social communication impairment observed in children with autism spectrum disorder, but they do not exhibit the restricted, repetitive, or stereotypic behaviors observed in autism spectrum disorder. Children with language disorders or language-based learning disabilities share the dissociated delays in language development observed in children with autism spectrum disorder (with an associated language disorder), and they may be hesitant to interact with other children due to their delayed language abilities. However, they tend not to exhibit the extent of dissociated verbal and nonverbal pragmatic language delays or the significant developmental deviation in language development observed in children with autism spectrum disorder. Children with language disorders also may attempt to communicate nonverbally to compensate for their verbal difficulties, and they do not exhibit the impairments in social reciprocity and development of social relationships nor the restricted, repetitive, and stereotypic behaviors observed in children with autism spectrum disorder. Similarly, children with nonverbal learning disabilities share the dissociated delays in nonverbal communication exhibited by children with autism spectrum disorder (without a language disorder) and may have difficulty with nonverbal communication and in "reading" social situations that negatively impact their social interactions, but they do not exhibit the severe impairments in social reciprocity and development of social relationships nor the restricted, repetitive, and stereotypic behaviors observed in children with autism spectrum disorder. Children with ADHD have deficits in attention and impulse control that impair their social interactions with peers, and they often have associated language-based or nonverbal learning disabilities, but they do not exhibit the impairments in social reciprocity and development of social relationships nor the restricted, repetitive, or stereotypic behaviors observed in autism spectrum disorder.

As a component of their globally delayed developmental milestones, children with intellectual disability often exhibit delays in their social skills. However, an additional diagnosis of autism spectrum disorder is given to children with intellectual disability only if their social communication and social interaction skills are discrepantly delayed (or dissociated) from their underlying cognitive abilities. Children with more severe to profound intellectual disability often engage in repetitive and stereotypic behaviors, but a diagnosis of autism spectrum disorder is given only if the child's social communication/social interaction is discrepantly delayed compared to his or her underlying cognitive abilities. In the continuum of developmental disabilities, the more delayed the development, the more frequently dissociation and deviation are also observed. Thus, children with intellectual disability are more likely to also have the developmentally dissociated and deviated developmental patterns observed in autism spectrum disorder. The Centers for Disease Control and Prevention reported that 32% of children with autism spectrum disorder also have intellectual disability and 56% of children with autism spectrum disorder have intellectual disability or borderline cognitive functioning.

Making a diagnosis of autism spectrum disorder is the first step for pediatric medical providers. Once a diagnosis is made, referral for early intervention or special education, medical laboratory workup, and longitudinal medical management, potentially including psychopharmacological management, remain the roles of the primary care pediatric medical provider.

EARLY INTERVENTION AND SPECIAL EDUCATION

As early identification and early intervention are critical for optimizing long-term neurodevelopmental outcome, children should be referred to their local early intervention or special education programs as soon as a diagnosis of autism spectrum disorder is even suspected. Behavioral interventions based on applied behavior analysis (ABA) have the strongest empiric support for autism treatment. A report of the Surgeon General of the United States (1999) affirmed that 30 years of research has demonstrated the efficacy of ABA in reducing inappropriate disruptive and maladaptive behavior and in increasing communication, learning, and appropriate social behavior in children with autism spectrum disorder. Families can access a local certified behavioral therapist to provide ABA services through the Behavior Analyst Certification Board (http://www.bacb.com/).

Once children with autism spectrum disorders reach 3 years of age, they should qualify for an Individualized Education Program through their local public school districts under a primary categorical label of autism spectrum disorder. Children with autism spectrum disorder benefit from intensive direct and consultative language, behavioral, and social skills interventions aimed at maximizing functional communication and social interaction abilities and at modifying atypical and maladaptive behaviors. Children with autism spectrum disorder also benefit from very highly structured and supervised inclusion into regular classroom settings and activities for exposure to socially and communicatively appropriate role models with whom they can practice the communication and social interaction skills that they learn through their special educational and therapeutic services. The National Research Council has published comprehensive recommendations for the education of children with autism spectrum disorders. These guidelines recommend that a minimum of 25 hours per week of services be provided year round in which the child is engaged in individualized, highly structured, systematically planned, and developmentally appropriate educational activities. The guidelines further suggest that the priorities of focus of educational services for children with autism spectrum disorders should include functional spontaneous communication, social instruction delivered throughout the day in various settings, cognitive development, play skills, and proactive approaches to atypical and challenging behaviors. It is also important that parents be included as integral members of the intervention team and extend therapeutic goals to the home environment. Generalization and maintenance of newly learned skills in natural environments should be considered as important as the acquisition of new skills.

Children with autism spectrum disorder should receive intensive direct and consultative language therapy services that include a pragmatic language therapy component to address their delayed and deviated language development and social communication impairment. Augmentative communication strategies (eg, picture exchanges, visual schedules, manual signing, communication boards/devices) should be considered as a component of speech/language therapy in an attempt to improve functional communication and decrease frustration with communication breakdowns. Children with autism spectrum disorder who exhibit associated gross motor incoordination or delays in activities of daily living with a fine-motor component might also benefit from direct physical therapy and occupational therapy services.

Families with children with autism spectrum disorder should also benefit from all social and community services available to children with developmental disabilities and their families in their local communities. These services might include case management services, supplemental medical insurance or other financial assistance programs, educational advocacy services, parent support groups, functional behavioral analysis/in-home behavior management counseling services, respite care services, personal care attendant services, counseling regarding long-term legal and financial planning issues, summer camps, and other extracurricular activities.

The response to intervention for children with autism spectrum disorders can be extremely variable, with some children making rapid, substantial gains and others making very slow gains. It has been reported that up to 25% of children diagnosed with autism spectrum disorder may lose their autism spectrum disorder diagnosis over time. Although it is very difficult to predict, those more likely to respond to intervention are those with higher IQs and language abilities, those with milder autism symptoms, and those who are diagnosed early and receive more intensive intervention using behavioral techniques.

MEDICAL LABORATORY WORKUP

As with all other children with delayed language development, all children with autism spectrum disorder with an associated language disorder need to undergo formal audiology evaluation to check their hearing status. In an attempt to establish an etiologic diagnosis to account for their autism spectrum disorder, all children with autism spectrum disorder should undergo a chromosome microarray analysis and DNA testing for fragile X syndrome. Further medical laboratory workup depends on the clinical presentation. If a child with autism spectrum disorder exhibits any metabolic signs or symptoms (eg, global developmental regression, decompensation with mild illness, tonal abnormalities, ataxia, recurrent vomiting), then a comprehensive metabolic workup to rule out inborn errors of metabolism would be indicated. If the physical examination reveals neurocutaneous findings (eg, tuberous sclerosis, neurofibromatosis), cleft palate and syndactyly (eg, Smith-Lemli-Opitz syndrome), or marked macrocephaly (eg, *PTEN* hamartoma syndromes), then targeted workup for these disorders should be pursued. In girls with deceleration of head growth, hand wringing, and loss of purposeful hand movements, testing for mutations in the *MECP2* gene should be pursued to rule out Rett syndrome.

MEDICAL COMORBIDITIES

As a component of their restricted, repetitive behaviors, children with autism spectrum disorders are often very picky eaters, and this is associated with frequent complaints of constipation or loose stools. The workup and management of gastrointestinal complaints in children with autism spectrum disorders should proceed just as it would for children without autism spectrum disorder.

Children with autism spectrum disorders also frequently present with sleep difficulties, and lack of an appropriate amount of sleep can exacerbate daytime behavioral problems and increase family stress. While these sleep difficulties are often behavioral in nature, children with autism spectrum disorder can have medical problems resulting in sleep disruption (eg, gastroesophageal reflux, abdominal pain related to constipation, sleep apnea) or primary sleep disorders (eg, restless legs syndrome), and further workup and/or referral to a sleep specialist should be considered just as in children without autism spectrum disorders.

Approximately one-third of children with autism spectrum disorder will develop epilepsy, usually during adolescence. Electroencephalograms (EEGs) should be considered if seizures are suspected. If a child exhibits a significant isolated loss of language after 3 years of age (as opposed to the more typical autism regression, which involves regression in both language and social skills before 2 years of age), a sleep-deprived EEG may be indicated to rule out acquired epileptic aphasia or Landau-Kleffner syndrome.

PSYCHOPHARMACOLOGY

There are no medications available to treat the core social communication/social interaction impairments or restricted/repetitive behaviors observed in children with autism spectrum disorders. However, children with autism spectrum disorders often present with secondary maladaptive behaviors, such as tantrums, aggression, and self-injury. Whenever there is an acute change in the behavior of a child with autism spectrum disorder, especially in children with limited verbal and nonverbal communication, it is important to rule out occult medical causes (eg, constipation, gastroesophageal reflux, lack of appropriate sleep, headaches, corneal abrasion, dental abscess, otitis media or externa, occult fractures) for the acute behavior change. Secondary maladaptive behaviors can also occur due to anxiety and frustration produced by demands and expectations for a child's performance at school or at home that are exceeding his or her underlying abilities. Such secondary maladaptive behaviors are also often inadvertently reinforced by parents or school personnel. Thus, behavioral interventions, including functional behavioral assessments and development of formal behavior management plans, along with parental behavior management training, should be considered the primary intervention for such maladaptive behaviors.

However, when maladaptive behaviors are preventing a child from taking advantage of therapeutic and educational interventions (tantrums), are causing significant harm to self (self-injurious behaviors) or others (aggression), or are becoming dangerous (elopement), then a careful trial of psychotropic medication can be considered. The only current medications with FDA approval for the treatment of irritability (including tantrums, aggressive behavior, and self-injury) in children with autism spectrum disorder are the atypical antipsychotics risperidone (for children > 5 years of age) and aripiprazole (for children > 6 years of age). As is true for children with intellectual disability, it is important to note that psychotropic medications for children with autism spectrum disorder typically are not as effective as when they are used in children without autism spectrum disorder, both in terms of the number of children with a positive response and in the magnitude of the positive response. In addition, children with autism spectrum disorder typically experience more psychotropic medication side effects compared to children without autism spectrum disorder. Thus, if a trial of psychotropic medication is being considered, it is important to start at a low dose and to increase the dose slowly to closely monitor both positive response and side effects.

NONSTANDARD THERAPIES

Families with children with autism spectrum disorder need to be very careful with regard to their potential choices of nonstandard and non–evidence-based interventions that may be advocated for children with autism spectrum disorders. Families will likely learn of many nonstandard and unproven therapeutic interventions and explanations for autism with questionable scientific validity or evidence of their effectiveness. Given the lack of any biomedical treatments or cures for autism spectrum disorder offered by traditional, evidence-based medicine, parents are understandably desperate for any kind of biomedical treatment that might help their children. Unfortunately, a plethora of unproven (and often costly and potentially dangerous) interventions are widely available to exploit this parental sense of desperation. Families need to be very careful about considering any proposed intervention that is supported exclusively by personal testimonials rather than by data from double-blind, placebo-controlled clinical trials that are published in the peer-reviewed medical literature. Families need to be very careful that any intervention they may decide to proceed with is not potentially harmful to their children from a medical standpoint (such as potential impurities in unregulated nutritional supplements; potential toxic effects of megadoses of vitamins or minerals; potential nutritional deficiencies derivative of special diets; inappropriate use of and side effects from hyperbaric oxygen therapy; antifungal, antiviral, or antibiotic medications; chelating agents; immunotherapies; or withholding immunizations) and would be neither a financial nor time-consuming burden to the family (such as with “facilitated communication," "auditory integration," or other "sensory therapies") nor prevent them from taking advantage of the educational and behavioral interventions that have been shown to be most effective for children with autism spectrum disorders. Families can review information about scientifically supported interventions for autism spectrum disorder through the National Autism Center's National Standards Project (http://www.nationalautismcenter.org/national-standards-project/phase-2/) or through the Association for Science in Autism Treatment (http://www.asatonline.org/).

CONCLUSION

Rather than the popular view of autism as a simple checklist of behaviors, given its prevalence in the pediatric population, pediatric medical providers need to understand autism within the entire spectrum and continuum of developmental disabilities. There is no such thing as an autism expert; autism and its differential diagnoses can only be understood through acquiring expertise across the entire spectrum and continuum of developmental-behavioral diagnoses. Concerns about autism should be treated no differently than any other chief complaint in pediatric medicine. The diagnosis of autism can be made through a comprehensive developmental history that is confirmed by observations made during a neurodevelopmental examination. This clinical approach will identify the delayed, dissociated, and deviated pattern of development observed in children with autism spectrum disorders, allowing pediatric medical providers to make the diagnosis; complete the medical laboratory workup in an attempt to establish an etiologic diagnosis; refer for evidence-based early intervention, special education, and therapeutic services; and longitudinally manage children with autism spectrum disorders and their families within their primary care medical home.

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SECTION 7: Pediatric Mental Health

92 Introduction to Pediatric Mental Health

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Emotional and behavioral health is an integral component of children's health, lifetime well-being and adjustment, educational attainment, and ability to thrive in all functional domains. It is strongly associated with physical health and resilience in the face of psychosocial or biological adversity. By contrast, behavioral and emotional problems predict or negatively influence the course of pediatric illness.

The significance of emotional and behavioral health problems in pediatrics is underscored by the prevalence of mental health disorders in children. One in 5 children presents with a serious and persistent mental health disorder. The prevalence increases to one-third for the nearly 25% of children in the United States exposed to 2 or more adverse childhood experiences (ACEs), which include physical, emotional, or sexual maltreatment, or chronic or severe medical conditions. (For further discussion of ACEs, see Chapter 28.)

Suicide, the most tragic correlate of mental health disorders, is the second leading cause of death in youth ages 15 to 24 years, accounting for 3 times more deaths than all childhood malignancies or heart conditions combined and more than 20 times the deaths attributable to diabetes in this age group. More ominously, the number of completed suicides appears to be increasing in at least some pediatric populations. Between 1999 and 2014, completed suicides in girls 10 to 14 years old tripled, the largest percent increase in deaths by suicide in any age group.

Other mental health disorders also show increases in prevalence, notably autism, which according to the Centers for Disease Control and Prevention, has gone from 1 in 100 at the beginning of this decade to 1 in 68. Similar increases are noted in adolescent depression and anxiety disorders.

Even more significantly, there is a growing appreciation of the profound lifetime consequences and the cost to both the individual and to society of childhood mental disorders in terms of poor educational attainment and significantly increased risk for substance abuse and dependency, incarceration, violence toward both self and others, and poor health. Pediatric mental health disorders shape dysfunctional and maladaptive developmental and neurodevelopmental trajectories not only by impairing the acquisition of cognitive and social skills but also, more subtly, by negatively impacting caretakers' ability to provide the very developmental opportunities that underlie healthy adjustment and resilience.

Thus, pediatric mental health disorders lead to chronic lifetime psychopathology and poor health. For example, Winning et al showed that even after 50 years, mental health disorders in childhood were still associated with significantly increased cardiovascular morbidity and mortality. Not surprisingly, the Institute of Medicine's 2015 report stated that "in the United States, mental and behavioral disorders are the largest contributors to years lived with disability. They are also co-morbid with medical conditions and costly to the health care system and society." This report, however, noted a "quality chasm" between the number of individuals afflicted with mental health disorders and the availability and appropriateness of the care they receive.

Among youth with mental health disorders, less than 30% receive any form of treatment. For those who do receive treatment, their care is often fragmented, mostly focused on psychopharmacology, and, not infrequently, involves treatment interventions unsupported by evidence.

While geography, race, and socioeconomic status factor heavily in limiting access to treatment, an overall dearth of mental health professionals, particularly child and adolescent psychiatrists, is a major impediment to the provision of appropriate care. Clearly, innovative healthcare models are urgently needed to address the mental health needs of the pediatric population. These models largely focus on integrating mental health care with overall health care.

The passage of the Patient Protection and Affordable Care Act of 2011 placed increased emphasis on the development of medical homes designed to address, in a collaborative and coordinated fashion, the physical and mental health needs of patients. The Health Parity Law of 2012 requires parity in access to services for both physical and mental health disorders, further adding to the pressure on payers to address mental health problems together with other health conditions.

The pressure to integrate mental health into general healthcare places pediatricians in a position to play a key role in meeting the mental health needs of children. Yet studies consistently find that pediatricians feel unsure they possess the skills, knowledge, and tools needed to address mental health problems in their practices. A random sample of 1600 members of the American Academy of Pediatrics concluded that pediatricians feel both responsible and comfortable assessing and managing attention-deficit/hyperactivity disorder (70% of respondents). Yet less than one-third of those sampled felt equally comfortable managing other mental health conditions. A study of recently graduated pediatricians points out that additional training in mental health and developmental pediatrics could allow pediatricians to assume a larger role in identifying and providing first-line treatment for the most common mental health disorders in a variety of settings, from stand-alone pediatric practices to integrated care models.

In sum, mental health is key to health. Pediatricians need to possess the attitudes, skill, and knowledge required to assume a vital role in assessing and treating pediatric mental health disorders and in addressing the impact of mental health problems on medical conditions and of medical conditions on their patients' mental health.

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93 Anxiety

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DIAGNOSIS AND PREVALENCE

Anxiety disorders are common throughout childhood and adolescence, and the anxiety disorders observed throughout the life span often emerge during childhood. Prevalence rates for pediatric anxiety disorders are as high as 15% to 20%. Although some fears and anxieties are developmentally normative and even adaptive, they must be differentiated from anxiety disorders, which entail considerable impairment in academic, social, and family functioning. Unfortunately, anxiety disorders may go unnoticed, particularly when compared to the disruptive behavior disorders (eg, attention-deficit/hyperactivity disorder [ADHD] and oppositional defiant disorder) that are more frequently referred for treatment. Early identification and assessment of pediatric anxiety disorders require an evaluation that includes clinical interview, multi-informant report (parent, child, and teacher), careful attention to developmental level and ability, an evaluation of the family environment and the response of the caregivers to the child's anxiety, and an assessment of functional impairment in daily life. Pediatric anxiety disorders are highly comorbid with one another, and affected youth are likely to meet criteria for multiple anxiety disorders. Moreover, anxiety disorders may serve as a gateway to developing major depressive disorder, suicidal ideation and behavior, and substance abuse in adolescence. Fortunately, anxiety disorders are highly treatable conditions, particularly when identified early. In this chapter, we review the various types of anxiety disorders, approaches to screening and evaluation of anxiety, and evidence-based treatment approaches.

Listed below are descriptions of anxiety disorders observed in children and adolescents, including diagnostic criteria and prevalence rates. For all of these disorders, significant distress and/or impairment is required for a diagnosis.

Separation Anxiety Disorder

Separation anxiety disorder is characterized by excessive anxiety or distress when anticipating or experiencing separation from caregivers or from home. Children suffering from separation anxiety disorder typically worry about harm befalling them or their caregivers when they are separated. They avoid attending social events or activities without their caregivers being present, and in severe cases, may try to avoid attending school to prevent separation. Children may also insist on sleeping near a caregiver and refuse to sleep away from home. Prevalence rates for separation anxiety disorder are estimated at 4%. Whereas some degree of separation anxiety is normative during the early toddler years, symptoms of separation anxiety disorder, interfering with the normal process of separation and individuation, may present as early as age 3. Initial onset is less common in adolescence.

Social Anxiety Disorder

Social anxiety disorder involves excessive fear of being embarrassed and of being negatively evaluated or rejected by others. To meet criteria for a diagnosis of social anxiety disorder, children and adolescents must fear negative evaluation or rejection by their peers (not only adults). Youth with social anxiety disorder typically avoid feared social situations, including participating in class, speaking informally with peers, performing or speaking in front of others, or initiating social plans. Prevalence rates for social anxiety disorder in youth are estimated at 7%. Prevalence is most often noted during early to mid-adolescence, although social anxiety disorder frequently presents in childhood as well.

Selective Mutism

Selective mutism is characterized by fear and unwillingness to speak with others. Youth suffering from selective mutism typically refuse to answer questions in class or speak when spoken to in informal settings, although they often feel comfortable speaking with parents, family members, or close friends. Selective mutism is highly comorbid with social anxiety disorder. Symptoms must persist for at least 1 month. Prevalence estimates are below 1% with median age of onset at 3 to 4 years of age. It is thought that selective mutism may represent an extension of early-onset social phobia.

Generalized Anxiety Disorder

Generalized anxiety disorder (GAD) is characterized by excessive and often uncontrollable worry that persists more days than not for at least 6 months. Typical areas of worry in childhood include school and relations with peers, family health and relationships, personal and family safety, world affairs, and the future (eg, going to college, adult responsibilities, death). To meet criteria for GAD, youth must also experience at least 1 of the following associated symptoms: muscle tension/tightness, difficulty sleeping, impaired concentration, irritability, fatigue, and increased restlessness. Approximately 1% to 2% of children and 1% to 4% of adolescents suffer from GAD.

Specific Phobia

A specific phobia is diagnosed when a child or adolescent exhibits an excessive fear of a particular situation or object. Common specific phobias in childhood and adolescence involve fear of certain animals, heights, injections, water, loud noises, vomiting, flying, choking, or dark areas. The fear must be considerably greater than would be expected of same-aged youth. Approximately 5% of children meet criteria for a specific phobia, with the prevalence rate increasing to at least 13% in adolescence. Phobias of animals and environmental situations, such as heights and water, have an earlier average age of onset, whereas situational phobias, including flights and elevators, are more common in adolescence.

Panic Disorder

Panic disorder is characterized by the occurrence of panic attacks and anticipatory anxiety about subsequent panic attacks or avoidance of places or situations in which a panic attack is feared. At least 1 panic attack must be untriggered, or unprovoked by an anxiety-provoking situation or thought. That is, panic disorder would not be diagnosed if all panic attacks occurred within the context of a feared social situation or around an animal that the youth fears. A panic attack is defined as an episode of intense anxiety and at least 4 physical symptoms, such as pounding or racing heart, sweating, shortness of breath, dizziness, nausea, or feelings of choking. When experiencing a panic attack, youth may also experience feelings of de-realization or fear of dying or losing control. Panic disorder is uncommon in children, with prevalence rates estimated at less than 0.4%; prevalence rates gradually increase through adolescence. However, these low rates may reflect children being unable to communicate the specific physical experiences and consequent anxiety associated with panic disorder.

While obsessive-compulsive disorder and post-traumatic stress disorder are organized under separate sections of the DSM-V, they are often categorized as anxiety disorders.

Obsessive-Compulsive Disorder

Obsessive-compulsive disorder (OCD) is characterized by the presence of both obsessions, which are unwelcome and recurring thoughts, impulses, or images that increase anxiety, and compulsions, which are behaviors or mental actions that are completed repeatedly to reduce the anxiety caused by obsessions. These compulsions are often performed to avoid feared outcomes, but are usually excessive or unreasonable. Youth may also avoid known triggers in an effort to avoid or suppress their obsessive thoughts. The prevalence of OCD is thought to be between 1% to 2%, with 25% of patients developing the disorder before age 14.

Post-Traumatic Stress Disorder

Post-traumatic stress disorder (PTSD) affects some children and adolescents who have undergone, witnessed, or indirectly experienced a traumatic event. To meet criteria for PTSD, youth must exhibit at least 1 re-experiencing symptom, including intrusive memories or nightmares, "flashbacks" about the event, or distress and/or physiological reactivity. Youth must exhibit avoidance of people, places, or thoughts and feelings associated with the event. Finally, youth experience distorted perceptions and emotions, such as negative beliefs about their future, diminished interest in activities, failure to remember details of the event, hypervigilance, feeling detached from others, or constricted affect. A modified set of criteria were included for children ages 6 years and younger in the *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition, with the major modification being that young children must experience either avoidance of stimuli associated with the trauma *or* distorted perceptions or emotions. These symptoms must persist for at least 30 days following the traumatic event. The prevalence of PTSD in children is thought to be less than 0.5%, while for adolescents, that rate may be as high as 5%.

ETIOLOGY

Anxiety disorders result from the interaction of genetic predisposition and environmental effects. Research has suggested an association between the 5-HTTLPR polymorphism and avoidance, but genetic studies overall have been limited and mixed. There is strong familial aggregation for anxiety disorders. Anxiety disorders are more prevalent in first-degree relatives of children with anxiety disorders compared to those of children without an anxiety disorder, although the relative contributions of genes and environmental influence are unclear. With regard to environmental effects, parental factors including modeling of anxious behavior and high parental control have been identified as potential contributors to increased anxiety in childhood. Low perceived control, or the belief that one has little control over one's environment, has also been associated with increased anxiety. More specifically, low perceived control has been found to mediate the relationship between high parental behavioral control and increased anxiety. Future research is needed to understand more clearly the factors that contribute to the onset of anxiety disorders.

APPROACHES TO SCREENING FOR CHILDHOOD ANXIETY DISORDERS

Identifying Symptoms of Anxiety Disorders

Identifying an anxiety disorder in a child or adolescent is often difficult because many hallmark symptoms of anxiety, including anxious cognitions, avoidance, and safety-related behaviors, are unobservable, completed privately, or understood by parents and other adults as "normal." Furthermore, young children often lack the capacity to identify and verbalize anxiety-related thoughts and feelings, and adolescents are frequently embarrassed to acknowledge symptoms of anxiety. As a result, pediatric anxiety disorders go unnoticed and untreated.

Often, the presence of physical symptoms may be the main indicator of a pediatric anxiety disorder. All anxiety disorders can present with a range of impairing physical symptoms (eg, headaches, gastric distress, insomnia, increased or decreased appetite, and increased activity). It is important that when these physical symptoms present themselves, and once medical causes are ruled out, a treating physician looks for the presence of an anxiety disorder. Parents and healthcare providers should consider potential signs, or "red flags" (**Table 93-1**), that may indicate the presence of each anxiety disorder in children and adolescents.

TABLE 93-1 **OBSERVABLE SIGNS OF ANXIETY DISORDERS IN CHILDREN AND ADOLESCENTS**

Disorder	Observable Signs
Separation anxiety disorder	• Child is overly "clingy" to parent • Child struggles to separate from parents at school arrival or social event • Child frequently calls parents when separated from them • Child requires parents to sleep with him/her
Social anxiety disorder	• Child avoids participating in class and avoids certain classes • Child sits alone at lunch, recess • Child does not make or attend many social plans • Child avoids speaking to unfamiliar people or attending events in which unfamiliar people will be present • Child refuses to answer telephone or order food from a restaurant
Selective mutism	• Child avoids speaking to teacher or classmates • Child does not answer questions asked of him/her • Child speaks to family members at home, but not if less familiar people are present
Generalized anxiety disorder	• Child repeatedly asks parents or other adults for reassurance • Child expresses worries that are not developmentally appropriate (eg, finances, terrorism) • Child endorses frequent stomachaches, headaches, muscle tension, or difficulty sleeping
Specific phobia	• Child avoids situations that most same-aged peers typically face • Child avoids eating due to fear of vomiting or choking • Child refuses to visit pediatrician due to fear of injection • Child refuses to sleep alone or attend movies due to fear of darkness • Child becomes highly anxious when anticipating feared situations or objects
Panic disorder	• Child complains of episodes of intense anxiety or physical symptoms (e.g., racing heart, sweating, nausea, dizziness) • Child avoids places in which he/she fears he/she will not be able to get help if anxious
Post-traumatic stress disorder	• Child exhibits marked change in mood or affect following a traumatic event • Child avoids all reminders of a traumatic event he/she experienced or witnessed • Child becomes more distant from family members or friends • Child loses interest in activities • Child demonstrates markedly increased vigilance or startle reflex
Obsessive-compulsive disorder	• Child expresses concerns that appear excessive or illogical, such as contracting germs from casual contact or causing harm to others • Child engages in actions repeatedly, such as washing/cleaning, checking, arranging, repeating tasks, or seeking reassurance, and finds it difficult to stop • Child becomes upset with parents or others who interrupt repeated actions, or who fail to assist in repeated actions

Normative Anxiety versus Anxiety Disorders

It is important for clinicians to distinguish between normal and pathological anxiety, keeping in mind that anxiety is an expected and typical part of development. Normal fear and anxiety follow a predictable course. It is largely understood that children's cognitive development serves to mediate the typical increase and decrease of certain fears and worries. In infancy, toddlerhood, and early childhood, fears may revolve around immediate and concrete threats. With infants exploring their world through direct sensory and motor contact, it follows that their fears may include loud noises and loss of physical contact from their caregivers. At 9 months of age, children are able to differentiate between familiar and unfamiliar faces and may present with typical stranger anxiety and separation concerns. As a child's cognitive and social capacities increase, their fears become more sophisticated and may incorporate anticipatory worries. Common preschool worries can include fears of thunder and lightning, the dark, or animals. Between the ages of 7 and 12, children develop an understanding of cause-and-effect relationships and anticipation of dangerous events. Concerns increasingly include health and harm as well as scrutiny and competence. Elementary school–aged children may present with worries about natural disasters, fears of traumatic events, school anxiety, and imaginary fears including ghosts and monsters. With adolescence, worries include fears of negative evaluations, rejection by peers, and anxiety about performance. Adolescents are also cognitively able to connect their physical symptoms to thoughts of anxiety. This may explain an increased awareness of panic cognitions for the adolescent, such as "fear of going crazy" when describing a panic attack.

Normative fears typically decrease with age and are relatively transient. Conversely, when anxiety becomes maladaptive, they are more persistent and typically interfere with a youth's academic, social, and/or family functioning. With regard to academic functioning, anxiety symptoms may lead to avoidance of schoolwork or even refusal to attend or stay in school. With regard to social functioning, anxiety symptoms may cause youth to avoid interacting with peers, attending social events, or stating their needs to others. With regard to family functioning, a youth's avoidance or need for reassurance may lead to family conflict or stress. Careful assessment of anxiety involves differentiating whether a child's fears are developmentally expected, and whether the worries present with increased frequency, severity, and persistent distress. **Table 93-2** summarizes common normative fears and anxiety disorders associated with developmental levels.

TABLE 93-2 COMMON NORMATIVE FEARS AND ANXIETY DISORDERS ASSOCIATED WITH AGE

Age	Common Fears	Anxiety Disorders
Infancy/toddlerhood, 0–3 years	Stranger anxiety Separation anxiety Fear of thunder, lightning, darkness, animals	Specific phobias Separation anxiety disorder
Early childhood, 4–5 years	Fear of death and dying Being kidnapped Blood	Specific phobias Separation anxiety disorder
Elementary school age, 5–10 years	Fear of specific objects (animals, monsters, ghosts) Fear of natural disasters (accidents, traumatic events) Germs, disease School anxiety	Generalized anxiety disorder Obsessive-compulsive disorder
Adolescence, 12–18 years	Rejection from peers	Social anxiety Panic disorder

Speaking with Children and Parents about Anxiety

When assessing for anxiety in youth, there are several factors to consider. First, as discussed earlier, it is important to obtain sufficient information to distinguish between normative fears and an anxiety disorder. The youth’s developmental level and distress and impairment caused by anxiety in all areas of his or her life should guide this distinction. Also important to keep in mind is that youth with elevated anxiety often have difficulty describing their symptoms in detail, or may be embarrassed to share their symptoms with others. Providers should expect that youth may be reluctant to discuss anxiety, and will often underreport anxiety symptoms. Furthermore, given the heritability of anxiety disorders, the provider should be mindful that parents may also suffer from anxiety, and that parental anxiety may impact the presentation of a youth’s symptoms.

Providers should ask open-ended questions of the child and parents to increase the likelihood of accurate responses, and to build a sense of collaboration between provider and the child and family. It is further recommended that providers normalize the experience of anxiety by explaining that all individuals experience anxiety, and that anxiety can often be helpful in keeping us safe from danger. A fire alarm analogy may benefit youth and parents: Just like a fire alarm may be set off by burning food or steam from the shower, our anxiety system can sometimes be set off by people or situations that are not actually dangerous. Finally, providers should keep in mind that pediatric anxiety disorders may manifest via physical symptoms, and they should consider an anxiety disorder if youth complain of physical symptoms that are not explained by a medical problem.

Providing basic psychoeducation to youth and families about anxiety can also aid discussion of anxiety symptoms and provide some relief to youth and parents. It is often helpful to share that (1) anxiety disorders are common among children and adolescents, (2) anxiety disorders are not the fault of the child or parents but instead are similar to a medical problem that requires appropriate treatment, and (3) there are helpful treatments for anxiety disorders, including psychosocial interventions and medication.

Available Screening Measures

Early and routine screening during primary care visits can assist with early detection and intervention of anxiety disorders. Agreement between child- and parent-reports of anxiety symptoms are typically low, although agreement is higher for observable symptoms (eg, presence of compulsions) compared to nonobservable symptoms (eg, worry). As a result, the gold standard assessment includes multi-informant feedback from the child, parent/caregiver, and teachers, as appropriate. Self-report and caregiver rating scales for anxiety can assist clinicians in screening for anxiety symptoms at baseline and monitor response to treatment. Furthermore, self-reports are helpful in identifying anxiety in children who are less inclined to discuss their symptoms during the clinical examination. Follow-up discussions that elaborate on endorsed items on the self-report rating scale invite a youth to further elaborate on a symptom they may have otherwise minimized or downplayed during an interview. **Table 93-3** provides a description of the most commonly used self-report and parent-report rating scales for assessment of anxiety symptoms in children and adolescents.

TABLE 93-3 MEASURES TO ASSESS ANXIETY IN CHILDREN AND ADOLESCENTS

Anxiety: Questionnaires

Instrument (Author)	Construct(s) Assessed	Recommended Age Range	Time Period Assessed	Time to Complete	Version: Items (Long/Short)	Response Format	Scoring/Reports	Any Cost?
Revised Children's Manifest Anxiety Scale, Second Edition (RCMAS-2); Reynolds & Richmond, 2008	Total anxiety; 3 subscales: physiological anxiety, worry, social anxiety; also assesses "defensiveness" (lie scale) and has "inconsistent reporting index"	6–19 years	Not specified; current; "true of you" trait	10–15 minutes	Youth: 49/10	Each item rated true or not true	Sum of items; total scores with cutoff points	Yes; published by WPS
Multidimensional Anxiety Scale for Children (MASC); March et al., 1997	Anxiety symptoms; 4 factors: physical symptoms, social anxiety, harm avoidance, separation anxiety	8–19 years	"Recently"	< 15 minutes	Youth: 50/10 Parent: 50	Each item rated on 4-point scale ranging from "never true about me" to "often true about me"	Sum of items; T-score cutoffs for different severity levels and for "caseness"	Yes; published by Multi-Health Systems
Screen for Child Anxiety Related Emotional Disorders (SCARED); Birmaher et al., 1999	Anxiety disorder, including school avoidance	9–19 years	Last 3 months	10–15 minutes	Youth and parent: 41/5	Each item rated on 3-point scale: "not true," "sometimes true," "very true"	Total scores with recommended cutoff points by age/gender	No; available at www.psychiatry.pitt.edu/research/tools-research/assessment-instruments
Spence Children's Anxiety Scale (SCAS); Spence, 1998	Anxiety disorders	7–17 years	Not specified; current (items written in present tense)	10 minutes	Youth and parent: 38	Each item rated on 4-point scale ranging from "never" to "always"; some items reverse scored	Sum of items; subscale scores for specific anxiety disorder given	No; available at www.scaswebsite.com
Revised Child Anxiety and Depression Scales (RCADS); Chorpita et al., 2000	Anxiety disorder; depression	Grades 3–12	Not specified; current (items written in present tense)	10 minutes	Youth and parent: 47	Each item comprises 4 statements of increasing intensity /severity (never to always)	Sum of items; T-score from appropriate grade level	No; available from authors, www.childfirst.ucla.edu

Anxiety: Clinician-Administered Measures

Instrument (Author)	Construct(s) Assessed	Age Range	Time Period Assessed	Time to Complete	Informant: Items (Long/Short)	Response Format	Scoring/Reports	Any cost?
Hamilton Anxiety Rating Scale (HAM-A); Hamilton, 1959	Psychic anxiety (mental agitation and psychological distress) and somatic anxiety (physical complaints related to anxiety)	Children, adolescents, adults	Current (not specified)	10–15 minutes	Youth only: 14 items	Each item has 4 statements of increasing intensity/severity (none to severe)	Sum of items; suggested cutoffs for different severity levels and for "caseness"	No, public domain
Pediatric Anxiety Rating Scale (PARS); RUPP Anxiety Study Group, 2002	Anxiety symptoms and impairment for social phobia, separation anxiety, generalized anxiety disorder	6–17 years	Last week	< 20 minutes	Youth and parent: 50 symptom items; 7 global items	Each symptom item scored as present or absent (yes/no); global ratings rated by clinician on 7 dimensions of severity using a 6-point scale of increasing severity	Total score is sum of 7 global severity items	No; obtain information and scale from Mark Riddle, MD, mriddle1@jhmi.edu

DSM, *Diagnostic and Statistical Manual of Mental Disorders*.

Differential Diagnosis

Anxiety disorders have been reported to have a significant degree of inter-anxiety (homotypic) and heterotypic comorbidity. By late adolescence and young adulthood, comorbidity is common for youth affected with an anxiety disorder. With respect to heterotypic comorbidity, pediatric anxiety is a significant predictor of adult anxiety disorders, depression, and substance use problems. There is also an association between a higher number of anxiety disorders and an increased risk for heterotypic comorbidity later in life. In a 21-year longitudinal study in New Zealand, significant associations were found between the number of anxiety disorders reported and the risk for anxiety disorders, major depression, substance use, and suicidal behavior. Additionally, the authors found an association between the number of anxiety disorders and educational underachievement and teen pregnancy. As individuals present with a higher number of comorbid anxiety disorders, there is an increased risk for depression. Interestingly, however, earlier onset of anxiety disorders has not been found to confer an increased risk for depression. With respect to treatment response, comorbid anxiety and depression in children and adolescents is often associated with increased symptom severity and treatment resistance. With respect to behavioral disorders, 20% to 40% percent of children and adolescents with ADHD have comorbid anxiety. Anxiety disorders are often more challenging to identify in this population, as the disruptive behaviors are often the chief complaint and reason for referral. Parent reports may be skewed with respect to the more outward and disruptive observable behaviors. Children with comorbid anxiety and ADHD may also uniquely present with increased working memory deficits and with less hyperactivity. Continued careful assessment that includes self-report of internalizing symptoms for children is important in capturing the full picture of ADHD and potential mood or anxiety symptoms.

TREATMENT APPROACHES

Psychotherapy

The front-line evidence-based psychotherapy for anxiety disorders in children and adolescents is cognitive behavioral therapy (CBT). Numerous studies have noted the efficacy of CBT. Approximately two-thirds of children treated with CBT report significant relief and improvement. Cognitive behavioral therapy is time-limited, with treatment typically lasting for 16 to 20 weeks, followed by monthly booster sessions. Cognitive behavioral therapy involves the use of specific strategies and practices to reduce anxiety. The clinician focuses on restructuring maladaptive cognitions, reducing avoidant or anxiety-related behaviors, and guiding the child or adolescent in more consistently facing feared situations. Concepts and skills are taught through active instruction, role-playing, and both in- and out-of-session practice.

The components of CBT for childhood anxiety disorders include (1) psychoeducation; (2) self-monitoring of anxiety symptoms; (3) the examination and restructuring of cognitive errors that occur as a result of feeling anxious (eg, "black-and-white thinking," "catastrophizing," and "mind reading"); and (4) repeated exposure to people, places, or situations that are feared and avoided (eg, engagement in social situations, separating from primary caregiver). Additional strategies include relaxation techniques such as deep breathing, imagery, and progressive muscle relaxation. Problem-solving and communication skills may also be taught, depending on the presentation and needs of the child.

Psychoeducation Psychoeducation involves teaching youth and caregivers about anxiety disorders and their treatment. A therapist typically reviews the role of normative anxiety (eg, fight-or-flight response) and distinguishes between adaptive and maladaptive levels of and responses to anxiety. Children are further informed that anxiety disorders are common and treatable. A therapist often reviews the interrelationships between thoughts, feelings, and behaviors, and spells out that treatment will target each of these components to reduce anxiety. This component of treatment should also focus on the role of avoidance in maintaining anxiety, and should provide a rationale for repeated exposure to feared situations. Finally, the goals for treatment are discussed, which generally relate to the overarching goal of more consistently facing feared situations to reduce overall anxiety.

Self-monitoring The next stage of treatment involves information gathering via self-monitoring: (1) Which situations trigger anxiety? (2) How does the child respond to that trigger? (3) What physical symptoms does the child experience? (4) What thoughts does the child experience when anxious? The therapist works with the child to identify the cognitions and behaviors that contribute to anxiety maintenance.

Cognitive Restructuring Based on information gathered from self-monitoring, the therapist and child identify common thoughts that the child experiences when feeling anxious. The therapist and child then generate more realistic or alternative thoughts, either by examining the evidence for and against the thoughts (eg, the likelihood that the event will happen), determining the function of the thoughts and how to better manage them (eg, distancing oneself from the thoughts, talking back to the thoughts, seeing the thoughts as false alarms), or playing out the thoughts to the worst outcome (eg, using a "downward arrow" technique to identify a feared outcome). Often, coping statements or more rational responses are created and rehearsed in response to different situations. For younger children, cognitive restructuring may be limited to generating simple coping statements.

Exposure to Feared Situations To modify learned associations with feared and avoided situations, youth are encouraged to begin gradually and repeatedly facing feared and avoided situations. The facing of a feared situation is referred to as an *exposure*. In preparation for completing exposures, the therapist and the youth (and the caregiver, if appropriate) collaboratively create a list of anxiety-provoking situations and rate the child's anxiety in each proposed situations. With guidance and support from the therapist, the youth begins to engage in exposures both in and out of therapy sessions. For instance, a child with social anxiety disorder will practice facing feared social situations, such as initiating conversations with others, asking questions of unfamiliar people, or engaging in mildly embarrassing behavior. Cognitions experienced during the exposure are also reviewed and challenged as necessary, and coping statements may be used to help encourage the youth to successfully complete the exposure. As treatment progresses, the youth gradually and repeatedly faces all situations typically avoided due to anxiety, making sure to monitor and eliminate any behaviors that seek safety (eg, compulsions, reassurance seeking), as these behaviors reduce the efficacy of the exposures.

Family Involvement Treatment almost always involves the use of family/caregivers; the extent of caregiver involvement depends on the anxiety presentation and age of the youth, with caregiver involvement being essential with younger children. At minimum, psychoeducation is provided to family members (including siblings if needed) to (1) increase understanding of the child's anxiety; (2) increase support of the child, particularly as the child progresses through treatment; and (3) discuss appropriate ways in which the family can respond to the child's anxiety. The therapist may also need to work in greater depth with the child's family. Caregiver goals may be to minimize accommodations to anxiety, to model approach behavior versus avoidance, and to structure the parent's/caregiver's role of co-therapist in the home. Another common form of family involvement includes the use of reinforcement strategies, to reward exposure to feared situations, which helps to engage children in treatment and support their efforts.

Medication

Well-delivered CBT is effective and recommended as a first-line treatment option for mild to moderate anxiety symptoms. However, it can be difficult to access CBT due to cost and/or lack of availability. Also, youth may be unable to successfully engage in CBT due to a number of factors, such as severity of symptoms, parental accommodation of anxiety symptoms, and comorbid diagnoses. Medication treatment, and specifically the class of antidepressants referred to as selective serotonin reuptake inhibitors (SSRIs), are considered first-line treatment as a monotherapy and in combination with CBT for pediatric anxiety disorders (**Table 93-4**). In cases of moderate to severe anxiety, CBT together with SSRI treatment is the optimal approach.

TABLE 93-4 MEDICATIONS APPROVED FOR ANXIETY DISORDERS IN CHILDREN AND ADOLESCENTS

Generic Name (Brand Name)	Subclass	Dosing	FDA-Approved Uses	Ages	Additional Information
Fluoxetine (Prozac)	SSRI	20–60 mg/d Start 10 mg/d, increase to 20 mg/d on day 14	OCD	7 y and older	Long half-life (4–6 d) Available in liquid form
Sertraline (Zoloft)	SSRI	25–200 mg/d Start at 25 mg/d and increase to 50 mg after 7 days; increase to 50 mg and then may increase every 4 weeks	OCD	6 y and older	Available in liquid form
Fluvoxamine (Luvox)	SSRI	50–200 mg/d Oral daily dosing to twice/day Start 25 mg at bedtime, increase by 25 mg/d q4–7 d; maximum dose is 200 mg/d up to 11 y, 300 mg/d for > 11 y	OCD	8 y and older	
Duloxetine (Cymbalta)	SNRI	30–60 mg/d Start 30 mg/d; may increase by 30 mg/d after 2 weeks	GAD	7 y and older	Used for treatment of neuropathic pain in adults
Clomipramine (Anafranil)	TCA	100–200 mg at bedtime Start 25 mg qd, increase by 25 mg/d q4–7 days; maximum 3 mg/kg/d up to 100 mg/d in first 2 wk and up to 200 mg/d maintainance	OCD	10 y and older	

GAD, generalized anxiety disorder; OCD, obsessive-compulsive disorder; q4–7, every 4 to 7; qd, once daily; SNRI, serotonin norepinephrine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

When a child begins a course of SSRI treatment, it is initially started at a low dose to assess tolerability. It is important for the provider to increase the dose to an adequate level to ensure treatment success while assessing carefully for adverse events. Dose increases should be made after the provider has assessed the child's level of response as well as tolerability of the medication. The adage "start low and go slow" is helpful to keep in mind when considering a course of SSRI treatment in youth. However, it is very important to achieve an adequate dose range, especially if the child remains symptomatic. It can be demoralizing for a child to be inadequately treated, leading to lack of response and delayed relief from symptoms. The most common adverse side effects associated with SSRI medications include headaches, upset stomach, diarrhea, and nausea. Weight gain is also a possible side effect, and weight should be monitored regularly. Adolescents should be made aware of the possibility of sexual side effects with SSRI treatment, and providers should inquire about sexual side effects during follow-up visits. Most side effects resolve over time, usually after 2 to 4 weeks, and it is helpful to maintain an inventory of physical symptoms that are reported at each visit. In addition, severity level of anxiety symptoms should be systematically assessed prior to medication initiation and subsequently at follow-up visits.

Family history of anxiety disorders may help guide medication choices based on treatment response by family members. Prescribers will also need to discuss the US Food and Drug Administration black box warning for antidepressants. It advises clinicians to monitor patients for increased risks of suicidality in children, adolescents, and young adults (under the age of 24 years) especially during the beginning of pharmacological treatment and during dose changes. It has been well documented that the benefits of treatment with antidepressant medications outweigh the risks. Another important recommendation is for the medication to be taken around the same time every day.

During the first 1 to 2 months of treatment with an SSRI, weekly or biweekly visits are recommended. If an actual visit is not possible, follow-up phone calls may be used. During follow-up visits, the prescriber should consider the following: (1) evaluate for side effects and medication tolerability, (2) assess for improvement and also signs of worsening, (3) assess adherence with medication and psychotherapy appointments, and (4) examine new and ongoing psychosocial stressors since last visit. The provider may also consider adjusting the timing of medication administration (eg, morning versus evening) based on side-effect occurrence, such as sedation or activation. Psychoeducation about developmentally normal fears and worries, anxiety disorders, and their treatment should be ongoing.

When the presenting anxiety is severe, or if insomnia due to anxiety is unresponsive to a behavioral sleep hygiene approach, youth may benefit from the short-term use of a low dose of diphenhydramine (Benadryl). If diphenhydramine is not effective, a benzodiazepine such as clonazepam is an alternative option, during the initial 4 to 6 weeks of treatment with an SSRI.

A therapeutic trial for SSRI medication is defined as a time frame of 8 to 16 weeks, during which treatment is provided at the maximum dose tolerated by the patient. If there is no improvement in symptoms after this time period, the provider should reassess for medication adherence, diagnosis, and comorbidities, and should consider the initiation of CBT if it has not been initiated previously. For patients who have not responded to an adequate medication trial, it is reasonable to consider a trial with a different SSRI. Another class of medications known as serotonin norepinephrine reuptake inhibitors (SNRIs; eg, duloxetine) are considered a third-line option following 2 adequate courses of an SSRI. For obsessive-compulsive disorder, clomipramine, which is a tricyclic antidepressant, is commonly used as a third-line option following 2 adequate trials of a SSRI. Clomipramine requires additional medical monitoring, including electrocardiograms at baseline, during titration, and at the maximum dose. Also, there is a therapeutic blood level that can be monitored.

It is recommended that an efficacious medication trial continue for 6 to 12 months after resolution of symptoms. Providers should consider delaying discontinuation if a youth is undergoing a life transition or potential stressor, such as a new academic year or moving away to college. Doses of SSRI medication should be tapered off gradually over several weeks to prevent discontinuation symptoms. Providers should schedule monthly follow-up visits for at least 1 to 2 months after the medication is discontinued in order to assess symptoms of discontinuation or relapse of symptoms.

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94 Pediatric Depression and Bipolar Spectrum Disorders

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INTRODUCTION

Affect regulation is a developmental acquisition that is impacted by environmental factors. For clinicians caring for children and adolescents, it is important to differentiate between normal mood fluctuations and emotions and pervasive mood disorders that significantly impact the child's global functioning. A careful clinical assessment is necessary to make the diagnosis of mood disorders in children and adolescents, as there are no laboratory or neuroimaging tests that will ascertain the diagnosis. The clinical diagnosis of mood disorders is supported by research data obtained in epidemiological and longitudinal studies, as well as by clinical observations.

This chapter discusses how clinical diagnosis of both depressive and bipolar spectrum disorders is made based on the best available evidence. Additionally, this chapter provides an overview of both psychopharmacologic and psychotherapeutic treatments of depressive and bipolar spectrum disorders.

EPIDEMIOLOGY

In the *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition (DSM-V), the pediatric depressive disorders include disruptive mood dysregulation disorder (DMDD), major depressive disorder (MDD), persistent depressive disorder, and premenstrual dysphoric disorder. Within the DSM-V, the bipolar disorders include bipolar I disorder, bipolar II disorder, cyclothymic disorder, and unspecified bipolar disorder.

Prevalence of MDD in children ranges between 0.5% and 1.4% and increases to 11.4% in adolescents, with a male-to-female ratio of 1:1 during childhood and 1:3 during adolescence. The risk for MDD increases significantly after puberty, particularly in females. Disruptive mood dysregulation disorder is a relatively new diagnosis in the DSM-V; hence, the prevalence of this illness has not been clearly established. However, epidemiological studies that used criteria of DMDD retrospectively found the prevalence to range between 0.8% and 3.3%. The rate of DMDD is shown to be higher in males and school-aged children compared to females and adolescents. Studies have found the rate of bipolar spectrum disorders to be as high as 6.7%, with lifetime prevalence of mania to be between 0.1% and 1.7% among adolescents. Bipolar spectrum disorders have a male-to-female ratio of 1:1 throughout the life cycle.

DIAGNOSING MOOD DISORDERS

Presentation of Mood Disorder Symptoms

The DSM-V has clearly defined criteria for diagnosing depression and bipolar disorders in adults, children, and adolescents. There are no separate criteria for mood disorders in the pediatric population. However, the manifestations and presentation of mood symptoms in children and adolescents differ from those of adults. These differences likely result from the developmental and neurodevelopmental differences between adults and children in the emotional, cognitive, and physical domains. Therefore, although the DSM-V criteria are utilized to make a diagnosis of a mood disorder in a child or adolescent, clinicians must be aware of the impact of development on the presentation of mood disorders in the pediatric population.

For instance, when children experience depression, parents, caregivers, and schoolteachers may notice the child being more withdrawn; crying more easily and more than usual; complaining of stomachaches, headaches, or other somatic complaints; refusing to go to school; having significant changes in peer relationships; and having new behavioral problems. Children may have a decline in school performance, which may be due to difficulty concentrating related to depressed mood, and they may complain of boredom, which may suggest anhedonia. Also, children may view themselves in a negative light. Therefore, to make an evaluation of depression in a child, the clinician needs to be aware of the changes that have been and are occurring in the child’s daily routine. This can assist the clinician in making a DSM-V diagnosis of a depressive disorder.

It is not uncommon for youth with mood lability, anger, and aggression to be considered for a diagnosis of bipolar disorder. Clinicians must explore the duration and quality of these symptoms while considering a diagnosis of a mood disorder in these youngsters.

DSM-V Diagnostic Criteria for Depression and Bipolar Spectrum Disorders

Criteria for Depressive Disorders The criteria for a major depressive episode requires the presence of 5 or more of the following symptoms during a 2-week period: depressed mood, loss of interest/pleasure, changes in appetite or weight, sleep disturbance such as insomnia or hypersomnia, psychomotor retardation/agitation, diminished energy, diminished concentration, feelings of guilt or worthlessness, and recurrent thoughts of death or suicidal ideations. One of these symptoms must be either depressed mood or loss of interest/pleasure. The symptoms also must lead to significant impairment in important areas of functioning, and must not be related to the effects of a drug/substance or another medical condition.

Disruptive mood dysregulation disorder is classified as a depressive disorder in which children or adolescents have persistently irritable or angry mood nearly every day, and in which temper outbursts occur 3 or more times per week. These outbursts are disproportionate to the situation and developmental level of the child. The above characteristics must all be present for 12 months, and there cannot be a 3-month or longer duration in which these symptoms have not occurred. These characteristics must also occur in at least 2 different settings (eg, school and home), and the diagnosis can only be made for children between the ages of 6 and 18 years. These symptoms must be unrelated to that of a hypomanic, manic, or major depressive episode, and must not be better accounted for by another mental disorder or be related to the effects of a drug/substance or medical condition.

Criteria for Bipolar I and II Disorders For a diagnosis of bipolar I disorder, the child needs to have an elevated, expansive, or irritable mood and increased goal-directed activity or energy that lasts for at least 1 week and is present almost every day for most of the day. Together with the change in mood and energy, 3 of the following symptoms (or 4, if mood is irritable) need to be present: inflated self-esteem or grandiosity; decreased need for sleep; increased talkativeness; flight of ideas; distractibility; increased goal-directed activity (eg, at school, outside of school in the form of hypersexual behaviors) or psychomotor agitation (that is, purposeless non–goal-directed activity); or excessive involvement in activities that can lead to poor consequences (eg, spending too much money, being involved in behaviors of a sexual nature). These mood changes lead to significant disturbance in social and occupational functioning and may require hospitalization. The mood changes are not due to the effects of a drug of abuse, medications, or another medical illness.

To make a diagnosis of bipolar II disorder, the child or adolescent needs to have met criteria for a hypomanic episode. The symptoms of mania and hypomania are the same as noted above, except that a hypomanic episode lasts for at least 4 days. A manic episode must last for at least a week. In hypomania, the mood disturbance is not severe enough to lead to a marked impairment in overall functioning or to require hospitalization.

If psychotic symptoms are present during the mood changes, then the episode is defined as a manic episode with psychotic features.

Differential Diagnosis and Comorbidity

Both depression and bipolar disorders have symptoms that can overlap with other psychiatric disorders. These include oppositional defiant disorder, substance use disorders, post-traumatic stress disorder, adjustment disorder, and bereavement. Irritability is a symptom that is present across psychiatric illnesses. Therefore, whenever irritability is prominent in a child or adolescent, it is imperative to explore whether it is due to anxiety, substance use, a mood disorder, or another psychiatric disorder.

Mood disorders can occur due to medical conditions. Depression secondary to medical illnesses includes hypothyroidism, mononucleosis, anemia, and autoimmune diseases. Hyperthyroidism may mimic manic symptoms. Additionally, certain medications such as stimulants, corticosteroids, and contraceptives can cause side effects that overlap with symptoms of depression. Therefore, a complete medical evaluation is always necessary to rule out medical conditions as the cause of mood symptoms. However, chronic conditions such as juvenile diabetes, leukemia, and mood disorders (commonly depression) may need treatment with psychotherapeutic or psychopharmacological interventions.

When a child or adolescent presents with mood symptoms, it can be difficult to differentiate between a depressive disorder, a bipolar depressive disorder, or even mania/hypomania. It is important to gather information about the course of illness to assess the presence of manic/hypomanic episodes, family history of psychiatric illnesses, and the level of functional impairment. For instance, youth with bipolar disorder are more likely to have a strong family history of bipolar disorder, greater levels of dysfunction in family relationships and occupational functioning, and prior psychiatric hospitalizations. Other indicators for bipolar disorder include a history of psychotic symptoms or medication-induced mania. Several symptoms of attention-deficit/hyperactivity disorder (ADHD) overlap with bipolar symptoms, such as distractibility, increased talkativeness, and increased energy levels. In ADHD, these symptoms are persistently present during the course of the illness. However, since bipolar disorder is an episodic mood disorder, in youth with bipolar disorder, these symptoms are also episodic in nature. On the other hand, depressed youth may present with symptoms of inattentiveness, such as difficulty with sustaining attention, absent-mindedness, and avoidance of tasks that require mental effort. In such cases, it is important to ascertain whether the youth met criteria for ADHD prior to the onset of the depressive episode.

Most cases of MDD are associated with psychiatric comorbidity. It is estimated that up to 40% to 90% of youth with a diagnosis of MDD will have a comorbid psychiatric disorder. Anxiety and disruptive behavior disorders are commonly associated with MDD, with a 4-fold increased risk. Additionally, ADHD and substance use disorders are associated with MDD, with a 3-fold increased risk. The rate of comorbidity is extremely high in youth with DMDD, which is most commonly associated with oppositional defiant disorder. Bipolar disorder frequently co-occurs with ADHD, with risk being as high as 69%. Other disorders that are common comorbidities in adolescents with bipolar disorder include conduct disorder and substance abuse.

PATHOGENESIS

The pathogenesis of affective illnesses is best conceptualized as an interaction between individual genetic vulnerabilities and exposure to adverse life events. The prevalence of a first-degree parent with depression is 30% to 50% in depressed youth, suggesting a strong genetic predisposition for this illness. Similarly, twin and family studies have shown high heritability of bipolar disorder. About 30% to 50% of patients with bipolar disorder have a first-degree relative with a bipolar spectrum illness. Other studies have suggested a heritability rate of 0.8.

Environmental factors, such as exposure to negative adverse childhood events, can influence the development of mood disorders. In fact, research has shown that genes are highly regulated by environmental signals throughout childhood development. Stressful events such as losses, abuse, neglect, and interpersonal conflict can moderate or mediate the onset and recurrence of depression. The effect of these stressors is an interplay between a child’s negative attributional styles for interpreting and coping with stress, the availability of a caregiver’s support, and genetic factors.

With the advances in brain imaging technology, there has been an increasing interest in the study of the neurobiological underpinnings of mood disorders.

Studies conducted by Rao have shown that depressed adolescents and those at high risk for depression have significantly smaller left and right hippocampal volumes in comparisons to healthy controls. Furthermore, smaller hippocampal volumes were associated with higher levels of early-life adversity. Imaging studies conducted on bipolar youth have shown decreased total cerebral and smaller hippocampal volumes.

ASSESSMENT OF YOUTH WITH MOOD DISORDERS

The assessment of a youth presenting with mood concerns can be summarized to include 3 main components. The first is screening for the presence of mood symptoms in a developmentally sensitive manner. When assessing the pediatric population for mental illness, it is important to obtain information from multiple sources to help develop an accurate picture of the illness and its course. The primary sources for information include (but are not limited to) the caretakers, the teachers, and the patient. Caretakers are better able to provide information pertaining to the onset of the illness and the external manifestations of mood disorders such as disruptive behaviors, sleep or appetite changes, decline in school performances, and increased isolation. It is important for the clinician to keep in mind the possibility that the caretaker's own psychopathology may influence their report of symptoms.

Adolescents should be interviewed alone to help develop rapport. Matters pertaining to confidentially should be discussed with adolescents at the beginning of the interview. Clinicians have the right to preserve confidentiality when working with the pediatric population as determined by state law. Children may have difficulty articulating their internal emotional state, reinforcing the importance that the clinician be attuned to the observable manifestations of mood disorders and the level of impairment.

Secondly, the diagnosis of mental illnesses is typically based on clinical presentation. Rating scales can be used to help assess the severity of symptoms and guide clinical management. Clinician-based rating scales include the Children's Depression Rating Scale, Revised, and the Young Mania Rating Scale. Parent-completed rating scales include the Child Behavior Checklist (which assesses for internalizing and externalizing disorders), the Child Mania Rating Scale, and the Quick Inventory of Depressive Symptoms. No laboratory tests are currently available to assist clinicians in making a diagnosis of a mood disorder.

Lastly, youth presenting with mood concerns should be assessed for the presence of suicidal ideation, self-injurious behaviors, and homicidal ideations (discussed further below). Preventing access to means of suicide is a vital suicide prevention strategy in adolescents. Studies have shown that the presence of firearms in households increases the risk of adolescent suicide, and restricting the access to firearms reduces this risk. It is important to differentiate suicidal behavior from other forms of self-injurious behaviors. Youth who engage in self-injurious behaviors may do so with the intent to find relief from a negative affective state versus intent to end their life.

TREATMENT OF DEPRESSION AND BIPOLAR SPECTRUM DISORDERS

Mental health disorders are among the leading causes of disability in the United States, making their recognition and treatment imperative. Approximately 20% of youth in America struggle with mental health concerns, and about 50% of all lifetime cases of mental health begin by age 14. Of note, mood disorders are a significant public health concern, as they account for extensive long-term morbidity. As primary care providers, pediatricians are required to screen for and address many issues within limited appointment times, often making mental health screening difficult. However, pediatricians are often the first line for families and patients struggling with mental health concerns, making it even more important for pediatricians to be well versed in recognizing depression and bipolar disorder, and knowing when to treat and when to refer.

Pediatricians may be hesitant to prescribe psychotropic medications to children and adolescents secondary to concerns over side effects or inexperience with treatment options, choosing instead to refer to a child and adolescent psychiatrist. However, due to the difficulty in accessing a child psychiatrist, it will be helpful for pediatricians to become familiar with the initial management of depressive and bipolar disorders in order to prevent delays of treatment and subsequent treatment resistance. Delays in treatment can lead to substantial sequelae such as worsening comorbid psychiatric issues, substance use, and decreased self-esteem. The content below provides guidelines for pediatricians aimed at the initial management of mood disorders.

Depression

The management of depression in children and adolescents involves medications, psychosocial interventions such as psychoeducation for parents and school personnel, and psychotherapy. Selective serotonin reuptake inhibitors

(SSRIs) are first-line medications for the treatment of depressive disorders. Their mechanism of action is thought to be primarily at the site of the serotonin transporter on the presynaptic terminal, inactivating it, thus preventing the reabsorption of serotonin back into the presynaptic terminal. This allows for higher levels of serotonin to bind to the postsynaptic serotonin receptor. There are 5 SSRIs used commonly in adults for depressive disorders; however, only 2 of them are approved for depression in children and adolescents. Fluoxetine, the oldest SSRI, is still considered the best initial choice for the treatment of depression in youth. Fluoxetine has Food and Drug Administration (FDA) approval for MDD in children and adolescents aged 8 to 18 years. Escitalopram has FDA approval for MDD in adolescents aged 12 to 17. Citalopram and sertraline have been shown to be helpful in pediatric depressive disorders but do not have FDA approval for this use. Paroxetine, although useful in adults, has had limited use in children and adolescents, as initial data regarding suicide risk led to an FDA statement proclaiming paroxetine to be unsafe in children and adolescents. Fluvoxamine, also an SSRI, is more commonly used for obsessive-compulsive disorder, for which it has a pediatric indication. Once a medication is started, pediatricians should be aware of potential side effects and the possibility of worsening thoughts (see below). The FDA recommends contact with patient and family on a weekly basis during the first 4 weeks of treatment, followed by biweekly assessments through week 12.

Psychosocial support plays a strong role in the treatment of depressed children and adolescents. Family support and education about the child's illness are a crucial aspect of treatment. As depression can impact social functioning and lead to symptoms such as lack of motivation, fluctuating energy, and decreased concentration, it is important to make school personnel such as teachers, daycare staff, and other adults involved in the child's life aware of the issues in order to allow for a better understanding of the child and enhanced support in those settings. Psychotherapy has an important role in the management of depression as well. For mild depressive disorders with time-limited stressors, psychotherapy is often the only treatment needed. Cognitive behavioral therapy (CBT) has robust evidence of effectiveness in the treatment of youth with depression. This therapeutic modality focuses on individuals' thoughts, feelings, and actions under the premise that changing maladaptive ways of thinking can change individuals' feelings and behavior patterns. Interpersonal therapy (IPT) is also effective in youth with depression, focusing on interpersonal issues and role transitions that are thought to lead to distress.

Studies have shown that combining medications with psychotherapy is the most effective way to treat pediatric depression. The Treatment for Adolescents with Depression Study (TADS) aimed to examine whether fluoxetine, CBT, or combination therapy was more efficacious. The 12-week data in this study showed response rates of 61% with fluoxetine monotherapy, 43% with CBT only, and 71% with combination therapy. The 36-week data showed that response rates were similar for all 3 interventions with 81% for fluoxetine monotherapy, 81% with CBT only, and 86% with combination therapy. These results suggest that intervention for depression in adolescents can be effective and that combination therapy may lead to a more rapid recovery than either medication or therapy alone. The Treatment of Resistant Depression in Adolescents (TORDIA) study enrolled adolescents aged 12 to 18 who had not responded to an initial trial with an SSRI. These subjects were then divided into the following 4 groups: switch to a different SSRI (paroxetine, citalopram, fluoxetine), switch to a different SSRI plus CBT, switch to venlafaxine, or switch to venlafaxine plus CBT. After 12 weeks, the combination of either an SSRI or venlafaxine with CBT showed the best response rate (54.8%), again providing evidence regarding the importance of combination treatment. The 24-week data from this study showed an overall remission rate of 38.9%, but more importantly, the data showed that the likelihood an adolescent would achieve remission of symptoms was based on how much improvement was noted within the first 12 weeks. This further illustrates the need for pediatricians to be able to detect depressive symptoms and initiate treatment while waiting for an appointment with a child psychiatrist. Treatment for depression can also be done in conjunction with consultation from a child psychiatrist.

Bipolar Disorder

Bipolar disorder can be a difficult diagnosis to make in children and adolescents. A careful history includes drawing a timeline of mood episodes, demonstrating episodic hypomanic or manic episodes, which can help establish a diagnosis of bipolar disorder. For pediatricians, recognizing manic/hypomanic symptoms and subsequently referring the patient to a child and adolescent psychiatrist for further management is most appropriate. It is still important for pediatricians to be knowledgeable about the treatment options, as a referral may be difficult.

Until recently, pharmacologic treatment of pediatric bipolar disorder had been largely off-label. However, recent FDA-approved medications have expanded treatment options (**Table 94-1**). Antidepressants are not recommended as monotherapy in patients with bipolar disorder due to the emergence of manic/hypomanic symptoms in these patients. They should be used with caution in any bipolar spectrum patient. Strong therapeutic alliance and rapport between patient and provider also play a significant role in outcomes.

TABLE 94-1 FDA-APPROVED MEDICATIONS FOR TREATMENT OF BIPOLAR DISORDER

Medication	Indication	Monotherapy or Adjunct	Age Range
Lithium	Bipolar I mania	Monotherapy	12–17 y
	Bipolar I maintenance		12–17 y
Asenapine	Bipolar I acute manic or mixed	Monotherapy	10–17 y
Aripiprazole	Bipolar I acute manic or mixed	Monotherapy	10–17 y
	Bipolar I acute manic or mixed	Adjunct to lithium or divalproex	10–17 y
	Bipolar I maintenance (prevent recurrence of bipolar disorder)	Both	10–17 y
Quetiapine ER	Bipolar I acute manic	Monotherapy	10–17 y
	Bipolar I acute manic	Adjunct to lithium or divalproex	10–17 y
Olanzapine	Bipolar I acute manic or mixed	Monotherapy	13–17 y
	Bipolar I acute depressive in conjunction with fluoxetine	Note: not considered first line due to weight gain and metabolic concerns	10–17 y
Risperidone	Bipolar I acute manic or mixed	Monotherapy	10–17 y

FDA , Food and Drug Administration.

As with depression, psychotherapy in combination with medications is a key component in the treatment of pediatric bipolar spectrum disorders. Several therapies have been shown to be effective in bipolar youth. Family-focused treatment for adolescents (FFT-A), interpersonal social rhythm therapy (IPSRT), child- and family-focused cognitive behavioral therapy (CFF-CBT), multifamily psychoeducational psychotherapy (MF-PEP), and dialectical behavioral therapy (DBT) all have evidence for being helpful in youth with bipolar spectrum disorders.

Adverse Effects of Medications

Side effects for SSRIs include nausea, gastrointestinal discomfort, constipation/diarrhea, drowsiness, insomnia, akathisia (inner restlessness and inability to sit still), prolonged bleeding time, and sexual dysfunction. The most concerning side effect is the possibility of increased suicidal thinking and behavior, for which the FDA has implemented a black-box warning on all classes of antidepressant medications. It is important to understand that although this is a significant risk and requires close monitoring, far more children and adolescents have shown decreased suicidality following treatment with antidepressants. This was clearly depicted during the initial years following release of the black-box warning, which saw an increase in youth suicide rates.

In addition, SSRIs carry discontinuation-related side effects if medication is stopped abruptly. This can include nausea, headaches, dizziness, sleep dysregulation, agitation, and worsening depression and anxiety. All of the SSRIs should be decreased slowly. Fluoxetine, because it has the longest half-life, can theoretically be stopped without dose reduction, as it is thought to self-taper; however, many child psychiatrists still recommend step-wise dose reduction to minimize chances of discontinuation effects.

Side effects of lithium include nausea, vomiting, constipation, diarrhea, headache, dry mouth, weight gain, tremors, confusion, memory impairment, hypothyroidism, electrocardiogram changes, and nephrogenic diabetes insipidus. Lithium is also teratogenic, leading to an increase in the development of Ebstein anomaly. For this reason, a pregnancy test is recommended in all females of child-bearing age prior to initiation of lithium. Other baseline laboratory tests necessary prior to starting treatment with lithium include the ratio of blood-urea-nitrogen (BUN) to creatinine (Cr), and thyroid-stimulating hormone (TSH), which should also be assessed longitudinally every 3 months initially and approximately every 6 to 12 months once stable. Lithium also carries a very narrow therapeutic index and therefore requires close monitoring of levels, as doses are adjusted. It is recommended that lithium levels be checked 5 days following initiation and after each dose change. Afterward, lithium levels may be monitored every 3 to 6 months or when clinically indicated. A safe lithium level is generally 0.6 mEq/L to 1.2 mEq/L.

As a class, antipsychotic medications can lead to extrapyramidal side effects such as dystonia, akathisia, bradykinesia and parkinsonism, and tardive dyskinesia (irregular and involuntary jerky movements). These symptoms should be assessed at every visit. Antipsychotic medications also impact metabolic parameters such as weight gain, dyslipidemia, and glucose metabolism, and should be assessed at baseline and at scheduled intervals.

SUICIDE

Suicide is the second leading cause of death in adolescents aged 15 to 19. According to the Youth Risk Behavior Surveillance System, the largest public health surveillance system in the United States, 17.7% of high school students had thoughts about suicide during the 12 months prior to the survey. During that same period, 14.6% of students had made a suicide plan, with 8.6% of students carrying out a suicide attempt.

Risk factors for suicide include previous attempt; witnessing or learning about a suicide attempt; mental health or substance use disorder; chronic medical disorder; family history of suicide; significant trauma, such as physical or sexual abuse; self-injurious behavior, such as cutting; and access to means, such as medications or firearms. Protective factors include family and peer support, cultural beliefs, access to and continued treatment, and positive coping skills. Pediatricians should make sure that parents are aware of local crisis hotlines and the nearest emergency rooms, and have done their best to remove means for suicide, such as access to medications and firearms.

Recent guidelines from the American Academy of Pediatrics have emphasized the importance of pediatricians being aware of warning signs that may signal suicidal thoughts, such as hopelessness, guilt, seeming withdrawn and isolated, being bullied, giving away valuables, substance use, and signs of self-harm, such as cutting. If these are noticed, it is important for the pediatrician to assess whether the child or adolescent is experiencing suicidal thoughts and/or behaviors. This sort of assessment may best be done with or without the parent/caretaker in the room. If the child or adolescent is assessed to be suicidal, the pediatrician must refer them to the nearest emergency room for an evaluation by a psychiatrist or send them directly to an inpatient psychiatric hospital for an evaluation. This same process must be followed if the pediatrician is unclear about the child's suicidal status.

CONCLUSION

Depressive and bipolar spectrum disorders in children and adolescents carry significant long-term consequences, making recognition and prompt treatment a key component to decreasing morbidity and mortality for these patients. As these disorders can present similarly to other mental health diagnoses, it is important to know how to differentiate these conditions. Depression may present differently in children and adolescents compared to adults, and youth may not always directly report being depressed, making it important for pediatricians to be aware of warning symptoms/signs, especially when it comes to suicide. Although many medications are used to treat depression and bipolar disorder, only a few have been FDA approved for these indications in children and adolescents. Psychosocial interventions and adjunctive therapy performed by trained therapists are an additional key component to treatment. Due to the dearth of child psychiatrists in the United States and the fact that many families present initially to primary care physicians for mental health concerns, initial management strategies of depression are an essential component of the pediatrician's repertoire. More severe depression and bipolar spectrum disorders are ideally an indication for referral directly to child and adolescent psychiatrists.

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95 Disruptive Behavior Disorders and Psychotic Disorders

Chadi Calarge

Disruptive, problematic behavior is ubiquitous and an integral aspect of normal development. In fact, the natural drive of children to increasingly assert their autonomy as they grow older and more independent inevitably sets them on a path leading to disagreement and conflict with their caregivers. Of course, children vary greatly in their frustration tolerance, their overall self-regulation, and their need to assert themselves. Such interindividual differences interact with environmental responses, particularly in the caregiving environment, which itself may vary greatly in terms of sensitivity and responsiveness to children's developmental needs, capacity to promote trust and social learning, and ability to provide effective models in such areas as prosocial behavior; coping with frustration, stress, and conflict; and problem solving. This complex interaction between biologically based motivational dispositions and environmental responses shapes, organizes, and reinforces developmental trajectories in which normative conflict can turn into disruptive behavioral disorder.

INDIVIDUAL DETERMINANTS OF BEHAVIOR

Temperament typically refers to the innate (ie, biological) aspects of one's personality. Children differ substantially in their temperamental traits, which can be observed as early as birth, although they are more reliably appreciated after the first few weeks of life. Traits include such characteristics as activity level, frustration tolerance, assertiveness, attention, emotional reactivity/intensity, impulsivity, fearfulness, and propensity to respond to soothing. A simple review of this list makes it apparent how temperament may play a key role in determining the child's inclination for engaging in behavior that could precipitate conflict. For instance, an impulsive, risk-taking child is more likely to break rules due to failure to plan his or her actions and consider their consequences. Of course, as executive functions are still developing during childhood, traits such as impulsivity must be evaluated in an age-appropriate manner.

ENVIRONMENTAL DETERMINANTS OF BEHAVIOR

A complex process such as disruptive behavior is virtually never the product of a single factor. Instead, it results from the interaction of a number of factors that may contribute, in varying degrees, to different behaviors in the same child over time and across different children at the same developmental stage. Importantly, for a behavior to persist, it must be rewarded or "reinforced"; otherwise, it will more likely be extinguished. In behavioral terms (ie, operant conditioning), contingencies of reinforcement consist of 3 components: antecedents, behaviors, and consequences (referred to as the *ABCs of behavior*). Antecedents are stimuli, settings, and contexts that precede and influence the onset of a behavior, while consequences are the events that follow. Consequences could have no impact on the recurrence of a behavior, but they can also increase it or decrease it. Positive reinforcers are events that follow a behavior and make it more likely to recur. For instance, when an infant is comfortable (ie, antecedent) and smiles (ie, behavior), even if exhibiting a reflexive social smile, the parents feel excited and respond with increased attention to and engagement with the child (ie, consequences). This positive attention is enjoyed by the infant, prompting him or her to learn that a smile leads to more parental attention and positive emotions. As a result, the infant will smile more when he or she sees the parent. Thus, the smile is the behavior that gets reinforced, and the parents' attention is the reward or, technically speaking, the positive reinforcer. (Note here that the parents' increased attention to and engagement with the child is not only a reinforcer for the child but also a behavior that is reinforced by the child's enjoyment, leading to more engagement by the parent. This highlights the interactive nature of these transactions.) In contrast, an example of a maladaptive positive reinforcement pattern is when a child throws a temper tantrum and the parents seek to console him or her by giving in to a demand made by the child (eg, "I want a candy") or dropping a demand placed on the child (eg, "Please finish your homework"). As a result, the child will learn that tantrums allow her to get what she wants. This is not to be confused with negative reinforcement, which refers to a response or behavior being strengthened by stopping, removing, or avoiding a negative outcome or aversive stimulus. As an example, having to deal with a tantrum or fighting about homework is certainly an unpleasant experience for a parent. If conflicts occur often, the parent learns that not placing demands on the child allows him or her to avoid these aversive interactions. Importantly, using the term *learn* does not imply that the process is necessarily conscious or deliberate. Rather, it highlights the fact that the process is repetitive, leading to the establishment of patterns of behavior. These examples illustrate how the parent's response shapes the child's behavior. However, they are also meant to emphasize the bidirectional nature of the interaction, with the child's behavior shaping that of the parent as well.

Of course, in reality, the child is interacting with many more individuals in his or her life than a single parent. These different relationships will influence the child's conduct in various ways, depending on multiple factors, not the least of which is the emotional value attached by the child to each particular relationship. As the child develops, the meaning and salience of these relationships will change. In fact, although in a Western society parents are the primary role models for younger children, peer relationships take on increasing influence during the transition into adolescence.

DETERMINANTS OF PATHOLOGICAL BEHAVIOR

Many individual and environmental factors have been implicated in the emergence of problematic disruptive behavior (**Table 95-1**). As noted earlier, significant impulsivity and emotional reactivity increase the risk for aggression. In addition, violent children exhibit deficits in social-cognitive or information-processing abilities (eg, they tend to interpret situations in a threatening and hostile manner). Intellectual disability and risk-taking behavior are additional risk factors, as is the presence of an anxiety or mood disorder, which may manifest in irritability, hostility, conflict, and even aggression. In utero exposure to alcohol, illicit drugs, malnutrition, or illness; perinatal complications; and lead toxicity have also been associated with disruptive behavior. Additionally, one's genetic endowment contributes to this risk not only through shaping temperamental traits but also by interacting with environmental factors, such as family-level risk factors that include poor family cohesion or high conflict, harsh or lax disciplinary practices, lack of positive parental involvement and supervision, and parental psychopathology.

TABLE 95-1 EXAMPLES OF DETERMINANTS OF PATHOLOGICAL BEHAVIOR

Individual Level

- Genetic background (eg, variants of the monoamine oxidase A gene)
- In utero exposure to alcohol, illicit drugs, malnutrition, or illness
- Perinatal complications
- Lead toxicity
- Temperamental traits (eg, impulsivity, risk taking, emotional reactivity)
- Intellectual disability
- Language delays
- Poor academic performance and low commitment to education
- Deficits in social, cognitive, or information processing abilities (eg, tending to interpret situations in a threatening and hostile manner)
- Comorbid psychopathology (eg, attention-deficit/hyperactivity disorder, depression, anxiety disorders)

Family Level

- Poor family cohesion or high conflict
- Divorce
- Large family size
- Harsh, lax, or inconsistent disciplinary practices
- Lack of positive parental involvement and supervision
- Poverty
- Poor education
- Parental substance use disorders or criminality
- Parental mental illness
- Teenage parenthood

Peer Level

- Peer rejection
- Association with delinquent peers
- Involvement with gangs

Society Level

- Enrollment in unsafe or failing schools
- Harsh school suspension and expulsion policies
- Severe punishment policies at schools
- Neighborhoods with high levels of crime and poverty
- Social disorganization in the community

Although aggressive children bully others, they are also victims of bullying and peer rejection. This pattern further isolates them, depriving them from the social incentives to adopt more appropriate behavior. In adolescence, as peers' influence increases, association with delinquent peers and involvement with gangs further encourages individuals to engage in disruptive behavior.

PROBLEMATIC BEHAVIOR

The spectrum of disruptive behaviors in childhood is broad and varies based on developmental stage. It ranges from the benign colic to more serious delinquency.

Colic

Although all newborns cry, as this is their primary way of expressing distress, about 15% develop colic, defined as crying more than 3 hours a day, more than 3 days a week, for more than 3 weeks in a well-fed infant. Colic starts around age 2 weeks and almost always resolves by the fourth month of life. The presence of colic has not been consistently associated with particular risk factors or unfavorable outcomes. As such, no specific interventions are indicated, but pediatricians can help parents arrange access to support, providing respite and reducing stress. This would alleviate parental fatigue and reduce the risk of untoward outcomes (eg, shaken baby syndrome).

Temper Tantrums

Temper tantrums are ubiquitous, reported by parents in as many as 80% of 2- to 4-year-old children. Tantrums occur daily in about 20% of 2-year-old children and 10% of 4-year-old children. Moderate to severe tantrums are reported in 5% of 3-year-old children. As would be expected, problem-solving skills are limited in toddlers. Therefore, they object to frustration with an emotional outburst, sometimes rising to the degree of a tantrum. Parents would be well-advised to avoid reinforcing the behavior, by doing the following:

1. Ensure the child is well-fed and rested, as hunger and sleep deprivation could contribute to irritability and noncompliance. In behavioral terms, hunger and sleep deprivation are a type of antecedents referred to as "setting events/establishing operations" because they increase the likelihood of engaging in behaviors to obtain or avoid consequences and alter the value of the reinforcer.

2. Identify triggers and establishing a routine that would give the toddler a sense of predictability.
3. Plan ahead by having access to activities that might engage the child and avoid triggering irritability (eg, having crayons or Play-Doh at a restaurant).
4. Distract the child to prevent initial frustration from escalating into a full-fledged temper tantrum. Many parents intuitively discover this technique and use it successfully.
5. Model for the child how to use words. Importantly, receptive language develops before expressive skills. Therefore, the parent can teach the child to use words like “I want,” “tired,” and “food.” Teaching short phrases or words actually helps build adaptive skills, as opposed to getting angry with the child, which teaches him or her what behavior is not acceptable but not necessarily what is acceptable.
6. Allow the child to take part in the decision-making process by providing choices that are all acceptable to the parent. This gives the child a sense of control.
7. Promote adaptive and prosocial behavior by providing a labelled praise. “You sound like a big boy when you use your words.” An unlabeled praise (eg, “good job”) is useful but not as descriptive in terms of highlighting to the child the actual behavior to be encouraged and repeated.

The astute reader will notice that these suggestions are essentially based on identifying antecedents that precede the behavior and influence its emergence, and on helping the parent to set up more favorable ones. They also include teaching and promoting (ie, reinforcing) more appropriate behavior by setting up consequences that deliberately provide positive attention (ie, a positive reinforcer). Once these basic concepts are understood, many minor behavioral challenges can be addressed.

If the temper tantrums become problematic or perhaps include violence, parents may need to remove the child from the situation and enforce a time-out. The parent must specify a time-out spot (eg, a chair in the living room), make it clear that the child must remain in it, and end the time-out when the child has calmed down. This may be followed by a brief discussion about the behavior that precipitated the time-out, but the child must be reintegrated back into family life without laboring excessively over what had happened. It is important to select a place for time-out that is not engaging and to avoid areas that are dedicated for other functions, such as bedrooms. In fact, bedrooms are often filled with toys and distractions, undermining the purpose of a time-out.

Moreover, using bedrooms could inadvertently associate them with aversive experiences, potentially leading to sleep problems. Also, it is important to remember that “time-out” is short for “time-out from positive reinforcement.” This implies that during time-outs there must not be an interaction with the child. This also implies that, for time-outs to be effective, the child must feel that he or she is missing out on enjoyable time with the parent/family, while in time-out. In families in which the interactions between parent and child are minimal or mostly aversive, time-outs will not be very effective. As such, it is important when using time-outs to evaluate the overall nature of the parent-child interaction and address any deficiencies. Finally, it is also important to make time-outs a successful experience, with the child able to complete them in as brief a period of time as is reasonable. For instance, many recommend that time-outs last about 1 minute per year of age, although the evidence supporting this suggestion is in fact limited.

Breath-Holding Spells

Breath-holding spells usually appear in the second year of life, occurring at the initiation of a tantrum. Up to 5% of children develop at least 1 breath-holding spell. A positive family history for breath-holding or fainting is common. Two types have been described: (1) a cyanotic form, in which cyanosis of the face occurs until breathing resumes, and (2) a pallid type, in which the face is pale secondary to a vasovagal response. Syncope occasionally develops. Apnea is brief and without sequelae. A minority of these children will have symmetric tonic-clonic movements before fully recovering. Although benign and without long-term sequelae, breath-holding spells may be frightening to parents.

Oppositional Defiant Disorder

As disruptive behavior increases in intensity, frequency, and duration, and becomes entrenched, it may reach a level where clinical attention by mental health specialists becomes necessary. According to the *Diagnostic and Statistical Manual for Mental Disorders*, 5th edition (DSM-5), a diagnosis of oppositional defiant disorder (ODD) is made when the child exhibits an established pattern of disruptive behavior, characterized by irritability, temper tantrums, argumentativeness, defiance, behavioral noncompliance, being easily annoyed by and annoying to others, and vindictiveness. The difficulties must be present for at least 6 months and must interfere with the child’s social, academic, or occupational functioning, hindering normal development.

Conduct Disorder

In conduct disorder, there is a persistent disregard of the rights of others and of societal norms and rules. Individuals with conduct disorder exhibit aggression (eg, bullying, threatening, initiating physical fights, cruelty, robbery, and rape), destructiveness (eg, deliberate destruction of property including by setting fires), deceitfulness (eg, breaking and entering, lying to obtain goods or avoid obligations, and theft), and serious violations of rules (eg, staying out past curfew and truancy, both starting before age 13 years, and running away from home). Again, the difficulties must be present for at least 6 months and impair the child's functioning (per DSM-5). Onset of conduct disorder before age 10 years (ie, childhood-onset type) is particularly pernicious, with a higher risk of delinquency and conversion into antisocial personality disorder in adulthood. In contrast, the adolescent-onset type appears more determined by peer influence, is time limited, and has a more favorable prognosis.

ASSESSMENT OF DISRUPTIVE BEHAVIOR DISORDERS

Disruptive behavior disorders are relatively common, with the prevalence of ODD ranging between 1% and 11% (average 3.3%) and that of conduct disorder ranging between 2% and 10% (median 4%). Thus, pediatricians are bound to see children with these conditions, since parents will turn to them for guidance when the problems reach a significant level.

A thorough evaluation of the predisposing, precipitating, and persisting factors (3 Ps) is a key first step. Individual-, family-, peer-, and society-level risk factors (Table 95-1) must be identified. In addition, it is informative to query about the most typical antecedents and consequences related to problematic behavior. As discussed earlier, a behavior occurs within a context and is perpetuated by reinforcers. Therefore, evaluating for marital conflict, parental substance use, inconsistent or harsh discipline, stressors that the child and family are undergoing, and child maltreatment is absolutely necessary. Establishing the duration, context, and progression of the difficulties often provides clues to the nature of the problem. For instance, a worsening but long-standing disruptive behavior is more likely to be consistent with a disruptive behavior disorder compared to difficulties only a few weeks old. A behavior problem restricted to a particular environment (eg, only at home, or with 1 parent) should prompt the clinician to work with the parent(s) on identifying triggers. For example, a child who becomes disruptive only in math class may suffer from a specific learning disorder (ie, difficulties mastering number sense or number facts).

Nonpsychosocial Causes

Subsequently, efforts should be made to rule out medical conditions (eg, sleep apnea, visual problems), iatrogenic causes (eg, steroid treatment), unhealthy lifestyle habits (eg, poor sleep hygiene), and environmental toxins (eg, exposure to lead). Although a number of neurobiological markers, such as lower heart rate or skin conductance activity, reduced basal cortisol reactivity, and abnormalities in the prefrontal cortex and amygdala, have been described in children with disruptive behavior disorders, laboratory tests should be targeted, based on the clinical history, in order to rule out suspected medical conditions. To date, there is no test to establish a disruptive behavior disorder diagnosis.

COMORBID PSYCHOPATHOLOGY

As noted earlier, temperamental traits and individual characteristics influence one's risk to develop disruptive behavior. When they reach an extreme, as is seen with attention-deficit/hyperactivity disorder (ADHD), impulsivity and disinhibition are associated with behavioral and emotional disinhibition. This accounts for the high comorbidity between ADHD and disruptive behavior disorders. For example, up to 95% of children with childhood-onset conduct disorder in clinical settings have ADHD. Determining what comorbid psychiatric conditions may be present is crucial to formulating a management plan. In fact, irritability is a nonspecific symptom that is part of the clinical presentation of virtually every psychiatric disorder in children (akin to fever in medical diseases). This includes major depressive disorder (in which irritability, rather than sadness or dysphoria, may be the sole affective symptom present), anxiety disorders, bipolar disorder, post-traumatic stress disorder, etc. Such comprehensive assessment is possible only when clinicians seek input from the child's teachers, daycare providers, or other care providers, in addition to the parents. This effort may be time-consuming but is necessary to establish the severity and pervasiveness of the symptoms and help determine the role comorbid conditions may be playing in the presenting problem. To that end, the use of rating scales (eg, National Institute for Children's Health Quality [NICHQ] Vanderbilt Assessment Scales, Pediatric Symptom Checklist-17, the Child Behavior Checklist) allows the clinician to collect this information from various informants in a reliable and time-efficient manner.

If the clinical picture appears complex, referrals to specialists may prove necessary. For instance, a referral for neuropsychological and academic testing may be indicated to evaluate for intellectual disability or specific learning disorders. A psychological evaluation may shed light on the inner emotional and psychological functioning of the child as well as on the family dynamics, and a psychiatric referral may also help determine the primacy of comorbid psychopathology, the presence of suicide risk, and whether pharmacotherapy is indicated.

TREATMENT OF DISRUPTIVE BEHAVIOR DISORDERS

Parent Management Training

Regardless of severity level, once the disruptive behavior reaches clinical attention, nonpharmacological interventions are a key component of the treatment. Mild forms may be successfully addressed with psychoeducation and self-help books on disruptive and problematic behavior. Many parents of younger children are capable of modifying the contingencies of reinforcement in their child's environment, leading to a resolution of the problem behavior. For chronic behavior problems, particularly as children become older, more formal professional help becomes necessary.

A variety of psychotherapeutic interventions and delivery methods (ie, group, family, individual settings) have been studied, some showing excellent efficacy. Most effective programs for disruptive behavior disorders are based on behavioral principles (ie, operant conditioning) and engage the parents directly in what is commonly referred to as *parent management training* (PMT). It must be made clear that the purpose of PMT is by no means to attribute the problematic behavior to inappropriate parenting. Many parents of children with disruptive behavior disorders are already burdened by guilt and a poor sense of self-efficacy. Rather, just as parents of a child with diabetes has to modify the diet to suit their child's special needs, parents of children with disruptive behavior are invited to try new methods to address their child's behavior challenges when the usual means are ineffective. The overarching goal of PMT interventions is to provide education about behavior principles (ie, ABCs of behavior), work with them on identifying antecedents (ie, triggers, contingencies, and reinforcers), encourage the use of rewards to reinforce adaptive behavior (eg, special time with parent, favorite activity, token economy), model prosocial behavior (eg, use words, complete chores, play gently), and withhold attention from (ie, ignoring) disruptive behavior. A common key component is promoting a positive, collaborative, warm parent-child relationship. The use of time-outs and withholding privileges may prove necessary but is not instrumental, and physical or aversive punishment is explicitly discouraged. The latter could impede the development of a healthy and warm parent-child relationship and serves as a poor model to help the child acquire adaptive behavioral skills.

Additional Services

As discussed earlier, comorbid conditions may either present as disruptive behavior or contribute to it and attenuate treatment response. Therefore, when indicated, the treatment plan should include interventions to address these conditions. For instance, if neuropsychological testing reveals slow processing speed or academic testing shows specific learning disorders, the school should be petitioned to make the necessary accommodations available to the child (eg, longer time to complete an in-school assignment or exam, tutoring). It may be necessary to develop an individualized education plan or 504 plan, which are formal documents that lay out the challenges facing the child and the school's plan to address them. Children with language disorders may benefit from speech therapy and those with developmental coordination disorders may require occupational or physical therapy. The presence of ADHD or mood and anxiety disorders also would need to be addressed, as indicated.

Pharmacotherapy for Disruptive Behavior Disorders

At times, the disruptive behavior reaches such a level of severity that it substantially interferes with the child's development (eg, academic failure, legal problems) or causes conflict with parents, caregivers, and peers (eg, school suspension, social isolation). Due to lack of access to effective psychotherapeutic interventions or limited response to them, or to address comorbid conditions, pharmacotherapy may prove necessary. A psychiatric referral or a consultation with a child and adolescent psychiatrist can be helpful to establish a treatment plan. Of course, any medication intervention will have to be tailored based on the child's needs, the family's preference, and the results of the assessment with regard to the presence of comorbid conditions and their contribution to the overall clinical picture. What follows will briefly describe some of the treatments available but is not meant to serve as an exhaustive summary.

Psychostimulants As noted earlier, ADHD is highly comorbid with disruptive behavior disorders, particularly when onset is in childhood. The impulsivity characteristic of ADHD contributes to behavioral and emotional disinhibition, associated with both ODD and conduct disorder. In fact, many studies have shown a substantial reduction in disruptive behavior when ADHD is successfully treated. A few studies have even shown a reduction in conduct problems with psychostimulants even in the absence of ADHD. The importance of treating comorbid ADHD cannot be overstated. Several studies in children with disruptive and aggressive behavior have shown that optimizing ADHD treatment obviates the need for any additional pharmacological interventions. Given that psychostimulants are the first-line medication treatment for ADHD, a trial may be indicated. While it may be cautious to initiate treatment with a short-acting formulation, particularly if the parents are nervous about adverse events, it is important to achieve symptom control in the evening and not simply during the school day in order to reduce conflict at home between the child and the parent. Importantly, children must be screened for cardiac abnormalities prior to initiating treatment and monitored for growth suppression and sleep disturbance afterwards.

α-2 Agonists and Atomoxetine Whether due to limited efficacy, intolerability, or parental objection, alternatives to psychostimulants may need to be considered. Guanfacine and clonidine are agonists of the presynaptic adrenergic α-2 receptor, and atomoxetine is an inhibitor of the presynaptic norepinephrine transporter. They are all approved by the United States Food and Drug Administration (FDA) for the treatment of ADHD and have been shown to reduce comorbid disruptive behavior. Guanfacine and clonidine may be particularly helpful if insomnia is problematic. Atomoxetine carries a black-box warning for emergence of suicidality.

Antipsychotic Medications A number of studies have found antipsychotic medications to be effective at controlling disruptive behavior, with a large effect size. In particular, risperidone and aripiprazole are 2 second-generation antipsychotics that are approved by the FDA for the treatment of irritability associated with autistic disorder. Over the last 2 decades, the newer second-generation antipsychotics have essentially replaced the older typical antipsychotics (eg, haloperidol or chlorpromazine). This was primarily due to concerns about irreversible movement disorders (ie, tardive dyskinesia) and the (unfounded) perception that they were more efficacious in psychotic disorders, as well as aggressive marketing practices. However, these newer medications are associated with substantial weight gain, insulin resistance, and dyslipidemia, as discussed later in this chapter. Therefore, while they may provide a fast and substantial reduction in aggression, clinicians must consider alternative options initially and reserve these drugs to more severe and treatment-resistant cases. Their initiation must be followed by close monitoring for cardiometabolic risk factors.

Other Medications Evidence also exists in support of antiepileptics, such as divalproex, in the treatment of aggression in children. Antidepressants, propranolol, buspirone, naltrexone, and many other medications, as well as supplements (eg, omega-3 fatty acids), have also been investigated with varying degrees of efficacy and evidence.

In summary, pharmacotherapy can be efficacious in the treatment of disruptive behavior, particularly reactive/impulsive (as opposed to premeditated or planned) aggression and disruptive behavior. However, nonpharmacological interventions should always be considered first and provided concurrently when pharmacotherapy proves necessary. In fact, studies have shown that, compared to pharmacotherapy alone, treatments combining pharmacotherapy with parent management training achieve better clinical response, use lower medication dosages, and are associated with higher parental satisfaction.

PSYCHOSIS

Whereas aggression is perceived to be linked to mental illness, in general, and psychosis, in particular, less than 5% of patients with psychopathology are violent, and most violent acts are committed by individuals without identifiable psychiatric disorders. Nonetheless, it remains imperative for clinicians to have the necessary skills to screen patients for the presence of risk factors, including psychosis, which will be briefly reviewed here.

Psychosis is characterized by an alteration in reality testing, with patients experiencing any number of symptoms including hallucinations, delusions, or disorganized thinking/speech or behavior. These symptoms are further explained here:

- Hallucinations are perception-like experiences, occurring without an external stimulus. Hallucinations can impact all the sensory systems, but auditory and visual hallucinations are the most common. The presence of other types of hallucinations usually raises concerns about neurological conditions (eg, seizures, tumors).
- Delusions are false fixed beliefs that may be persecutory, referential (ie, when events or cues in the surroundings are believed to be directed at oneself), somatic, grandiose, or religious. Importantly, the beliefs must not be shared by others within the same cultural, social, or religious group. Some delusions

may be bizarre or clearly implausible. Examples include thought insertion or withdrawal, when one's thoughts are believed to be put in or removed by outside forces.

- Disorganized thinking or speech refers to instances in which the individual is unable to maintain a train of thought or be coherent (eg, loose associations or word salad).
- Disorganized behavior may include immature or inappropriate behavior (eg, walking outside naked on a cold night) as well as catatonia, characterized by a marked decrease in reactivity to the environment.
- Primary psychotic disorders, like schizophrenia, are also often characterized by negative symptoms, which often account for much of the disease's morbidity. These are referred to as negative symptoms to capture the fact that the impacted area is subtracted from one's functioning or life. The 2 symptoms most prominent in schizophrenia are diminished emotional expressiveness affecting facial expressions, eye contact, intonation of speech, and body gestures, and avolition (ie, lack of desire to engage in activities, apathy). Additional negative symptoms include alogia (reduced speech output), anhedonia (reduced ability to experience pleasure), or asociality.

Of note, all psychotic symptoms may vary greatly in severity. Moreover, establishing the symptoms in children and adolescents can be particularly challenging, as hallucinations and delusions may not have become fully formed and crystallized. In addition, visual hallucinations, which occur more commonly in children, must be distinguished from normal fantasy play. This is especially true given that up to 15% of children may experience hallucinations, and while their presence is associated with a substantially increased risk for later psychosis, the vast majority resolve without sequelae.

SCHIZOPHRENIA

The prototypical primary psychotic disorder is schizophrenia. It is referred to as childhood-onset schizophrenia when symptoms are established before 13 years of age and early-onset schizophrenia when onset is in adolescence (13–18 years old). While childhood-onset schizophrenia is fortunately rare, with a prevalence of about 1:10,000, up to 50% of cases of adult schizophrenia may have their onset in adolescence. Schizophrenia has been described in all countries and cultures, with estimates for its lifetime prevalence ranging between 0.3% and 0.7%, depending on race, ethnicity, geographic region, and sex. Cases with more negative symptoms and longer duration appear more common in males.

The diagnosis of schizophrenia is made based on criteria set in the DSM-5. According to the criteria, patients must exhibit 2 or more of the following: (1) delusions, (2) hallucinations, (3) disorganized thinking or speech, (4) grossly disorganized or catatonic behavior, or (5) negative symptoms. At least 1 of the symptoms must consist of 1 of the first 3. These symptoms must have been present for at least 1 month (less if successfully treated). In addition, continuous signs of the disturbance, including disruption of the child's or adolescent's developmental trajectory in interpersonal, academic, and occupational functioning, must persist for at least 6 months. Outside the periods of acute psychosis, participants may have prodromal or residual symptoms with the symptoms present in a more attenuated form. This is often the case in the initial stage of the disease, when delusions may simply be ideas that are overvalued but still disputable by facts (eg, feeling excluded by friends) and hallucinations merely consist of minor unusual perceptual experiences (eg, hearing one's name being called). As the condition progresses, symptom severity worsens (eg, feeling like friends are plotting against him or her or hearing voices making derogatory statements). The presence of symptoms for less than 6 months or the co-occurrence of mood disorders for a substantial period may lead to other diagnoses (eg, schizophreniform disorder or schizoaffective disorder) that would similarly require clinical attention but may have different prognoses.

ETIOLOGY

Schizophrenia is a heterogeneous syndrome with no single etiology but, rather, the result of a complex interaction of multiple factors during development (ie, neurodevelopmental model). Genetic factors are certainly involved, as evidenced by the higher rate of schizophrenia among first-degree relatives of affected individuals and among monozygotic as opposed to dizygotic twins. Specific genes have in fact been identified, including the major histocompatibility complex, MIR137, and ZNF804a. Environmental factors include in utero exposure to maternal famine, paternal age, prenatal infections, obstetric complications, and immigrations. These are thought to cause direct cellular damage in the brain and alter genetic expression or induce epigenetic changes or de novo mutations in brain cells.

Of interest, recent research has focused on examining whether schizophrenia may be a manifestation of autoimmune encephalitis affecting the limbic system, caused by autoantibodies targeting the NR1 subunit of the N-methyl-D-aspartate receptor, the voltage-gated potassium channel, and the gamma aminobutyric acid receptor ($GABA_{B1}$), among others. However, the findings to date have been inconsistent.

LABORATORY ABNORMALITIES

To date, no single laboratory, neuroimaging, or neuropsychological test is available to confirm or rule out the diagnosis of schizophrenia. However, a variety of structural and functional brain abnormalities have been associated with schizophrenia and include increased lateral ventricle volumes and decreases in hippocampus, thalamus, and frontal lobe volumes. Youth with childhood- and early-onset schizophrenia exhibit decreases in gray matter volumes and cortical folding that may plateau by early adulthood.

PROGNOSIS

Onset of schizophrenia may be abrupt, but most children and adolescents exhibit an insidious course with symptoms gradually increasing in severity. Children with schizophrenia are more likely to have had psychopathology, including disruptive behavior, intellectual and language disabilities, and subtle motor delays, prior to disease onset. Earlier age of onset is associated with poorer prognosis, educational and cognitive achievement, and overall functioning. In addition, up to 20% of individuals with schizophrenia attempt suicide, with 5% to 6% of those attempts being successful. While suicide risk is elevated throughout life, it may be especially high for younger males with comorbid substance use disorders.

DIFFERENTIAL DIAGNOSIS AND COMORBIDITY

The negative symptoms of schizophrenia overlap with depressive symptoms (eg, anhedonia, social withdrawal, psychomotor retardation, and concentration problems). Moreover, depressive disorders may be associated with psychosis, particularly when severe. In children, some schizophrenia symptoms may be characteristic of autism spectrum disorder, which is sometimes associated with mannerisms, bizarre behaviors, and social withdrawal. Obsessive compulsive disorder, eating disorder, and post-traumatic stress disorder all can be associated with perceptions or beliefs that may border on psychosis. Furthermore, several medical conditions and drugs of abuse, including psychostimulants, hallucinogens, and cannabinoids (whether it is marijuana or the increasingly used synthetic marijuana/cannabinoids), may be associated with psychosis. A thorough history with specific attention to developmental aspects as well as a thoughtful medical workup often helps establish the diagnosis.

Because multiple medical conditions may be associated with psychosis (**Table 95-2**), youth with suspected schizophrenia should be carefully evaluated with routine laboratory testing assessing blood counts, toxicology screen, and liver, kidney, and thyroid functions. A metabolic screen (lipid profile, glucose, insulin) is indicated at baseline to monitor treatment tolerability. More focused testing is based on the presence of specific indications (eg, neuroimaging studies or electroencephalogram in the presence of neurologic symptoms or seizures). Genetic testing is indicated if there are associated dysmorphic or syndromic features.

TABLE 95-2 CONDITIONS OR SUBSTANCES ASSOCIATED WITH PSYCHOSIS

Conditions	Examples (Nonexhaustive List)
Central nervous system infections	Herpes encephalitis, human immunodeficiency virus infection, Lyme disease, syphilis
Hormonal abnormalities	Adrenocortical insufficiency, thyroid and parathyroid disease, panhypopituitarism
Electrolyte imbalances	Elevated or reduced concentrations of sodium, potassium, calcium, magnesium, phosphorus
Metabolic disorders	Hypoglycemia, lactic acidosis, liver failure and hepatic encephalopathy, porphyria
Vitamin deficiencies	Folate, niacin, vitamin B_{12}, vitamin D
Neurologic disorders	Seizures, encephalitis (including autoimmune encephalitis), traumatic brain injury, neoplasms, cerebrovascular disease, multiple sclerosis, Sydenham's chorea
Genetic	Velocardiofacial syndrome, Wilson disease, metachromatic leukodystrophy
Medications	Psychostimulants, dopamine receptor agonists, corticosteroids, anticholinergics, antiepileptics, antidepressants, antihypertensives, antihistaminergics, fluoroquinolone and cephalosporin antibiotics, procaine derivatives (procainamide, procaine, penicillin G), nonsteroidal anti-inflammatory drugs
Recreational substances	Alcohol (particularly delirium tremens), psychostimulants, hallucinogens, cannabinoids, opioids
Toxins	Contaminants in supplements and herbal remedies, anabolic steroids, heavy metal poisoning (eg, mercury, lead), carbon monoxide poisoning

TREATMENT OF SCHIZOPHRENIA

Antipsychotic medication is a primary treatment for schizophrenia. Even though several large-scale studies have failed to show any advantage of some of the newer, second-generation antipsychotics over the older, typical antipsychotics, the former are nevertheless used as first-line treatments. Several second-generation drugs, including risperidone, aripiprazole, quetiapine, paliperidone, and olanzapine, are approved by the FDA for use in youth with schizophrenia. Due to its propensity to cause significant weight gain without additional therapeutic advantage in youth, the FDA has advised against using olanzapine as first-line treatment in this age group. In treatment-resistant schizophrenia, clozapine has shown efficacy. However, it is associated with serious adverse events, including neutropenia, and requires mandatory monitoring.

Antipsychotics may cause a variety of adverse effects. Most notably, second-generation antipsychotics lead to substantial weight gain (several kilograms in 2 to 3 months) and cardiometabolic abnormalities. Therefore, recommendations for close monitoring of blood pressure, body mass index, waist circumference, lipids, and glucose/insulin have been made. Concerns about iron deficiency have also been recently raised. In addition, all antipsychotics may cause extrapyramidal side effects, including parkinsonian tremor, acute dystonia, and akathisia. Following long-term use, tardive dyskinesia could develop and may be irreversible, although the risk appears lower in children and adolescents than in adults. Nonetheless, patients should be periodically assessed while medicated.

Nonpharmacological interventions, such as psychoeducation, social skills training, and cognitive remediation, can help optimize the functioning of children and adolescents with schizophrenia, maintain high treatment adherence, and empower and support families. Due to the presence of cognitive and functional deficits, accommodations at school (eg, specialized educational programs) and vocational training may also be beneficial for some youth.

In summary, disruptive behavior is common in children, and pediatricians are uniquely positioned to guide families to seek and secure the necessary care. This is particularly true given that early interventions may improve long-term outcomes. While psychosis is fortunately rare in children, the limited access to mental health services due to the severe shortage of child psychiatrists and the emergence of cardiometabolic adverse events secondary to treatment also require pediatricians to lend their expertise to address the children's needs and minimize long-term sequelae.

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SECTION 8: Acute and Critical Illness

PART 1 THE PATHOPHYSIOLOGY OF ACUTE AND CRITICAL ILLNESS

96 Acute Respiratory Dysfunction

Bhushan Katira and Brian Kavanagh

INTRODUCTION

Acute respiratory dysfunction in children warrants prompt diagnosis and management, particularly in neonates and infants, as decompensation can be fast and respiratory arrest is the most common cause of cardiac arrest in children. Their smaller airways, increased metabolic demands (relative to body mass), and decreased respiratory reserve may put children at a disadvantage compared to adults.

Severe respiratory dysfunction necessitates admission to the intensive care unit (ICU). Since the polio epidemic of the 1950s, ICUs have developed in parallel with our increasing understanding of respiration in health and disease. Respiratory support is now extensively used and has become progressively more sophisticated.

In this chapter, we will explore the approach to a child with acute respiratory dysfunction, the initial considerations about the differential diagnosis of those conditions, and the broad principles of management strategies. In order to provide perspective, we first review some essentials of normal respiratory function.

NORMAL RESPIRATION

The principal purposes of respiration are to provide oxygen and remove carbon dioxide, and this normal function involves the following elements.

Control of Respiration

Respiratory centers in the medulla receive stimulatory input from central respiratory pacer cells, central and peripheral chemoreceptors, upper airway receptors, and volitional pathways. These signals are integrated into a combined output to respiratory muscles in order to regulate breathing frequency, as well as the durations of inspiration and expiration.

Neuromuscular Function

Central signals are carried by the cervical and thoracic nerves to the neck, thoracic muscles, and diaphragm. These signals maintain patency of the upper airway and drive the thoracic bellows to provide ventilation. Downward movement of the diaphragm and outward movement of internal intercostal muscles lead to negative intrathoracic pressure and movement of air into the lungs (inhalation). The relaxation of these muscles leads to passive recoil of the lung and moves the air out (exhalation). The more negative the intrathoracic pressure, the greater the tidal volume generated.

Resting Lung Volume

At rest, *elastic recoil* tends to move the thoracic cage outward and the lung inward. The balance of these opposing forces, along with a surfactant in the alveoli that reduces the surface tension (ie, lessens the force required to distend the lung). Elastic recoil determines the resting volume of the lung at the end of expiration, also known as functional residual capacity (FRC). This is the effective volume available for gas exchange.

Airflow in the Large and Small Airways

Airflow generated by mechanical movements passes through the airways. The nasal passages, nasopharynx, larynx, trachea, and main-stem bronchi constitute the large airways. The bronchial tree, from segmental to subsegmental bronchi and down to each alveolar unit, comprise the lower airways. Large airways offer less resistance and airflow is mostly laminar except at bifurcations and in the upper trachea near the glottis, where it tends to be turbulent. The airways humidify and warm the air.

Alveoli

Air reaches the *alveoli*, which are small air-filled sacs covered with a surface-active agent called surfactant. Each alveolus is lined with epithelium and has an adjacent endothelial surface that lines the nearby capillaries. The interface is termed the alveolar-capillary membrane. Oxygen (O_2) and carbon dioxide (CO_2) are exchanged here along their respective concentration gradients. CO_2 moves from the pulmonary arterial blood (high concentration, endothelial) to the airspace (low concentration, epithelial), and oxygen moves in the reverse direction. Thus, adequate surfactant function and an intact alveolar-capillary membrane are keys to gas exchange.

Pulmonary Vessels

The pulmonary artery brings blood rich in CO_2 and low in O_2 into contact with the alveoli, and the pulmonary veins deliver blood rich in O_2 and lower in CO_2 to the left atrium, for systemic distribution. Movement of blood flow along the capillaries is necessary for effective gas exchange. At alveolar level, the microvessels are either in the alveoli (alveolar) or in the interstitial space (extra-alveolar). At FRC, alveoli and vessels are optimally open.

Ventricular Function

Normal ventricular function assures good pulmonary blood flow in the presence of enough preload. Low preload or suboptimal pump function leads to lower pulmonary blood flow and may affect gas exchange. The negative intrathoracic pressure also moves more blood into the lungs during inspiration and keeps it out of thorax during expiration, thus keeping a good balance throughout the respiratory cycle.

RESPIRATORY FAILURE

The respiratory system can fail if any of the above components of respiration is dysfunctional. Infants and children are at increased risk of developing airways obstruction when compared with adults. This is in part due to anatomic differences, particularly of the upper airways (see **Fig. 96-1**), but also due to conditions that more frequently affect younger patients. Large (or upper) airway obstruction usually occurs in children because of inhalation of a foreign body or airway inflammation such as croup or epiglottitis; rare causes include external compression from tumours or vascular malformations. Usually, large airway obstruction causes inspiratory stridor; in contrast, peripheral airway obstruction from asthma or bronchiolitis causes expiratory wheezing. Bronchiolitis is caused by small airway inflammation and narrowing, leading to expiratory obstruction. If severe, it can lead to trapping of air in the alveoli and hyperinflation. Alveolar damage mostly occurs from infection, edema, alveolar collapse, or contusion or alveolar hemorrhage. Such alveoli have diminished ventilation but are perfused; thus, shunting of deoxygenated blood occurs from the pulmonary artery to the pulmonary vein. Hypercapnia and hypoxemia increase respiratory drive, which coupled with reduced compliance from an injured lung, increases the work of breathing.

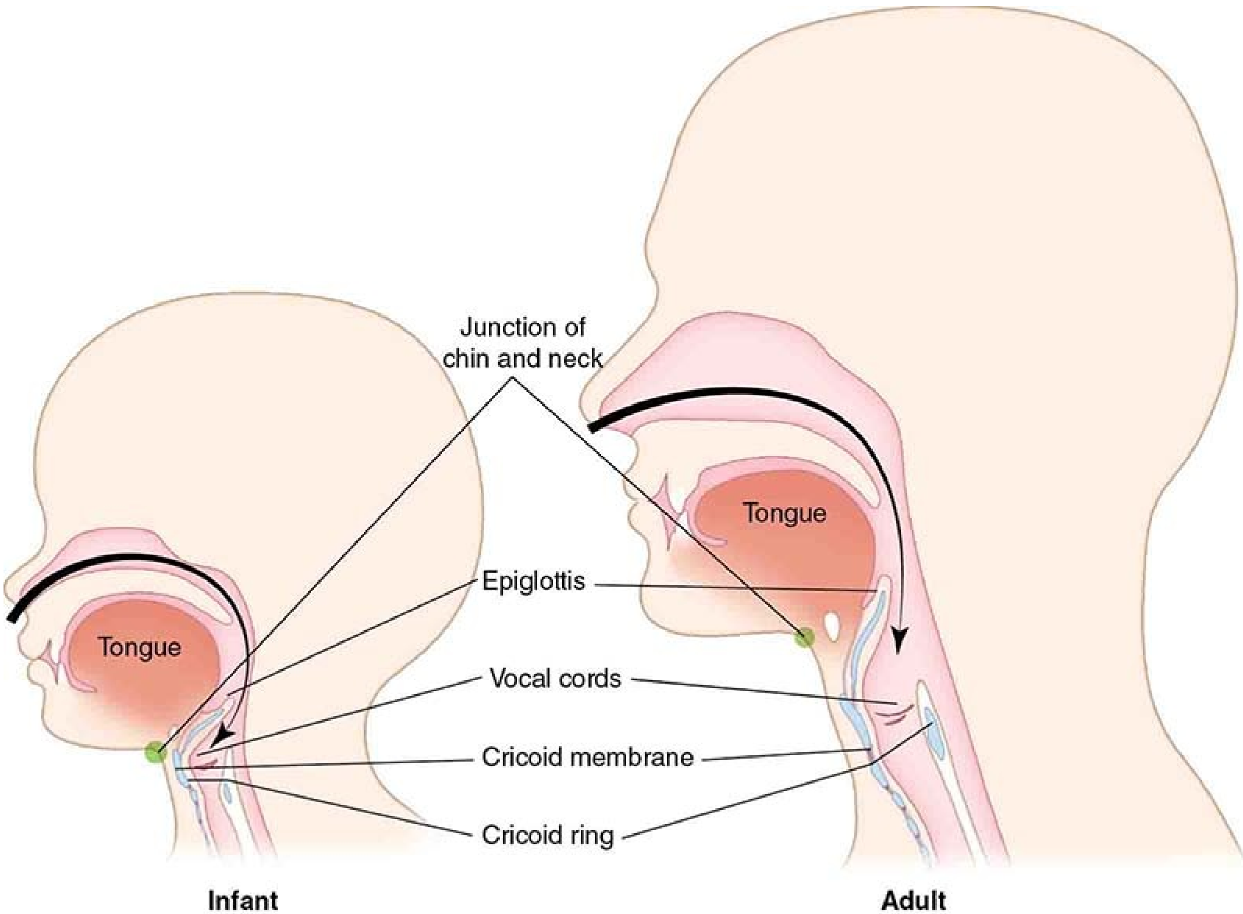

FIGURE 96-1 Differences between infant and adult airways. (From Brown: *The Walls Manual of Emergency Airway Management*, 5th edition, Philadelphia, Wolters Kluwer, 2017 and The Difficult Airway Course [www.theairwaysite.com]. Used with permission.)

Chest wall restriction (eg, circumferential burn scars, kyphoscoliosis) limits tidal volume and necessitates rapid, shallow breathing. Diaphragm paralysis results in the diaphragm being pulled up (passively) by the inspiratory effort, creating a paradoxical pattern of breathing and mechanical disadvantage.

Neurological dysfunction, if severe or associated with elevated intracranial pressure, can depress respiration; in contrast, hyperventilation is characteristic of many congenital or acquired encephalopathies. Hyperventilation is also a typical finding as a compensation for metabolic acidosis (eg, ketoacidosis, salicylate ingestion). Sepsis increases utilization of oxygen, which elevates the work of breathing, and systemic inflammatory disorders (eg, systemic lupus erythematosus) can cause alveolar or pleural inflammation.

Finally, mechanical ventilation can damage the lungs; this is called ventilator-induced lung injury (VILI). Use of high tidal volume, suboptimal end-expiratory pressure, high driving pressure, and spontaneous breathing in injured lungs are some of the mechanisms that may contribute.

ACUTE RESPIRATORY FAILURE

Based on the underlying pathophysiology, acute respiratory failure can be either hypoxemic or hypercapnic.

- Hypoxemic. Alveolar airspace disease (eg, acute respiratory distress syndrome, or ARDS) and/or pulmonary vascular disease (eg, persistent pulmonary hypertension, pulmonary embolism) result primarily in oxygenation failure.
- Hypercapnic. Airway obstruction and disordered control of breathing lead to low tidal volume or reduced flow, resulting in primary ventilation failure with elevation of CO_2.

CHRONIC RESPIRATORY FAILURE

Chronic inflammation of alveoli, airways, or both (eg, cystic fibrosis, bronchopulmonary dysplasia), and chronic dysregulation of control of breathing (eg, central hypoventilation syndrome) can lead to long-term gas exchange failure, resulting in a baseline increase in CO_2 and ongoing O_2 dependence. Compensatory mechanisms of metabolic alkalosis to counter respiratory acidosis and re-establish a normal pH, and polycythemia to counter chronic hypoxemia become evident over time.

EPIDEMIOLOGY OF RESPIRATORY FAILURE

The epidemiology of respiratory failure in children is complex because of developmental issues, variable severity of illnesses in the pediatric ICU, and classification based on syndromes (eg, ARDS, sepsis) versus disease entities.

Croup

Croup is common in children aged 6 to 36 months, mostly in males. Family history is a risk factor, and it occurs most commonly in the fall and winter. Approximately 5% of children with croup in the emergency department (ED) are admitted to the hospital, and of those discharged home, approximately 5% return to the ED in 48 hours (see also Chapters 118 and 503).

Respiratory Syncytial Virus

Respiratory syncytial virus (RSV) is typically seasonal, with outbreaks usually occurring between October and April in the northern hemisphere and April and September in the southern hemisphere. It is the most common cause of lower respiratory tract infection in children less than 1 year of age, and hospitalization is most common in those aged less than 3 months. The mortality from RSV is about 1% in children less than 4 years of age. Risk factors include age less than 6 months, daycare, older siblings, underlying lung disease, gestational age less than 35 weeks, congenital heart disease, secondhand smoke, and Down syndrome.

Asthma Requiring ICU Admission

Sixty percent of asthmatic children have acute exacerbations, and the risk is greater in younger children or in those with inadequate interval asthma control. ICU admission is rare, but previous admission and exposure to multiple allergens are predictors (see also Chapter 505).

Pneumonia

Approximately 156 million children under 5 years of age develop pneumonia each year worldwide, and about 20 million cases require hospital admission. Mortality in the developed world is low (1/1000) but high in the developing world. Risk factors include winter months, malnutrition, crowding, and chronic pulmonary, cardiac or neurologic conditions.

Acute Respiratory Distress Syndrome

Acute respiratory distress syndrome in children is similar in its clinical, radiological, and pathological (see **Fig. 96-2**) characteristics to ARDS in adults. However, the incidence (8 vs 15/1000 ICU admissions) and the mortality (less than 20%) are lower in children than in adults. In addition, death due to pediatric ARDS is usually due to hypoxemia, whereas in adults it is more commonly from multiorgan failure.

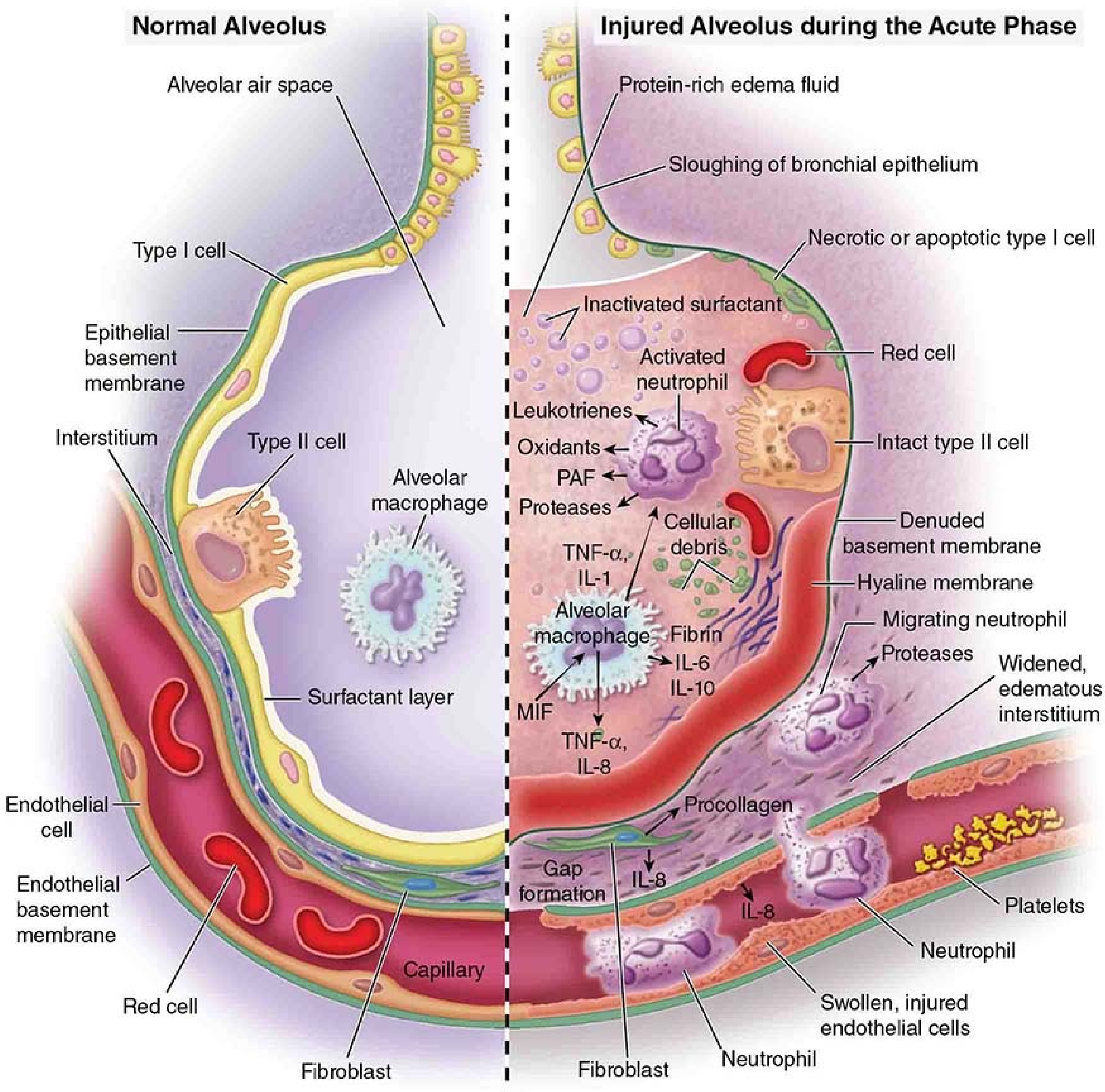

FIGURE 96-2 Molecular biology and pathogenesis of ALI. (Reproduced with permission from Ware LB, Matthay MA. The acute respiratory distress syndrome, *N Engl J Med*. 2000 May 4;342(18):1334-1349.)

INITIAL EVALUATION AND MANAGEMENT

A prompt evaluation can identify and facilitate the early stabilization of most sick children with respiratory failure. Hypopnea, severely increased work of breathing, dusky or mottled skin, pallor, and unresponsiveness can all herald imminent respiratory arrest.

Airway

Immediate airway assessment is essential for any patient presenting in actual or pending respiratory failure. The airway is patent if there are good chest movements and no obstructive sounds. If necessary, airway patency can be achieved with suction to clear airway contents and mechanical maneuvers such as a jaw thrust or chin lift, repositioning, placement of an oropharyngeal airway, or endotracheal intubation.

Breathing

Normal chest wall movement and breath sounds usually indicate adequate ventilation, although they cannot in themselves rule out hypercapnia or hypoxia. *Respiratory distress* is recognized by the use of accessory muscles, nasal flaring, subcostal and intercostal retraction. Severe distress is indicated by substernal, sternal, and suprasternal retraction. Auscultation may reveal crackles (airspace disease) or wheeze (airway disease), and the absence of cyanosis (confirmed by pulse oximetry) indicates adequate oxygenation. The unilateral absence of, or diminished, breath sounds indicates ipsilateral pneumothorax, major atelectasis, or pleural effusion.

UNDERLYING DIAGNOSIS

History

The history should include details of the course of the current symptoms, as well as any previous history of breathlessness, wheezing, cough or restricted activity, or failure to thrive. Access to or opportunities for ingestion of small objects, such as toys, or small pieces of food, should be specifically discussed with the caregivers, particularly if a foreign body is part of the differential diagnosis in a child with acute obstruction (see also Chapters 118 and 503).

Investigations

Multiple investigations are available to help establish the underlying diagnosis, but the key lies in narrowing down the selection to a definitive test. A chest radiograph can confirm hyperinflation, consolidation, asymmetry, collapse, and pleural collections. Blood gas analysis will provide arterial O_2 and CO_2 tensions, and the pH may be useful as a guide to chronicity of a respiratory acidosis, or may determine whether there is a metabolic acidosis driving the process.

Ultrasound of the chest is a useful tool to characterize pleural collections (single or loculated), to assist with drainage, or in recent years to confirm atelectasis or consolidation. Fiber-optic bronchoscopy can be used in the evaluation of central airway obstruction and sampling of alveolar fluid using the technique of bronchial alveolar lavage (BAL), and in children is usually conducted under general anesthesia.

Diagnosis of Specific Conditions

- *Croup*. The acute onset of stridor, fever, barking cough, and hoarseness in older children strongly suggest a diagnosis of croup. Throat pain, drooling, or swallowing difficulty may suggest *epiglottitis* rather than croup. Differential diagnoses in which the patients are typically toxic, with high fevers on presentation, include acute epiglottitis, retro- and parapharyngeal abscesses, paratonsillar abscesses, and bacterial tracheitis (see also Chapter 118). Additional possibilities include foreign body aspiration, acute angioneurotic edema, and upper airway injury from burns.
- *Bronchiolitis*. A 1- to 3-day history of upper respiratory infection–like symptoms followed by fever, cough, and respiratory distress (eg, tachypnea, retractions, wheezing, and rales) in children younger than 2 years of age suggest bronchiolitis. Cyanosis, lethargy, dehydration, or apnea would indicate severe bronchiolitis, and children presenting in this way should be hospitalized.
- *Asthma*. Acute exacerbations of asthma can be characterized by a dry cough with or without specific exposure (eg, cold air, pollens, dust). A history of atopy, positive family history for asthma, and seasonal symptoms indicate the diagnosis. Expiratory wheezes and hyperinflation are common. When severe, air entry may be almost absent with severe fatigue and altered sensorium (see also Chapter 505).
- *Pneumonia*. High-grade fever, cough, breathlessness, chest pain with increased work of breathing, presence of crackles, and bronchial breathing make a clinical diagnosis of pneumonia very likely. The chest radiograph typically shows patchy or homogenous consolidation. Hypoxemia usually is present, and hypercapnia also may be present if the pneumonia is severe or if the patient becomes fatigued. Elevated white counts, high inflammatory marks (eg, C-reactive protein and/or procalcitonin) point toward a bacterial etiology over viral. Nasal swab, sputum cultures, or bronchoscopic cultures are the most specific tests to identify the microbial etiology.
- *Pneumothorax*. Respiratory distress with unilateral absent air entry, a resonant

percussion note, and a shift of the mediastinum (trachea and heart sounds) make the diagnosis, and chest x-ray is confirmatory.

- *Pleural effusion*. Reduced air entry with a dull percussion note, and bronchial breath sounds at the upper levels with a shift of the mediastinum to the opposite side (if ipsilateral atalecatsis is not marked) indicate pleural effusion. Chest x-ray is diagnostic in most cases and chest ultrasound provides additional confirmation and permits image-guided aspiration.

RESPIRATORY SUPPORT

Supplemental Oxygen

Supplemental oxygen can be provided by nasal cannula or via a regular facemask or a nonrebreather mask. Oxygen flow is generally maintained at 3 to 4 times the minute ventilation so that all the inspiratory flow is provided by oxygen (100% O_2). Heated and humidified high-flow nasal cannula (HFNC) can also deliver high inspired O_2 with an additional component of positive pressure, which may assist with lung recruitment, though the level achieved is highly variable. The constant airflow displaces CO_2 from the anatomical dead space, thereby creating an oxygen reservoir. A HFNC also improves mucociliary function due to humidification and high flows.

Positive Pressure Ventilation

Noninvasive ventilation is routinely used to provide respiratory support in modern ICUs and can be applied in various ways. Continuous positive airway pressure (CPAP) opens the alveoli and increases FRC, thus improving oxygenation while the patient breathes spontaneously above this pressure to maintain ventilation. If efforts are inadequate, then bilevel positive airways pressure (BiPAP) can be applied. This provides an additional level of inspiratory support, as well as the option of providing mandatory breaths to further assist inspiratory effort and gas exchange. If problems persist or deteriorate, invasive ventilation should be instituted. In the case of a hypoxemic patient, an optimal positive end-expiratory pressure (PEEP) derived from a recruitment maneuver helps to open the alveoli and keep them open, thus increasing FRC. A tidal volume set at 6 mL/kg or adjusted to allow permissive hypercapnia (high CO_2 and pH > 7.2) reduces ongoing VILI. Adequate minute ventilation is achieved by adjusting the rate and providing good patient-ventilator synchrony. The inspiratory-to-expiratory time ratio is set such that it allows maximal oxygenation without reducing ventilation. High inspired O_2 concentrations are toxic to lung tissue and moderating the goals of oxygenation with a degree of permissive hypoxia might be beneficial.

Prone Positioning

Prone positioning has been used to improve oxygenation and ventilation in patients with ARDS. Prone positioning shifts the ventilation from nondependent to dependent parts of the lungs and moves abdominal contents away from the diaphragm, thus reducing impedance to ventilation in the injured area (**Fig. 96-3**).

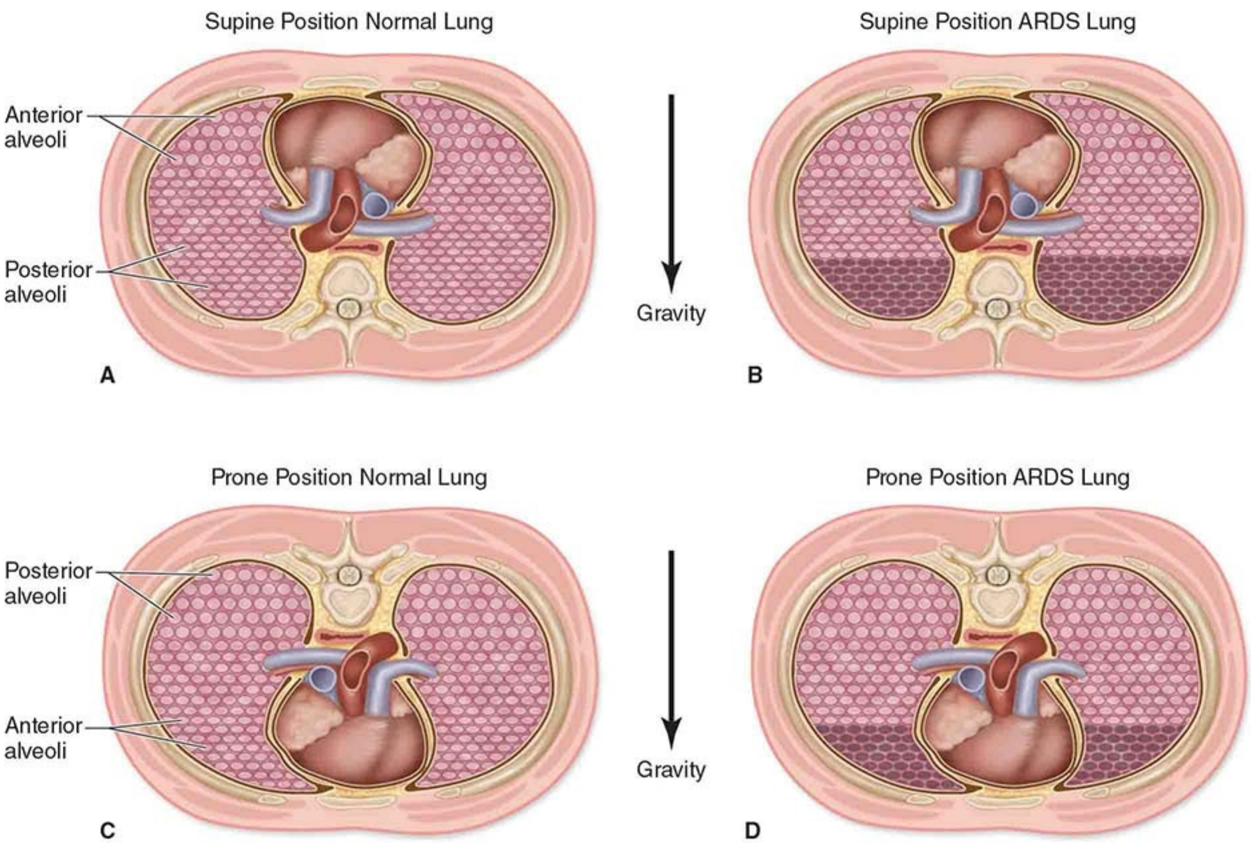

FIGURE 96-3 Physiologic benefits from prone positioning.

High-Frequency Oscillatory Ventilation *High-frequency oscillatory ventilation* (HFOV) was designed with the goal of keeping the lungs open and reducing tidal shear stress. It ventilates the lungs at the optimal compliance and moves air/oxygen flow continuously in a central column instead of in bulk flow. It achieves ventilation directly in relation to the amplitude of the percussions and inversely in relation to the frequency applied.

Inhaled Nitric Oxide *Inhaled nitric oxide* (iNO) is a potent, selective pulmonary vasodilator. It is primarily indicated in pulmonary hypertension, especially in newborns. It is also used as an adjunct in hypoxic respiratory failure refractory to improve ventilation/perfusion matching, and therefore improve oxygenation. However, it is important to note that there is little evidence to support improved overall outcomes associated with the use of iNO in patients with ARDS. Inhaled nitric oxide delivery should be carefully monitored, as it can be toxic in high concentrations and may produce methemoglobinemia, though this is rare in the therapeutic dose range (usually up to 20 parts per million). It can also be cytotoxic to alveolar and vascular tissues.

Extracorporeal Membrane Oxygenation

Hypoxia unresponsive to all above therapies needs to be managed with extracorporeal life support. Use of venovenous extracorporeal membrane oxygenation (VV ECMO) helps the lungs rest and recover while oxygenation is achieved by an artificial membrane. Coexistent right heart failure or the presence of circulatory shock warrants the use of venoarterial (VA) ECMO. See also Chapter 108.

SPECIFIC THERAPIES

Antimicrobials

Specific choices of primary or adjunctive antimicrobial agents are largely dictated by the disease severity; factors relating to the underlying host (immunocompetent or otherwise); whether the infection may have been nosocomial or community acquired; and local pathogens and likely susceptibility. Optimal broad-spectrum antibiotics, started early and narrowed by sensitivity results from culture, are the standard of care for the treatment of pneumonia. This may be achieved with single agents, or if specific organisms such as *Chlamydia*, *Mycoplasma*, or methicillin-resistant *Staphylococcus aureus* (MRSA) are potential pathogens, or in severe cases requiring ICU admission, then 2 or more empiric antibiotics should be used. Antiinfluenza therapy using oseltamivir should be considered for severe pneumonia needing ICU admission until influenza is ruled out.

Surfactants

Ongoing surfactant replacement in *primary surfactant deficiency* in neonates should be instituted if transplantation is an ultimate goal. Outside of the newborn period, there is otherwise little evidence to support the use of surfactant in acute lung injury.

Bronchodilators

Rescue albuterol nebulization given continuously initially, and potentially followed by intravenous (IV) infusion in unresponsive cases of asthma, has become the standard of care. Magnesium sulphate and aminophylline are potential adjuncts to β_2-agonist therapy, and in the most severe cases, administration of inhalational anesthetic agents such as isoflurane or sevoflurane can be considered if appropriate delivery systems and monitoring exist.

Corticosteroids

Systemic steroids are strongly recommended in acute asthma exacerbations needing admission. Methylprednisolone 1 to 2 mg/kg/dose given every 6 to 8 hours either enterally or IV is the standard practice, though recent evidence suggests that a shorter course of dexamethasone may be a viable alternative. Steroids may also be useful to help reduce inflammation of the alveoli in cases of ARDS if given after a serious infection has been excluded.

PREVENTION

Prevention is the key to reduce the burden of morbidity and mortality from childhood respiratory illnesses, and to prevent hospitalization.

Asthma

Outpatient stabilization and well-controlled, long-term treatment and action plans may prevent acute exacerbations from requiring hospital admissions, and from reaching the ICU. Prompt use of inhalers, allergen identification and control, and patient education and follow-up become keys to the management of childhood asthma (see also Chapter 505).

Viral and Bacterial Infections

Timely vaccinations, avoiding overcrowding, minimizing travel during the peak season, and early visits for timely diagnosis and management are important to avoid progression to severe illness and ICU admission.

Hospital-Acquired Infection

Adherence to strict protocols of hand hygiene, ventilator-associated pneumonia

(VAP) prevention bundles, and ongoing healthcare team education are imperative to reduce hospital-acquired infections.

Minimizing Ventilator-Induced Lung Injury

Use of lung protective strategy, minimizing repeated opening and closing of alveoli, and judicious use of PEEP are some of the measures to reduce VILI.

SUMMARY

Acute respiratory dysfunction is the most common cause for emergency and ICU visits in childhood. Preventive measures, prompt diagnosis, and early interventions with optimal respiratory support and diagnosis-driven therapies are all important mechanisms to reduce morbidity and mortality, in addition to potentially reducing healthcare costs.

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97 Pathophysiology of Sepsis

Trung C. Nguyen and Joseph A. Carcillo

INTRODUCTION AND EPIDEMIOLOGY

Sepsis continues to be the leading cause of death among children worldwide, accounting for more than 5.9 million deaths per year according to data from the World Health Organization and the United Nations Children's Fund. In the United States, sepsis poses a significant healthcare burden with approximately 75,000 children admitted annually to hospitals with severe sepsis and an associated cost of $4.8 billion. Understanding the pathophysiology of systemic inflammatory response syndrome (SIRS) and sepsis syndrome is essential for recognition and management of this deadly disease.

PATHOGENESIS OF SIRS AND SEPSIS

In the 1990s, Bone and colleagues set forth the notion that systemic inflammation due to an infection can lead to SIRS, sepsis syndrome, multiple organ dysfunction syndrome (MODS), and multiple organ failure (MOF). Typically, after sensing an invading pathogen, the host's innate immune system is locally activated in order to eradicate the infection. This immune activation is mediated by pathogen-associated molecular patterns (PAMPS), such as lipopolysaccharides from bacterial cell walls that interact with the pattern recognition receptors (PRRS) such as toll-like receptors (TLRs), which are present on local immune cells. This initial interaction between PAMPS and PRRS will lead to a signaling cascade, resulting in inflammatory cytokine and chemokine synthesis and release. Injured and dying cells caused by this inflammation go on to release damage-associated molecular patterns (DAMPS) such as the intracellular protein high-mobility group box 1. DAMPS released from tissue injury amplify the cytokine response to PAMPS that can progress into a vicious cycle of tissue and organ injuries (**Fig. 97-1**). The inflammatory mediators synthesized and released during sepsis syndrome are meant to help the host in combating the invading pathogen. However, if unregulated, these inflammatory mediators can induce a pathologic state of shock in the host. Shock is defined as a state of cellular and tissue hypoxia due to reduced oxygen delivery and/or increased oxygen consumption or inadequate oxygen utilization. For example, the inflammatory cytokines tumor necrosis factor alpha (TNF-α) and interleukin-1beta can both (1) depress myocardial function and (2) pathologically vasodilate blood vessels partly through mechanisms of nitric oxide generation, both of which cause low blood perfusion to tissues. TNF-α also causes vascular leak syndrome by triggering the shedding of glycocalyx, which disrupts the endothelial cell lining. Glycocalyx, a multicomponent layer consisting of proteoglycans and glycoprotein that lines the luminal membrane of the endothelium, is essential in regulating vascular permeability and barrier function, hemostasis, vasomotor control, and immunological function. Thus, sepsis can cause significant disturbance in all organs, which can result in cardiovascular collapse, respiratory failure, immune dysregulation, acute kidney injury, coagulopathy, thrombotic microangiopathy, ischemic hepatitis, endotheliopathy, and mitochondrial dysfunction. The severity of these disturbances depends on many factors, including the genetic and environmental factors of both the host and the pathogen.

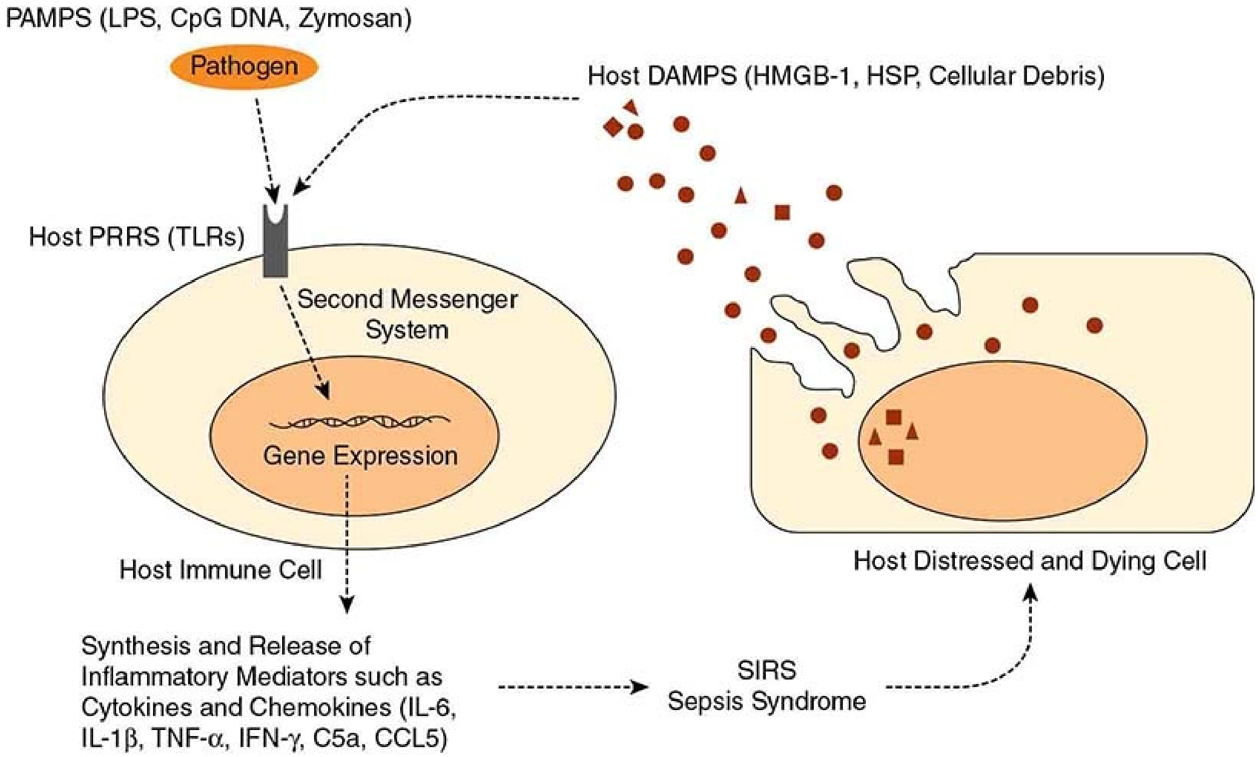

FIGURE 97-1 Pathogen-associated molecular patterns (PAMPS) activate immune cells, and damage-associated molecular patterns (DAMPS) amplify the cytokine response to PAMPS via interaction with pattern recognition receptors (PRRS). These interactions lead to intracellular signal transduction cascades, synthesis and release of inflammatory mediators, and SIRS or sepsis syndrome. C5a, complement component C5; CCL5, chemokine ligand 5; HMGB-1, high-mobility group box 1; HSP, heat shock protein; IL-1β, interleukin-1beta; IL-6, interleukin 6; IFN-γ, interferon gamma; LPS, lipopolysaccharide; SIRS, systemic inflammatory response syndrome; TLR, toll-like receptor; TNF-α, tumor necrosis factor alpha.

It is important to note that pathogenic microbes double every 30 minutes. Therefore, early source control with appropriate antibiotics and nidus removal is the key to stopping the progression from sepsis to severe sepsis or septic shock to MODS or MOF to death.

DEFINITIONS

Bone and colleagues first proposed clinical definitions of SIRS, sepsis, severe sepsis, and septic shock. These definitions with modifications are still being used by pediatricians, though the term SIRS is no longer used in adults.

CLINICAL MANIFESTATIONS OF SEPSIS AND SEPTIC SHOCK

Responding to the request from the Institute of Medicine for establishment of best practice guidelines across medicine, in 2002, the American College of Critical Care Medicine (ACCM) published its first guidelines, "Clinical Practice Parameters for Hemodynamic Support of Pediatric and Neonatal Patients in Septic Shock." This document reported the best clinical practices for pediatric sepsis that were associated with the best outcomes. By then, sepsis mortality had already decreased from 97% to 9% over past decades due to early recognition and innovations. The ACCM/PALS guidelines, were updated in 2007 and again in 2014. Many studies have evaluated the outcomes associated with implementation of the 2002 and 2007 guidelines. In resource-rich settings with intensive care units available, studies have found adherence to these ACCM/PALS guidelines have resulted in improved outcomes. The ACCM/PALS guidelines stress the importance of early sepsis recognition in order to implement the time-sensitive and goal-directed recommendations of septic shock resuscitation. Thus, a clear and easy clinical definition of septic shock is essential.

Septic pediatric patients can present to the doctor's office with a vague constellation of soft signs and symptoms including fever, runny nose, irritability, not eating well, not acting playful, and warm hands and feet. Others may present with overt septic shock with high fever, lethargy, and cold hands and feet, or with obvious vasodilatory shock with warm peripheries, bounding pulses and a high-output state. **Table 97-1** lists the definition of clinical pediatric shock from the 2007 ACCM/PALS guidelines.

TABLE 97-1 DEFINITIONS OF SHOCK

Cold or warm shock	Decreased perfusion manifested by altered decreased mental status, capillary refills > 2 sec (cold shock) or flash capillary refill (warm shock), diminished (cold shock) or bounding (warm shock) peripheral pulses, mottled cool extremities (cold shock), or decreased urine output < 1 mL/kg/min
Fluid-refractory/ dopamine-resistant shock	Shock persists despite ≥ 60 mL/kg fluid resuscitation (when appropriate) and dopamine infusion to 10 mcg/kg/min
Catecholamine-resistant shock	Shock persists despite use of the direct-acting catecholamines: epinephrine or norepinephrine
Refractory shock	Shock persists despite goal-directed use of inotropic agents, vasopressors, vasodilators, and maintenance of metabolic (glucose and calcium) and hormonal (thyroid, hydrocortisone, insulin) homeostasis

Reproduced with permission from Brierley J, Carcillo JA, Choong K, et al: Clinical practice parameters for hemodynamic support of pediatric and neonatal septic shock: 2007 update from the American College of Critical Care Medicine, *Crit Care Med*. 2009 Feb;37(2):666-88.

The ACCM/PALS guidelines suggest that septic shock should be diagnosed by clinical signs and symptoms including (1) hypothermia or hyperthermia; (2) altered mental status; and (3) peripheral vasodilation, bounding peripheral pulses, and wide pulse pressure (warm shock) or vasoconstriction with capillary refill > 2 seconds, diminished pulses, and mottled cool extremities (cold shock) before hypotension. Once the septic shock is recognized, shock resuscitation should be rapidly initiated with the goals of restoring normal (1) mental status, (2) threshold heart rates, (3) peripheral perfusion (capillary refill < 3 seconds), (4) palpable distal pulses, and (5) blood pressure for age. The ACCM/PALS guidelines recommend implementing 5 key elements during the first hour of managing pediatric septic shock: (1) recognition, (2) establishing intravenous access, (3) starting intravenous fluids and resuscitation as needed, (4) administering antibiotics, and (5) starting vasoactive agents as needed (see also Chapter 106).

WARM VERSUS COLD SHOCK

The majority (58%) of community-acquired fluid refractory pediatric septic shock presents as cold shock. On examination, the patient is difficult to arouse and exhibits cold hands and feet, weak peripheral pulses, and capillary refills delayed to > 2 seconds. In cold shock, the main defect is the significant reduction in oxygen delivery to tissues due to intravascular hypovolemia and/or myocardial dysfunction. The patient feels cold to the touch because the blood vessels vasoconstrict as a physiologic compensatory mechanism to maintain appropriate perfusion pressure and oxygen delivery to vital organs. If not appropriately resuscitated, the tissues will be starved of oxygen and energy substrates leading to tissue hypoxia, metabolic acidosis, and organ dysfunction. Thus, pediatric cold septic shock frequently responds well to volume resuscitation and to inotropic support, especially with peripheral epinephrine infusion being an excellent first choice for myocardial dysfunction. These patients commonly require mechanical ventilation to reduce cardiopulmonary work requirements.

In contrast, warm shock is the typical presentation in septic adults but only in 20% of septic children and is predominantly seen in older children or those with long-term central access. On examination, the patient again is difficult to arouse and has warm hands and feet, bounding peripheral pulses, and flash capillary refills. The main circulatory defect in warm shock is vasomotor paralysis (vasoplegia) with significant vasodilatation. Cardiac output is typically normal or increased though overall oxygen balance is deranged. Of note, adults with warm shock typically have a defect in oxygen extraction rather than oxygen delivery. Warm shock frequently responds to volume resuscitation and vasoconstrictor agents. Over time, a minority (10%) of patients may have a complete change in the hemodynamic state between warm and cold shock. In general, persistent pediatric septic shock will progress to cold shock with significant decrease in cardiac function.

SEPSIS-INDUCED MULTIPLE ORGAN DYSFUNCTION SYNDROME AND FAILURE

Even with the best efforts in septic shock resuscitation and early control of the infective source, a subset of patients still progresses to significant tissue injury and MODS. Individual patients respond very differently from each other to septic shock resuscitation, and their response depends on the time to source control first and foremost, but also to the genetic makeup and environmental factors of both the host and pathogen. Current clinical and experimental evidence suggests that these genetic and environmental factors can impede the resolution of systemic inflammation in sepsis-induced MODS. Investigators propose that there is a spectrum of MODS pathobiology phenotypes such as thrombocytopenia-associated multiple organ failure (TAMOF), immunoparalysis, sequential MODS, hemophagocytic lymphohistiocytosis (HLH), macrophage activation syndrome (MAS), and others. The common denominator in these MODS phenotypes is immune dysregulation. Thus, understanding and recognizing these MODS pathobiology phenotypes may lead to better therapeutic strategies for the leading reversible cause of late death in sepsis.

TAMOF is a clinical syndrome characterized by new onset thrombocytopenia in a setting of evolving MODS. TAMOF represents a spectrum of mixed thrombotic microangiopathies and coagulopathies including thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, and disseminated intravascular coagulation. The drop in platelet count in this syndrome suggests the pathologic involvement of platelets as they form disseminated microvascular thromboses in the vascular beds of tissues, which leads to ischemia and injury resulting in the observed MODS. Persistent SIRS in sepsis induces extensive endothelial and immunologic activation leading to a prothrombotic and antifibrinolytic state. In TAMOF, tissue factor, von Willebrand factor and platelets, and/or complement pathways are dysregulated. For example, patients with a genetic mutation in plasminogen activator inhibitor type 1, ADAMTS-13, or complement factors are at higher risk of developing TAMOF during sepsis.

Immunoparalysis is a state of impaired innate and/or adaptive immunity that occurs after the initial life-threatening proinflammatory insult in sepsis. The host immune system responds to the initial SIRS with a compensatory anti-inflammatory response syndrome (CARS) to prevent uncontrolled SIRS. If CARS persists, it will lead to a form of acquired immune deficiency. These patients are unable to clear the invading pathogens and will succumb to unresolved infection. Genetic predisposition such as an increase in anti-inflammatory cytokine production or other pathologic immune downregulation mechanisms places the patient at risk for immunoparalysis. Immunosuppressive medications such as steroids, antirejection drugs and chemotherapies can also induce a state of immunoparalysis.

Pathologic immune activation is an immune dysregulated state that leads to hyperinflammation. This occurs when the host immune system is unable to shut down the initial SIRS. The pathologic mechanism can be caused by a failure to shut down activated immune cells, such as a defect in the Fas/Fas ligand apoptotic pathway or in perforin or granzyme B-mediated cytolysis. Persistent presence of activated antigen presenting cells or persistent stimulation of TLR9 by pathogen can also lead to pathologic immune activation and hyperinflammation. Sequential MODS phenotypes have characteristic respiratory dysfunction followed by hepatic and renal failure evolving over the initial 3 days following sepsis presentation. These patients have an increased soluble Fas level, which inhibits apoptosis of activated immune cells after a viral infection. The clinical manifestations of this, HLH and MAS include a characteristic severe hepatitis and disseminated intravascular coagulation or TAMOF after an infection. HLH and MAS patients are typically found to have either an inherited or acquired defect in the ability of cytotoxic lymphocytes to induce cytolysis of target cells such as antigen presenting cells.

Currently, there is no single “magic bullet” agent to cure sepsis or abate the sequelae of septic shock. Nevertheless, in the United States, the overall sepsis mortality has significantly decreased over the past 4 decades thanks in part to early recognition and application of the best practice guidelines of septic shock resuscitation. However, sepsis-induced MODS is still the leading reversible cause of late death in pediatric sepsis with mortality increasing with each additional dysfunctional organ. Current active research is being directed at identifying common MODS pathobiology phenotypes and identifying therapeutic strategies for each pathologic mechanism. For example, anti–C5 monoclonal antibody and/or therapeutic plasma exchange are being studied to reverse TAMOF. Immune modulation therapies such as granulocyte-macrophage colony-stimulating factor are being studied to reverse immunoparalysis. Steroids, intravenous immunoglobulin, and interleukin-1 receptor antagonists are being studied to reverse MAS. Steroids, etoposide, and anti–interferon-gamma antibody therapies are agents that are currently being investigated in the treatment of HLH.

CONCLUSION

Sepsis remains a healthcare burden worldwide with enormous human and economic toll. Significant progress has been made with septic shock resuscitation in the resource-rich developing and developed settings. National sepsis initiatives are in place to improve septic shock outcomes. Research targeting a role for personalized medicine strategies in pediatric sepsis-induced

MODS and MOF is underway.

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98 Acute Neurological Dysfunction

Ken Brady and John Beca

ACUTE NEUROLOGICAL DYSFUNCTION

Altered consciousness can be a manifestation of a variety of disease processes. It is commonly indicative of life-threatening illness that requires urgent stabilization, diagnosis, and institution of disease-specific therapy to prevent ongoing neurologic injury and long-term neurodevelopmental disability. Although the anatomic structures related to consciousness have been defined, depressed consciousness is usually poorly localized to specific nervous system pathways, and often represents the consequence of a failed non-neurologic organ system.

The goals of this chapter are to review the pathophysiology and age-specific epidemiology of altered mental status, to delineate the current framework used to assess the presentation of depressed consciousness, to give a practical overview of its differential diagnosis, and finally to outline a strategy for stabilization, diagnostic evaluation, and emergent treatment options. A full discussion of the clinical approach to altered consciousness, including a comprehensive review of the interpretation of clinical signs, can be found in standard texts on coma.

PATHOPHYSIOLOGY AND EPIDEMIOLOGY

Consciousness has 2 components, wakefulness and awareness (of self and of the surrounding environment). Awareness cannot occur without wakefulness but wakefulness can occur without awareness. Consciousness is dependent on the function of the reticular activating system (RAS) that promotes widespread cortical activation. The core areas for maintaining wakefulness are thought to be ascending glutamatergic and cholinergic neurons in the dorsal tegmentum of the midbrain and pons. These activate the central thalamus and basal forebrain, which in turn activate the cortex.

The level of arousal is a function of the overall state of activity of the brain. Awareness is a more complex and integrated process involving the cerebral cortex and thalamus. Coma is a state in which a child has eyes closed and is unaware and unresponsive to any external stimuli, except for reflex responses. It results from either damage to the RAS or profound global cortical dysfunction. While lesions of the RAS can result in coma, detailed knowledge of this anatomic pathway is generally not required as global cortical dysfunction is by far the most common cause of altered consciousness. Focal lesions that affect the RAS do occur, especially those that produce pressure in the posterior fossa.

Regulation of Cerebral Blood Flow

Cerebral blood flow is tightly matched to the metabolic need of brain tissue, which is dependent on a continuous supply of oxygen and glucose for obligate-aerobic metabolism. Neither excessive nor inadequate cerebral blood flow can be tolerated for prolonged duration without causing brain injury. Multiple layers of adaptive mechanisms work simultaneously to adjust and constrain cerebral blood flow to an optimized level during disrupted physiological states such as shock or disturbed delivery of substrate.

Systemic Vasoconstriction The first adaptive mechanism to preserve cerebral blood flow is the systemic vasoconstrictor response to shock or hypotensive states, which is characteristically associated with neurohormonal activation. However, vessels in the brain have diminished responses to this neurohormonal state, so the net effect is to maintain arterial blood pressure with preserved cerebral blood flow and diminished blood flow to other systemic organ beds. This mechanism of cardiac output redistribution is highly relevant to the child in shock from low cardiac output. Vasoactive agents that modulate systemic vascular resistance will preserve cerebral perfusion pressure and cerebral blood flow in this setting, but do so at the expense of visceral perfusion.

Pressure Autoregulation When systemic vasoconstriction fails to maintain arterial blood pressure, vessels of the brain actively dilate to preserve cerebral blood flow. Modulation of cerebrovascular resistance in response to changes in arterial blood pressure is called *pressure autoregulation*, and is known to fail at a lower limit of arterial blood pressure. Although descriptions of autoregulation in hypertensive adults and women treated for pre-eclampsia suggested a lower limit of autoregulation at a mean arterial blood pressure of 50 mm Hg, lower limits of acceptable arterial blood pressure for pediatric patients are not known, but they are likely much lower and influenced by pathologic states, especially those that increase intracranial pressure.

Neurovascular Coupling When the global control of cerebral blood flow is at undisturbed, regional control is also active to provide heterogeneous distribution of flow to areas of high metabolic activity. This mechanism is called metabolic autoregulation or neurovascular coupling and is mediated by neurovascular units with astrocyte projections connecting neuronal synapses and resistance arterioles. Neurovascular coupling is useful clinically when it is desirable to reduce cerebral blood flow (and therefore blood volume and intracranial pressure). The induction of electrical silence with barbiturate coma reduces cerebral metabolism and cerebral blood flow by as much as 60% without causing a deficit of oxygen delivery.

Carbon Dioxide Reactivity Carbon dioxide diffuses freely into the cerebral spinal fluid, but bicarbonate buffers are unable to cross the blood-brain barrier. When minute ventilation and arterial carbon dioxide tension are normal, pH reactivity facilitates the regulation of constant carbon dioxide and pH levels in the spinal fluid. Acute hypercarbia lowers the pH of cerebrospinal fluid and this induces a profound vasodilatory effect on the resistance vessels of the brain. Acute hypocarbia raises the pH and induces a vasoconstrictive effect. These pathologic increases or reductions of cerebral blood flow are not necessarily matched to the metabolic activity of the brain and can cause significant injury. Hyperventilation to control intracranial pressure in children with traumatic brain injury is thought to cause ischemic injury and may be harmful outside of transient use to control acute spikes in intracranial pressure and impending herniation syndromes.

Hypoxic and Hypoglycemic Vasodilation Cerebral blood flow is increased (secondary to vasodilation) when oxygen or glucose delivery is reduced below the normal physiologic range. In patients with coma or acute brain injury, it is important to maintain hematocrit, systemic oxygenation, and blood glucose levels in the normal range.

Herniation Syndromes

Brain herniation occurs if intracranial pressure is not controlled. There is displacement of part of the brain from 1 compartment in to another (see Chapter 105) and compression of adjacent structures. The syndromes of herniation include subfalcial, transtentorial, and tonsillar.

Subfalcial herniation is when unilateral hemisphere swelling leads to herniation of the cingulate gyrus under the falx. There is a direct relationship between the amount of midline shift and the degree of impaired consciousness. In adults, shift of 3 to 6 mm is associated with drowsiness, 6 to 9 mm with stupor, and > 9 mm with coma. This may lead to compression of branches of the anterior cerebral artery and infarction of the medial surface of the frontal lobes.

Transtentorial herniation occurs when there is downward displacement of the brain through the tentorial opening. It is classically divided into uncal herniation, in which the uncus of the temporal lobe herniates over the free edge of the tentorium compressing the 3rd cranial nerve, and central herniation, in which the diencephalon and brain stem are directly compressed. In earlier diencephalic stages, there is decorticate posturing with small, reactive pupils. As more rostral compression occurs, the midbrain–upper pons is compressed with decerebrate posturing and fixed mid-position pupils. As the lower pons and medulla are compressed, motor responses are lost and pupils become dilated and fixed.

Tonsillar herniation occurs when the cerebellar tonsils are forced into the upper spinal canal and there is compression of the caudal medulla. Respiratory and cardiovascular function may be impaired. Tonsillar herniation is commonly fatal.

Epidemiology of Neurological Dysfunction

Traumatic brain injury is thought to be the leading cause of altered mental status in pediatrics. While the majority of cases are mild and do not result in hospital admission, the incidence of hospitalization for moderate to severe traumatic brain injury is close to 40 per 100,000. Hospitalization or death for nontraumatic pediatric coma occurs at an annual incidence of 30 per 100,000, with a reported mortality of nearly 50%. The incidence of this varies significantly with age: Children under 1 year of age have the highest incidence of nontraumatic coma, 160 per 100,000 per year. The most common causes of coma in children resulting in hospital admission or death are infection (38%), intoxication (10%), seizure (10%), complications of congenital malformations (8%), accidental causes such as inhalation or drowning (7%), and metabolic disorders (5%) (**Fig. 98-1**). The cause of coma was unknown in 15% of cases.

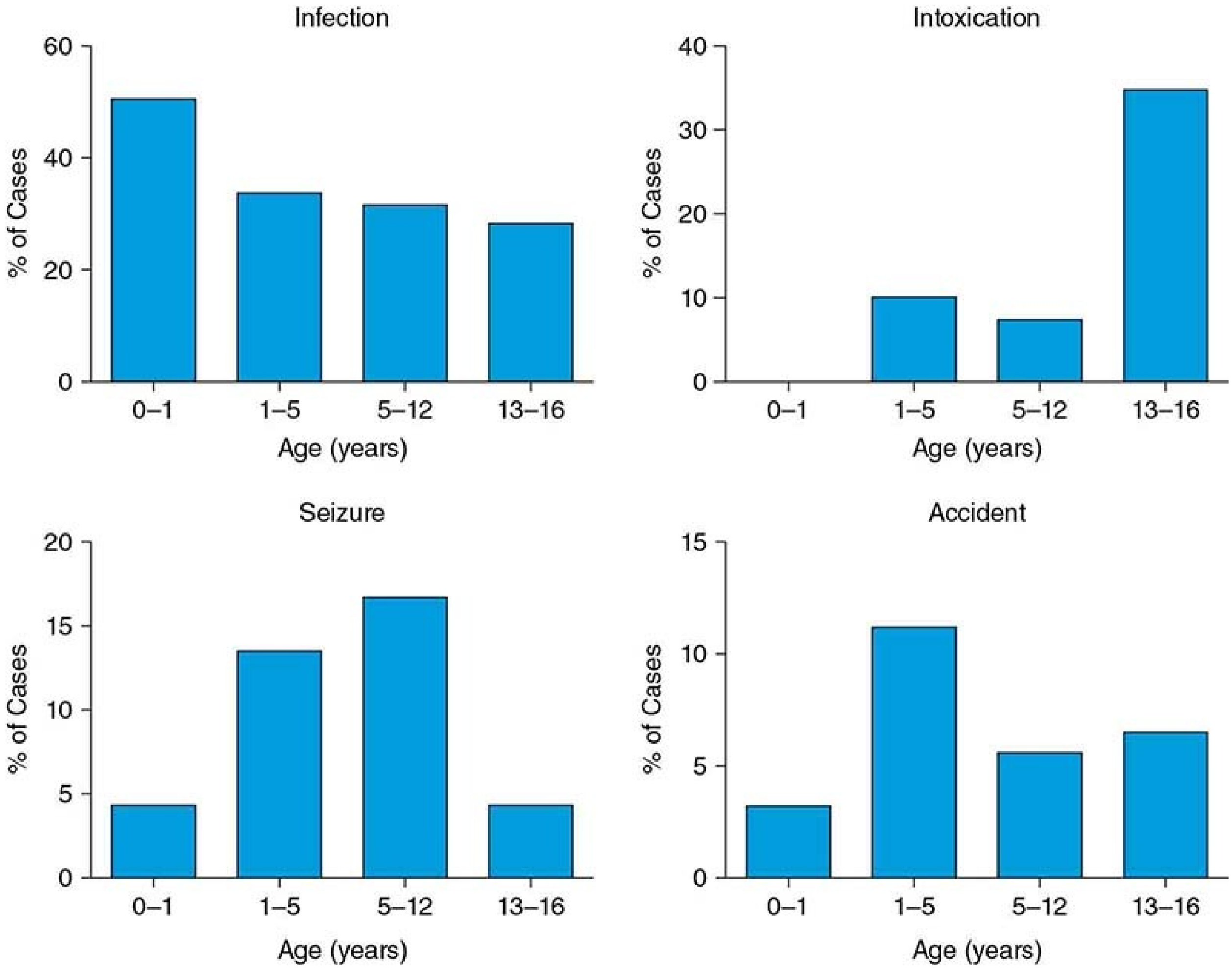

FIGURE 98-1 Four common nontraumatic causes of depressed consciousness shown as a percentage of incidences by age. Infectious causes of coma are the most common at any age, with the highest incidence in infancy. Intoxication is most commonly seen in teenage populations. Seizures present with prolonged coma are mostly seen in the preteen and toddler years, and toddlers have the highest rates of accidents causing coma. Data from Wong CP, Forsyth RJ, Kelly TP, Eyre JA. Incidence, aetiology, and outcome of non-traumatic coma: a population based study. Archives of disease in childhood 2001;84:193-9.

Etiologies of Depressed Level of Consciousness

The full differential diagnosis of depressed consciousness in pediatrics is too large to list in this chapter, as neurologic dysfunction can be a common manifestation of any severe systemic illness, but key causes are listed in **Table 98-1**. They are commonly classified as those conditions with structural changes in the brain and those with a diffuse nonlocalizing etiology. In practice, the distinction is artificial because many of the pathologies may coexist in the same patient. For example, status epilepticus or electrolyte disorders may complicate most causes of coma. Trauma, stroke, poisoning, electrolyte (glucose, sodium, calcium, magnesium) disorders, and hypoxic ischemic encephalopathy will often have obvious features in the history, clinical presentation, or initial laboratory and head CT assessment, which lead promptly to the diagnosis. Some further considerations about less immediately obvious causes are discussed below.

TABLE 98-1 **ETIOLOGY OF ACUTE NEUROLOGICAL DYSFUNCTION**

Category	Examples
Trauma	Diffuse axonal injury
	Hemorrhage (epidural/subdural/subarachnoid/parenchymal)
	Contusion
Infection	Bacterial, fungal, or viral meningitis/encephalitis
	Intracranial abscess
	Tuberculous meningitis/encephalitis
	Rickettsial encephalitis
Postinfectious/autoimmune	Acute disseminated encephalomyelitis (ADEM)
	Acute necrotizing encephalopathy (ANE)
	Multiple sclerosis
	Lupus cerebritis
Toxins	Common medications: narcotics, benzodiazepines, antidepressants, aspirin, acetaminophen, hypoglycemics
	Recreational drugs
	Environmental toxins
Seizure disorders	Febrile seizures
	Post-ictal state
	Secondary seizures
	Status epilepticus
	N-methyl-D-aspartate receptor–mediated epilepsy
Hypoxic encephalopathy	Shock
	Sepsis
	Drowning
	Cardiopulmonary arrest
Mass lesions	Neoplasms
	Hydrocephalus
	Hemorrhage
Electrolyte and endocrine disorders	Hypoglycemia
	Diabetic ketoacidosis
	Hyperosmolar nonketotic diabetes
	Hyponatremia and hypernatremia
	Uremia
	Hyperammonemia
Inborn errors	Urea cycle defects
	Amino and organic acidemias
	Mitochondrial disorders
Vascular	Vascular malformations
	Aneurysmal rupture
	Stroke, thrombosis

Infective Causes The range of organisms (eg, bacteria, virus, fungus, protozoa) that can cause central nervous system (CNS) infection vary with geographic region, age, immune status (eg, immunizations, immunocompromise), season, and underlying comorbidities. CNS infection can be difficult to diagnose, and in up to 60% a causative agent is not identified. Commonly identified causes in developed economies include *Mycoplasma* pneumonia, herpes simplex virus 1 (HSV1), enterovirus, varicella, and bacterial meningoencephalitis. Arboviruses (eg, West Nile, eastern and western equine encephalitis) have occurred in seasonal outbreaks in parts of North America. In developing countries, tuberculous meningitis and vaccine preventable causes (eg, hemophilus influenza, measles, varicella, mumps, rubella) remain a frequent and important cause. Cerebral malaria, dengue, and Japanese encephalitis occur especially when mosquito populations are high. Cerebral abscess is associated with chronic sinus and middle ear infections and cyanotic congenital heart disease and is suggested by fever, focal neurological signs, and signs of raised intracranial pressure (ICP).

Immune-Mediated and Demyelinating Causes Acute demyelinating encephalomyelitis (ADEM) is 1 of the most common causes of encephalitis. It most commonly affects prepubertal children (mean age 5–8 years), is often preceded by infection, and typically involves the white matter tracts of the cerebral hemispheres, brain stem, optic nerves, and spinal cord. While ADEM typically has a monophasic course and a favorable prognosis, up to 20% to 25% of children have at least 1 relapse.

Immune-mediated encephalitides are increasingly being recognized in children. A now well-established example is anti- N-methyl-D-aspartate (NMDA) receptor encephalitis that is characterized by a mixture of psychiatric manifestations, seizures, hyperkinetic movement disorder, dysautonomia, and coma. In the California Encephalitis Project, anti-NMDA receptor encephalitis was more common than any single infectious cause. In about half of cases, it is associated with an ovarian teratoma. There is often an infectious prodrome, and an association with herpes virus as a trigger has been postulated.

Other conditions suspected of having an autoimmune basis are febrile infection-related epilepsy syndrome (FIRES) and acute necrotizing encephalopathy (ANE). Both typically follow a febrile illness. FIRES is an epileptic encephalopathy presenting with acute refractory status epilepticus, usually in previously well children. The seizures are commonly focal in onset and may have secondary generalization, and an EEG typically shows multiple bilateral frontal and temporal foci. ANE is characterized by altered consciousness and bilateral thalamic and external capsule lesions on MRI, which resolve between attacks. It is classically triggered by influenza A in children with an autosomal dominant mutation in *RANBP2*. The differential diagnosis includes Leigh Disease.

Inborn Errors of Metabolism Coma is an important and sometimes presenting feature of a number of inborn errors of metabolism (IEM). These include maple syrup urine disease, urea cycle enzyme defects, disorders of fatty acid oxidation, nonketotic hyperglycinemia, mitochondrial cytopathies, congenital lactic acidosis, and certain leukodystrophies. Minor preceding infections may precipitate decompensation and coma.

CLINICAL MANIFESTATIONS AND DIAGNOSIS

History

The initial history should be focused in nature, and a full history may be delayed by procedures to stabilize the patient. The goal of the initial history, therefore, is to efficiently gather sufficient preliminary information to direct immediate management, including rapid identification of elevated intracranial pressure or a structural lesion that requires urgent neurosurgical intervention. Knowledge of the timing of onset can be helpful: Sudden loss of consciousness is more likely with acute hemorrhage, seizure, and stroke, while a slower or fluctuating onset is more likely with hydrocephalus, an expanding intracranial mass, metabolic process, or infection. A history of vomiting, headache, fever, and seizures should be sought. Headache with positional changes or Valsalva maneuver suggest raised ICP. Fever suggests infection, although the absence of fever does not exclude it, especially in infants under 3 months of age or in immunocompromised children. A history of recent infectious illness is common in autoimmune and demyelinating conditions. A history of trauma may or may not be forthcoming if it was nonaccidental. Exposure to and availability of common toxins such as pesticides and prescription and recreational drugs can be rapidly assessed. Failure to thrive, developmental delay and parental consanguinity may each suggest an inborn error of metabolism in a young infant. Preexisting medical conditions, such as cancer, epilepsy, congenital heart defects, or metabolic or endocrine abnormalities, may also direct the workup. Recurrent coma may suggest recurrent nonconvulsive status epilepticus, inborn error of metabolism, or nonaccidental poisoning. A history of travel may indicate exposure to infections prevalent in specific areas.

Examination

The goals of the physical examination are to (1) to identify, for immediate treatment, any compromise in airway, breathing, and circulation; (2) determine the level of consciousness; (3) assess for signs of raised intracranial pressure and/or localizing neurological signs; and (4) identify stigmata of any underlying cause.

The physical exam begins with basic vital signs to assess the status of oxygen delivery to the brain, including adequacy of ventilation and arterial oxygen saturation. Circulation is assessed with standard electrocardiogram (ECG) monitoring and findings should include an arterial blood pressure that is normal for age and extremities that are normothermic, without pallor or cyanosis, and without delayed capillary refill. Abnormalities of the basic vital assessment are addressed while further examination is performed, including a basic trauma survey.

Stereotypic patterns of spontaneous respiration may be observed while assessing vital signs, and these can help identify the etiology of the insult. A summary of stereotypic pathologic breathing patterns is given in **Table 98-2**.

TABLE 98-2 BREATHING PATTERNS OBSERVED IN THE SETTING OF ACUTE NEUROLOGICAL DYSFUNCTION

Breathing Pattern	Definition	Possible Etiologies
Apnea	Absence of respiratory effort	Brain stem/spinal cord injury, brain death, neuromuscular blockade
Bradypnea	Decreased respiratory rate	Opioids, metabolic disorder, brain stem injury
Cheyne-Stokes	Alternating apnea and hyperpnea in a gradual crescendo-decrescendo pattern	Bilateral hemispheric or diencephalon injury
Biot's respiration (ataxic breathing)	Irregular rate and tidal volume	Medullary injury
Central neurogenic hyperventilation	Rapid and deep breathing	Pontine or midbrain injury
Apneustic respiration	Prolonged inspiration with a pause before a brief exhalation	Pontine injury
Kussmaul breathing	Tachypnea and hyperpnea	Systemic acidosis (eg, diabetic ketoacidosis, renal failure)

Quantifying the Level of Consciousness

Full consciousness has 2 components: wakefulness and awareness of self and the surrounding environment. *Coma* is a state of loss of wakefulness and complete lack of awareness. Between arousal and coma, the following terms have been applied in an inconsistent gradation: lethargy, obtundation, and stupor. *Lethargy* is a pathologic state of mild sleepiness and confusion. *Obtundation* is blunted alertness with diminished sensation of pain and diminished interaction with the environment. *Stupor* is a state of unresponsiveness but vigorous stimulation can produce some limited arousal. *Delirium* is also a type of impaired consciousness, both in wakefulness and awareness. The main characteristics are impaired attention, altered alertness, disorganized thinking and an acute onset with a fluctuating course. Other features such as disordered sleep-wake cycles, memory impairment, and hallucinations may also be present.

The most well-known scale that is used to quantify the arousal to stimuli is the Glasgow Coma Scale (GCS; **Table 98-3**). The GCS records the best response achieved during assessment, which includes efforts to arouse the patient with verbal, tactile, and painful stimuli. The GCS is the sum of scores in 3 simple domains: eye opening, verbal response, and motor response. It has been adapted for pediatric use, requiring mostly modifications in the verbal scoring. The individual components as well as the total score should be recorded. The motor response is the most powerful predictive component, so that particular care should be taken with performing and recording this accurately. In traumatic brain injury (TBI) in children, the predictive power of the motor score alone was as good as the full GCS and had a linear relationship with mortality.

TABLE 98-3 GLASGOW COMA SCORE

	Adult Glasgow Coma Score	Pediatric Glasgow Coma Score	Nonverbal Glasgow Coma Score
Best eye-opening response			
1	No eye opening		
2	Eye opening to pain		
3	Eye opening to speech		
4	Spontaneous eye opening		
Best verbal response			
1	No verbal response	No vocal response	No response to pain
2	Incomprehensible sounds	Occasional whimpers or moans	Mild grimace to pain
3	Inappropriate words	Cries inconsolably	Vigorous grimace to pain
4	Confused	Inappropriate interactions, cries but consolable	Less than usual ability or response to touch only
5	Oriented	Age-appropriate verbalization, smiles, orients to sound, tracks, interacts	Spontaneous facial and oromotor activity
Best motor response			
1	No motor response		
2	Abnormal extension to pain		
3	Abnormal flexion to pain		
4	Withdraws from pain		
5	Localizes pain or withdraws from touch		
6	Obeys commands or performs spontaneous appropriate movement		

Presenting Neurological Signs Associated with Depressed Level of Consciousness

A targeted neurologic examination, focusing on cranial nerves and the motor system, can quickly identify signs of raised intracranial pressure and any focal signs. Core brain stem reflexes and the cranial nerves involved are the following: direct and consensual light reflex (II, III); oculocephalic ("doll's eyes") reflex (III, VI, VIII); oculovestibular ("caloric") reflex (III, VI, VIII); corneal reflex (V, VII); cough and gag reflexes (IX, X). Completely normal pupillary function and eye movements suggest that the lesion causing coma is above the midbrain. Pupillary abnormalities with correlates of the associated anatomic abnormalities are listed in Table 98-4. To properly assess the pupils, baseline pupil size is established in dim light and activity is assessed with a bright light directed at the eye. In the case of asymmetric pupils, the abnormality (constriction or dilation) is assigned from the baseline dim light assessment. When the pupils are more asymmetric in the dark, the smaller pupil is usually abnormal (Horner syndrome), while when they are more asymmetric in the light, the larger pupil is usually abnormal (3rd cranial nerve).

TABLE 98-4 PUPILLARY RESPONSES TO LIGHT

Size	Activity	Causes
Pinpoint	–	Pontine injury
Small (myosis)	Symmetric, reactive	Bilateral hemispheric dysfunction ("diencephalic pupils"), global metabolic disturbance, opiates, cholinergics
	Asymmetric	Horner's syndrome (sympathetic innervation of spinal cord, brachial plexus, and carotid)
Mid-dilation	Fixed	Midbrain lesion
Large (mydriasis)	Symmetric, reactive	Sympathomimetics, anticholinergics
	Symmetric, fixed, hippus (rhythmic spasm of the iris)	Dorsal midbrain lesion
	Asymmetric, fixed	Cranial nerve III, uncal herniation syndrome

Conjugate lateral deviation of the eyes occurs with an ipsilateral hemisphere lesion, a contralateral hemisphere seizure focus, or a lesion of the contralateral pontine region. Tonic downward eye deviation occurs with injury or compression of the thalamus or dorsal midbrain (eg hydrocephalus), while tonic upward gaze has been associated with bilateral hemispheric damage. Ocular bobbing suggests a pontine lesion and rapid horizontal eye movements usually suggest seizures. Spontaneous roving eye movements indicate that the brain stem is intact.

The motor response is assessed in the awake child with a verbal command in the verbal child and spontaneous observation in the preverbal child. When consciousness is impaired, it is assessed by a painful stimulus. A sternal rub can be applied and the response observed. If there is any sort of flexion response, pain should be applied in the cranial nerve distribution (eg squeezing ear lobes, supraorbital pressure) as otherwise it is difficult to clearly distinguish flexor posturing, withdrawal, and localization. Elevated intracranial pressure is strongly suspected when abnormal flexion or extension or flaccidity is present. Lack of a motor response to painful stimulus with either increased or decreased tone is concerning for a low brain stem or spinal cord lesion. Flexion of the arms with extension of the legs has been classically labeled decorticate posturing, implying pathology above the tentorium, leaving the posterior compartment reflexes unopposed. Extension and internal rotation of the arms with leg extension has been similarly labeled decerebrate posturing, implying disruption of the infratentorial (brain stem) reflexes. These anatomic localizations, however, are not as clear-cut; flexor arm responses suggest less severe dysfunction, while extensor arm responses suggest more severe dysfunction but may still be supratentorial. Arm extension with leg flexion suggests pontine injury, and generalized flaccidity implies injury below the pontomedullary junction. At initial presentation, the distinctions are largely academic, as the presence of any of these abnormalities suggests life-threatening severity and prompts rapid evaluation with brain imaging. Asymmetries of motor response suggest lateralization and make a structural cause of altered consciousness more likely.

Initial Management of a Child with Depressed Level of Consciousness

A depressed level of consciousness is a manifestation of either severe systemic disease or significant central nervous system disease and so requires urgent management. Coma is a medical emergency because of the potential for life-threatening complications and preventable secondary brain injury. Initial management must be rapid and systematic with immediate priorities being (1) resuscitation and maintenance of cardiorespiratory stability, (2) treatment of reversible causes (eg, hypoglycemia), (3) treatment of complications (eg, intracranial hypertension, seizures), and (4) investigation and treatment of underlying causes. In practice, many of these can be achieved concurrently.

Resuscitation and Immediate Treatment

Children with a depressed level of consciousness are at risk of airway obstruction and respiratory depression. Intubation and mechanical ventilation are indicated if any of the following is present (1) GCS < 9 or deteriorating, (2) signs of intracranial hypertension, (3) airway obstruction or loss of airway protection, (4) hypoxia (SpO_2 < 92% despite oxygen therapy), (4) hypoventilation, and (5) shock requiring ongoing fluid resuscitation and vasoactive therapy. The goals of mechanical ventilation are normal values for PaO_2 and $PaCO_2$. There is no indication for hyperventilation to reduce $PaCO_2$ below normal apart from the emergency management of life-threatening intracranial hypertension (until more definitive therapy is established). Immediate interventions should be aimed at correcting hypovolemia, hypoxemia, hypotension, hypoglycemia, and any other factors that may further exacerbate existing injury to the vulnerable brain. Hypotonic fluids can lead to hyponatremia and worsening of cerebral edema and should not be given; all intravenous fluids should be isotonic.

Early monitoring should include continuous ECG, oxygen saturation, and respiratory monitoring, and at least hourly measurement of blood pressure. If the child is critically ill, with shock or coma, an arterial line should be placed, for continuous blood pressure measurement and blood gas analysis.

Early Investigation

After initial stabilization, a head computed tomography (CT) scan will almost always be required, apart from children with known pre-existing conditions having typical presentations of their illness (eg, epilepsy). This scan can detect intracranial hemorrhage, space-occupying lesions (eg, tumor, abscess), hydrocephalus, cerebral edema, and in some cases focal hypodensities (eg, infarct, ADEM, herpes simplex encephalitis). Contrast is rarely needed in the acute setting. Signs of raised ICP on CT scan include compression of sulci and lateral ventricles, loss of gray/white differentiation, lateral midline shift, and effacement of basal and quadrigeminal cisterns. However, a normal CT scan does not exclude intracranial hypertension, particularly if early in the clinical course. It is also less sensitive than a magnetic resonance imaging (MRI) scan for many pathologies. Once a patient has been stabilized and had initial investigations, an MRI scan is indicated if the cause of coma is still unclear or if there is a cause for which the MRI will provide significantly more information, particularly for injuries involving subcortical structures, the brain stem, and the spinal cord and for detecting ischemia (eg, stroke, early hypoxic ischemic encephalopathy), demyelinating diseases (eg, ADEM), hypertensive encephalopathy, metabolic brain disease, diffuse axonal injury, and venous thrombosis.

TREATMENT OF COMPLICATIONS

Complications are discussed next as they may also need urgent management before further investigations and specific treatments are instigated.

Intracranial Hypertension

Intracranial hypertension may be caused by mass lesions (eg, hemorrhage, tumor, abscess), hydrocephalus, or cerebral edema. Cerebral edema is common in trauma, infection, hepatic encephalopathy, inborn errors of metabolism, status epilepticus, and strokes and tumors. Intracranial hypertension should be considered in all children with impaired consciousness and especially those with a GCS < 9. Other suggestive signs are a bulging or tense fontanelle, a "setting-sun" sign (a persistent downward gaze), dilated pupils, and papilledema. Severe intracranial hypertension with impending herniation is suggested clinically by (1) hypertension with bradycardia and abnormalities of breathing pattern (Cushing's triad); (2) abnormalities of posture: decorticate, decerebrate, or flaccid (ie, motor response of the arms to pain is flexor posturing, extensor posturing, or absent, respectively); (3) abnormal pupils: unilateral or bilateral dilated or unreactive; (4) abnormal pattern of breathing; and (5) abnormal doll's eye (oculocephalic) or caloric (oculovestibular) reflexes. In addition, there are features on the head CT scan that suggest raised intracranial pressure (see above).

Severe intracranial hypertension is a medical emergency. Immediate treatment to lower ICP includes hyperosmolar therapy (hypertonic saline, mannitol), intubation and mechanical ventilation (if not performed already), and acute hyperventilation, together with anesthesia (narcotic and benzodiazepine, barbiturate) to reduce cerebral oxygen demand and muscle relaxation. Urgent CT scan may reveal pathology amenable to neurosurgical intervention to reduce ICP, including evacuation of mass lesions and diversion of cerebrospinal fluid (CSF) in hydrocephalus. Other therapies and neurosurgical interventions will be dictated by the underlying cause.

Standard neuroprotective measures should be undertaken in all patients with coma. These include a neutral head position; head of the bed elevated to 30° above horizontal; sedation and analgesia; maintenance of normal PaO_2, $PaCO_2$, and blood pressure/cardiac output; maintenance of normothermia (36°C–37.5°C); maintenance of normal to slightly high sodium (140–150 mmol/L); and avoidance of hyper or hypoglycemia. When intracranial pressure is known to be raised, either clinically, radiologically, or based on direct measurement, a tiered approach to reduction is taken. Additional interventions that have been considered include CSF drainage, hyperosmolar therapy, barbiturate coma, and decompressive craniectomy. Full discussion of these therapies can be found in Chapter 105.

Seizures

Seizures are common in comatose children. They may be the cause of the coma or they may be a complication of the underlying condition. If coma is persistent 1 hour after a seizure, the seizure should not be assumed to be the cause of the coma and management should be that of an unknown cause of coma. In children presenting with status epilepticus, an underlying acute condition is present in 30% to 50% of cases. For example, almost half of children with encephalitis will have at least 1 seizure. Prolonged seizures (> 30 minutes) may cause or exacerbate brain injury and can also increase ICP. They should therefore be treated aggressively and metabolic derangements (eg, low glucose, sodium, calcium, magnesium) should be rapidly identified and treated. The longer a seizure lasts, the less likely it is to resolve spontaneously, the harder it is to control, and the greater the risk is of morbidity and mortality. Most (80–92%) seizures that stop spontaneously do so within 5 minutes. Seizures should be treated with first-line agents if they have not stopped spontaneously after a maximum of 5 minutes (sooner if they are occurring secondary to an underlying condition or in the presence of raised ICP), and the cycle should be repeated after a further 5 minutes if the seizure does not stop. Intravenous lorazepam and diazepam are equally efficacious in terminating seizures acutely. If seizures do not stop after a second dose of 1 of these agents, a second-line agent should be used. Second-line agents include phenytoin or fosphenytoin, levetiracetam, and sodium valproate. Fosphenytoin is better tolerated than phenytoin and is preferred where it is available. Intravenous (IV) phenobarbital may also be used if other agents are not available, but it is more likely to cause respiratory depression. If the seizure persists beyond 30 to 40 minutes, options include another second-line agent or anesthetic doses of midazolam, pentobarbital, or propofol. In this setting, intubation and mechanical ventilation are typically required as hypoventilation or apnea will typically ensue. Where there is a severe underlying condition, especially if the child is in an intensive care unit, the treatment algorithm may be delivered faster.

An electroencephalogram (EEG) is indicated for any refractory seizures or when the cause of coma is unclear. EEG is principally useful to confirm the presence or absence of seizures, to exclude nonconvulsive seizures as a cause of persistent coma, and to differentiate nonconvulsive motor phenomena from electrographic seizures. Electroencephalogram patterns may also suggest a cause (eg, HSV encephalitis). Continuous EEG, where available, is indicated in children with refractory clinical or electrographic seizures, especially if intracranial hypertension is present or if muscle relaxant drugs are being used.

Investigation and Treatment of Underlying Causes

A lumbar puncture (LP) to test for infection should be performed if the patient has a fever, if meningitis or encephalitis is suspected, or if there is no clear cause of impaired consciousness. Cerebrospinal fluid should be analyzed for cell count, protein and glucose levels, Gram stain, viral polymerase chain reaction (PCR), bacterial culture, and other stains/PCR/culture (eg, tuberculosis, fungal) as indicated. The interpretation of CSF results is shown in Table 98-5. Additional CSF may be collected and stored for later testing for metabolic and immune mediated conditions as clinically indicated. The Gram stain may be negative in up to 60% of cases of bacterial meningitis; neither a normal Gram stain nor lymphocytosis excludes a bacterial cause. An abnormal CSF, even if typical for a viral infection, should be treated with antibiotics until culture results are known.

TABLE 98-5 CEREBROSPINAL FLUID ANALYSIS

	White Cell Count		Biochemistry	
	Neutrophils ($\times 10^6$/L)	Lymphocytes ($\times 10^6$/L)	Protein (g/L)	Glucose (CSF: blood ratio)
Normal (> 1 month of age)	0	≤ 5	< 0.4	≥ 0.6 (or ≥ 2.5 mmol/L)
Normal term neonate	0	< 20	< 1.0	≥ 0.6 (or ≥ 2.5 mmol/L)
Bacterial meninigitis	100–1000 (may be normal)	Usually < 100	> 1.0 (may be normal)	< 0.4 (may be normal)
Viral meningitis	Usually < 100	10–1000 (may be normal)	0.4–1.0 (may be normal)	Usually normal
Tuberculous meningitis	Usually < 100	50–1000 (may be normal)	1–5 (may be normal)	< 0.3 (may be normal)

An LP is contraindicated or should be deferred in the presence of (1) shock, (2) clinical diagnosis of meningococcemia, (3) respiratory compromise, (4) coagulopathy or thrombocytopenia (platelets < 50,000), (5) local infection at the site where LP would be performed, (6) papilledema or clinical or CT signs of raised ICP, (7) focal neurological signs, (8) GCS < 9 or deteriorating, and (9) prolonged seizure and GCS ≤ 12. Opinions differ as to whether the last 3 of these are absolute contraindications. Decisions as to whether to perform an LP should be made by the most experienced clinician, taking into account the risks of herniation and benefits (making a definite diagnosis, knowing the antimicrobial sensitivity of an identified bacteria). If an LP is deferred, antibiotics that reflect the local sensitivity pattern of likely bacteria, and antiviral agents should be given. Herpes simplex encephalitis may be suggested by the presence of focal neurological signs, a fluctuating GCS of greater than 6 hours duration, and the presence of or contact with herpetic lesions.

A toxicology screen should always be performed if the cause of coma is unclear. Standard urine screens differ in terms of the drugs covered, so that knowledge of the local test is required. Additional blood or urine testing will be dictated by the clinical toxidrome (see Chapter 114). Drugs present in the house (especially in cases involving toddlers) and possible drugs of abuse (especially in cases involving teenagers) should be specifically sought out in the history.

Hypertension is common in the setting of coma and can be difficult to interpret. Impaired consciousness may be caused by hypertensive encephalopathy or hypertension may be a protective physiological response to raised intracranial pressure. The former requires urgent treatment to reduce blood pressure, while in the latter, such treatment may cause cerebral ischemia and exacerbate intracranial hypertension. Hypertensive encephalopathy may be caused by renal, cardiovascular (eg, coarctation of the aorta), vasculitic, and endocrine conditions (eg, pheochromocytoma) and also drugs (recreational and prescribed). It is suggested by a history of hypertension, headaches, visual disturbance, and seizures. It may be associated with posterior reversible encephalopathy syndrome (PRES) that has a typical pattern on CT/MRI scan with edema typically affecting the parietal-occipital regions, has most commonly been reported in children after hematopoietic stem cell and solid organ transplantation. PRES is thought to be caused by loss of autoregulation to rapid changes in blood pressure in the areas affected. However, 20% to 30% of children with PRES have normal or only slightly high blood pressure and it is speculated that endothelial dysfunction and inflammation may play an important role in the pathogenesis. Hypertensive encephalopathy should be managed with intravenous vasodilators and/or β-blockers with the goal of initial reduction in blood pressure of approximately 25% and then further reduction more slowly over 24 to 48 hours.

Metabolic emergencies such as diabetic ketoacidosis or severe hyperammonemia associated with acute liver failure or inborn errors of metabolism should be managed according to established guidelines that are disease specific. Demyelinating and autoimmune encephalopathies may be treated with steroids, immunoglobulin, or plasmapheresis (see Chapter 548).

An extensive evidence-based guideline and an abbreviated algorithm for the investigation and management of a child with impaired consciousness have been published online by the Pediatric Accident and Emergency Research Group of the Royal College of Pediatrics and Child Health and the British Association of Emergency Medicine (http://www.nottingham.ac.uk/paediatric-guideline/index2.htm).

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PART 2 STABILIZATION AND MANAGEMENT OF THE ACUTELY ILL INFANT AND CHILD

99 Physiologic Monitoring

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INTRODUCTION

Children with serious illness or injury with the potential for rapid deterioration invariably require close observation to detect changes in function or state. Careful bedside assessment provides the basis for appropriate triage and early awareness of clinical deterioration. Electronic monitoring complements this assessment by providing (1) repetitive or continuous assessment that does not disturb the patient, (2) a means for detecting the effectiveness of interventions, and (3) warning signals for physiological disturbances that permit staff to observe multiple patients simultaneously. The ideal monitoring system has been described by many as one that is accurate, able to predict deterioration, reproducible with a rapid response, operator independent, easy to use, safe, continuous, inexpensive, and integrative with other medical systems. To date, no such monitoring currently exists in clinical use.

RESPIRATORY MONITORING

Respiratory Rate

The respiratory rate (RR) and *tidal volume* (V_T; volume of each breath) together determine minute ventilation ($\dot{V}_E$; volume of air movement into and out of the lungs in 1 minute). Minute ventilation is product of RR × V_T. Since arterial blood carbon dioxide levels ($PaCO_2$) are inversely proportional to $\dot{V}_E$, decreases in RR and/or V_T will lead to increases in $PaCO_2$. Therefore, determination of RR is an essential component for the evaluation of respiratory function.

There are several methods for determining RR, ranging from counting visual chest movement and auscultated breaths to using devices that measure changes in electrical bioimpedance, air movement at the nares, or inductance plethysmography. These methods each have advantages and limitations but none measure carbon dioxide (CO_2) levels. Auscultation provides the best estimation of RR and depth of breathing. It provides the practitioner the opportunity to listen for air movement within the lungs and to differentiate between problems related to airway obstruction (upper vs lower) and problems with poor lung/chest wall compliance. The drawback is that auscultation is time consuming and a static measure.

Transthoracic impedance monitoring is the technique most commonly used to continuously measure respiratory rate. A small current is passed between 2 electrodes across the chest (similar to those used for electrocardiography). During inhalation and chest rise, the electrodes move further apart, increasing the impedance between them. During exhalation, the electrodes move closer together and the impedance decreases. The sinusoidal changes in impedance are displayed as the RR. Transthoracic impedance monitoring is safe, relatively cheap, and readily available to all patients. It is important, however, to recognize its limitations. First and foremost, the interpreted signal is dependent on chest movement and not breathing, per se. Therefore, chest movement unrelated to breathing may be mistaken for breathing, overestimating RR. Next, chest movement may negate impedance measurements and also may result in the underestimation of RR. Finally, transthoracic impedance monitoring does not accurately measure the depth of breathing nor does it rely on actual air flow. Thus, patients who have airway obstruction may have significant chest movement but little to no ventilation of the lungs. This may lead to under-recognition of airway obstruction.

Other methods of monitoring RR are respiratory inductance plethysmography, airflow thermistor, and capnography (discussed below). Respiratory inductance plethysmography tends to be used in specialized settings, such as during polysomnography (sleep studies). This method utilizes elastic bands that encircle the chest and abdomen, and assumes that the thoracic cavity is a cylinder that expands and constricts circumferentially during inhalation and exhalation, respectively. Electrodes within the bands measure the changes in circumference of the chest and abdomen in order to determine a RR as well as to estimate a tidal volume. When used in conjunction with other measures of respiration, inductance plethysmography can be useful to differentiate different disorders of breathing, such as central versus obstructive sleep apnea. Caveats to using this device are the relatively specialized equipment needed for accurate measurement, correct placement of the bands, and tolerability of wearing the device.

Respiratory thermistors are devices that detect changes in air temperature at the nares in order to estimate RR. The thermistor is placed at the patient's nares and detects a difference between the warm gas breathed out during exhalation versus the cooler inhaled air. The changes in temperature are then used to determine RR. The relative changes in temperature do not relate to the volume of each breath, however; thus this device can only detect RR. Limitations in its use include inaccuracies during mouth breathing and underestimation of hypopnea.

Capnography

$\dot{V}_E$ is inversely proportional to $PaCO_2$. Although RR is important, it is often inadequate to assess ventilation due to the inability to accurately take into account depth of breathing and physiologic changes within the lung including "dead space" ventilation ($\dot{V}_D$) which refers to air movement without the exchange of gases between the lung and the blood. Therefore, during situations in which accurate measurements of ventilation are desired, measurement of CO_2 levels may be necessary. The gold standard for measurement of ventilation is the arterial blood gas (discussed below). Although it is the truest measure of ventilation, it is invasive and is usually not continuous. In situations where continuous monitoring is necessary, capnography has become a standard of care, especially in the emergency department, intensive care units (ICUs), and operating rooms, and during procedural sedation.

Quantitative capnography is based on the ability of CO_2 to absorb infrared light. When gas is sampled within the path of airflow (mainstream) or from the side (sidestream), it passes through a detector that emits infrared light and detects the absorbance by CO_2. Alternatively, some devices use spectrometry to detect CO_2 levels. As the detector measures the exhaled CO_2 (P_{ECO2}), a tracing appears on the monitor (**Fig. 99-1**); this is the standard time-based capnograph. During a respiratory cycle, the partial pressure of carbon dioxide (PCO_2) of inhaled air is 0 mm Hg, accounting for the minimal amount of atmospheric CO_2 in relation to nitrogen and oxygen. This air enters the trachea, bronchi, and then the respiratory lung units (alveoli and respiratory bronchioles). Carbon dioxide quickly diffuses from the blood to the respiratory lung units, but not into the dead space. This dead space ventilation can be areas of normal, anatomic dead space (ie, trachea, mainstem bronchi, and nonrespiratory bronchioles) or pathologic areas of the lung that are poorly perfused (eg, pulmonary embolus and during cardiac arrest). During exhalation, the first volume of gas is from the anatomic dead space (trachea and bronchi); therefore, the P_{ECO2} is 0 mm Hg (phase 1). As gas moves from the respiratory lung units into the larger airways, the P_{ECO2} increases (phase 2). In normal lungs, this increase in expired CO_2 is rapid. In the latter phase of exhalation, the P_{ECO2} plateaus (phase 3), resulting in the typical square-shaped waveform. At the end of exhalation, the end-tidal CO_2 (P_{ETCO2}) is quantified and displayed. As inhalation begins again (phase 4), the P_{ECO2} quickly falls due to the movement of atmospheric CO_2 through the detector. Therefore, based on the capnograph waveform and P_{ETCO2} value, the adequacy of ventilation can be continuously assessed in a reliable fashion.

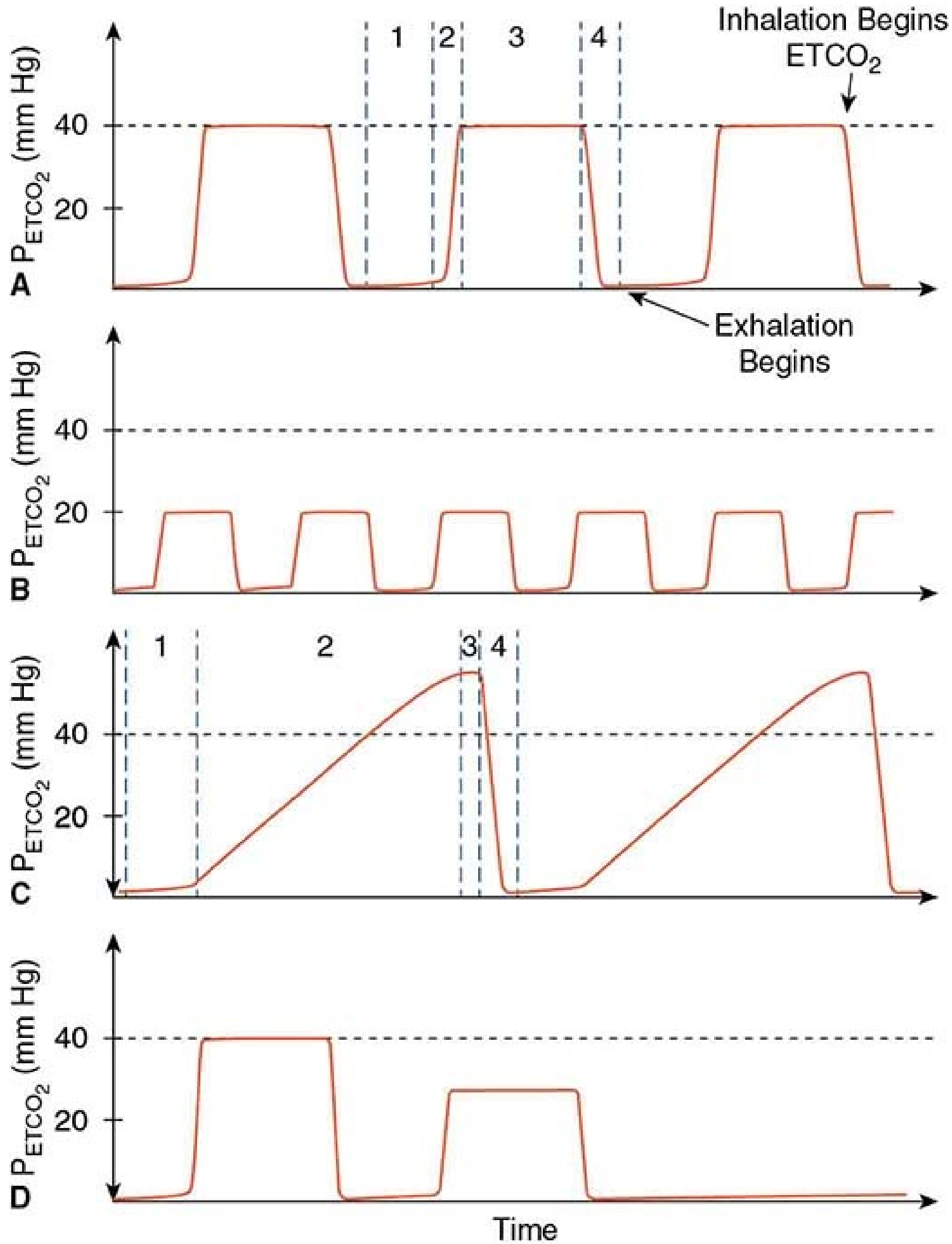

FIGURE 99-1 Quantitative capnography. **A:** In a normal capnograph, there is a prototypical square waveform for exhaled CO_2 (P_{ECO2}) that consists of 4 phases: (phase 1) exhalation of gas from the anatomic dead space; (phase 2, ascending phase) rapid addition of CO_2-containing alveolar gas; (phase 3, plateau phase) equalization of alveolar gas; and (phase 4, inhalation phase) rapid decline in P_{ECO2}. The intersection between phases 3 and 4 is where inhalation begins and is the point where the end-tidal CO_2 (P_{ETCO2}) is quantified. **B–D:** The waveform

shape can provide insight to changes in the patient or equipment malposition. **B:** An increased respiratory rate with decreased P_{ETCO2} may signify hyperventilation. **C:** Prolongation of phase 2 typically signifies bronchospasm. **D:** A previously normal waveform that quickly disappears may indicate detector dislodgement, complete airway obstruction, or cardiac arrest.

Quantitative time-based capnography is used in situations where the adequacy of ventilation needs to be continuously monitored. The most common scenario includes one in which there may be disordered control of breathing, such as during procedural sedation and anesthesia or if patients have depressed sensorium or neuromuscular weakness. Additionally, capnography should routinely be used as a safety device in endotracheally intubated patients since disconnection from the ventilator, accidental extubation, or other mechanical problems will be detected by a loss of capnographic signal. To that end, qualitative (or colorimetric) capnography is used to quickly confirm endotracheal tube placement since the litmus paper within the device changes color when the expired CO_2 reaches a certain threshold. Capnography is also useful for making adjustments to a patient's ventilator settings, especially if rapid titration is necessary (such as to manage increased intracranial pressure) or if a blood gas measurement is not possible or warranted. Finally, an increasing recommendation is the use of quantitative capnography to assess the quality of chest compressions during cardiopulmonary resuscitation. Since the P_{ETCO2} depends on blood flow through the lungs in addition to air flow, during cardiac arrest, P_{ETCO2} will drop toward 0 due to lack of pulmonary blood flow. As chest compressions are instituted, P_{ETCO2} will start to rise. As the effectiveness of chest compressions improve, the P_{ETCO2} should also increase, given the same $\dot{V}_E$, due to improved pulmonary blood flow (Fig. 99-1E).

The utility of quantitative capnography lies not only in the P_{ETCO2} value, but also with the shape of the waveform. Typically, the P_{ETCO2} value is less than the $PaCO_2$ (~0–5 mm Hg) in normal lungs, owing to the anatomic dead space. In patients with higher degrees of lung disease, the difference between $PaCO_2$ and P_{ETCO2} (dead space ventilation; $\dot{V}_D$) increases and can be followed to determine disease progression in critically ill patients, especially if they are tracheally intubated with a cuffed endotracheal tube. One of the most useful attributes of the capnographic waveforms is the ability to detect the degree of airway obstruction and bronchospasm. In patients with bronchospasm, CO_2 from the alveoli enters the large airways at variable rates due to increased resistance in the small airways. This is manifested as a slow, upward ramping of the capnographic waveform instead of the normal square waveform. If the obstruction is severe enough, the P_{ETCO2} may never reach the true level of the $PaCO_2$ since inhalation begins prior to complete exhalation.

Capnography, like other devices, has its limitations and caveats. In nonintubated patients, capnography typically uses sidestream detectors via nasal cannula. In these cases, a decrease in P_{ETCO2} may be caused by other environmental factors, such as dislodgement of the cannula, mouth breathing, or entrainment of atmospheric air (such as from the concomitant use of a high-flow nasal cannula for oxygenation). Additionally, appropriate interpretation of the capnographic waveform and P_{ETCO2} can be confusing, at times. As an example, decreased ventilation will increase P_{ETCO2}, whereas decreased perfusion to areas of the lungs will decrease P_{ETCO2}. If a patient has both decreased ventilation and perfusion, then it will be difficult to interpret the true nature of the findings. The interpretation of changes in mainstream quantitative time-based capnography also assumes consistent (or no) breath-to-breath leaks in the system.

Pulse Oximetry

Pulse oximetry is a mainstay in the monitoring of oxygen saturation. To recognize the benefits and limitations of pulse oximetry, it is important to understand the principles on which it is designed. Pulse oximetry depends on light absorption in a tissue bed, depending on the relative ratio of oxyhemoglobin (HbO_2) and deoxyhemoglobin (Hb) at 2 different wavelengths of light (**Fig. 99-2**). At red wavelengths, Hb has a higher light absorption than HbO_2; whereas at infrared wavelengths, HbO_2 has a higher light absorption than Hb. For standard pulse oximeters, light in the red and infrared wavelengths is emitted from a light emitting diode (LED) and travels through a tissue bed (eg, finger, earlobe, toe) to a detector on the opposite side. The ratio of the emitted (known) and the detected (measured) light intensities is calculated for each wavelength to determine the relative level of absorbance of each. Based on correlation curves using blood O_2 saturation levels from healthy human subjects, the O_2 saturation of the tissue is estimated. To estimate the arterial saturation, the pulse oximeter uses the pulsatility of blood in the tissue to its advantage. During systole, more blood is in the arterial system; therefore, more light is absorbed as there is more Hb and HbO_2 in the path of light. During diastole, there is less blood to absorb light in the pathway; therefore, the absorption of light is caused by the combination of tissue, venous blood, and non-pulsatile arterial blood. The difference in absorption during systole and diastole is then assumed to be caused by the contribution of arterial blood alone. The device then displays the oxygen saturation of pulsatile blood (SpO_2).

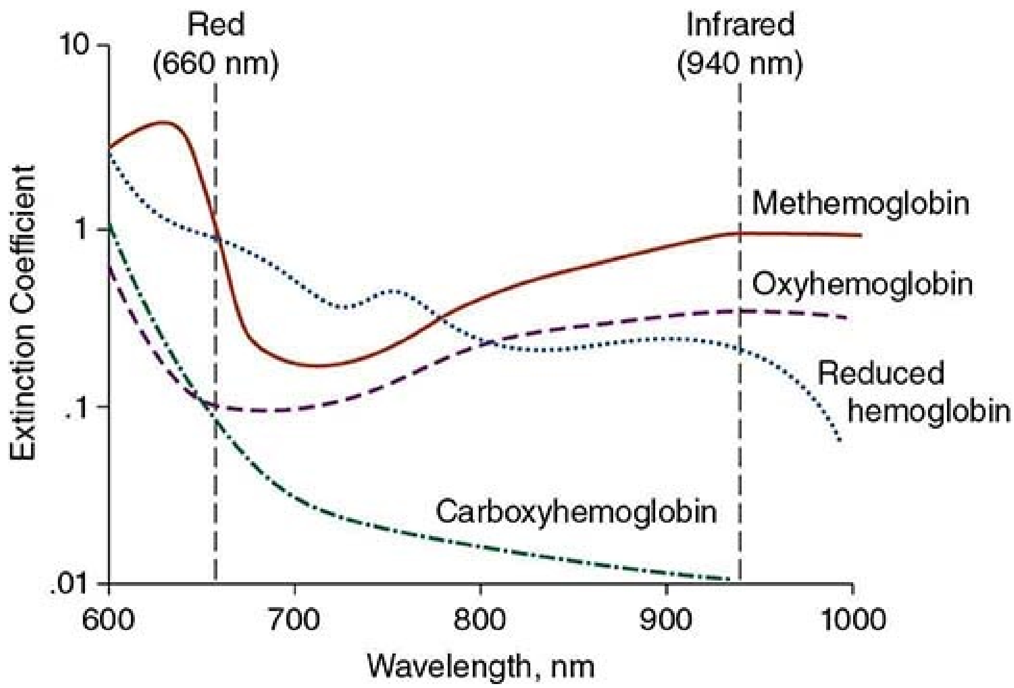

FIGURE 99-2 Absorption of light by various types of hemoglobin. Note that at the 2 wavelengths used in standard pulse oximetry, the absorption of light of methemoglobin is equal. (Reproduced with permission from Jubran A: Pulse oximetry, *Crit Care*. 2015 Jul 16;19:272.)

An understanding of the basis of pulse oximetry helps clinicians to recognize its benefits and limitations as well as how to troubleshoot problems. Clearly, the main benefit of pulse oximetry is the ability to estimate arterial O_2 saturation in a continuous and noninvasive manner. For the vast majority of patients, SpO_2 measurements have replaced arterial oxygen saturation (SaO_2) measurements. Although newer devices have improved the reliability of SpO_2 measurements, there are still limitations to the accuracy of current pulse oximeters. It is important to remember that pulse oximeters average the signal over the preceding 20 beats; therefore, the displayed SpO_2 is actually reflective of changes that occurred up to 20 cardiac cycles ago. This is important during critical periods, such as during resuscitation or intubation. Since pulse oximetry relies on the device to distinguish between pulsatile blood and non-pulsatile blood, there are a number of conditions that may reduce the accuracy and/or reliability of measurement: patients with poor perfusion to the extremity used for pulse oximetry (eg, in severe shock, arterial occlusion, or hypothermia), excessive movement, and venous pulsation (eg, in severe tricuspid regurgitation). Since standard pulse oximeters typically rely on only the red and infrared wavelengths of light, certain dyshemoglobinemias, such as carboxyhemoglobinemia (overestimates SpO_2) and methemoglobinemia (overestimates SpO_2 and trends toward 85% as MetHb levels rise), may result in inaccurate measurements. In these instances, co-oximeters, which utilize multiple wavelengths of light, should be used in place of pulse oximeters. Finally, excessive ambient light interference or alterations in absorption (eg, from fingernail polish, dyes, pigments) can affect the accuracy of measurements. Another caveat is that there is a lower limit for accurate SpO_2 measurements with most devices. At lower O_2 saturations (below 80–85%), SpO_2 readings are considered to reflect a trend rather than an accurate saturation. This is of particular relevance in patients with cyanotic congenital heart disease, or in the setting of severe hypoxemic respiratory failure, when permissive hypoxia is a part of routine intensive care.

Blood Gas Measurement

As described above, the gold standard to determine oxygenation and ventilation in patients is through the measurement of arterial blood gases (ABG), which characterizes efficiency of gas exchange and also disturbances in the metabolic and respiratory control of acid–base balance.

All modern blood gas analyzers take 4 measurements: sample temperature, pH, PO_2, and PCO_2. The pH, PO_2, and PCO_2 are directly measured, whereas most analyzers calculate the following using various equations and nomograms: bicarbonate (HCO_3^-), O_2 saturation (SO_2), and acid–base excess (ABE). This is important to take into consideration when there is a need for exact SO_2 values, such as in the presence of dyshemoglobinemias and when used to estimate cardiac output using the Fick principle. It is also important to understand the different areas of sampling available for blood gas determination. Although arterial sampling is the gold standard, blood gases can be obtained from the venous and capillary vessels, especially for outpatient monitoring of ventilation, but they cannot be used as a guide to adequacy of oxygenation. Blood venous samples tend to be approximately 0.05 to 0.1 pH units lower and have PCO_2 levels approximately 5 mm Hg higher than simultaneous arterial samples owing to the diffusion of CO_2 from the tissues into the capillary beds prior to returning to the venous system. In patients with poor perfusion due to shock, hypothermia, or the use of a tourniquet, these discrepancies may be greater. Therefore, when obtaining samples from these areas, it is important that the blood is free-flowing without the use of a tourniquet because restricted blood flow during sampling may artificially decrease the pH and increase the PCO_2. Venous PO_2 levels are a reflection of cellular O_2 uptake, and high levels may be indicative of impairment in mitrochondrial O_2 uptake from processes such as cyanide toxicity. The utility of capillary samples is entirely questionable, and many practitioners believe they do not have any role in blood gas monitoring. Finally, when obtaining any blood sample for blood gas analysis, it is important to ensure that there are no air bubbles in contact with the sample as this will affect the measurements, due to diffusion of gas between the blood and air.

The arterial pH (normal 7.35–7.45) reflects the acid-base balance within the system. If the arterial pH is less than 7.35, the blood is deemed *acidemic*. If the arterial pH is greater than 7.45, the blood is *alkalemic*. The 2 main determinants of pH in the blood are the PCO_2 (normal 35–45 mm Hg; a volatile acid released by the lungs) and serum bicarbonate levels (HCO_3^-; normal 22–26 mM beyond early postnatal life; primary buffering system), which are related to each other by the following chemical reaction:

$$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$$

and are related to pH by the Henderson-Hasselbalch equation:

$$pH = 6.1 + \log \frac{[HCO_3]}{(0.03)(PCO_2)}$$

Other determinants are the levels of various proteins, ions, and other weak buffers within the blood and are out of the scope of this chapter. Since PCO_2 is a volatile acid that is exhaled by the lungs, as PCO_2 increases, the blood becomes more acidic and pH falls. For acute changes, if PCO_2 changes by 10 mm Hg, pH will change in the *opposite* direction by 0.08. Conversely, changes in HCO_3^-, a base, result in changes in pH in the *same* direction, such that an increase in HCO_3^- by 10 mM increases the pH by 0.15. Of note, the HCO_3^- result on standard blood gas analyzers is a *calculated* result based on the Henderson-Hasselbalch equation and the value should be compared to the serum total CO_2 (tCO_2; normal 23–30 mM) that is measured by serum chemistry. Taking these factors into account, basic acid–base disorders can be determined by proceeding in a step-wise fashion (**Fig. 99-3**), with more complex evaluations discussed in other resources.

Basic Algorithm for Acid-Base Disorders:

I) Is the pH high (alkalemic), low (acidemic), or normal?
- a. *N.B.* If the pH is 7.40 but pCO_2 and/or HCO_3^- are abnormal, the patient has a mixed acid-base disorder

2) What is the primary disorder: respiratory, metabolic, or mixed?
- a. Evaluate pCO_2 and HCO_3^-
  - i. pCO_2 moves in the opposite direction as pH
  - ii. HCO_3^- moves in the same direction as pH
- b. If there is metabolic acidosis (low pH with low HCO_3^-), calculate the anion gap (AG)
  - i. AG = $s[Na^+] - s[Cl^-] - s[tCO_2]$, normal 12 ± 2 mM
    - 1. If elevated, think MUDPILES:
      - a. Methanol
      - b. Uremia
      - c. Diabetic (or other) ketoacidosis
      - d. Propylene glycol, paracetamol (acetaminophen) toxicity
      - e. Iron, ibuprofen, isoniazid, inborn errors of metabolism
      - f. Ethylene glycol
      - g. Salicylates (aspirin)
    - 2. If normal, think renal or gastrointestinal losses using urinary electrolytes: UAG = $u[Na^+] + u[K^+] - u[Cl^-]$
      - a. If UAG < 0, consider GI losses of bicarbonate
      - b. If UAG ≥ 0, consider renal losses of bicarbonate

3) Is there compensation of the pH?
- a. Calculate the expected change in pH for abnormal pCO_2 and $[HCO_3^-]$
  - i. Changes in pCO_2 by 10 mm Hg change pH by 0.08 in the **opposite** direction
  - ii. Changes in $[HCO_3^-]$ by 10 mM change pH by 0.15 in the **same** direction
- b. Calculate the expected compensation by pCO_2 or HCO_3^-:
  - i. For primary metabolic disorders, pCO_2 should compensate as follows:
    - I. Metabolic acidosis: $pCO_2 = (1.5 \times [HCO_3^-]) + 8 \pm 2$
    - 2. Metabolic alkalosis: $pCO_2 = (0.7 \times [HCO_3^-]) + 20 \pm 5$
  - ii. For primary respiratory disorders, there are both acute and chronic compensatory mechanisms due to the changes in $[HCO_3^-]$ by buffering in the blood versus by the kidneys:
    - 1. Respiratory acidosis:
      - a. Acute: $[HCO_3^-] = 24 + \left(\frac{pCO_2 - 40}{10}\right)$
      - b. Chronic: $[HCO_3^-] = 24 + 3.5 \times \left(\frac{pCO_2 - 40}{10}\right)$
    - 2. Respiratory alkalosis:
      - a. Acute: $[HCO_3^-] = 24 - 2 \times \left(\frac{40 - pCO_2}{10}\right)$
      - b. Chronic: $[HCO_3^-] = 24 - 4 \times \left(\frac{40 - pCO_2}{10}\right)$
- c. If the measured pH, pCO_2, and $[HCO_3^-]$ are not as expected, the patient has a mixed acid-base disorder.

FIGURE 99-3 Basic algorithm for arterial blood gas analysis. pCO_2, partial pressure of carbon dioxide; UAG, urinary anion gap.

The other component use of the arterial blood gas analysis is PaO_2 (normal 75–100 mm Hg). Although the majority of oxygen is carried by hemoglobin in the erythrocytes and reliably monitored using pulse oximetry, determination of PaO_2 is useful in patients with cyanotic heart disease in which SpO_2 is lower and less reliable, as well as when differentiating low SpO_2 in neonates using the hyperoxia test. Finally, PaO_2 levels are useful for characterizing the degree of lung injury in acute lung injury and acute respiratory distress syndrome for prognosis and research.

CARDIOVASCULAR MONITORING

Heart Rate and Rhythm

The heart rate and rhythm and their trends over a finite period of time are a fundamental part of the evaluation and treatment of critically ill patients. Electrocardiograms (ECGs) can be monitored continuously from electrodes placed on the chest, and a cardiotachometer determines heart rate from the frequency of the QRS signal. There are important sources of error to consider with these methods, however. When the signal is large, the T wave and QRS may both be detected, and the cardiotachometer may show a value that is twice normal. Failure to detect a signal can occur when the QRS amplitude changes—for example, with respiration, repositioning of the leads, or changes in posture. It is quite easy to misinterpret rhythm disturbances or misdiagnosis ST-segment abnormalities when reviewing a single lead rhythm or rhythm strip. Whenever a dysrhythmia or ECG abnormality is suspected based on the monitor, a 12 to 15-lead ECG should be obtained for true diagnostic purposes. Finally, one must remember that ECG only measures electrical activity and not actual cardiac muscle contractions, which is an important distinction in patients with pulseless electrical activity (PEA). Therefore, a pulse check is imperative in these patients to assess their systemic perfusion.

Bedside arrhythmia analysis and bed-to-bed switching (so that a central monitoring station was not required) allowed widespread use of electrocardiographic monitoring in adult ICUs in the 1960s and 1970s. Widespread adoption in pediatric ICUs did not occur until the late 1980s with the advent of modular units that allowed input from various devices. At this time, the potential for arrhythmia in postoperative cardiac surgical patients and other organ dysfunction states was becoming appreciated. Focused attention to ST-segment changes has been commonplace after operations that involve manipulation or reimplantation of the coronary origins such as the arterial switch operation and the Ross operation. The theoretical advantage of automated ST-segment analysis is balanced by the limitations of algorithms used by currently available software for this purpose in the pediatric populations.

Blood Pressure

Noninvasive Blood Pressure Most commonly, noninvasive blood pressure assessment is performed using intermittent methods. Determining pressure using a sphygmomanometer with a cuff that occupies about two-thirds of the length and almost completely (80–100%) encircles the circumference of the limb segment remains the standard approach for initial patient assessment. The cuff is inflated to a pressure well above that occluding the arterial systolic pulse and is gradually deflated while listening for Korotkoff sounds. The first sound corresponds to arterial systolic pressure, and the muffling of the second sound corresponds to the arterial diastolic pressure. The systolic pressure can also be determined by palpating the artery distal to the cuff and noting the pressure at which the pulse is first felt when the cuff is deflated.

Alternatively, most automated blood pressure cuffs utilize oscillometry to determine blood pressure. These devices detect pressure changes caused by turbulence in arterial blood flow as the cuff is deflated. These devices record the peak oscillations as the mean arterial blood pressures (MABP) and then extrapolate the systolic and diastolic blood pressure measurements. The most important factor that determines the accuracy of these oscillometric techniques in measuring MABP is cuff width–to–arm circumference ratio which ideally should be in the range of 0.44 to 0.55. A smaller cuff will overestimate the true MABP. It is also important to note that the mean bias of noninvasive methods become unreliable at both the higher and lower limits of physiologic blood pressure measured invasively (ie, severe hypertension or clinical shock, respectively).

Principles of Invasive Pressure Monitoring Briefly, once a vascular compartment of interest has been accessed by a catheter (eg, arterial, central venous, left atrial), the pulsations generated by the space are transmitted via a continuous column of fluid to a transducer that converts the analog signal into a digital one. When setting up a direct pressure monitoring system, it is important to ensure that the system has a baseline reference value; this is usually set to atmospheric pressure at the anatomic level of interest (such as the right atrium for vascular pressures). Other facets of calibration, such as damping, are important but out of the scope of this text.

Central Venous Pressures and Atrial Pressures

Central venous pressure (CVP) is used as a proxy for central vascular volume. In general, as the volume in the vasculature increases, the CVP rises. More accurately, the measured CVP reflects the back pressure of venous return from the peripheries (this driving pressure from the peripheries is known as *mean* *circulatory filling pressure*). Changes in intravascular volume status, ventricular compliance, or intrathoracic pressure will affect the value of the CVP. The pressure from a catheter in the right atrium or a large central vein is used to monitor cardiac filling or, more precisely, right ventricular preload. Many assumptions are made when using CVP as a measure of intravascular volume, including lack of downstream obstruction, normal lung compliance, and a consistent relationship between pressure and volume (known as *ventricular elastance*). A more accurate measure of this is the *left atrial pressure*, which can be directly measured with a left atrial line, or indirectly using a Swan Ganz catheter with measurement of pulmonary artery wedge pressure. Left atrial lines are still placed in selected patients undergoing open heart surgery. However, the use of Swan Ganz catheters in pediatric patients, is now at most, rare and, in many units, obsolete.

With these constraints in mind, CVP can serve as a useful guide for determining how interventions are affecting cardiac filling in patients in whom responses are not very predictable (eg, the child with congestive circulatory failure in whom positive pressure ventilation is initiated or adjusted) or in patients in whom there is a narrow margin of safety (eg, the child with elevated intracranial pressure and low cardiac output). Normal values for CVP range from –3 to +3 mm Hg in the infant to 5 to 10 mm Hg in the child and young adult. These values, however, can be greatly affected by the use of positive-pressure breathing. Accordingly, it is usually more useful to judge relative changes rather than to rely on absolute targets for therapy. In addition to pressure monitoring, additional information regarding the adequacy of systemic oxygen delivery is provided by central venous saturations ($ScvO_2$) drawn from an appropriately positioned central catheter with the tip in the superior vena cava–right atrial junction and without the confounding effects of a significant left-to-right atrial level shunt. By using the oxygen extraction ratio $(\mathrm{OER} = [\mathrm{SaO_2} - \mathrm{ScvO_2}/\mathrm{SaO_2}];\ \text{normal } 0.2 - 0.3)$, an estimation of adequacy of cardiac output can be made, with OER > 0.3 as concerning for inadequate cardiac output for the current oxygen consumption state.

Continuous venous pressure tracings, and the appearance of its component *a*, *c*, and *v* waves, can also provide useful information as to nature of the underlying cardiac rhythm and changes therein (**Fig. 99-4**).

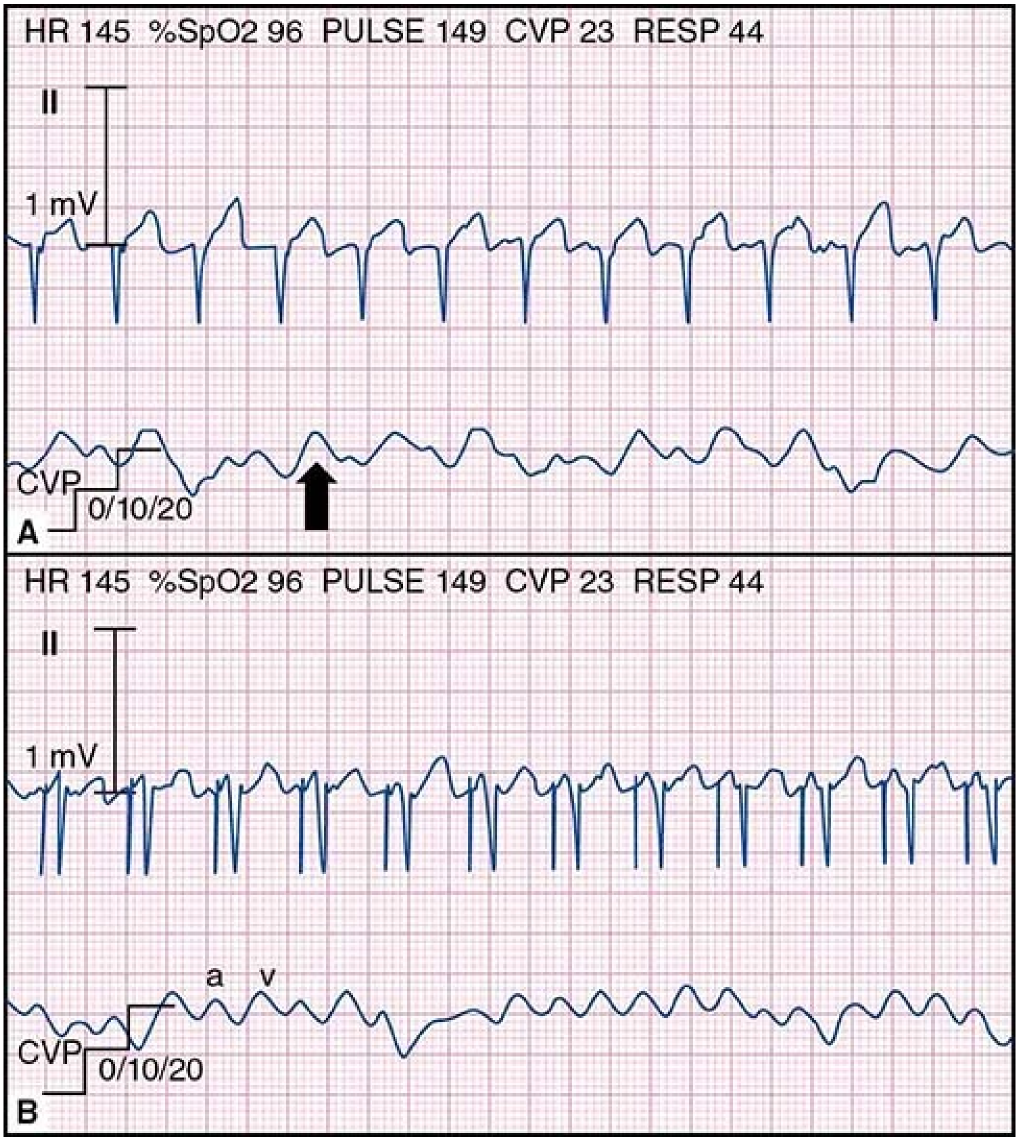

FIGURE 99-4 Simultaneous electrocardiography and central venous pressure monitoring waveforms. **A:** The electrocardiogram has no easily discernible P-waves and the corresponding central venous pressure (CVP) trace shows canon *a*-waves (*arrow*), which is consistent with an atrioventricular (AV) dissociated rhythm and is mechanically reflective of atrial contraction against a closed AV valve apparatus. **B:** The same patient with atrial pacing demonstrates a CVP trace showing *a*- and *v*-waves related to coordinated atrial and ventricular contraction.

Arterial Pressures

The use of invasive arterial pressure monitoring is reserved for intraoperative, perioperative, or intensive care monitoring. The most common reasons for arterial line placement are for frequent blood gas measurement in patients with respiratory failure, for frequent blood sampling in unstable patients, and the need to monitor hemodynamic changes during periods of critical illness and in response to interventions and titrations of therapies. The choice of location of invasive arterial pressure monitoring is based on ease of access and limitation of complications, with important exceptions. These include not attempting catheterization of an artery that is known to have been completely or partially occluded, or in a circulation where there may already be compromised flow (a systemic-to-pulmonary shunt affecting flow to that arm, or dialysis fistula) or poor potential collateralization. In the case of surgery on the aortic arch, a right arm arterial line (typically radial) is used to guide cerebral perfusion both prior to and during cardiac surgery. In smaller infants and neonates, especially in the context of shock, cannulating peripheral arteries can be challenging. In these situations, the femoral artery may be the next choice. As a rule, the longer term consequences of vascular occlusion should always be considered when deciding on the optimal location of an arterial line, and the operator performing the procedure.

It is important to recognize that there are regional variations in blood pressure and that these can be accentuated in pathological states. Normally, mean blood pressures are equal throughout the large arteries, although systolic pressure is higher and diastolic is lower when comparing the large arteries of the legs to the brachial or radial artery. However, in states of low cardiac output, the systolic and mean blood pressures from a large central artery often are higher than values from more peripheral arteries depending on the degree of impairment of the perfusion, and peripheral vasoconstriction. The central systolic blood pressure may be sustained near normal levels even in the presence of shock, owing to severe vasoconstriction of peripheral arteries and arterioles.

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100 Vascular Access

Andreas A. Theodorou and Robert A. Berg

INTRODUCTION

Rapid establishment of vascular access is necessary for aggressive fluid resuscitation and administration of medications such as catecholamines, antibiotics, narcotics, and sedatives during emergencies. However, attaining vascular access in a child during a life-threatening illness is difficult and often consumes precious time. An organized approach to vascular access can minimize this potentially life-threatening delay in treatment. This chapter discusses the priorities in vascular access during emergent, urgent, and stable situations. It also reviews various techniques for achieving vascular access and covers the relative indications and potential complications.

PRIORITIES OF VASCULAR ACCESS

Time is critical when attaining vascular access in life-threatening emergencies such as cardiopulmonary arrest or shock. Of course, any pre-existing intravenous catheter should be utilized in the initial resuscitation efforts, regardless of how small such a catheter might be. If such access is not available during a life-threatening emergency, intraosseous access should be attained as rapidly as possible, especially in children under 6 years old. A practical approach is to pursue this and peripheral venous access simultaneously. Time should not be wasted waiting for attempts at peripheral venous catheterization before attempting intraosseous access, which can be attained more rapidly and more reliably. Similarly, skilled clinicians may attempt placing central venous catheters during life-threatening emergencies, but such attempts should not preclude simultaneous attempts at intraosseous access. After attaining intraosseous access for initial fluid resuscitation and infusion of emergency medications, peripheral or central venous catheterization is the next priority in order to ensure a more reliable, long-lasting vascular access.

For urgent situations, such as fluid resuscitation of a child with compensated shock or dehydration, the risk–benefit ratio shifts. Generally, it is most appropriate to initially insert a peripheral venous over-the-needle catheter. If multiple attempts are unsuccessful, or if the child requires fluids or medications that cannot be given safely in a peripheral vein, central venous catheterization should be attempted by a qualified individual. Of course, if the child's clinical condition deteriorates prior to achieving vascular access, priorities should be reassessed and it may be necessary to attain intraosseous access.

Relatively stable children may need vascular access for maintenance fluids or intravenous medications. Generally, peripheral venous cannulation with an over-the-needle catheter is adequate. If vascular access is necessary for more than 2 to 3 weeks, or if solutions to be infused can cause serious tissue injury if extravasated, a central venous catheter may be necessary. However, central venous catheterization entails added risks, and its inherent risks and benefits deserve consideration (Table 100-1).

TABLE 100-1 VASCULAR ACCESS TECHNIQUES

Route	Indications	Benefits	Risks and Disadvantages
Peripheral	Short-term infusion of fluids or medications	Minimal training required Minimal risk	Local Hematoma, infiltrations, infections Infection Time-consuming (especially if vascular collapse)
Intraosseous	Emergent vascular access Cardiac arrest Hypotensive shock	Attained rapidly and reliably	Adverse effects *rare* Osteomyelitis Compartment syndrome Fracture Soft-tissue necrosis
Central venous—acute	Emergent vascular access Hemodynamic monitoring Infusion of: Irritating medications and vesicants Vasoactive agents	More secure then peripheral Safer for certain infusions Vasoactive agents	Advanced skills necessary Often time-consuming Pneumothorax Chylothorax Arrhythmia Thrombosis Embolus (air, catheter, wire) Hemothorax Hematoma Malposition Infection
Central venous—chronic (eg, Broviac, Hickman, peripherally inserted catheter)	Long-term parenteral nutrition Prolonged or frequent administration of intravenous medications (eg, antibiotics, chemotherapy)	Compared to central venous—acute: More secure Lower infection risk Lower thrombosis risk	Same as for central venous—acute

PERCUTANEOUS PERIPHERAL VEIN CANNULATION

Small-caliber plastic catheters have allowed for increasingly easy and reliable peripheral venous cannulation. Small over-the-needle catheters (22–24 gauge) are available to cannulate even the small veins of the hand, foot, or scalp of neonates and premature babies. Such peripheral venous catheters are generally all that are necessary for short-term delivery of intravenous fluids or medications.

The most common complication of peripheral venous cannulation is catheter displacement and infiltration of the tissues with the infusing fluid. Most intravenous solutions are isotonic or hypotonic and easily absorbed from the tissues. Infiltration with these solutions is generally treated by removing the catheter and elevating the limb. However, solutions that are very hypertonic (eg, those containing greater than 12.5% dextrose, 3% sodium chloride, or 8.4% sodium bicarbonate) or that contain irritating substances such as calcium or potassium salts, some antibiotics (eg, erythromycin), or medications with an extreme pH (eg, phenytoin) can cause substantial tissue injury leading to necrosis of the skin or subcutaneous tissues. Importantly, extravasation of vasoconstricting agents such as dopamine, epinephrine, and norepinephrine can result in profound local vasoconstriction and subsequent substantial tissue injury. Even without extravasation of the previously noted medications and fluids, local vascular injury can result in aseptic thrombophlebitis. This condition may serve as a nidus for suppurative thrombophlebitis. The risks of aseptic and suppurative thrombophlebitis increase with the duration of the indwelling catheter.

Selection of catheter size and peripheral venous site are important issues. For a patient in shock, the widest and shortest catheter is optimal, because longer, narrower catheters result in more resistance to flow. For the trauma victim, it is preferable to insert the catheter into an uninjured limb or at least a limb apparently free from major vascular trauma. The greater saphenous vein, median cubital vein, and external jugular vein are often used, because they are relatively large and consistent in their anatomy. In older children, veins in the back of the hands and forearms are commonly used. Avoiding the dominant hand or the hand with the finger or thumb that the infant prefers to suck allows for improved patient comfort and function, although such choices are not always available.

Before the vein is cannulated, the operator should wash his or her hands well and use universal precautions, including protecting the operator’s hands with gloves. The extremities should be adequately immobilized, and the site should be cleansed with alcohol or iodine-containing solutions and allowed to dry. A tourniquet should be applied in order to engorge and distend the vein. The skin can be stretched taut with the operator’s nondominant hand in order to immobilize the vein. The operator should puncture the skin at a 15° to 30° angle, 5 to 10 mm distal to the expected entrance site into the vein. Next, the needle punctures the vein, usually resulting in blood return into the catheter hub. When this blood return occurs, the catheter is advanced a few millimeters to ensure that the catheter tip and the needle are in the lumen of the vein. The catheter is then advanced over the needle into the vein, and the needle is removed. The tourniquet is released, and saline is flushed intravenously to ensure patency of the catheter and vein. After the needle is removed, it should generally not be reinserted while the catheter is in the subcutaneous space or in the vein, because this could shear the tip of the catheter and generate a small plastic embolus.

Adequate immobilization of the extremity is an important aspect of successful peripheral vein cannulation. Moreover, it is important to adequately secure and protect the catheter after successful cannulation. Notably, ultrasound or devices using near-infrared have been used for difficult peripheral vein access.

INTRAOSSEOUS INFUSION

Although emergency intraosseous (IO) infusions were common in the 1940s and 1950s, the arrival of butterfly needles and plastic catheters and the widespread use of venesection led to a virtual extinction of this technique for several decades. However, there has been a resurgence in the use of intraosseous infusions since the early 1980s. The principal value of intraosseous access is that it can be performed rapidly and reliably. Medical personnel can usually attain access in less than 1 minute during emergencies. The bone marrow cavity is effectively a noncollapsible vascular space, even in the setting of shock or cardiac arrest. Therefore, intraosseous access is the initial vascular access site of choice in patients with life-threatening problems such as cardiopulmonary arrest or decompensated shock.

Almost any medication that can be administered into a central or peripheral vein can be safely infused into the bone marrow, including crystalloid solutions, colloid solutions, blood products, and hypertonic solutions. In particular, all the medications the American Heart Association recommends for pediatric advanced life support can be safely and effectively administered via this route, including catecholamine infusions. Pharmacokinetic variables, such as onset of action and plasma concentrations, are similar with intraosseous or peripheral venous administration in cases of circulatory shock or cardiac arrest with cardiopulmonary resuscitation (CPR). Emergency medications should be followed by a saline flush to ensure rapid delivery into the circulation. In addition, the initial bone marrow aspirate is a reliable specimen for venous pH and PCO_2, blood typing and cross-matching, serum glucose, electrolytes, and blood cultures. Due to stasis in the bone marrow, the results of such studies may be less reliable after administering drugs through the intraosseous needle during CPR.

The greatest obstacle to successful intraosseous cannulation is psychological and originates from the natural reluctance by any inexperienced provider to force a needle into a child's bone. Experienced providers may at times wrongly forfeit this quick and reliable alternative to undertake venous cannulation via rapid percutaneous central venous catheterization or peripheral vein cutdown, sometimes resulting in dangerous delays in starting therapy. Intraosseous cannulation is generally a safe procedure but can result in complications, including osteomyelitis, fractures, subcutaneous or intramuscular extravasation of toxic medications (eg, epinephrine, calcium), and compartment syndrome. Compartment syndrome is easily avoided by monitoring the insertion site for swelling and discontinuing the infusion if significant swelling occurs. Microvascular pulmonary fat and bone marrow emboli have been demonstrated but do not appear to be a clinically significant problem. The risk of such complications is acceptable in the dire circumstances of hypotensive shock or cardiac arrest. Such risks, and the pain involved, may not be acceptable when the clinical indications are less stringent.

The most commonly utilized site for intraosseous access is the medial surface of the tibia, 1 to 3 cm below the tibial tuberosity (**Fig. 100-1**). Alternative sites include the distal tibia above the medial malleolus, the distal femur, and the anterior superior iliac spine. There are various styles of intraosseous needles. They are all designed to easily penetrate the bone cortex, and they all have a stylet to keep bone core from obstructing the needle during insertion. Aseptic technique and universal precautions should be followed. The needle should be twisted into, rather than pushed through, the bone marrow. Evidence for successful entrance into the marrow includes (1) the lack of resistance (or a "give") after the needle passes through the cortex, (2) the ability of the needle to remain upright without support, (3) aspiration of the bone marrow into a syringe, and (4) free flow of the infusion without significant subcutaneous infiltration. Aspiration of bone marrow into the intraosseous needle is not always possible, especially in a very dehydrated patient. An alternative method of IO insertion involves using a battery-operated drill that allows for quick and effective insertion of the IO needle. Regardless of the insertion method used, infiltration of fluid into tissues around the bone is common when this route of infusion is used for a prolonged period of time or if the fluid is infused under great pressure. Infiltration will manifest as an enlarging leg circumference, fluid leaking out from the skin insertion site, or resistance to flow of the infusate.

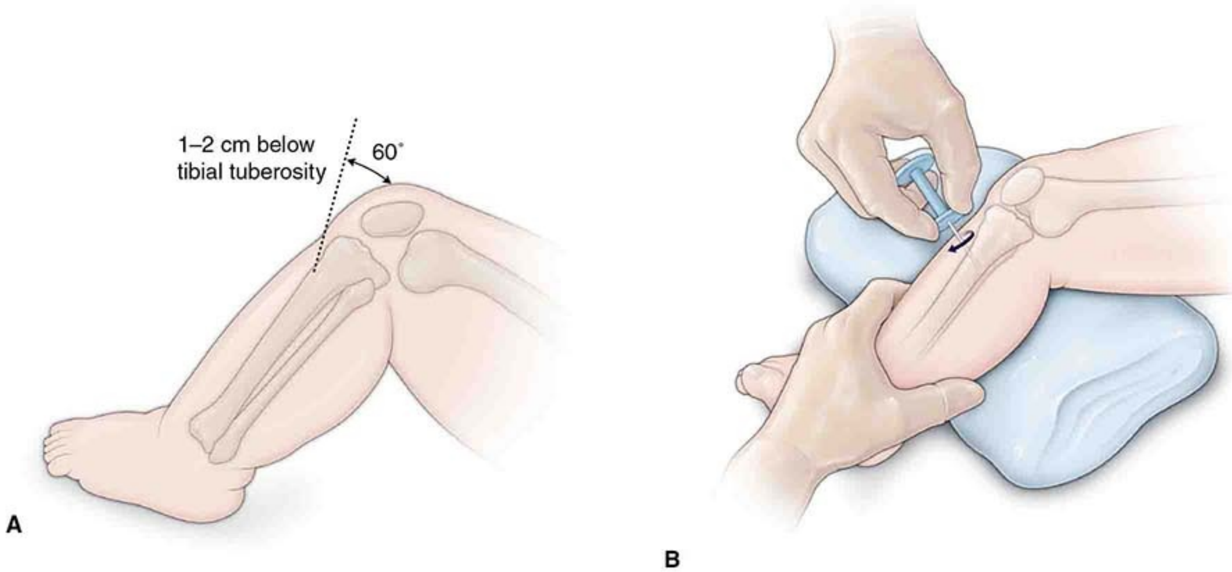

FIGURE 100-1 Intraosseous cannulation. **A:** Preferred site for intraosseous cannulation in infants. **B:** Technique for insertion of the trocar. Note that the trocar is advanced with a twisting movement and in a caudal direction to avoid injuring the tibial growth plate.

CENTRAL VENOUS CATHETERIZATION

Central venous catheterization provides more reliable and longer term vascular access than peripheral venous catheterization. In addition, central venous catheters permit hemodynamic monitoring and laboratory sampling of central venous blood. However, the convenience of central venous catheters must be balanced against the added risks.

Central venous cannulation is particularly suited for administering irritating or vasoconstrictive medications, which are diluted in the high blood flow of central veins, thereby limiting their contact with the vascular endothelium. Central venous cannulation may also offer a more expeditious alternative to peripheral venous access, particularly for patients who are in circulatory shock or cardiac arrest. Especially in these cases, a central venous catheter provides a means to monitor central venous pressure. In some circumstances, specialized central venous catheters may be used to monitor cardiac output, mixed venous oxygen saturation, pulmonary artery pressure, and pulmonary artery occlusion pressure.

In critically ill infants and children, polyurethane central venous catheters are generally inserted percutaneously through a large central vein such as the internal jugular, the subclavian, or the femoral vein. For chronic vascular access, particularly in patients requiring long-term chemotherapy, total parenteral nutrition, or antibiotics, Silastic catheters are often preferred, because they are less thrombogenic and have lower infection rates. It is now widely recommended that these catheters are placed using ultrasound guidance if this is available.

Peripherally inserted central catheters (PICC) have gained enormous popularity, because they can be inserted easily at the bedside, usually through a peripheral vein (eg, brachial vein), and advanced into a central location, using ultrasound and in some cases fluoroscopic guidance. The Broviac and Hickman catheters are placed in a central vein and tunneled out through a distant exit site in the skin. The Hickman catheters have Dacron cuffs that are very fibrogenic. The resultant fibrous scar around the Dacron allows for the catheter to be more securely anchored and decreases the risk of catheter infection from the skin. A last category of Silastic catheters is equipped with a proximal port that is embedded in the subcutaneous tissue (Port-A-Cath). These totally implantable venous access devices reduce the chance of microbial propagation through the catheter track and are most useful when only intermittent infusions are necessary.

There are unavoidable risks associated with all central venous catheters. During attempts to cannulate the internal jugular or subclavian veins, inadvertent needle puncture of the lungs can result in pneumothorax. Perforation of a large vessel with resultant leak into the pleural space can lead to hemothorax, and mechanical damage to the thoracic duct or thrombosis of the left brachiocephalic vein downstream from the insertion of the thoracic duct can lead to chylothorax. Central vein thrombosis and its attendant risks (venous congestion and edema, chylothorax) are particularly common in newborns and small infants because of the small size of their central vessels. In addition, injuries to nerves and arteries (including inadvertent cannulation) near the insertion site have been well documented.

When the central venous catheter is open to the atmosphere and the patient creates negative intravascular pressure with a spontaneous breath, it is possible for air embolism to occur. This can be minimized by keeping the needle or catheter exit site in a dependent position (eg, Trendelenburg position during internal jugular or subclavian catheterization). The Seldinger technique, or guidewire technique, is commonly used to simplify catheter placement. A needle or trocar is used to puncture the vessel, followed by passage of a wire through the trocar. The trocar is then removed, and the catheter is then introduced over the wire. This technique is convenient but is associated with both an increased risk of vessel perforation by the guidewire, and catheter malposition because of the potential for the guidewire to take an unexpected course in the vascular system.

The most significant risk of central venous catheterization is infection. The risks of both thrombosis and suppurative thrombophlebitis increases with time and is especially high in children who suffer from hypercoagulability and/or immune deficiencies. The possibility of a hypercoagulable state should be considered in critically ill children who develop catheter-related central venous thrombosis. Newer catheter materials (polyurethane, Silastic), heparin bonding, and low-dose heparin administration through the catheter may minimize the risks of thrombosis. Similarly, antibiotic bonding has been used to decrease the risk of infection. However, the most effective preventative measures are based on the conventional wisdom of placing central venous catheters only when indicated and removing them as soon as possible.

Local hemorrhage is common when removing the central venous catheters. Applying direct pressure as the catheter is removed can minimize the size of the hematoma. Air embolism is a rare, life-threatening complication associated with catheter removal. As during insertion, the risk of air embolism can be minimized by placing the patient in the Trendelenburg position, removing the catheter at the end of deep inspiration, and immediately covering the exit site with petroleum jelly gauze.

The most commonly used sites for central venous catheterization (**Fig. 100-2**) in acutely ill children are the internal jugular (less frequently, the external jugular vein is cannulated) or femoral veins. The subclavian veins are also often used by experienced clinicians, although the risk of pneumothorax or hemothorax is higher than with the other techniques. The femoral vein is the safest approach for less-experienced providers and, unlike in adults, does not seem to carry a higher risk of infection. Femoral cannulation offers clear anatomical landmarks, ease of hemostasis, and a safe distance from vital organs and sites of activity during resuscitation. Prior to catheterization of the neck veins, the child should be placed in the Trendelenburg. Conversely, if the femoral vein is used, a slight reverse Trendelenburg position will engorge the veins and improve the success rate.

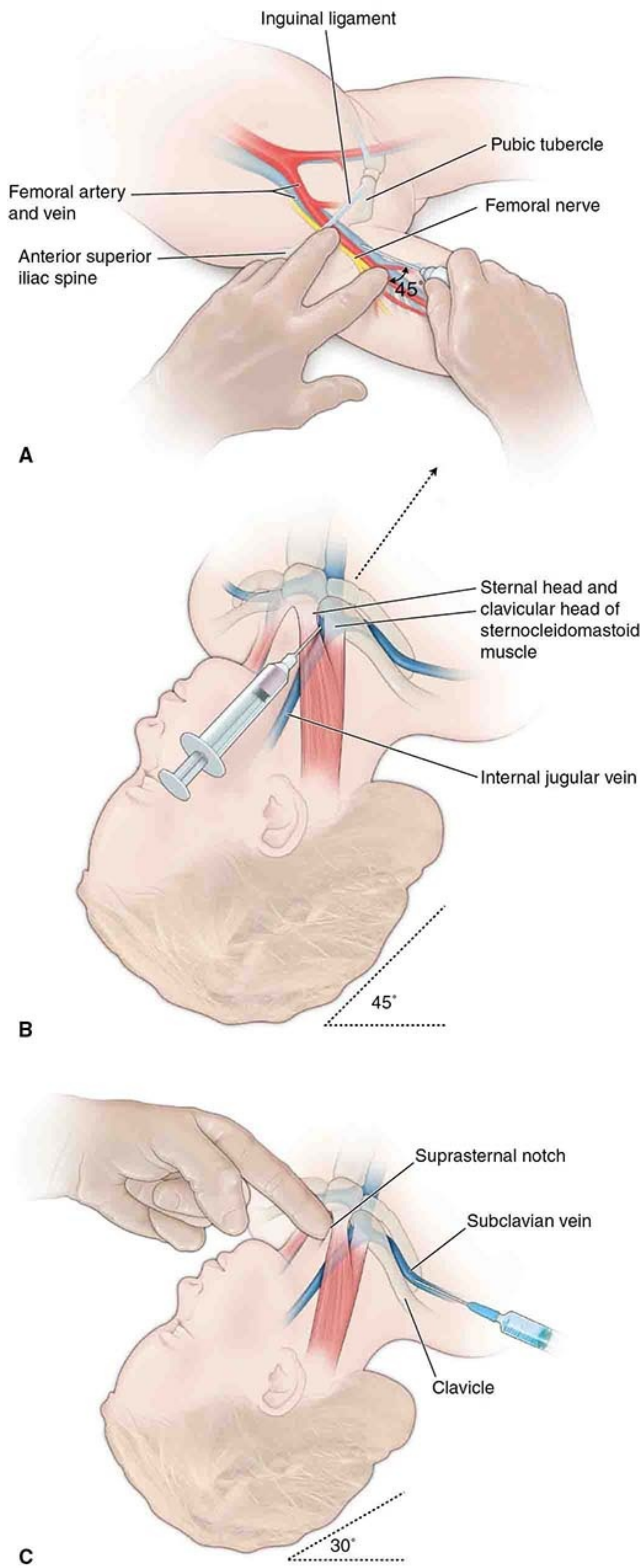
Inguinal ligament
Pubic tubercle
Femoral artery and vein
Femoral nerve
Anterior superior iliac spine
45°
A
Sternal head and clavicular head of sternocleidomastoid muscle
Internal jugular vein
45°
B
Suprasternal notch
Subclavian vein
Clavicle
30°
C

FIGURE 100-2 Three commonly used sites for venous cannulation in children. **A:** Femoral vein. **B:** Internal jugular vein (the middle approach is shown here; the vein can be accessed via other alternative approaches). **C:** Subclavian vein.

Ultrasound-guided central venous access is now the preferred approach to clarify anatomy of nearby arteries and veins, identify the vein for cannulation, determine that the vein is patent, assure that the insertion needle tip is in the vessel prior to advancing a wire through the needle, and decrease the risk of complications, such as inadvertent arterial cannulation. The technique requires additional training and equipment but increases safety and effectiveness. The overall procedure is otherwise similar to the traditional approach utilizing landmark identification.

Local anesthesia with 1% lidocaine should be provided subcutaneously unless the patient is anesthetized. A syringe and needle are used to locate the vein and aspirate blood. For femoral vein catheterization, the vein is located immediately medial to the artery, 1 to 2 fingers' breadths below the inguinal ligament. The introducer needle is inserted into the vein at a 45° angle. When blood is aspirated freely, the syringe is removed and the hub of the needle is covered with the thumb of the nondominant hand. A guidewire is then placed through the needle, and the needle is removed (Seldinger technique). A dilator is advanced over the wire to dilate the subcutaneous space and the entrance into the vein. The dilator is then removed, and the catheter is placed over the guidewire. After the catheter is introduced into the vein, the guidewire is removed. The electrocardiogram should be monitored during the procedure, because the guidewire can trigger arrhythmias when advanced into the heart.

Once the procedure is complete, it is essential to aspirate all of the lumens of the line to check that blood flows freely, and then check the position of the central line using 1 or more of the following confirmatory tests (depending on availability): x-ray to confirm the position of the line tip; a blood gas to distinguish from an inadvertent arterial cannulation (although this can be confounded by the presence of arterial hypoxemia); and observation of the pressure waveform of the cannulated vessel after zeroing the line (again, to rule out arterial cannulation). Continuous monitoring of the pressure during the procedure is essential in the setting of pulmonary artery catheterization.

Central line-associated bloodstream infections (CLABSI) are key quality indicators for not only intensive care units, but for hospitals as a whole and are associated with increased morbidity and increased mortality across multiple populations. Many institutions have central line checklists, or "bundles" that incorporate key interventions to minimize the risk of any adverse events related to central line insertion, in particular CLABSI. These include proper hand hygiene; preparing the skin with chlorhexidine; using a full-body sterile barrier, surgical masks, gowns, caps, and gloves for those performing the procedure; and removing unnecessary catheters. The introduction of central line bundles has been associated with a marked decrease in the rate of CLABSI.

Transduction of vascular pressure during the procedure provides a more effective means of confirmation and is essential if the catheter is to be inserted into the pulmonary artery. A radiograph can also be used to confirm position, but sometimes a single projection is insufficient to determine the precise distal site of the catheter (eg, distinguishing right atrium from right ventricle).

ANALGESIA AND SEDATION FOR PROCEDURES

An important yet frequently neglected issue for any invasive procedure on a child is that of pain management and sedation. In the nonemergency situation, topical or local anesthetics should be used, and oral sucrose has been shown to lessen distress in younger infants. The psychological milieu during the procedure should also be optimized. Music, toys, and interaction with family members can be important tools for improving comfort, thereby increasing the likelihood of cooperation and technical success. Moreover, minimizing discomfort during the procedure generally encourages better cooperation during the next potentially painful procedure for that child. However, when sedation is provided, personnel qualified to handle sedation-induced respiratory failure or shock should be present.

SUGGESTED READINGS

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101 Fluid Management of the Acutely Ill or Injured Child

Trevor Duke

INTRODUCTION

Children with acute or critical illness may be hypovolemic, euvolemic, or hypervolemic. They may have lost blood (eg, hemorrhagic shock), plasma (eg, severe burns), extracellular fluid (eg, gastroenteritis or diarrhea), or electrolytes. They may have internal compartment fluid redistribution (eg, ascites or capillary leak from sepsis or Dengue shock syndrome) and despite being edematous may still have intravascular volume depletion. Children with acute or critical illness may have renal impairment or high antidiuretic hormone (ADH) levels that results in fluid retention. Decisions about fluid management need to be based on clinical features and understanding of pathophysiology, thus systematic and repeated clinical evaluations are essential to good quality care. In this chapter, several acute clinical syndromes that require careful fluid management will be discussed.

SHOCK AND FLUID MANAGEMENT

Cells require oxygen for the production of adenosine triphosphate (ATP), the principal cellular energy source. In pathophysiological terms, shock occurs when cardiac output is insufficient to provide oxygen delivery to tissues for ATP production, leading to cellular dysoxia, with resulting anaerobic metabolism, lactic acidosis, and cellular dysfunction. The assumption behind fluid bolus therapy for shock is that it will improve cardiac output and in turn augment oxygen delivery to tissues. There are several clinical definitions of shock. Hypovolemic, low cardiac output shock manifests as “cold shock” with vasoconstriction, cold limbs, and tachycardia. This type of shock requires urgent fluid resuscitation. Cold shock occurs in severe dehydration (the classic examples being from cholera or gastroenteritis), hemorrhagic shock, and in some (generally late) stages of septic shock.

Fluid resuscitation for children with fever and signs of shock has been controversial. This is partly because many clinical definitions of shock encompass a continuum from adaptive physiological changes to fever, to states of severe hypotension and dysoxia. Fluid therapy for shock has also been controversial because fluid therapy alone will not deal with shock apart from that which occurs solely from extracellular fluid losses. Therefore, in settings where intensive care support is unavailable, fluid therapy alone in some forms of shock will be insufficient. Many children with fever and 1 or 2 clinical cardiovascular signs of shock do not have hypovolemia or dysoxia. They have high levels of adrenaline and renin-angiotensin, leading to tachycardia and vasoconstriction, and raised levels of ADH, which leads to fluid retention, thus protecting them from hypovolemia and shock. The assumption that fluid boluses will improve cardiac output and in turn augment oxygen delivery to tissues is not certain if the cardiac output is normal or the child is not hypovolemic. In this situation, the benefits of bolus fluid therapy are less and the costs of excess fluid will be tissue edema and perhaps a blunting of the adaptive cardiovascular responses. A transient increase in stroke volume after fluid bolus therapy, because of an increase in venous return and left ventricular end-diastolic (filling) pressure, is often accompanied by a reduction in heart rate. Thus, the effect on cardiac output, which is the product of the 2, in some circumstances will be small and transient. Large amounts of intravenous (IV) fluid will tend to accumulate in children with high ADH levels, and edema, reduced lung compliance, and other negative consequences may occur. The effect of fluid boluses on dampening adaptive responses like the hyperadrenergic state has been proposed but needs further study.

Revised guidelines for the management of shock, incorporating the World Health Organization’s new fluid guidelines for shock, are in Appendix 101-1.

APPENDIX 101-1 GUIDELINES FOR THE FLUID MANAGEMENT OF CHILDREN WITH SIGNS OF IMPAIRED

CIRCULATION OR SHOCK

Children Who Have Some Signs of Circulatory Impairment but Do Not Have Shock

Children with only 1 or 2 signs of impaired circulation, either cold extremities or a weak and fast pulse or capillary refill > 3 seconds, but who do not have the full clinical features of shock (ie, all 3 signs present together), should not receive any rapid infusions of fluids but should still receive maintenance fluids appropriate for age, weight, and disease process.

In the absence of shock, rapid IV infusions of fluids may be particularly harmful to children who have severe febrile illnesses, severe pneumonia, severe malaria, meningitis, severe acute malnutrition, severe anemia, congestive heart failure with pulmonary edema, congenital heart disease, or renal failure.

Children with any sign of impaired circulation (ie, cold extremities or weak and fast pulse or prolonged capillary refill), should be prioritized for full assessment and other treatments (treatment of the underlying cause and other possible treatments below), and reassessed within 1 hour.

Children Who Have Shock

Children who have shock (ie, who have all of the following signs: cold extremities and a weak and fast pulse and capillary refill > 3 seconds) should receive IV fluids and consideration for other treatment as follows:

- Give high-flow oxygen
- Give 10–20 mL/kg of isotonic crystalloid fluids over 30–60 min
- If severe anemia (severe palmar or conjunctival pallor), give blood as soon as possible and do not give other boluses of IV fluid
- Check blood glucose and correct if low
- Monitor the effect of fluid; fully assess to look for an underlying cause of shock

If the child is still in shock after initial fluid therapy, then consider a further infusion of 10 mL/kg over 30 min, and at the same time assess the need for other emergency treatments.

- Inotrope (adrenaline, dopamine) or vasoactive agent (noradrenaline) if hypotensive or persistent shock
- Antibiotics for bacterial sepsis +/– antimalarial if in malaria endemic area
- Diuretic (furosemide 1 mg/kg IV) if signs of fluid overload or heart failure
- Positive pressure respiratory support (such as continuous positive airway pressure) if hypoxemia or severe respiratory distress despite oxygen
- Hydrocortisone if hypotensive despite inotrope or vasopressor
- Adrenaline IV or intramuscular if any sign of anaphylaxis
- Antivenom if signs of snake bite
- Thiamine if beriberi is suspected, or if the child has severe malnutrition
- Echocardiogram to assess heart function and exclude tamponade
- Tranexamic acid and urgent blood transfusion if there is traumatic hemorrhage
- Surgical review if abdominal emergency or trauma

Reassess airway, breathing, circulation; monitor for the effects of emergency treatments; and further review the underlying cause of shock by history and examination.

If shock has resolved, then provide fluids to maintain normal hydration status only (maintenance fluids).

Decide on the best location of ongoing care and monitoring: intensive or high dependency area, pediatric unit. Order frequency of monitoring, reassessment, and other supportive care.

SEVERE DEHYDRATION AND FLUID MANAGEMENT

Children with hypovolemic shock from severe dehydration should be given parenteral fluid immediately: administer 20 mL/kg of 0.9% sodium chloride (NaCl) or Ringer's lactate (or similar fluid) repeatedly until plasma volume is restored, and pulses are present (20–100 mL/kg may be required). Fluid can be given intravenously, or if an IV cannula cannot be inserted, into the bone marrow through an intraosseous needle. Nasogastric fluid is effective in severe dehydration, but once shock has developed, the splanchnic circulation is less able to absorb fluid. Intraperitoneal fluid is not effective in shock. Any child with shock who requires more than 40 mL per kg as a bolus should be urgently reviewed to consider the need for vasopressor or inotropic support, and to consider the adverse effects of more fluid on lung function, and where available intensive care review should be considered.

In calculating fluid requirements for children with dehydration, 3 fluid types need to be considered:

- Existing fluid deficit
- Ongoing fluid losses
- Maintenance fluid requirements

Fluid Deficit

The fluid deficit is most reliably estimated from the loss of body weight if a recent pre-illness weight is available. The loss of weight is equal to the fluid deficit. (For example, if 600 grams have been lost, there is a fluid deficit of 600 mL, which represents 6% dehydration in a 10-kg child). The deficit may be overestimated in infants whose most recent weight was months ago, as the weight difference does not take account of normal growth, which may be 100 g to 150 g per week.

Clinical signs may be used to determine approximate fluid deficit. These are useful but not precise.

1. Mild dehydration (< 4%)
   - Usually no clinical signs; the child is thirsty

2. Moderate dehydration (4–6%)
   - Delayed capillary refill time
   - Increased respiratory rate
   - Mildly decreased tissue turgor

3. Severe dehydration (≥ 7%)
   - Very delayed capillary refill, > 3 seconds; cold, mottled skin
   - Other signs of shock (irritable or reduced consciousness level, hypotension, tachycardia)
   - Deep, acidotic breathing
   - Decreased tissue turgor

In children, hypotension is a late sign, but is often preceded by a narrowing of the pulse pressure from vasoconstriction. Other markers of mild to moderate dehydration include the following:

- A history of oliguria
- Lethargy

- Sunken eyes
- Dry mucous membranes
- Sunken fontanelle
- Absence of tears

In hypernatremic dehydration, the clinical signs of dehydration are less marked because the extracellular compartment is relatively preserved, and the degree of dehydration is underestimated. A child with hypernatremic dehydration may look only mildly unwell until they suddenly collapse with shock.

In severe dehydration, the existing deficit should be replaced over the first 12 to 24 hours, using the IV (or nasogastric) route. Faster deficit replacement has been shown to be safe in many children, but confers no advantage apart from earlier discharge from an emergency department. For replacement of deficit, an isotonic fluid should be used.

In severe dehydration electrolytes should be checked; hypokalemia is common in dehydration from diarrhea, but hyperkalemia can occur if there is prerenal failure resulting from hypovolemia, or severe metabolic acidosis. Hyper- and hyponatremia, hypoglycemia, and hypomagnesemia are commonly seen in children with severe dehydration.

Hypernatremic dehydration is most frequently caused by a water deficit (increased losses or reduced intake). In some cases, it is caused by an increase in total body sodium (such as ingestion of large quantities of sodium, incorrect preparation of milk formula, ingestion of hyperosmotic feeds in the setting of diarrhea). For hypernatremic dehydration, after the correction of shock if present, the aim is to lower the serum sodium slowly at a rate of no faster than 12 mmol/L in 24 hours (0.5 mmol/L per hour). This rate of normalization of serum sodium should be even slower if hypernatremia is chronic (weeks). If sodium is corrected too rapidly, cerebral edema, seizures, and permanent brain injury may occur. In moderate hypernatremic dehydration, Na^+ 150 to 169 mmol/L after initial resuscitation, replace the deficit plus maintenance over 48 hours. Nasogastric oral rehydration salts should be used whenever possible, remembering that these have a low sodium concentration (60–90 mmol/L). Monitor electrolytes every 4 to 6 hours. If requiring IV rehydration, then Ringer's lactate +5% dextrose, 0.9% NaCl + 5% dextrose, or PlasmaLyte with 5% dextrose should be given, adding maintenance potassium chloride once urine output has been established. In severe hypernatremic dehydration, $Na^+ > 169$ mmol/L, the child should be admitted to intensive care. After initial resuscitation from shock if present, the fluid deficit should be replaced using 0.9% NaCl + 5% dextrose over 72 to 96 hours. In severe hypernatremic dehydration, electrolytes and glucose should initially be checked hourly. If serum sodium is falling faster than 0.5 mmol/h, then the infusion rate should be slowed down by 20%, and the serum sodium should be rechecked after 1 hour. If, after 6 hours of rehydration therapy, the sodium is decreasing at a steady rate, then the electrolytes and glucose can be checked every 4 hours. If there are persistent neurological signs, cerebral imaging should be considered. Venous sinus thromboses can occur from severe hypernatremic dehydration.

Ongoing Losses

It is vitally important to document fluid balance carefully and, ideally, hourly in patients during active fluid resuscitation. Fluid balance charts should accurately record—whenever possible—output volumes, including urine, gastric aspirates, vomitus, diarrhea, drainage from fistulae, and other fluid losses. These will determine the volume and type of fluid and electrolyte replacement. In practice, it can be difficult to accurately measure all losses. A rule of thumb is to add 100 mL for every diarrheal stool in the previous 12 hours to make up the ongoing losses that are added to the total fluid intake for the next 12 hours.

Maintenance Fluid

For over 5 decades, the standard maintenance IV fluid for children in hospitals throughout the world was hypotonic saline and glucose. A commonly used solution was 4% dextrose and 0.18% NaCl (containing 30 mmol/L sodium and chloride), and another fluid used in some countries was 3.3% dextrose and 0.3% NaCl. The recommended flow rates of IV maintenance fluid followed the 100/50/20 rule: 100 mL/kg/day for the first 10 kg body weight, 50 mL/kg/day for the next 10 kg, and 20 mL/kg/day for each kg thereafter. This roughly equates with the 4/2/1 rule, which is in mL/kg per hour. This is approximately the amount of water that a well, active child who has a normal metabolism and normal renal function requires to excrete nitrogenous wastes in an isosmotic urine, that is, urine that is neither overly concentrated nor overly dilute. The composition of the recommended IV fluids was based on the calculation that, if a child receives "normal maintenance volumes" for weight using 4% dextrose and 0.18% NaCl, he or she would receive 2 to 3 mmol/kg/day sodium (the amount needed for metabolism and growth), and 3.5 mg/kg/min of glucose (enough to avoid hypoglycemia in a patient who is nil orally).

The problem lies in the assumptions underlying these recommendations. Sick children are not the same as well, active children. Hospitalized children with infectious processes, such as pneumonia, bronchiolitis, sepsis, or central nervous system infections, and those with other conditions including surgical emergencies and those who are postanesthesia, commonly have increased levels of ADH, which causes water retention. Depending on how sick they are, such children may also have impaired renal function, increased capillary permeability (including in the brain, lungs, and soft tissues), poor lung compliance, and hypoproteinemia. The retention of water by ADH is a protective physiological mechanism for fasting sick patients. However, the administration of "normal maintenance IV fluids" to a child who cannot excrete a free water load risks fluid overload and edema of tissues, lungs, brain, and other organs. The administration of an IV solution with a sodium content substantially lower than plasma in a child who has impaired ability to excrete free water leads to hyponatremia, and this can lead to brain cell edema. On an unknown number of occasions, this has resulted in seizures, cerebral edema, and death. While some risk factors for these complications are known, it is not easy to predict to which patients this will occur, and hyponatremia and cerebral edema can also occur in patients without known factors. Hyponatremia is not always preventable.

Probably even more frequent than neurological complications are those arising from excessive volumes of IV fluid relative to true requirements for fluid homeostasis. Tissue edema leads to poor lung compliance and worsening hypoxemia in children with respiratory infections, soft tissue edema, immobility and pressure areas in children in coma, and impaired kidney function. There are now many case series of iatrogenic hyponatremia and its consequences; patent safety alert warnings by national and coronial authorities from the UK, Northern Ireland, Canada, and elsewhere; and randomized trials showing a lower risk of hyponatremia in children who receive isotonic fluids.

In 2013, the World Health Organization (WHO) recommended that "children who require IV fluids for maintenance should be managed with Ringers lactate with 5% dextrose, or 0.9% normal saline with 5% dextrose or half-normal saline (0.45% NaCl) with 5% glucose."

Fluid management of critically ill children requires careful assessment of disease process, pathophysiology, hydration, and intravascular volume status and responses to administered fluid. Both standardized guidelines and an individualized approach are needed for optimal fluid management. Guidelines for prescribing maintenance fluid are detailed in Appendix 101-2, and the water and electrolyte contents of commonly used fluids are given in **Table 101-1**.

TABLE 101-1 WATER AND ELECTROLYTE CONTENT OF COMMONLY USED IV FLUIDS

Fluid Type	Na^+ mmol/L	K^+ mmol/L	Cl^- mmol/L	Other Anions mmol/L	Osmolality	Electrolyte-Free Water/L	Comments
0.9% NaCl	154	0	154		308	0	
0.45% NaCl	77	0	77		154	500	
0.9% NaCl 5% dex	154	0	154		560	0	
5% dex 0.45% NaCl	77	0	77		406	500	
5% dex 0.2% NaCl	34	0	34		321	780	**Not safe to be used as maintenance fluid**
4% dex 0.18% NaCl	31	0	31		284	800	
5% dex	0	0	0		252	1000	
Ringers lactate	130	4	109	lactate 28	272	130	**Suitable as IV maintenance fluids**
Hartmann's solution	131	5.4	112	lactate 28	274	100	
PlasmaLyte	140	5	98	acetate 27, gluconate 23	294	60	
3% saline	**513**	**0**	**513**		**1027**	**0**	

Cl, chloride; dex, dextrose; IV, intravenous; K, potassium; Na, sodium

APPENDIX 101-2 GUIDELINES FOR PRESCRIBING MAINTENANCE FLUID IN SICK CHILDREN

Does This Child Need IV Fluid?

Whenever possible, give enteral milk feeds to sick children, as this provides nutrition and reduces gastrointestinal complications and infections. If the child is too sick to take fluids orally, consider milk feeds via a nasogastric tube. Standard IV fluids provide no effective nutrition.

Maintenance Fluid Composition

For maintenance fluid in children older than 1 month of age, use isotonic fluids:

- Hartmann's solution with 5% dextrose (or Ringer's lactate with 5% glucose)
- PlasmaLyte with 5% dextrose where this is approved and available
- 0.45% NaCl with 5% dextrose
- 0.9% NaCl with 5% dextrose[a]

Do *not* use 0.18% NaCl, or 0.3% NaCl

Fluid Volumes and Rates

Calculate the total fluid intake (TFI) the child requires (oral, IV) for the day.

Full maintenance IV fluid:

- 100 mL/kg/day for first 10 kg of child's weight
- 50 mL/kg/day for each kg of child's weight between 10 and 20 kg
- 20 mL/kg/day for every kg of child's weight > 20 kg
- max: 2500 mL/day

This concept of full maintenance, based on the 100/50/20 (or 4/2/1 per hour) rule relates to the *maximum* volumes that are usually required for children receiving IV fluids. *For most sick children* less *IV fluid is required to maintain normal volume status*. It is generally best to start with no more than two-thirds of the calculated full maintenance IV fluid.

For children with central nervous system infections (meningitis, encephalitis, or coma) or acute lower respiratory infection (pneumonia or bronchiolitis), even less IV fluid may be required to maintain normal volume status. Correct dehydration if it is present, then continue with reduced TFI at approximately 50% to 70% maintenance and monitor closely.

Infants on full enteral milk feeds often need more than this for optimal growth: up to 150 mL/kg/day.

As enteral/oral feeds increase, reduce IV fluids. If enteral feeds cease, increase IV fluids.

Monitoring

1. **Monitor the TFI.** Monitor the fluid the child actually receives, not just the volumes that are ordered or prescribed. Experience shows that the TFI is often in excess of the volumes that are prescribed.
2. **Children receiving IV fluids should be weighed every day.** An increase in weight of 5% or more over 24 hours indicates fluid overload. The IV fluids should be stopped and the serum sodium measured. A decrease in weight of 5% or more indicates dehydration. Weigh on the same scales in similar lightweight clothing for more accurate monitoring.
3. **Check for edema every day.** Check for puffy eyes and lower limb swelling. If either if these is present, stop IV fluids and reassess.
4. **Every child receiving 50% or more of maintenance volumes in IV fluids should have the serum electrolytes checked daily.** If the serum sodium is below 130 mmol/L, or has fallen by more than 5 mmol/L since admission, reassess the type and volume of IV fluid given. If the serum sodium is 150 mmol/L or above, or has risen by 5 mmol/L from the time of admission, review the possible causes (eg, dehydration, sodium excess).
5. **Measure blood glucose every 6 to 12 hours in infants under 6 months of age on IV fluid.** If the blood glucose is less than 3 mmol/L (54 mg/dL), change the IV fluid from that containing 5% to one containing 10% glucose, or feed the child.

[a]Normal saline (0.9% NaCl) is not ideal maintenance fluid for children, as it contains high chloride levels (150 mmol/L), which can lead to metabolic acidosis when large volumes are administered. Also, it may lead to sodium accumulation in children with malnutrition. Ringer's lactate, Hartmann's solution, and PlasmaLyte have lower chloride (110 mmol/L) and sodium (130–140 mmol/L) levels, and therefore are not likely to lead to hyperchloremic metabolic acidosis.

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102 Cardiopulmonary Resuscitation

Mary Fran Hazinski

INTRODUCTION

Successful closed-chest cardiopulmonary resuscitation (CPR), the fundamental life-saving skill, was first reported in 1960, and the American Heart Association (AHA) has published pediatric and neonatal resuscitation guidelines since 1980. Although CPR has been widely taught to healthcare providers (HCPs) and the public, it is only recently that survival from cardiac arrest has improved. This improvement has occurred in association with an emphasis on teaching, monitoring, and improving CPR quality; widespread implementation of lay-rescuer CPR and automated external defibrillation (AED) programs; increased frequency of dispatcher-guided lay-rescuer CPR; and improvements in postcardiac arrest care, including targeted temperature management. Published studies from both out-of-hospital and in-hospital registries have provided additional information about the epidemiology, presentation, and outcome of pediatric cardiopulmonary arrest (CPA) at different ages and in different settings. The AHA Guidelines for Pediatric Basic Life Support (PBLS) and Pediatric Advanced Life Support (PALS) are now based on a continuous, structured international evidence-based review process. The reviews, sponsored by the AHA and the International Liaison Committee on Resuscitation (ILCOR), are posted on ILCOR's Web site (http://www.ilcor.org./home/), in the Systematic Evidence Evaluation and Review System.

Neonatal resuscitation guidelines are jointly created by the AHA and the American Academy of Pediatrics. These guidelines target resuscitation in the delivery room, emphasizing establishment of effective ventilation, because inadequate airway and ventilation are the most common problems requiring newborn resuscitation. Neonatal resuscitation guidelines are typically applied during the initial hospitalization of the newborn. The AHA infant Basic Life Support (BLS) and CPR guidelines typically apply to the infant after the initial hospitalization until 1 year of age. If a newborn remains hospitalized for treatment of heart disease, it is appropriate to apply infant CPR guidelines to provide a higher compression-to-ventilation ratio consistent with focus on establishment of adequate blood flow as well as oxygenation and ventilation. The AHA PBLS and CPR guidelines apply to children from 1 year of age to puberty. Adult BLS guidelines apply to adolescents. For ease of teaching and practice, providers in a unit such as a pediatric cardiovascular intensive care unit (ICU) may elect to apply the infant and child CPR guidelines to all patients in the unit.

In 2010, the AHA recommended a change in the initial sequence of CPR from ABC (airway, breathing, circulation/compressions) to CAB (circulation/compressions, airway, breathing). This change was made because the initial steps of opening the airway and delivering breaths are relatively complicated and often created long delays to initiating chest compressions. Although support of airway, oxygenation, and ventilation is especially important during pediatric cardiac arrest, this change in sequence should delay ventilation only by a few seconds, and is designed to shorten the overall time to initiation of CPR for all victims of cardiac arrest.

Every step in resuscitation is important, including treating prearrest conditions when possible; immediately identifying the arrest itself and providing high-quality CPR; and seamlessly integrating shock delivery with CPR when needed. Skilled, evidence- and protocol-supported postcardiac arrest care, including targeted temperature management is critical.

PATHOGENESIS AND ETIOLOGY OF CARDIAC ARREST

The causes and types of cardiac arrest during infancy and childhood vary by age and arrest location and will affect the priorities of resuscitation. In the past, much of the evidence used to characterize pediatric cardiac arrest was based on extrapolation from adult series; however, in recent years, several pediatric cardiac arrest registries have provided a wealth of information about many characteristics of pediatric CPA.

Data from Registries

Multiple large pediatric out-of-hospital cardiac arrest series have been published from registries such as the Cardiac Arrest Registry to Enhance Survival (CARES; https://mycares.net/), the Resuscitation Outcomes Consortium (ROC; https://roc.uwctc.org) registry, and the all-Japan Utstein Registry. In the United States, the Get with the Guidelines-Resuscitation (GWTG-R) registry, formerly the National Registry of Cardiopulmonary Resuscitation (http://www.heart.org/HEARTORG/Professional/GetWithTheGuidelines-Resuscitation/Get-With-The-Guidelines-Resuscitation_UCM_314496_SubHomePage.jsp) has published multiple pediatric series. These registries collect data based on the reporting templates known as the Utstein criteria. Publications from registries such as these contribute substantially to our knowledge of pediatric prearrest, arrest and postarrest conditions, further informing the emphasis of the AHA PBLS and PALS Guidelines.

Asphyxial Versus Sudden Cardiac Arrest

There are 2 major types of CPA: those that occur secondary to progression of respiratory failure or shock (so-called *asphyxial* or *hypoxic* arrest) and those that occur as an abrupt cessation of cardiac function, called *sudden cardiac arrest* (SCA). These types of CPA can be further identified by the terminal cardiac rhythm. Bradycardia and pulseless electrical activity (PEA) are the most common terminal rhythms associated with asphyxial arrest; these rhythms are called “nonshockable” because shock delivery is not indicated to treat the arrest. Pulseless ventricular tachycardia (pVT) and ventricular fibrillation (VF) are the terminal rhythms associated with SCA; these rhythms are known as “shockable” rhythms because shock delivery is required. The incidence of each type of CPA varies with the age and condition of the child, and with the location and circumstances of the arrest. Resuscitation priorities vary with arrest type, the number of rescuers, and the location of the arrest.

At birth, approximately 5% to 10% of newborns require some resuscitation, ranging from simple stimulation or clearing of the airway to support of ventilation and occasionally to chest compressions. Most neonates in distress demonstrate bradycardia, and most will respond to the establishment of effective ventilation.

Asphyxial arrest is the most common type of CPA in infants and children. However, this type of arrest can occur in patients of any age with conditions such as respiratory failure, trauma, drowning, drug overdose, poisonings, metabolic disorders, and shock. In the United States, the most common causes of infant death include congenital malformations, disorders related to prematurity, sudden unexpected infant death (including sudden infant death syndrome [SIDS], suffocation, and strangulation) and injury. Injuries are the most common causes of childhood and adolescent death. See also Chapters 113 and 116.

The infant or child often develops bradycardia as a preterminal event after asphyxia arrest. Bradycardia in children may also result from vagal stimulation (eg, induced by suctioning or gagging), from other causes of hypoxia, or from heart block. In the minutes before an asphyxial or bradycardic arrest, oxygen delivery to the tissues is compromised by low arterial oxygen content, by inadequate blood flow, or both, and major organ ischemia is likely to be present even before the arrest occurs. Resuscitation requires conventional CPR because support of effective oxygenation and ventilation, and compressions are needed to support oxygen delivery.

The infant or child with SCA has pVT/VF that causes the abrupt cessation of cardiac function. This type of cardiac arrest is most common in adults but is less common in children, particularly before adolescence. SCA has been reported in young athletes following a blow to the chest (commotio cordis) and in infants and children with genetic anomalies of impulse formation and conduction (eg, channelopathies such as prolonged QT syndrome), congenital heart disease, coronary artery abnormalities, cardiac inflammation, cardiomyopathy, and drug toxicity. Because arterial oxygen content is normal at the time of SCA, immediate support of blood flow with chest compressions may maintain oxygen delivery for the first few minutes of the arrest. However, beyond the first minutes of arrest, arterial oxygen content declines rapidly, so conventional CPR is needed.

Characteristics of Out-of-Hospital Arrest

Estimates of the incidence of asphyxial arrest versus SCA are based on extrapolation from mortality data collected by the Centers for Disease Control and Prevention and from published case series and registry data. In general, two-thirds of out-of-hospital pediatric cardiac arrests are asphyxial, about two-thirds of victims are infants who arrest at home, most are unwitnessed, and about half of the victims receive bystander CPR. These characteristics provide important information for exploring causes and prevention of CPR, targeting CPR education to parents and caregivers, and establishing resuscitation priorities for dispatcher-guided CPR.

Cardiac arrest in schools, particularly among student athletes, has stimulated grassroots efforts to establish CPR and AED programs in schools. Extrapolation from voluntary US databases suggests that approximately 1 per 100,000 high school athletes suffers from SCA annually, with the incidence higher in males and lower in females, and lower among nonathletes. In the United States, more than half of the SCA deaths in athletes are attributed to hypertrophic cardiomyopathy, commotio cordis (a blow to the chest that triggers ventricular tachycardia or fibrillation), or coronary artery anomalies. Some athletes or children with arrhythmias such as those caused by channelopathies may have syncopal episodes prior to the cardiac arrest event. Vigorous exercise can trigger lethal ventricular arrhythmias; many episodes of SCA in athletes occur during sporting events and are witnessed.

Characteristics of In-Hospital Arrest

Publications from the nearly 10,000 pediatric arrests in the GWTG-R registry have enabled the characterization of in-hospital arrest in children. Most children with in-hospital cardiac arrest have prearrest respiratory insufficiency, shock, or congestive heart failure. The most common intermediate causes of arrest are hypotension, acute respiratory insufficiency, and arrhythmias. PEA or asystole is the terminal rhythm in two-thirds of the arrests. Of the children with initial pVT/VF, approximately half have a cardiac illness or surgical condition, and prearrest vasoactive infusions have been identified as proarrhythmic factors. Although pVT/VF is not a common initial arrest rhythm, VF is present at some time during resuscitation in approximately 25% of pediatric in-hospital arrests, so providers must be facile at integrating shock delivery with high-quality CPR.

RESPIRATORY AND CARDIAC ARREST: CLINICAL MANIFESTATIONS

Respiratory Arrest

Respiratory arrest is defined as the cessation of spontaneous respiratory activity in the presence of spontaneous circulation (palpable central pulses). The development of hypoxemia and tissue hypoxia will produce bradycardia and poor perfusion. Detection and immediate treatment of respiratory arrest and symptomatic bradycardia often prevent progression to CPA.

Cardiopulmonary Arrest

Cardiopulmonary arrest is defined as the cessation of effective cardiac mechanical function and systemic perfusion. The victim is unconscious/unresponsive, with apnea or agonal gasps and with no palpable central pulses. Agonal gasps may be mistaken for effective spontaneous breathing, delaying the identification of cardiac arrest. They are much more common in the first minutes of SCA but are uncommon in children with asphyxial cardiac arrest. Agonal gasps can increase pulmonary gas exchange if the airway is patent. Gasps can also enhance venous return and create some coronary and carotid perfusion during the first minutes of arrest.

Once cardiac arrest develops, oxygen delivery ceases. Unless CPR is provided and spontaneous rhythm and perfusion are restored, the myocardium, brain, and all tissues will become progressively ischemic, and the likelihood of successful resuscitation diminishes with every passing minute. Generation of lactic acid, increased cell membrane permeability, production of free oxygen radicals, and activation of inflammatory mediators contribute to progressive organ and tissue destruction.

Pulseless Electrical Activity and Asystole

Pulseless electrical activity is a term used to describe cardiac electrical activity that fails to produce sufficient mechanical function to generate a palpable central pulse. Although this term could encompass agonal (eg, bradyasystolic) rhythms, it is typically reserved for patients with narrow QRS complexes that fail to produce a palpable pulse. Reversible causes of PEA in children include tension pneumothorax, cardiac tamponade, hypovolemia, and, rarely, pulmonary embolus. If PEA is not promptly treated, the rhythm will ultimately deteriorate to asystole.

Asystole is the ultimate terminal rhythm, characterized by electrical silence. Asystole is likely to be present for any unwitnessed or prolonged arrest, and it is the most common rhythm reported in pediatric prehospital and in-hospital arrest. Although overall survival with asystole or PEA is poor, survival is higher among children than adults who present with these rhythms in in-hospital arrests.

Ventricular Tachycardia and Fibrillation

Just as in adults, children in cardiac arrest who present with pVT/VF in the out-of-hospital or in-hospital setting typically have higher survival rates than those presenting with PEA or asystole. Untreated pVT will rapidly progress to VF, so treatment of these rhythms is identical and the rhythms are considered together in treatment algorithms.

During the first few minutes of VF, the amplitude of the VF will be high (so-called good VF), indicating that the myocardium initially has adequate oxygen and substrates. High-quality CPR can help maintain myocardial oxygenation and the amplitude and duration of VF (before it deteriorates to asystole). Rapid shock delivery is likely to result in elimination of the VF, enabling return of spontaneous cardiac rhythm and return of spontaneous circulation (ROSC). It is important to note that typically there is a delay between elimination of VF and ROSC, so CPR is needed until the shock can be delivered and after shock delivery until ROSC occurs.

If the VF remains untreated (ie, no CPR or shock delivery) for several minutes after arrest, the VF amplitude will gradually decrease (so-called bad VF), indicating severe myocardial ischemia. At this point, shock delivery is less likely to eliminate the VF; multiple shocks may be required. Even if VF is eliminated, return of spontaneous rhythm and return of spontaneous cardiac rhythm are less likely.

Untreated VF will ultimately progress to asystole. Once asystole is present, shock delivery will not be effective. High-quality CPR may result in the reappearance of VF (so-called secondary VF) that may respond to shock delivery, although the survival of patients who present with secondary VF is much lower than that of patients who present with VF as their initial rhythm.

RESPIRATORY AND CARDIAC ARREST: MANAGEMENT

Respiratory Arrest and Bradycardia

The child with respiratory failure, shock, or respiratory arrest typically develops bradycardia with poor perfusion before developing pulseless arrest. If rescuers detect and treat respiratory arrest and bradycardia before the development of pulseless arrest, survival is typically 75% or higher in both out-of-hospital and in-hospital settings.

Symptomatic bradycardia (ie, bradycardia with signs of poor perfusion) and respiratory arrest are both treated with support of airway, oxygenation, and ventilation through provision of bag-mask ventilation with oxygen. If the heart rate and perfusion improve but ventilation remains inadequate, mechanical ventilation is indicated. If the child's heart rate remains at or below 60 beats per minute with signs of poor perfusion despite support of adequate oxygenation and ventilation, chest compressions are added to the bag-mask ventilation (ie, CPR is provided).

When symptomatic bradycardia persists despite bag-mask ventilation and compressions, epinephrine is the drug of choice. Rescuers should consider using atropine for symptomatic bradycardia with increased vagal tone or when primary heart block is present. Cardiac pacing may also be needed.

Opiate overdose has emerged as a common cause of death in young adults. If the lay first responder or HCP encounters a victim of respiratory arrest with suspected drug overdose, naloxone administration is added to interventions such as bag-mask ventilation and CPR as indicated.

Cardiopulmonary Arrest

Once *cardiopulmonary arrest* is present, immediate high-quality CPR is needed because CPR maintains a small but critical amount of blood flow (estimated at 10% to 33% of normal) and oxygen and substrate delivery to the heart, brain, and other organs. Performing CPR can prolong VF and can maintain or increase VF amplitude. This increases the window of opportunity for shock delivery and increases the likelihood that the shock delivery will be followed by return of spontaneous cardiac rhythm and ROSC.

Cardiopulmonary Resuscitation Sequence

The appropriate sequence for pediatric resuscitation is determined by the type and location of the arrest and the type of rescuers and equipment available. A lone HCP must choose a sequence of actions, while multiple rescuers are able to accomplish several resuscitation steps simultaneously.

For the out-of-hospital arrest of a child or likely victim of asphyxial arrest, CPR focuses on initiation of high-quality conventional CPR. The lone HCP (with no nearby help or mobile phone) delivers about 5 cycles or about 2 minutes of CPR before leaving the victim to activate the emergency response system and retrieve an AED. The rescuer then returns to the victim to resume CPR and use the AED. If multiple bystanders are present during the out-of-hospital arrest, 1 rescuer remains with the victim to provide high-quality CPR while others activate the emergency response system and retrieve an AED.

During an in-hospital arrest, multiple rescuers are generally present to accomplish many tasks at the same time. The first HCP to identify cardiac arrest calls for nearby help and initiates resuscitation. As additional rescuers arrive with equipment, they will perform tasks such as providing bag-mask ventilation, attaching the adhesive defibrillator electrode pads and operating the defibrillator, establishing vascular access, serving as an alternate compressor, recording the resuscitation events, and functioning as team leader.

High-Quality Cardiopulmonary Resuscitation Technique

The elements of *high-quality CPR* are compressions of adequate rate and depth, complete chest recoil after each compression, minimal number and duration of any interruptions in chest compressions, and avoidance of excessive ventilation.

During CPR, the child should be placed supine on a firm, flat surface. A backboard is generally used as this surface.

Chest Compressions of Adequate Rate and Depth *Chest compressions* create blood flow by increasing intrathoracic pressure and by directly compressing the heart. Myocardial blood flow is determined by the coronary perfusion pressure present between compressions. Coronary perfusion pressure is calculated as the difference between aortic end-diastolic pressure (during CPR, this is the aortic relaxation pressure) and the mean right atrial pressure; the mean CVP may be used as a surrogate for right atrial pressure. In adults with cardiac arrest, a coronary perfusion pressure greater than 15 mm Hg has been linked with ROSC. Although a similar threshold has not been established in children, if arterial and right atrial or central venous monitoring catheters are in place, rescuers should attempt to optimize coronary perfusion pressure by maximizing aortic relaxation pressure and minimizing right atrial and central venous pressure (see “Monitoring During Cardiopulmonary Resuscitation,” below).

During CPR, blood flow, mean aortic relaxation pressure, and mean coronary perfusion pressure are increased by increasing the rate and force of chest compressions, and these variables are reduced when compressions are of either inadequate or excessive rate or inadequate depth. If the compression rate is too slow, blood flow generated by the compressions is inadequate. If the chest compression rate is too rapid, the depth of compressions may be compromised and chest recoil between compressions may be incomplete. The AHA-recommended chest compression rate for infants, children, and adults is 100 to 120 per minute.

The sternum is depressed approximately one-third the depth of the chest in the newborn and at least one-third the depth of the chest (about 1.5 in, or 4 cm) in the infant. The lone rescuer uses 2 fingers to compress the newborn or infant sternum, just below the intermammary line. When 2 HCPs are delivering CPR to the newborn or infant, the compressor uses the 2-thumb-encircling hands chest compression technique. The compressor's 2 thumbs are placed together on the sternum to depress the sternum (the thumbs can be placed side by side or they may overlap), while the fingers of both hands spread to encircle the thorax. The 2-thumb-encircling hands technique generates better coronary perfusion pressure and enables creation of more accurate depth of compression, and it may produce higher systolic and relaxation pressures with less rescuer fatigue than the 2-finger compression method.

Chest compressions in the child 1 year of age to puberty may be accomplished with 1 or 2 hands, whichever allows compression of the lower third of the sternum to a depth of at least one-third the depth of the chest and about 2 inches (5 cm). For chest compressions in the adolescent (after puberty) the traditional adult 2-hand technique is used to compress over the lower third of the sternum to a depth of 2 to 2.4 inches (5–6 cm). While excessive compression depth in adults can cause injuries such as rib fractures, inadequate compression depth is the more common problem and clearly reduces blood flow, coronary perfusion pressure, ROSC, and even survival from cardiac arrest.

Many CPR feedback devices are commercially available to provide visual and auditory indication and documentation of rate and depth of chest compressions and other resuscitation variables. These devices may be used for training and in actual resuscitations (see "Continuous Quality Improvement," below).

Complete Chest Recoil after Each Compression If rescuers fail to allow the chest wall to recoil completely after each compression, venous return to the heart is decreased, elevating right atrial pressure and reducingcoronary perfusion pressure. In addition, it prevents venous return to the heart and reduces the blood flow generated by the next compression. Rescuers who become fatigued while providing compressions are more likely to lean on the chest, failing to allow complete chest wall recoil. For this reason, during CPR, rescuers should rotate compressors about every 2 minutes (or more often if the compressor becomes fatigued or compression quality declines), and should monitor for and reduce leaning.

Minimal Interruptions in Chest Compressions Aortic relaxation pressure and coronary perfusion pressure fall whenever chest compressions are interrupted (eg, when breaths are provided or when compressions are interrupted to check rhythm or deliver a shock). Frequent or prolonged interruptions in chest compressions reduce mean coronary perfusion pressure, ROSC, and survival. Interruptions should, therefore, be minimized during CPR and rescuers should monitor the frequency and duration of any interruptions and strive to reduce them. The chest-compression fraction (CCF) is the total CPR time spent delivering compressions. The CCF should be at least 60% of total CPR time and the resuscitation team should strive for a CCF goal exceeding 80% of total CPR time.

Avoid Excessive Ventilation It is important to avoid excessive ventilation during CPR. When cardiac arrest is present, pulmonary blood flow, carbon dioxide delivery to the lungs, and oxygen uptake from the lungs are approximately 10% to 33% of normal, so victims need only a fraction of normal minute ventilation to match ventilation to perfusion during CPR. Excessive ventilation is harmful because it will reduce coronary perfusion pressure and blood flow during CPR. It will create positive intrathoracic pressure, reduce venous return to the heart, and reduce coronary perfusion and systemic and cerebral blood flow and survival.

The optimal compression-to-ventilation ratio for pediatric resuscitation is unknown. For pediatric CPR, the AHA recommends a 30:2 compression-to-ventilation ratio for single rescuers and a 15:2 ratio for 2 or more rescuers. Once an advanced airway is placed, rescuers should no longer deliver cycles of compressions and breaths. Instead, the rescuer performing compressions should deliver continuous chest compressions, and the rescuer providing ventilation should deliver approximately 10 breaths per minute (1 breath every 6 seconds).

Continuous Chest Compressions versus Conventional CPR Since 2008, the AHA has recommended Hands-Only/compression-only CPR for treatment of adult sudden witnessed cardiac arrest, particularly if the bystander is untrained or unwilling to provide mouth-to-mouth ventilation. The purpose of this recommendation was to increase the portion of victims of out-of-hospital cardiac arrest who receive some bystander CPR before the arrival of EMS providers. Because most pediatric cardiac arrest is asphyxial in origin, the AHA continues to recommend conventional CPR for infants and children with out-of-hospital cardiac arrest. This recommendation has been supported by pediatric data from out-of-hospital registries, with conventional CPR associated with higher survival than compression-only CPR, particularly in infants and particularly when asphyxial arrest is present.

Advanced Cardiopulmonary Resuscitation Techniques

Open-chest CPR can produce nearly normal blood flow but is impractical for most resuscitation situations. This technique is often used in the immediate postoperative period for pediatric cardiovascular surgical patients, when the chest is left open or the sternotomy can be opened quickly.

Extracorporeal cardiopulmonary resuscitation (ECPR) is the use of extracorporeal membrane oxygenation (ECMO) during attempted resuscitation. Extracorporeal cardiopulmonary resuscitation has been successful for pediatric in-hospital resuscitation, particularly among children with heart disease (see Chapter 108). The timely implementation of ECMO may also be effective for preventing cardiac arrest among children with low cardiac output (eg, following cardiovascular surgery), septic shock, myocarditis, and cardiomyopathy.

Monitoring during Cardiopulmonary Resuscitation

Cardiopulmonary Resuscitation Feedback Devices Several elements of CPR quality may be monitored and recorded, with devices that provide visual and auditory signals. These feedback devices range from relatively simple accelerometers to more complicated feedback defibrillators. The thresholds provided on the defibrillator screens and in recordings are typically consistent with a minimum 2-inch (5-cm) depth of acceptable chest compressions. These feedback defibrillators can be used to monitor the ECG, shock delivery, depth and rate of compressions, frequency and volume of breaths, and frequency and duration of any interruptions in chest compressions. This information is critical for team debriefing following every arrest as well as to set goals and monitor progress for improving CPR quality and survival in any resuscitation system.

Exhaled or End-Tidal Carbon Dioxide Tension Under normal conditions, the carbon dioxide tension in exhaled gas should be approximately equal to the carbon dioxide tension in both alveolar gas and pulmonary capillary blood. In very low flow states, if minute ventilation is fixed, the end-tidal carbon dioxide (CO_2) pressure (P_{ETCO2}) will vary with cardiac output. When cardiac arrest develops, blood flow to the lungs stops, so the CO_2 tension in the pulmonary capillaries, alveoli, and exhaled gas approaches zero. When effective chest compressions generate adequate pulmonary blood flow, the CO_2 delivered to the lungs and the partial pressure of CO_2 in exhaled gas should rise. ROSC is associated with a sharp and sustained rise in the P_{ETCO2}.

In intubated adult victims of cardiac arrest, a P_{ETCO2} less than 10 mm Hg despite 20 minutes of resuscitation is associated with very low survival. Although a similar threshold has not been identified in children, if the P_{ETCO2} remains less than 10 mm Hg, providers should make additional efforts to optimize CPR quality. Waveform capnography should be used for these measurements rather than intermittent colorimetric exhaled CO_2 monitors.

Estimating Coronary Perfusion Pressure Most hospitalized children who develop cardiac arrest are in the ICU (rather than in the general care ward), and invasive monitoring catheters may already be in place. If an intra-arterial and a central venous or right atrial catheter are in place, the coronary perfusion pressure can be estimated by subtracting the right atrial or central venous pressure from the aortic relaxation pressure (it will appear as the aortic diastolic pressure). The CPR technique should be optimized to maximize the estimated coronary perfusion pressure.

Defibrillation Because AED algorithms have been shown to be accurate in interpreting pediatric rhythms, the AHA recommends AED use for children ages 1 to 8 years in cardiac arrest, particularly for out-of-hospital arrest. Ideally, rescuers should use an AED with pediatric pads and a pediatric dose attenuator to deliver a shock dose appropriate for a small victim. However, if an AED with pediatric pads and dose attenuator is not available, rescuers should use an AED with adult pads. The pads should be placed according to the illustrations on the pads. They should not touch or overlap.

When an infant requires defibrillation, the use of a manual defibrillator is preferred to the use of an AED, because the dose can be more closely adjusted. If a manual defibrillator is not available, an AED with pediatric pads and a pediatric dose attenuator should be used. If such an AED is not available, an adult AED (one that administers an adult dose through adult pads) should be used.

When both pediatric and adult pads are available with the AED, it is important to use pediatric pads *only* for victims less than 8 years of age. If the pediatric pads are used for an older child, an adolescent, or an adult, they will likely attenuate the shock dose to one that is too small to eliminate VF in the larger victim.

For in-hospital manual defibrillation, rescuers should use infant pads or paddles for attempting defibrillation in infants up to 1 year of age and should use larger pads or paddles for patients 1 year of age and older. Rescuers should use the pads recommended by the defibrillator manufacturer.

Most commercially available defibrillators now use a biphasic waveform, while older defibrillators used a monophasic waveform. There are very limited data available regarding optimal manual biphasic waveform shock dose for defibrillation of children. In pediatric data from the GWTG-R in-hospital registry, an initial dose of 1 to 3 J/kg was associated with higher ROSC than a dose of more than 3 to 5 J/kg. The AHA recommends an initial shock of 2 J/kg. If the 2 J/kg initial dose fails to eliminate VF, rescuers should use a 4 J/kg dose for the second shock. For subsequent shocks, a dose of 4 J/kg is reasonable and a higher dose (up to a maximum of 10 J/kg) may be considered.

More research regarding the optimal defibrillation dose is needed. In general, weight-based defibrillation dosing may not be optimal, because the dose or current delivered to the myocardium is influenced not only by paddle or pad size and energy dose but also by transthoracic impedance, and transthoracic impedance is not linearly related to body weight. In the future, current-based defibrillation would make more sense than energy-based defibrillation.

Organization of Pediatric Resuscitation

Resuscitation is organized around 2-minute periods of uninterrupted CPR. Once a defibrillator is attached, the rhythm is checked every 2 minutes and a shock is delivered if pVT/VF is present. When CPR is interrupted for shock delivery, rescuers should have the defibrillator charged and should deliver the shock within 10 seconds or less of the last chest compression, then should resume chest compressions immediately after shock delivery. Shock effectiveness (ie, likelihood of termination of VF and ultimate ROSC) decreases with every 10 seconds that elapse between the last compression and the shock delivery. When a shock eliminates VF, the most common rhythm for 30 to 60 seconds after shock delivery is asystole or PEA, so compressions are needed immediately after shock delivery.

Advanced Airway

During resuscitation, the potential benefits of inserting an advanced airway should be weighed against the detrimental effects of the intubation attempt itself. Bag-mask ventilation for short periods may be as effective as ventilation through an advanced airway, but this form of ventilation requires training and frequent retraining. Inserting an advanced airway will enable delivery of uninterrupted chest compressions and may reduce gastric insufflation (and its attendant risks of regurgitation and aspiration), but it does require training and experience and will require interruption of chest compressions.

During resuscitation, the most experienced provider should perform intubation, with careful preparation and coordination of rescuer activities to minimize interruptions in chest compressions. Once the advanced airway is inserted, rescuers should also confirm placement using clinical examination plus a device such as a quantitative waveform capnography. If waveform capnography is not available, a qualitative colorimetric exhaled carbon dioxide detector may be used after several positive pressure breaths have been given. If the child weighs more than 20 kg, rescuers may use esophageal detector devices to confirm tracheal tube placement. These devices use suction to determine the ease with which gas can be withdrawn from the tube; the esophagus collapses easily under suction, preventing gas withdrawal, but the trachea is rigid and allows easy withdrawal of gas. Rescuers should verify correct tube position when the patient is transported and, during postcardiac arrest care, whenever the intubated child suddenly deteriorates.

Drug Therapy

Although no drug has been shown to increase survival from pediatric cardiac arrest, the use of vasoconstrictors has been shown to increase blood pressure and coronary and cerebral blood flow and return of spontaneous cardiac rhythm in animals. Drugs with beta-adrenergic effects also increase spontaneous myocardial depolarization and contractility. The AHA recommends administering a standard dose of intravenous (IV) epinephrine (0.01 mg/kg) every 3 to 5 minutes during cardiac arrest.

During resuscitation, intravenous or intraosseous drug administration is preferable to endotracheal administration (see Chapter 100). Although lipid-soluble drugs can be administered by the endotracheal route, drug absorption is poor and unpredictable, and optimal drug doses for this route of administration are unknown. In fact, the lower blood concentrations of epinephrine resulting from endotracheal administration could produce undesirable vasodilatory β-2 adrenergic effects.

Intravenous high-dose epinephrine (HDE) is no longer recommended for routine use in pediatric resuscitation, because it is associated with decreased survival and neurological outcomes. Although HDE can increase the return of spontaneous cardiac rhythm, it can also increase postcardiac arrest myocardial oxygen consumption, myocardial dysfunction, and hemodynamic instability.

When shock-refractory VF is present, either amiodarone (5 mg/kg dose, to a maximum of 3 doses) or lidocaine (1 mg/kg dose) may be given as an antiarrhythmic. Because amiodarone can produce hypotension and arrhythmias, expert consultation is advised when the drug is considered for treatment of prearrest or postcardiac arrest arrhythmias.

Magnesium administration is indicated for treating *torsades de pointes* and when hypomagnesemia is documented or strongly suspected. However, it is no longer used routinely during resuscitation.

Termination of Resuscitation Efforts

There are no intra-arrest factors that reliably predict poor resuscitation outcomes. In the out-of-hospital setting, age less than 1 year, unwitnessed arrest, no bystander CPR, and nonshockable rhythm on EMS arrival are associated with poor outcome. In the in-hospital setting, age greater than 1 year and longer duration of cardiac arrest have been associated with poor outcome. However, no single factor predicts outcome, and recent emphases on CPR quality, and improved postcardiac arrest care could all have a significant positive impact on CPR outcomes.

Postcardiac Arrest Care

Following ROSC, ischemic and reperfusion injuries can cause cardiorespiratory instability, perfusion abnormalities, and organ dysfunction. *Reperfusion injury* is characterized by calcium entry into cells, activation of inflammatory mediators, and cell death. This inflammatory response includes the development of endothelial injury, capillary leak, neutrophil activation, platelet aggregation, and increases in mediators such as tumor necrosis factor and interleukins. Increased production of free oxygen radicals and decreased production of nitric oxide can cause vasoconstriction and further ischemic injury. Hyperglycemia is common in children after an arrest, and both extreme hyperglycemia and hypoglycemia have been linked with increased mortality in children.

Skilled postcardiac arrest care in adults has more than doubled neurologically intact survival to hospital discharge. Such care requires evidence-based and protocol-driven bundled care, including targeted temperature management and support of organ system function.

Following ROSC, rescuers must begin targeted temperature management, optimize hemodynamic support, titrate inspired oxygen administration, and support of ventilation to maintain normoxemia (unless cyanotic heart disease is present) and partial pressure of carbon dioxide ($PaCO_2$) appropriate for that patient, and support end-organ function. The use of protocols will support consistent care that can then be evaluated and modified as needed.

Children often demonstrate a brief period of spontaneous hypothermia followed by the development of hyperthermia after cardiac arrest. While hyperthermia is associated with worse neurological outcome, the recent Therapeutic Hypothermia to Improve Survival After Cardiac Arrest in Pediatric